

GUIDELINES FOR THE USE OF ANTIRETROVIRAL AGENTS IN ADULTS AND ADOLESCENTS WITH HIV



Developed by the HHS Panel on Antiretroviral Guidelines for
Adults and Adolescents—A Working Group of the NIH
Office of AIDS Research Advisory Council (OARAC)

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It is emphasized that concepts relevant to HIV management evolve rapidly. The Panels have a mechanism to update recommendations on a regular basis, and the most recent information is available on the Clinicalinfo website (<https://clinicalinfo.hiv.gov/>).

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WHAT'S NEW IN THE GUIDELINES

Updated: September 25, 2025

Reviewed: September 25, 2025

These guidelines were updated by the U.S. Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) based on key new clinical evidence on the use of antiretroviral therapy (ART) for the treatment of people with HIV.

New Chapter Added to the Guidelines

Cardiovascular and Metabolic Complications in People With HIV

With the success of ART, the life expectancy of people with HIV is approaching that of people without HIV. As a result, many people with HIV are experiencing age-related comorbidities. This new chapter aims to provide guidance on the prevention and management of cardiovascular disease and metabolic complications in people with HIV and includes the following sections:

- [Introduction](#)
- [Immune Activation and Inflammation Among People With HIV Receiving Antiretroviral Therapy](#)
- [Cardiovascular Complications in People With HIV](#)
- [Statin Therapy as Primary Prevention of Atherosclerotic Disease in People With HIV](#)
- [Weight Gain in People With Treated HIV](#)

Key Revisions to the Guidelines

Laboratory Testing for Initial Assessment and Monitoring of People With HIV Receiving Antiretroviral Therapy

Updates made to the [Laboratory Testing](#) section include the following:

- Updated recommended monitoring frequencies for CD4 T lymphocyte (CD4) cell count, basic metabolic panel, liver enzymes, and lipid profile (see [Table 3](#)).
- A new table ([Table 4. Laboratory Monitoring If ART Is Deferred](#)) provides guidance on initial laboratory assessments and recommended follow-up frequency if ART is deferred.

Plasma HIV-1 RNA (Viral Load) and CD4 Count Monitoring

Updates made to the [Plasma HIV-1 RNA \(Viral Load\) and CD4 Count Monitoring](#) section include the following:

- The Panel previously considered repeat HIV-1 RNA monitoring optional when ART was deferred. To support ongoing discussions about the benefits of ART, for both HIV treatment and the prevention of transmission to others, the Panel now recommends testing every 3 to 6 months when ART is deferred.
- Updated the rationale for the changes in recommended CD4 count monitoring frequency, as noted in [Table 3](#)

Initiation of Antiretroviral Therapy

Two new subsections have been added to the [Initiation of Antiretroviral Therapy](#) section:

- **Early (Acute/Recent) HIV Infection:** This subsection focuses on the importance of initiating ART as soon as possible, as ART can provide immunologic benefits for people with early HIV infection. Additionally, early HIV is a period of high infectivity, and ART substantially reduces the risk of transmission.
- **Antiretroviral Therapy Initiation in Hospitalized People With HIV:** This subsection discusses the benefits of initiating ART in people who receive an HIV diagnosis during hospitalization, while acknowledging that this may not be possible in all settings. Based on the discussion, the Panel recommends initiating ART during hospitalization whenever possible **(BIII)**. The Panel notes that if ART is initiated before hospital discharge, clinicians should ensure sufficient ART supply is available until the outpatient clinic appointment.

The Elite HIV Controllers subsection was updated with two recommendations based on discussion of new observational studies regarding clinical outcomes of this population:

- The Panel strongly recommends ART for elite controllers with evidence of HIV-related complications, declining CD4 counts, intermittent detectable viral loads, or comorbidities (e.g., cardiovascular disease, cancer, hepatitis B virus/hepatitis C virus coinfection) or for those who are pregnant **(AIII)**.
- The Panel also recommends initiation of ART for all other HIV elite controllers **(BII)**.

Suboptimal CD4 Cell Recovery Despite Viral Suppression

This section—previously titled “Poor CD4 Cell Recovery and Persistent Inflammation Despite Viral Suppression”—has been renamed following the relocation of content on persistent inflammation to the new [Immune Activation and Inflammation](#) section. Additional updates to the [Suboptimal CD4 Cell Recovery](#) section include the following:

- Updated evidence on the clinical consequences of suboptimal CD4 recovery, including the increased risk of AIDS and non-AIDS events.
- The Panel notes that promptly initiating ART in people diagnosed early with HIV provides the best opportunity for maximal CD4 recovery.
- The Panel emphasizes that to date, there is no effective therapeutic intervention to improve CD4 count in people with suboptimal CD4 recovery.
- The Panel notes that efforts to decrease morbidity and mortality should focus on preventive care, addressing modifiable risk factors for chronic disease, and optimizing management of comorbidities.

Cost Considerations and Antiretroviral Therapy

Several new subsections have been added to the [Cost Considerations and Antiretroviral Therapy](#) section:

- Health Care Coverage to Maximize Access to Antiretroviral Therapy
- Medicare Beneficiaries With HIV
- Long-Acting Injectables

- 340B Drug Pricing Program, Prescription Drug Affordability, and Program Savings for People With HIV

Table 22a (Insurance and Health Program Prescription Drug Pricing and Access) and Table 22b (Monthly Average Prices of Commonly Used Antiretroviral Drugs) have been updated with the most recent information.

Other Updates

Minor updates have been made to the following sections of the guidelines:

- [Antiretroviral Therapy to Prevent Sexual Transmission of HIV \(Treatment as Prevention\)](#)
- [Adverse Effects of Antiretroviral Medications](#)
- [Appendix A: Drug Characteristics Tables](#)
- [Appendix B: Antiretroviral Dosing Recommendations in Adults With Renal or Hepatic Insufficiency](#)

PANEL ROSTER

Updated: September 25, 2025

Reviewed: September 25, 2025

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These guidelines were developed by the U.S. Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents (a working group of the National Institutes of Health Office of AIDS Research Advisory Council).

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FINANCIAL DISCLOSURE

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U.S. Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents: Financial Disclosure for Companies Related to HIV Treatment or Diagnostics for the Period of April 1, 2024, to March 31, 2025

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Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents: Financial Disclosure for Companies Related to HIV Treatment or Diagnostics for the Period of April 1, 2024, to March 31, 2025

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Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents: Financial Disclosure for Companies Related to HIV Treatment or Diagnostics for the Period of April 1, 2024, to March 31, 2025

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Virginia Sheikh	M	None	N/A
Serena Spudich	M	None	N/A

Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents: Financial Disclosure for Companies Related to HIV Treatment or Diagnostics for the Period of April 1, 2024, to March 31, 2025

Panel Member	Role	Financial Disclosure	
		Company	Relationship
Kimberly Struble	M	None	N/A
Virginia Triant	M	None	N/A
Netanya Utay	M	Gilead Sciences, Inc.	Grant Evaluator
Steven Vargas	M	Merck & Co., Inc.	Honoraria

Key: AC = Acting Co-chair; C = Co-chair; **CC = Community Consultant**; ES = Executive Secretary; M = Member; N/A = not applicable; PC = Pharmacology Consultant

INTRODUCTION

Updated: September 21, 2022

Reviewed: September 21, 2022

Antiretroviral therapy (ART) for the treatment of HIV infection has improved steadily since the advent of potent combination therapy in 1996. ART has dramatically reduced HIV-associated morbidity and mortality and has transformed HIV infection into a manageable chronic condition, with life expectancy approaching that for people without HIV.^{1,2} ART is also highly effective at preventing sexual transmission of HIV in patients who have adequately suppressed viral loads.³⁻⁵ Lack of viral load suppression is mostly due to undiagnosed HIV infection and failure to link or retain patients with HIV in care.

The U.S. Department of Health and Human Services (HHS) Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) is a working group of the Office of AIDS Research Advisory Council (OARAC). The Panel's primary goal is to provide HIV care practitioners with recommendations that are based on current knowledge of the antiretroviral (ARV) drugs that are used to treat adults and adolescents with HIV in the United States. The Panel reviews new evidence and updates recommendations when needed. These guidelines include recommendations on baseline laboratory evaluations, treatment goals, benefits of ART and considerations when initiating therapy, choice of the initial regimen for ART-naïve people with HIV, ARV drugs or combinations to avoid, management of treatment failure, optimizing ARV regimens, management of adverse effects and drug interactions, and special ART-related considerations in specific populations. This Panel works closely with the HHS Panel on Antiretroviral Therapy and Medical Management of Children Living with HIV to provide recommendations for adolescents at different stages of growth and development. Recommendations for ARV regimens in these guidelines are most appropriate for postpubertal adolescents (i.e., those with [sexual maturity ratings](#) [SMR] of 4 and 5). Clinicians should follow recommendations in the [Pediatric Antiretroviral Guidelines](#) when initiating ART in adolescents with an SMR of 3 or lower. For recommendations related to pre-exposure prophylaxis (PrEP) and post-exposure prophylaxis (PEP) for people who do not have HIV, clinicians should consult recommendations from the Centers for Disease Control and Prevention.⁶

These guidelines represent current knowledge regarding the use of ARV drugs. Because the science of HIV evolves rapidly, the availability of new agents and new clinical data may change therapeutic options and preferences. Information included in these guidelines may not always be consistent with approved labeling for the specific drugs or indications, and the use of the terms “safe” and “effective” may not be synonymous with the U.S. Food and Drug Administration–defined legal standards for drug approval. The Panel frequently updates the guidelines (current and archived versions of the guidelines are available on the [Clinical Info](#) website). However, updates to the guidelines may not keep pace with the release of new data, and the guidelines cannot offer guidance on care for all patients. Patient management decisions should be based on clinical judgement and attention to unique patient circumstances.

The Panel recognizes the importance of clinical research in generating evidence to address unanswered questions related to the optimal safety and efficacy of ART and encourages both the development of protocols and patient participation in well-designed Institutional Review Board–approved clinical trials.

HIV Expertise in Clinical Care

Several studies have demonstrated that overall outcomes in patients with HIV are better when care is delivered by clinicians with HIV expertise (e.g., those who have cared for a large group of patients

with HIV),⁷⁻¹¹ reflecting the complexity of HIV transmission and its treatment. Appropriate training, continuing education, and clinical experience are all components of optimal care. Providers who do not have this requisite training and experience should consult HIV experts when needed.

Guidelines Development Process

Table 1. Outline of the Guidelines Development Process

Topic	Comment
Goal of the guidelines	Provide guidance to HIV care practitioners on the optimal use of antiretroviral (ARV) agents for the treatment of HIV in adults and adolescents in the United States.
Panel members	The Panel is composed of approximately 50 voting members who have expertise in HIV care and research and includes at least one representative from each of the following U.S. Department of Health and Human Services (HHS) agencies: Centers for Disease Control and Prevention (CDC), U.S. Food and Drug Administration (FDA), Health Resource and Services Administration (HRSA), and National Institutes of Health (NIH). Approximately two-thirds of the Panel members are nongovernmental scientific members. The Panel also includes four to five community members with knowledge of HIV treatment and care. The U.S. government representatives are appointed by their respective agencies; other Panel members are selected after an open call for nominations. Each member serves on the Panel for a 4-year term, with an option for reappointment for an additional term. See the Panel Roster for a list of current Panel members.
Financial disclosure	All members of the Panel submit a written financial disclosure annually, reporting any association with manufacturers of ARV drugs or diagnostics used to manage HIV infection. The latest version of the Financial Disclosure list is available on the Clinical Info website.
Users of the guidelines	HIV treatment providers
Developer	Panel on Antiretroviral Guidelines for Adults and Adolescents—a working group of the Office of AIDS Research Advisory Council (OARAC)
Funding source	Office of AIDS Research, NIH
Evidence collection	The recommendations in the guidelines are based on studies published in peer-reviewed journals or data available in FDA drug labels. On some occasions, particularly when new information may affect patient safety, unpublished data presented at major conferences or prepared by the FDA and/or manufacturers as warnings to the public may be used as evidence to revise the guidelines.
Recommendation grading	As described in Table 2 below
Method of synthesizing data	Each section of the guidelines is assigned to a working group of Panel members with expertise in the section's area of interest. The working groups synthesize available data and propose recommendations to the Panel. The Panel discusses all proposals during monthly teleconferences. Recommendations endorsed by the Panel are included in the guidelines.

Table 1. Outline of the Guidelines Development Process

Topic	Comment
Other guidelines	These guidelines focus on antiretroviral therapy (ART) for adults and adolescents with HIV. For a more detailed discussion on the use of ART in children and prepubertal adolescents (those with sexual maturity ratings of 1 to 3), clinicians should refer to the Pediatric Antiretroviral Guidelines. These guidelines also include a brief discussion on the management of women of childbearing potential and pregnant women. For more details on the use of ARV drugs during pregnancy, see the Perinatal Guidelines.
Update plan	The Panel meets monthly by teleconference to review data that may warrant modification of the guidelines. Updates may be prompted by new drug approvals (or new indications, dosing formulations, or frequency of dosing), new safety or efficacy data, or other information relating to ARV drugs that may have an impact on the clinical care of people with HIV. In the event of new data of clinical importance, the Panel may post an interim announcement with recommendations on the Clinical Info website until the guidelines can be updated with the appropriate changes. Updated guidelines are available on the Clinical Info website.
Public comments	A 2-week public comment period follows the release of the updated guidelines on the Clinical Info website. The Panel reviews comments to determine whether additional revisions to the guidelines are indicated. The public also may submit comments to the Panel at any time at HIVinfo@NIH.gov .

Basis for Recommendations

Recommendations in these guidelines are based on scientific evidence and expert opinion. Each recommendation statement includes a letter (**A**, **B**, or **C**) that represents the strength of the recommendation and a Roman numeral (**I**, **II**, or **III**) that represents the quality of the evidence that supports the recommendation (see Table 2 below).

Table 2. Rating Scheme for Recommendations

Strength of Recommendation	Quality of Evidence for Recommendation
A: Strong recommendation for the statement B: Moderate recommendation for the statement C: Weak recommendation for the statement	I: One or more randomized trials with clinical outcomes and/or validated laboratory endpoints II: One or more well-designed, nonrandomized trials or observational cohort studies with long-term clinical outcomes III: Expert opinion

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BASELINE EVALUATION

Updated: September 21, 2022

Reviewed: September 21, 2022

Every patient with HIV entering into care should have a complete medical history, physical examination, and laboratory evaluation and should be counseled regarding the implications of HIV infection. The goals of this initial evaluation are to confirm the diagnosis of HIV infection, obtain appropriate baseline historical and laboratory data, ensure patient understanding about HIV infection and its transmission, and initiate care as recommended in the HIV Medicine Association of the Infectious Diseases Society of America's (HIVMA/IDSA) [Primary Care Guidance for Persons with HIV](#)¹ and the [Adult and Adolescent Opportunistic Infections Guidelines](#).² The initial evaluation also should include discussion of the benefits of antiretroviral therapy (ART) for the patient's health and to prevent HIV transmission, as well as strategies to optimize care engagement and treatment adherence **(AIII)**. Information obtained in this baseline evaluation then can be used to define treatment management goals and plans. The Panel on Antiretroviral Guidelines for Adults and Adolescents recommends initiating ART at the time of diagnosis (when possible) or as soon as possible afterward to increase the uptake of ART, decrease the time required to achieve linkage to care and virologic suppression, improve the rate of virologic suppression, and reduce HIV transmission **(AII)**.

The following laboratory tests performed during initial patient visits can be used to stage HIV progression and to assist in the selection of antiretroviral (ARV) drug regimens:

- HIV antigen/antibody testing (if prior documentation is not available or if HIV RNA is below the assay's limit of detection) **(AI)**
- CD4 T lymphocyte (CD4) cell count **(AI)**
- Plasma HIV RNA (viral load) **(AI)**
- Complete blood count; chemistry profile, including glucose, blood urea nitrogen and creatinine, liver enzymes and bilirubin, urinalysis, and serologies for hepatitis A, B, and C viruses **(AIII)**
 - If random blood glucose level is abnormal, repeat fasting
- Serum lipids (if random levels are abnormal, fasting lipids should be obtained)
- HLA-B*5701 test (if abacavir is being considered) **(AI)**
- Genotypic drug-resistance testing **(AII)**. Based on current rates of transmitted drug resistance to different ARV medications, standard genotypic drug-resistance testing in ARV-naïve people should focus on testing for mutations in the reverse transcriptase (RT) and protease (PR) genes. If transmitted integrase strand transfer inhibitor (INSTI) resistance is a concern in people with newly diagnosed HIV or in people who acquired HIV after receipt of long-acting cabotegravir (CAB-LA) as pre-exposure prophylaxis (PrEP), testing for mutations in the integrase gene also should be performed.
- For patients who have HIV RNA levels <1,000 copies/mL, viral amplification for drug-resistance testing should still be performed; however, it may not always be successful **(BII)** (see [Drug-Resistance Testing](#)).

In addition, other tests (including screening tests for sexually transmitted infections, opportunistic infections, and cancer) should be performed as recommended in the HIVMA/IDSA's [Primary Care Guidance for Persons with HIV](#)¹ and the [Adult and Adolescent Opportunistic Infections Guidelines](#).²

Many clinics have adopted a rapid start policy to initiate ART on the day of HIV diagnosis in order to increase ART uptake and engagement in care and to accelerate the time to viral suppression. Rapid ART initiation also reduces the time during which people with newly diagnosed HIV can transmit HIV. Prior to ART initiation, HIV infection should be confirmed. HIV RNA and CD4 count also should be obtained, but results need not be available before starting ART. CD4 count will determine the need for prophylaxis for certain opportunistic infections.

- If available, results of safety testing—such as complete blood count, renal function tests, and liver enzymes—should be reviewed. If safety test results are not available, ART can still be started, but a clinician should review the results as soon as possible.
- Genotypic resistance testing for RT and PR (and INSTI resistance testing if patient has a history of CAB PrEP use or if INSTI transmission is suspected) should be obtained before ART initiation. It is not necessary to delay ART until results are available, but results should be reviewed as soon as possible in order to make adjustments to the regimen, if needed.
- Screening for viral hepatitis should be done before starting ART, and if ART initiation occurs before results are available, a regimen that has activity against hepatitis B virus should be selected.
- In patients who do not have reliable methods of contact, rapid ART may be initiated, with a plan for a return clinic visit soon after ART initiation to review test results.
- Screening for sexually transmitted infections should, ideally, occur at the initial visit, but results do not need to be available before starting ART.

For previously treated patients who present for an initial evaluation with a new health care provider, it is critical to obtain a complete ARV history (including drug-resistance testing results, if available), preferably through the review of past medical records. A complete immunization history (including for SARS-CoV-2) also should be obtained. Newly diagnosed patients also should be asked about any prior use of ARV agents for prevention of HIV infection.

People with HIV often must cope with many social, psychiatric, and medical issues that are best addressed through a patient-centered, multidisciplinary approach. The baseline evaluation should include consideration of the patient's readiness for ART, including an assessment of substance use (including tobacco use), social support, mental health, medical comorbidities, economic factors (e.g., unstable housing, food instability), medical insurance status and adequacy of coverage, and other factors that are known to impair adherence to ART and increase the risk of HIV transmission. Once evaluated, these factors should be managed accordingly. The baseline evaluation also should include a discussion of risk reduction and disclosure to sexual and/or needle-sharing partners, especially with untreated patients who are still at high risk of HIV transmission. People with HIV should be informed that maintaining a plasma HIV RNA of <200 copies/mL, including any measurable value below this threshold, with ART prevents sexual transmission of HIV to their partners (AII). Patients may recognize this concept as Undetectable = Untransmittable or U=U.

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LABORATORY TESTING

Laboratory Testing for Initial Assessment and Monitoring of People With HIV

Updated: September 25, 2025

Reviewed: September 25, 2025

Several laboratory tests are important for initial evaluation of people with HIV upon entry into care. Some tests should be performed before and after initiation or modification of antiretroviral therapy (ART) to assess the virologic and immunologic efficacy of ART and to monitor for laboratory abnormalities that may be associated with antiretroviral (ARV) drugs.

Two surrogate markers have been used to monitor people with HIV: (1) plasma HIV RNA (viral load) to assess the level of HIV viremia and (2) CD4 T lymphocyte (CD4) cell count to assess immune function. Standard (reverse transcriptase and protease) genotypic drug-resistance testing should be used to guide selection of an ARV regimen; if integrase strand transfer inhibitor (INSTI) resistance is a concern due to potential transmitted resistance or previous use of an INSTI for pre-exposure prophylaxis (e.g., long-acting cabotegravir), post-exposure prophylaxis, or treatment, **providers should ensure that genotypic resistance testing also includes the integrase gene** (see [Drug-Resistance Testing](#)). For guidance on the choice of ARV regimens before drug-resistance testing results become available, clinicians should consult the [What to Start](#) and [Early \(Acute or Recent\) HIV Infection](#) sections. A viral tropism assay should be performed before initiation of a CCR5 antagonist or at the time of virologic failure if a person is receiving a CCR5 antagonist. HLA-B*5701 testing should be performed before initiation of abacavir (ABC) to reduce the risk of hypersensitivity reaction, and HLA-B*5701-positive people should not be prescribed ABC. People should be screened for hepatitis B virus (HBV) and hepatitis C virus infections before initiating ART and, if indicated, periodically after ART initiation, because treatment of these coinfections may affect the choice of ART and likelihood of drug-induced hepatotoxicity. **Before initiating nucleoside reverse transcriptase inhibitor (NRTI)–sparing or NRTI-limited ARV regimens, HBV serology testing should be performed in people who are not known to have chronic HBV** (see [Hepatitis B Virus/HIV Coinfection](#)). The rationale for and utility of some of these laboratory tests are discussed in the corresponding sections of the guidelines.

Currently available newer ARVs are generally easy to take and typically do not cause significant laboratory abnormalities. Most people with HIV achieve and maintain viral suppression and have improvement in CD4 count within 1 year after starting ART. [Table 3](#) provides general guidance on the frequency of laboratory monitoring before initiation of and throughout treatment with ART. However, more frequent monitoring of selected tests may be needed as clinically indicated, based on issues such as responses to therapy, underlying medical conditions, drug-related adverse effects, engagement in care, and adherence to therapy. ART is recommended for all people with HIV (AI); [Table 4](#) provides guidance for laboratory monitoring in the rare occasions when ART initiation is deferred.

Table 3. Laboratory Testing Schedule Before and After Antiretroviral Therapy Initiation^a

This table provides general guidance for laboratory monitoring for people with HIV before and after antiretroviral therapy (ART) initiation. Clinicians should use their clinical judgment to determine the monitoring frequency for an individual person based on clinical needs. This schedule is not intended to be a guide on how often people with HIV should undergo clinical evaluation, as some individuals may need to be seen more frequently to accomplish other aspects of clinical care. Please refer to [Table 4](#) for laboratory monitoring guidance if ART initiation is deferred.

Laboratory Test	Timepoint or Frequency of Testing							
	Entry Into Care	ART Initiation ^b or Modification	After ART Initiation or Modification	Every 3–4 Months	Every 6 Months	Every 12 Months	Treatment Failure	Clinically Indicated
HIV Antigen/Antibody Test	✓ If HIV diagnosis not confirmed							
HIV Viral Load	✓	✓	✓ ^c At 4–8 weeks	✓ ^d	✓ ^d		✓	✓
CD4 Count	✓	✓	✓ 3 months (after ART initiation only)	✓ ^e If CD4 count is <300 cells/mm ³	✓ During the First 1–2 Years on ART and With Viral Suppression, if CD4 Count Is ≥300 Cells/mm ³ : • Every 6 months After 1–2 Years on ART With Consistently Suppressed Viral Load and CD4 Count ≥300 Cells/mm ³ : • Optional unless clinically indicated		✓ Monitor CD4 count every 3–6 months.	✓

Table 3. Laboratory Testing Schedule Before and After Antiretroviral Therapy Initiation

Laboratory Test	Timepoint or Frequency of Testing							
	Entry Into Care	ART Initiation ^b or Modification	After ART Initiation or Modification	Every 3–4 Months	Every 6 Months	Every 12 Months	Treatment Failure	Clinically Indicated
Genotypic Resistance Testing (PR/RT Genes) ^f	✓	✓					✓	✓
Genotypic Resistance Testing (Integrase Genes) ^f	✓ If transmitted INSTI resistance is suspected or if there is a history of CAB-LA use for PrEP or INSTI use for PEP	✓ If transmitted INSTI resistance is suspected or if there is a history of INSTI use					✓ If there is a history of INSTI use for treatment or prevention	✓
Tropism Testing		✓ If considering a CCR5 antagonist					✓ If considering a CCR5 antagonist, or upon virologic failure on a CCR5 antagonist	✓

Table 3. Laboratory Testing Schedule Before and After Antiretroviral Therapy Initiation

Laboratory Test	Timepoint or Frequency of Testing							
	Entry Into Care	ART Initiation ^b or Modification	After ART Initiation or Modification	Every 3–4 Months	Every 6 Months	Every 12 Months	Treatment Failure	Clinically Indicated
HLA-B*5701 Testing		✓ If considering ABC (perform once; keep result in health record)						
Hepatitis B Serology (HBsAb, HBsAg, HBcAb total) ^{g,h,i} Also see Hepatitis B Virus/HIV Coinfection	✓	✓ In people not immune to HBV, repeat testing if switching to a regimen that does not contain TAF or TDF.						✓ Before starting HCV DAA In people not immune to HBV and at high risk for HBV, ⁱ periodic testing may be considered.
Hepatitis C Screening (HCV antibody or, if indicated, HCV RNA) ^k	✓					✓ Repeat HCV screening for at-risk people. ^j		✓

Table 3. Laboratory Testing Schedule Before and After Antiretroviral Therapy Initiation

Laboratory Test	Timepoint or Frequency of Testing							
	Entry Into Care	ART Initiation ^b or Modification	After ART Initiation or Modification	Every 3–4 Months	Every 6 Months	Every 12 Months	Treatment Failure	Clinically Indicated
Basic Metabolic Panel ^{l,m}	✓	✓	✓ At 4–8 weeks For people with preexisting conditions or at risk of laboratory changes after ART initiation		✓ Every 6–12 months			✓
ALT, AST, Total Bilirubin	✓	✓	✓ At 4–8 weeks For people with preexisting conditions, at risk for laboratory changes after ART initiation, or with HBV coinfection		✓ Every 6–12 months			✓
CBC With Differential ⁿ	✓	✓		✓ When monitoring CD4 count (if required by lab)	✓ When monitoring CD4 count (if required by lab)	✓ When no longer monitoring CD4 count		✓

Table 3. Laboratory Testing Schedule Before and After Antiretroviral Therapy Initiation

Laboratory Test	Timepoint or Frequency of Testing							
	Entry Into Care	ART Initiation ^b or Modification	After ART Initiation or Modification	Every 3–4 Months	Every 6 Months	Every 12 Months	Treatment Failure	Clinically Indicated
Lipid Profile ^o	✓	✓	At 3–6 months once viral suppression is reached			✓ If aged ≥40 years or on a statin (Every 1–3 years if aged <40 years and not on a statin)		✓ If there are changes in CV risk factors ^p
Random or Fasting Glucose ^q	✓	✓						✓
Urinalysis ^{m,r}	✓							✓ e.g., in people with CKD or DM
Pregnancy Test ^s	✓	✓						✓

^a This table pertains to laboratory tests done to select an ARV regimen and monitor for treatment responses or ART toxicities. Please refer to the HIV Medicine Association of the Infectious Diseases Society of America's (HIVMA/IDSA) [Primary Care Guidance for Providers Who Care for Persons With HIV](#) for other laboratory tests generally recommended for primary health care maintenance of HIV people.¹

^b If ART is initiated soon after HIV diagnosis and entry into care, repeat baseline laboratory testing is not necessary.

^c If HIV RNA is detectable at 4–8 weeks, repeat testing every 4–8 weeks until viral load is suppressed to <50 copies/mL. Thereafter, repeat testing every 3–6 months.

^d For people on ART, viral load is typically measured every 3–6 months. More frequent monitoring may be considered in individuals having difficulties with ART adherence or at risk for nonadherence. However, for adherent people with consistently suppressed viral load and stable immunologic status for more than 1 year, monitoring can be extended to 6-month intervals.

^e After 1–2 years of consistently suppressed HIV RNA, less frequent monitoring (e.g., every 6 months) may be considered.

^f Standard genotypic drug-resistance testing in ARV-naïve persons should focus on testing for mutations in the PR and RT genes. If transmitted INSTI resistance is a concern, or if a person may have a history of INSTI use as PrEP, PEP, or treatment, or a person presents with viremia while on an INSTI, providers also should

Table 3. Laboratory Testing Schedule Before and After Antiretroviral Therapy Initiation

test for resistance mutations in the integrase gene. In people who are ARV-naïve and who do not immediately begin ART, repeat testing before initiation of ART is optional if drug-resistance testing was performed at entry into care. In people with virologic suppression who are switching therapy because of toxicity or for convenience, viral amplification will not be possible; see the [Drug-Resistance Testing](#) section for a discussion of the potential limitations and benefits of proviral DNA assays in this situation. Results from prior drug-resistance testing should be used in constructing a new regimen.

^g If a person has HBV infection (as determined by a positive HBsAg or HBV DNA test result), TDF or TAF plus either FTC or 3TC should be used as part of the ARV regimen to treat both HBV and HIV infections (see the [Hepatitis B Virus/HIV Coinfection](#) section).

^h If HBsAg, HBsAb, and HBcAb test results are negative, HBV vaccine series should be administered. Refer to the HIVMA/IDSA's [Primary Care Guidance for Persons With HIV](#) and the [Adult and Adolescent Opportunistic Infection Guidelines](#) for detailed recommendations.^{1,2}

ⁱ Most people with isolated HBcAb have resolved HBV infection with loss of HBsAb. Consider performing an HBV viral load test for confirmation. If the HBV viral load test is positive, the person may be acutely infected (and will usually display other signs of acute hepatitis) or chronically infected. If the test is negative, the person should be vaccinated. Refer to the HIVMA/IDSA's [Primary Care Guidance for Persons With HIV](#) and the [Adult and Adolescent Opportunistic Infection Guidelines](#) for more detailed recommendations.²

^j Injection drug users, people with a history of incarceration, men with HIV who have unprotected sex with men, and people with percutaneous/parenteral exposure to blood in unregulated settings are at risk of **HBV or** HCV infection.

^k The HCV antibody test may not be adequate for screening in the setting of recent HCV infection (acquisition within the past 6 months) or advanced immunodeficiency (CD4 count <100 cells/mm³). HCV RNA screening is indicated in persons who have been successfully treated for HCV or who spontaneously cleared prior infection. HCV antibody-negative people with elevated ALT may need HCV RNA testing.

^l Serum Na, K, HCO₃, Cl, BUN, creatinine, glucose, and Cr-based eGFR. Serum P should be monitored in people with CKD who are on TDF-containing regimens.³

^m Consult the HIVMA/IDSA's [Clinical Practice Guideline for the Management of Chronic Kidney Disease in People Infected With HIV](#) for recommendations on managing people with renal disease.³ More frequent monitoring may be indicated for people with evidence of kidney disease (e.g., proteinuria, decreased glomerular dysfunction) or increased risk of renal insufficiency (e.g., people with diabetes, hypertension).

ⁿ CBC with differential should be done when a CD4 count is performed. When CD4 count is no longer being monitored, the recommended frequency of CBC with differential is once a year. More frequent monitoring may be indicated for people receiving medications that potentially cause cytopenia (e.g., TMP-SMX).

^o If random lipids are abnormal, fasting lipids should be obtained. Consult the American College of Cardiology/American Heart Association's 2018 Guideline on the Management of Blood Cholesterol for diagnosis and management of people with dyslipidemia.⁴

^p Changes in CV risk factors such as new diagnosis of hypertension, weight gain, or new medications (including ART) that may raise lipid levels or overall CV risk.

^q If random glucose is abnormal, fasting glucose should be obtained. HbA1C is no longer recommended for diagnosis of diabetes in people with HIV on ART (see the [American Diabetes Association Guidelines](#)).⁵

^r Urine glucose and protein should be assessed before initiating TAF- or TDF-containing regimens and monitored during treatment with these regimens.

^s For women of childbearing potential.

Key: 3TC = lamivudine; ABC = abacavir; ALT = alanine aminotransferase; ART = antiretroviral therapy; ARV = antiretroviral; AST = aspartate aminotransferase; BUN = blood urea nitrogen; CAB-LA = cabotegravir long-acting; CBC = complete blood count; CD4 = CD4 T lymphocyte; CKD = chronic kidney disease; Cl = chloride; Cr = creatinine; CV = cardiovascular; DAA = direct-acting antiviral; DM = diabetes mellitus; eGFR = estimated glomerular filtration rate; FTC = emtricitabine; HbA1C = hemoglobin A1c; HBcAb = hepatitis B core antibody; HBsAb = hepatitis B surface antibody; HBsAg = hepatitis B surface antigen; HBV = hepatitis B virus; HCO₃ = bicarbonate; HCV = hepatitis C virus; INSTI = integrase strand transfer inhibitor; K = potassium; Na = sodium; P = phosphorus; **PEP = post-exposure prophylaxis**; PR = protease; PrEP = pre-exposure prophylaxis; RT = reverse transcriptase; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate; TMP-SMX = trimethoprim-sulfamethoxazole

Table 4. Laboratory Monitoring If ART Is Deferred

Antiretroviral therapy (ART) is recommended for all people with HIV **(AI)**. People with HIV should be advised to begin ART as soon as feasible to reduce morbidity and mortality and to prevent transmission of HIV to others. This table provides guidance for laboratory monitoring in the rare occasions when ART initiation is deferred.

Laboratory Test	Initial Assessment	Monitoring Frequency If ART Is Deferred	Comments
CD4 Count	✓	Every 3–6 months	To assess for urgency of ART initiation and the need for OI prophylaxis
HIV Viral Load	✓	Every 3–6 months	
Genotypic Resistance Testing (PR/RT +/- integrase gene) See Drug Resistance Testing for more information.	✓		Repeat genotypic resistance testing at the time of ART initiation is optional.
Hepatitis B Serology (HBsAb, HBsAg, HBcAb) See Hepatitis B in the Adult OI Guidelines for more detailed guidance.	✓	Clinically indicated	Before starting HCV DAA If HBsAg Positive: - ART with HBV-active drugs that include TAF or TDF should be initiated. If Not Immune at Baseline: - Repeat after vaccination or before starting an NRTI-sparing (or limited) ART regimen.
Hepatitis C Screening (HCV antibody, or, if indicated, HCV RNA)	✓	Every 12 months for at-risk people, or if clinically indicated	
Basic Metabolic Panel	✓	Every 6–12 months or if clinically indicated	
ALT, AST, Total Bilirubin	✓	Every 6–12 months or if clinically indicated	
CBC With Differential	✓	When monitoring CD4 count or if clinically indicated	
Pregnancy Test (when pregnancy is possible)	✓	If clinically indicated	ART should be initiated as soon as possible after confirming pregnancy to prevent perinatal HIV transmission.

Key: ALT = alanine aminotransferase; ART = antiretroviral therapy; AST = aspartate aminotransferase; CBC = complete blood count; CD4 = CD4 T lymphocyte; DAA = direct-acting antiviral; HBcAb = hepatitis B core antibody; HBsAb = hepatitis B surface antibody; HBsAg = hepatitis B surface antigen; HBV = hepatitis B virus; HCV = hepatitis C virus; NRTI = nucleoside reverse transcriptase inhibitor; OI = opportunistic infection; PR = protease; RT = reverse transcriptase; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate

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Plasma HIV-1 RNA (Viral Load) and CD4 Count Monitoring

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HIV RNA (viral load) and CD4 T lymphocyte (CD4) cell count are the two surrogate markers of antiretroviral therapy (ART) responses and HIV disease progression that have been used for decades to manage and monitor HIV infection.

Viral load is a marker of response to ART. A person's pre-ART viral load level and the magnitude of viral load decline after initiation of ART provide prognostic information about the probability of disease progression.¹⁻⁴ The key goal of ART is to achieve and maintain durable viral suppression. Thus, an important use of the viral load is to monitor the effectiveness of therapy after initiation of ART. The use of viral load testing to confirm the attainment and maintenance of viral suppression is instrumental in the operationalization of Undetectable = Untransmittable (or U=U)/Treatment as Prevention (see the [Antiretroviral Therapy to Prevent Sexual Transmission of HIV \[Treatment as Prevention\]](#) section).

It should be noted that commercially available HIV-1 RNA assays do not detect HIV-2 viral load. For further discussion on HIV-2 RNA monitoring in people with HIV-1/HIV-2 coinfection or HIV-2 mono-infection, see the [HIV-2 Infection](#) section.

CD4 count provides information on the overall immune function of a person with HIV. Measurement of CD4 count at entry into care and **before** initiation of ART is particularly useful in establishing the need to initiate opportunistic infection (OI) prophylaxis. Monitoring **after** initiation of ART is important for assessing immunologic response and establishing the need to discontinue OI prophylaxis.

The management of people with HIV has changed substantially with the availability of newer, more potent, **easier-to-use**, and less toxic antiretroviral (ARV) drugs. ART is recommended for all people with HIV regardless of their viral load or CD4 count (**AI**) (see the [Initiation of Antiretroviral Therapy](#) section). Because **almost all people** with HIV in care now receive ART and achieve good immunologic responses, the rationale for frequent CD4 count monitoring is weaker. The roles and usefulness of these two tests in clinical practice are discussed below.

Plasma HIV-1 RNA (Viral Load) Monitoring

Viral load is the most important indicator of initial and sustained response to ART and should be measured in all people with HIV at entry into care (**AI**), at initiation of therapy (if treatment is delayed) (**AI**), and on a regular basis thereafter. For those people in whom therapy is delayed, repeat viral load testing is recommended approximately every 3 to 6 months (**BIII**), aligning with CD4 count testing. Repeating viral load testing at these intervals provides an opportunity for discussions about the benefits of ART, both for the treatment of HIV and prevention of HIV transmission. Pre-treatment viral load level is also an important factor in the selection of an initial ARV regimen because some ARV drugs or regimens have been associated with poorer responses in people with high baseline viral load (see the [What to Start](#) section).

Several systematic reviews of data from clinical trials involving thousands of participants **in the early era of potent ART** have established that decreases in viral load following initiation of ART are associated with reduced risk of progression to AIDS or death.¹⁻⁴ Thus, viral load testing is an established surrogate marker for treatment response. The minimal change in viral load considered to be statistically significant (2 standard deviations) is a threefold change (equivalent to a 0.5 log₁₀ copies/mL change). Optimal viral suppression is defined as a confirmed HIV RNA level below the lower limit of detection of available assays (generally <20 copies/mL, depending on the assay used). After virologic suppression, an isolated detectable HIV RNA level that is followed by a return to virologic suppression, known as a "blip," may occur in successfully treated people and is not usually predictive of virologic failure.⁵ Furthermore, the data on

the association between persistently low-level but quantifiable viremia (HIV RNA <200 copies/mL) and virologic failure is conflicting. Several studies showed an increased risk of subsequent failure, **drug resistance, morbidity, and mortality** at this level of viremia⁶⁻¹²; however, the association was not observed in other studies.^{10,11,13-17} These guidelines now define virologic failure as the inability to achieve or maintain suppression of viral replication to HIV RNA levels of <200 copies/mL—a threshold that eliminates most cases of apparent viremia caused by viral load blips or assay variability¹⁸ (see the [Virologic Failure](#) section).

Individuals who are adherent to their ARV regimens and do not harbor resistance mutations to the component drugs can generally achieve viral suppression 8 to 12 weeks after ART initiation or after modification due to virologic failure; rarely, it may take longer in some people. Recommendations on the frequency of viral load monitoring are summarized below:

- **After initiation of ART:** Plasma viral load should be measured when ART is initiated **(AI)** and within 4 to 8 weeks after treatment initiation **(AIII)**. The purpose of these measurements is to confirm an adequate virologic response to ART, indicating appropriate regimen selection and individual adherence to therapy. Repeat viral load measurement should be performed every 4 to 8 weeks until the level falls below the assay's limit of detection **(BIII)**.
- **In people with viral suppression, with ART modification because of regimen simplification or drug toxicity:** Viral load measurement should be performed within 4 to 8 weeks after changing therapy **(AIII)**. The purpose of viral load monitoring at this point is to confirm the effectiveness of the new regimen.
- **In people on a stable, suppressive ARV regimen:** Viral load measurement should be repeated every 3 to 4 months **(AIII)** or as clinically indicated to confirm continuous viral suppression. Clinicians may extend the interval to 6 months for adherent people whose viral load has been suppressed for more than a year, whose clinical and immunologic status is stable, and who are not at risk for inadequate adherence **(AIII)**.
- **In people with virologic failure who require a change in ARV regimen:** Plasma viral load should be measured before ART change and within 4 to 8 weeks after treatment modification **(AIII)**. The purpose of the measurements is to confirm an adequate virologic response to the new regimen. Repeat viral load measurement should be performed every 4 to 8 weeks until the level falls below the assay's limit of detection **(BIII)**. If viral suppression is not possible, repeat viral load measurement every 3 months or more frequently if indicated **(AIII)**.
- **In people with suboptimal response:** The frequency of viral load monitoring will depend on clinical circumstances, such as adherence and availability of further treatment options. In addition to viral load monitoring, several other factors—such as individual adherence to prescribed medications, suboptimal drug exposure, or drug interactions—should be assessed. People who fail to achieve viral suppression should undergo drug-resistance testing to aid in the selection of an alternative ARV regimen (see the [Drug-Resistance Testing](#) and [Virologic Failure](#) sections).

CD4 Count Monitoring

The CD4 count is the most important laboratory indicator of immune function in people with HIV. **Low pre-treatment CD4 count** is the strongest predictor of disease progression and mortality according to findings from clinical trials and cohort studies.^{4,19} CD4 counts are highly variable; a significant change (2 standard deviations) between two tests is approximately a 30% change in the absolute count, or an increase or decrease in CD4 percentage by three percentage points. Monitoring of lymphocyte subsets other than CD4 (e.g., CD8, CD19) **is not recommended (BIII)** because it has not been proven to be clinically useful and is more expensive than monitoring CD4 count alone.

Use of CD4 Count for Initial Assessment

CD4 count should be measured in all people with HIV at entry into care **(AI)**. It is the key factor in determining the need to initiate OI prophylaxis (see the [Adult and Adolescent Opportunistic Infections Guidelines](#))²⁰ and the urgency to initiate ART **(AI)**, even though ART is recommended for all people with HIV, regardless of CD4 count **(AI)** (see the [Initiation of Antiretroviral Therapy](#) section). Although most OIs occur in people with CD4 counts <200 cells/mm³, some OIs can occur in people with higher CD4 counts.²¹

Use of CD4 Count for Monitoring Response to ART

The CD4 count is used to assess a person's immunologic response to ART. It is also used to determine whether prophylaxis for OIs can be discontinued (see the [Adult and Adolescent Opportunistic Infections Guidelines](#)).²⁰ For most people, an adequate response is defined as an increase in CD4 count in the range of 50 cells/mm³ to 150 cells/mm³ in the first year of ART, generally with an accelerated response in the first 3 months of treatment. Subsequent increases average approximately 50 cells/mm³ to 100 cells/mm³ per year until a steady state level is reached.²² People who initiate therapy with a low CD4 count^{23,24} or at an older age²⁵ may have a blunted increase in their counts despite virologic suppression.

Frequency of CD4 Count Monitoring

ART is recommended for all people with HIV. For people with HIV and pre-treatment CD4 count <300 cells/mm³, repeating CD4 count 3 months after ART initiation will provide information regarding the magnitude of immune reconstitution **(AIII)**. This repeat measurement is most important in people who initiate ART with more advanced disease and require OI prophylaxis or treatment. In these individuals, the magnitude and duration of CD4 count increase can be used to determine whether to discontinue OI prophylaxis and/or treatment as recommended in the [Adult and Adolescent Opportunistic Infections Guidelines](#). After ART initiation, CD4 count should be repeated every 3 to 4 months for the first 1 to 2 years of suppressive ART for those with CD4 counts <300 cells/mm³ **(BII)** and every 6 months if CD4 count is ≥300 cells/mm³ with viral suppression **(BIII)**. After 1 to 2 years of suppressive ART, reducing CD4 count monitoring to every 6 months may be considered for some people whose CD4 counts remain at <300 cells/mm³ **(BIII)**; for those with CD4 counts ≥300 cells/mm³, CD4 count monitoring is optional unless clinically indicated **(BIII)**.

The CD4 count response to ART varies widely, but a poor CD4 response in a person with viral suppression is rarely an indication for modifying an ARV regimen (see the [Suboptimal CD4 Recovery](#) section). In people with consistently suppressed viral loads who have already experienced ART-related immune reconstitution (i.e., CD4 count ≥300 cells/mm³), the CD4 count provides limited information. Frequent testing is unnecessary because the results rarely lead to a change in clinical management. One retrospective study found that declines in CD4 count to <200 cells/mm³ are rare in people with viral suppression and CD4 counts >300 cells/mm³.²⁶ Similarly, the ARTEMIS trial found that CD4 count monitoring had no clinical benefit in people who had suppressed viral loads and CD4 counts >200 cells/mm³ after 48 weeks of therapy.²⁷ Furthermore, the risk of *Pneumocystis jirovecii* pneumonia is extremely low in people on suppressive ART who have CD4 counts between 100 cells/mm³ and 200 cells/mm³.²⁸ Although uncommon, CD4 count declines can occur in a small percentage of virologically suppressed people and may be associated with adverse clinical outcomes, such as cardiovascular disease, malignancy, and death.^{29,30}

An analysis of costs associated with CD4 count monitoring in the United States estimated that reducing CD4 count monitoring in treated people from every 6 months to every 12 months could result in annual savings of approximately \$10 million.³¹

CD4 count should be monitored more frequently, as clinically indicated, when there are changes in a person's clinical status that may decrease CD4 count and thus prompt OI prophylaxis. Examples of such changes include the appearance of new HIV-associated clinical symptoms or initiation of treatment known

to reduce CD4 count (e.g., chronic corticosteroids, antineoplastic agents) **(AIII)**. In people who do not maintain viral suppression while on ART, CD4 count monitoring should be done every 3 to 6 months **(AIII)**.

In people who remain untreated for whatever reason, CD4 count should be monitored every 3 to 6 months to monitor for disease progression and assess the urgency of ART initiation and the need for OI prophylaxis **(AIII)**.

Factors That Affect Absolute CD4 Count

The absolute CD4 count is a calculated value based on the total white blood cell (WBC) count and the percentages of total and CD4 T lymphocytes. This absolute number may fluctuate in individuals or may be influenced by factors that may affect the total WBC count and lymphocyte percentages, such as use of bone marrow-suppressive medications, chronic corticosteroids, or the presence of acute infections. Splenectomy^{32,33} or coinfection with human T-lymphotropic virus type I³⁴ may cause misleadingly elevated CD4 counts. In all these settings, CD4 percentage remains stable and may be a more appropriate parameter to assess a person's immune function.³⁵

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Drug-Resistance Testing

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Panel's Recommendations

For Initial Treatment of HIV

- HIV drug-resistance testing is recommended at entry into care for people with HIV to guide the selection of the initial antiretroviral (ARV) regimen **(AII)**. If antiretroviral therapy (ART) is deferred, repeat testing may be considered at the time of ART initiation **(CIII)**.
- The Panel on Antiretroviral Guidelines for Adults and Adolescents recommends genotypic, rather than phenotypic, testing as the preferred resistance testing to guide therapy in ARV-naïve people **(AIII)**.
- In people with early (acute and recent) HIV infection, in pregnant women with HIV, or in people who will initiate ART on the day of or soon after HIV diagnosis, ART initiation should not be delayed while awaiting resistance testing results; the regimen can be modified once results are reported **(AIII)**.
- Standard genotypic drug-resistance testing in ARV-naïve people involves testing for mutations in the reverse transcriptase and protease genes. If transmitted integrase strand transfer inhibitor (INSTI) resistance is suspected, if the person has ever used long-acting cabotegravir (CAB-LA) as pre-exposure prophylaxis, **or if the person has ever received an INSTI-based regimen for post-exposure prophylaxis**, providers should ensure that genotypic resistance testing also includes the integrase gene **(AIII)**.

For Antiretroviral Therapy–Experienced People

- HIV drug-resistance testing should be performed to assist the selection of active drugs when changing ARV regimens in—
 - People with virologic failure and HIV RNA levels >200 copies/mL **(AI for >1,000 copies/mL, AIII for 501–1,000 copies/mL, CIII for confirmed HIV RNA 201–500 copies/mL)**. For people with confirmed HIV RNA levels >200 copies/mL but <500 copies/mL, drug-resistance testing may be unsuccessful but should still be considered.
 - People with suboptimal viral load reduction **(AII)**.
- Reverse transcriptase and protease genotypic resistance testing should be performed on people with virologic failure. Integrase resistance testing should be performed on individuals who have virologic failure and **have a history of prior use of an INSTI (for prevention or treatment) or are currently receiving an INSTI-based regimen (AII)**.
- For people taking a non–long-acting ARV regimen, drug-resistance testing in the setting of virologic failure should be performed while the person is still taking their ARV regimen or, if that is not possible, within 4 weeks after discontinuing their ARV regimen **(AII)**. If more than 4 weeks have elapsed since the non–long-acting agents have been discontinued, resistance testing may still provide useful information to guide therapy; however, it is important to recognize that previously selected resistance mutations can be missed due to lack of drug-selective pressure **(CIII)**.
- **For people who previously received long-acting cabotegravir and rilpivirine (LA CAB/RPV) and present with virologic failure, resistance testing (including INSTI genotypic testing) should be performed regardless of the time since the last dose of LA CAB/RPV (AIII).**
- Genotypic testing is preferred over phenotypic-resistance testing to guide therapy in people with suboptimal virologic response or virologic failure while on first- or second-line regimens and in people in whom resistance mutation patterns are known or not expected to be complex **(AII)**.
- The addition of phenotypic to genotypic resistance testing is recommended for people with known or suspected complex drug-resistance mutation patterns **(BIII)**.

- All prior and current drug-resistance test results, when available, should be reviewed and considered when constructing a new regimen **(AIII)**.
- When the use of a CCR5 antagonist is being considered, a co-receptor tropism assay should be performed **(AI)**.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Genotypic and Phenotypic-Resistance Assays

Genotypic and phenotypic-resistance assays are used to assess viral strains and select treatment strategies. These assays provide information on resistance to nucleoside reverse transcriptase inhibitors (NRTIs), non-nucleoside reverse transcriptase inhibitors (NNRTIs), protease inhibitors (PIs), and integrase strand transfer inhibitors (INSTIs). In some circumstances, INSTI-resistance tests may need to be ordered separately, and clinicians should check this with the testing laboratory. INSTI-resistance testing is particularly important in people who experience virologic failure while taking an INSTI-containing regimen or in those with prior use of injectable long-acting cabotegravir (CAB-LA) (either for treatment of HIV or as pre-exposure prophylaxis [PrEP]). Testing for fusion inhibitor resistance can be ordered separately when needed. There is currently no commercially available resistance test for the CD4 T lymphocyte post-attachment inhibitor ibalizumab, the gp120 attachment inhibitor fostemsavir, **or the capsid inhibitor lenacapavir**. For a description of co-receptor tropism testing, see [Co-Receptor Tropism Assays](#).

Genotypic Assays

Genotypic assays detect drug-resistance mutations in relevant viral genes; in general, these assays require a plasma viral load of at least 500 to 1,000 copies/mL. Most genotypic assays involve conventional Sanger sequencing of the reverse transcriptase (RT), protease (PR), and integrase (IN) genes of circulating RNA in plasma to detect mutations that are known to confer drug resistance. A genotypic assay that assesses mutations in the gp41 (envelope) gene associated with resistance to the fusion inhibitor enfuvirtide is also commercially available. Genotypic assays can be performed rapidly, and results are available within 1 to 2 weeks of sample collection. Interpreting these test results requires knowledge of the mutations selected by different antiretroviral (ARV) drugs and of the potential for cross-resistance to other drugs conferred by certain mutations. The [International AIDS Society–USA](#) maintains an updated list of significant resistance-associated mutations in the RT, PR, IN, and envelope genes. The [Stanford University HIV Drug Resistance Database](#) also provides helpful guidance for interpreting genotypic resistance test results.¹ Various additional tools are available to assist providers in interpreting genotypic test results.^{2–5} Clinical trials have demonstrated that consulting with specialists in HIV drug resistance improves virologic outcomes.⁶ Clinicians are thus encouraged to consult a specialist to interpret genotypic test results and design new, optimal ARV regimens.

A next-generation sequencing genotypic resistance assay that analyzes HIV-1 proviral DNA in host cells is now commercially available. This test aims to detect archived resistance mutations in people with HIV RNA below the limit of detection or with low-level viremia.

Phenotypic Assays

Phenotypic assays measure the ability of a virus to grow in different concentrations of ARV drugs. RT, PR, and, more recently, IN and envelope gene sequences derived from a person's plasma HIV RNA are inserted into the backbone of a laboratory clone of HIV or used to generate pseudotyped

viruses that express that person's HIV genes of interest. Replication of these viruses at different drug concentrations is monitored by expression of a reporter gene and is compared with the replication of a reference HIV strain. The drug concentration that inhibits viral replication by 50% (i.e., the median inhibitory concentration [IC₅₀]) is calculated, and the ratio of the IC₅₀ of test and reference viruses is reported as the fold increase in IC₅₀ (i.e., fold resistance).

Automated phenotypic assays that can produce results in 2 to 3 weeks are commercially available, but they cost more to perform than genotypic assays. In addition, interpreting phenotypic assay results can be complicated by incomplete information regarding the specific resistance level (i.e., fold increase in IC₅₀) associated with drug failure, although clinically significant fold increase cutoffs have been described for some drugs.⁷⁻¹¹ Again, consulting with a specialist to interpret test results can be helpful.

Limitations of Genotypic and Phenotypic Assays

Limitations of both genotypic and phenotypic assays include lack of uniform quality assurance testing for all available assays, relatively high cost, and insensitivity to minor viral species. Drug-resistant viruses that constitute <10% to 20% of the circulating virus population will probably not be detected by commercially available assays. This limitation is important to note because a wild-type virus often re-emerges as the predominant population in the plasma after discontinuation of drugs that exert selective pressure on drug-resistant populations. As a consequence, the proportion of virus with resistance mutations can decrease to below the 10% to 20% threshold.¹²⁻¹⁴ In the case of some oral ARV drugs, this reversion to predominantly wild-type virus can occur in the first 4 to 6 weeks after the drugs are discontinued. However, with injectable agents that have prolonged half-lives (e.g., long-acting cabotegravir and rilpivirine [LA CAB/RPV]), drug pressure may persist for prolonged periods. Prospective studies have demonstrated that despite plasma reversion, reinitiating the same ARV agents (or those with similar resistance pathways) often leads to early treatment failure; the virus responsible for this failure is derived from a previously archived resistant virus.¹⁵ Therefore, for people taking a non-long-acting ARV regimen, drug-resistance testing in the setting of virologic failure should be performed while the person is still taking their ARV regimen or, if that is not possible, within 4 weeks after discontinuing their ARV regimen **(AII)**. If more than 4 weeks have elapsed since the non-long-acting agents were discontinued, resistance testing may still provide useful information to guide therapy; however, it is important to recognize that previously selected resistance mutations can be missed due to lack of drug-selective pressure **(CIII)**. Given the long half-lives of the long-acting injectable ARV drugs, resistance testing (including testing for resistance to INSTIs) should be performed in all people who have experienced virologic failure on a regimen of LA CAB/RPV or acquired HIV after receiving CAB-LA as PrEP, regardless of the amount of time since drug discontinuation **(AIII)**. However, the absence of detectable resistance in people not currently on antiretroviral therapy (ART) must be interpreted with caution when designing subsequent ARV regimens. Importantly, in addition to considering prior ART history, prior genotypic- or phenotypic-resistance test results should be obtained from old records when possible. Because the most current drug-resistance test may not be able to detect resistance mutations that were previously detected, these prior test results are clinically important and should be reviewed and considered when designing a new ARV regimen **(AIII)**.

A next-generation sequencing genotypic assay that analyzes HIV-1 proviral DNA may provide additional information on drug resistance in people with low levels of plasma HIV RNA or in people whose levels are below the limit of detection **(CIII)**. However, these assays might miss some or all previous drug-resistance mutations, and they should be interpreted with caution. The usefulness of these assays in the clinic is still under investigation and has yet to be fully determined.

Use of Resistance Assays in Clinical Practice

Note: See [Table 5](#) below for recommendations for the use of drug resistance assays.

Use of Resistance Assays in Determining Initial Treatment

Transmission of drug-resistant HIV strains is well documented and associated with suboptimal virologic response to initial ART.¹⁶⁻¹⁹ The risk of acquiring drug-resistant virus is related to the prevalence of drug resistance in people with HIV who engage in activities that may result in HIV transmission within a given community. The prevalence of resistance and mutations by ARV drug class depends upon the population being studied (e.g., people with previous ARV exposure versus ARV-naïve people), geography, and ARV class available in the region.²⁰ Pre-existing HIV drug resistance before initiation of ART is recognized as an issue for both high- and low-income countries. The prevalence of transmitted drug resistance (TDR) in high-income countries ranges from 9% to 14% and varies by country.²¹⁻²³ Pre-treatment drug resistance—which includes people exposed to ARVs prior to initiating first-line therapy (e.g., for PrEP or for the prevention of perinatal transmission), as defined by the World Health Organization—exceeds 10% in many countries.²⁰ In most TDR surveys, NNRTI resistance and NRTI resistance are the most common mutation class types detected, followed by PI- and INSTI-resistance mutations, respectively.²¹⁻²³

Resistance testing is recommended to guide therapy selection to optimize virologic response in all people starting ART **(AII)**. A genotypic assay is preferred for this purpose **(AIII)**. In early (acute and recent) HIV infection, in pregnant women with HIV, or in people willing and able to initiate ART on the day or soon after HIV diagnosis, treatment initiation should not be delayed pending resistance testing results. Once results are reported, the regimen can be modified if warranted (see [Early \[Acute and Recent\] HIV Infection](#)) **(AIII)**. In the absence of ART, resistant viruses may decline over time to less than the detection limit of standard resistance tests. However, when ART is eventually initiated, even low levels of resistant viruses may still increase the risk of treatment failure.²⁴⁻²⁶ Therefore, if ART is deferred, resistance testing should still be performed at the time of entry into care to optimize the chance of capturing transmitted resistance **(AIII)**. In this situation, the genotypic resistance test result should be used for regimen selection in the future when the person begins ART. If a person received CAB-LA as part of ART or PrEP, genotypic resistance testing should include the IN gene.

The rate at which transmitted resistance-associated mutations revert to wild-type virus has not been completely delineated, but mutations present at the time of HIV transmission are more stable than those selected under drug pressure. It is often possible to detect resistance-associated mutations in viruses that were transmitted several years earlier.²⁷⁻²⁹ Though no prospective trial has directly addressed whether drug-resistance testing before initiation of therapy confers benefit in this population, data from several studies, including one prospective clinical trial, suggest that virologic responses in people with baseline resistance mutations are suboptimal.^{16-19,30-34} In addition, an analysis of early RT and PR genotypic resistance testing in ARV-naïve people suggests that baseline testing in this population is cost-effective and should be performed.³⁵ Therefore, resistance testing in people with chronic HIV is recommended at the time of entry into HIV care **(AII)**.

Although no definitive prospective data exist to support the choice of one type of resistance testing over another, genotypic testing is generally preferred over phenotypic testing because of lower cost, faster turnaround time, greater sensitivity for detecting mixtures of wild-type and resistant virus, and easier interpretation of test results **(AIII)**. If therapy is deferred, repeat testing shortly before initiating ART may be considered, because the person may have acquired drug-resistant virus (i.e., superinfection) **(CIII)**.³⁶ Standard genotypic drug-resistance testing in ARV-naïve people involves testing for mutations in the RT and PR genes. Although reports of transmission of INSTI-resistant virus are rare, as use of INSTIs increases, the potential for transmission of INSTI-

resistant virus also may increase. The prior use of CAB-LA for PrEP also may increase the risk of INSTI resistance at the time of HIV diagnosis. When INSTI resistance is possible, providers should supplement standard, baseline, genotypic resistance testing with genotypic testing of the IN gene, which may need to be ordered separately **(AIII)**.

The next-generation sequencing genotypic resistance assay that analyzes proviral DNA in host cells can be considered in people with baseline HIV RNA <1,000 copies/mL or when conventional HIV RNA drug-resistance testing is unsuccessful **(CIII)**. As outlined above, the results should be interpreted with caution, as this assay might miss some or all previously existing drug-resistance mutations.

Use of Resistance Assays in the Event of Virologic Failure

Resistance assays are important tools to inform treatment decisions for people who experience virologic failure while on ART. Several prospective studies have assessed the utility of resistance testing to guide ARV drug selection in people who experience virologic failure. These studies have involved genotypic assays, phenotypic assays, or both.^{6,37-43} In general, these studies have found that changes in therapy based on resistance test results produce better early virologic response to salvage regimens than regimen changes guided only by clinical judgment.

In addition, one observational cohort study found that the use of genotypic drug-resistance testing in ART-experienced people with detectable plasma HIV RNA was independently associated with improved survival.⁴⁴ Thus, resistance testing is recommended as a tool for selecting active drugs when changing ARV regimens because of virologic failure in people with HIV RNA >200 copies/mL (**AI** for >1,000 copies/mL, **AIII** for 501 to 1,000 copies/mL, **CIII** for confirmed HIV RNA 201 to 500 copies/mL) (see [Virologic Failure](#)). In people with HIV RNA >200 copies/mL but <500 copies/mL, testing may be difficult to obtain outside of a research setting but still should be considered. Conventional drug-resistance testing in people with plasma viral loads ≤200 copies/mL **is not recommended** because there are unclear benefits and drug-resistance assays cannot be consistently performed at very low HIV RNA levels **(AIII)**.

Resistance testing can also help to guide treatment decisions for people with HIV who have suboptimal viral load reduction **(AII)**. Virologic failure in the setting of ART is, for certain people with HIV, associated with resistance to only one component of the regimen.⁴⁵⁻⁴⁷ In this situation, substituting individual drugs in a failing regimen may be an option, but this concept will require clinical validation (see [Virologic Failure](#)).

Genotyping is preferred for resistance testing in people with HIV who experience virologic failure or suboptimal viral load reduction while on a first or second ARV drug regimen and in people for whom resistance mutation patterns are known or not expected to be complex (i.e., mutations that are straightforward, usually limited in number, or those that have clear significance) **(AII)**. Often in these situations, the mutation patterns detected can be interpreted by algorithms used to predict the impact of subsequent regimens on virologic response. For people with HIV who have extensive treatment history, complex mutational patterns may occur. In such situations, the interpretation of complex genotypes and the impact of the mutation pattern on subsequent treatment regimens can be challenging. For these individuals, phenotypic-resistance testing may provide additional helpful information **(BIII)**. Rather than only predicting the impact of the detected mutations, these assays can measure *in vitro* the actual fold change in drug susceptibility, as well as the actual impact of mutation combinations and interactions on each drug under consideration.

When compared with phenotypic testing, genotypic testing costs less to perform and has a faster turnaround time and greater sensitivity for detecting mixtures of wild-type and resistant virus. In addition, observations show that genotypic and phenotypic assays are comparable predictors of

virologic response to subsequent ARV regimens.⁴⁸ In people with HIV who experience virologic failure while on INSTI-based regimens or in those with prior INSTI exposure, including to CAB-LA for HIV treatment or prevention, testing for INSTI resistance should be performed to determine whether to include drugs from this class in subsequent regimens **(AII)**. In this circumstance, clinicians should confirm that, when they order a resistance test, their laboratory is testing for INSTI resistance in addition to NNRTI, NRTI, and PI resistance. If INSTI-resistance testing needs to be ordered separately (as is the case in some laboratories), clinicians should request this assay in addition to standard drug-resistance testing. The addition of phenotypic to genotypic testing is generally indicated for people with known or suspected complex drug-resistance mutation patterns **(BIII)**.

The next-generation sequencing genotypic resistance assay that analyzes proviral DNA can be considered for people with HIV who are experiencing treatment failure and for whom conventional HIV RNA genotypic drug-resistance testing is unsuccessful or unavailable due to low HIV RNA levels **(CIII)**. As outlined above, results should be interpreted with caution, as these assays might miss some or all previously existing drug-resistance mutations.

When the use of a CCR5 antagonist is being considered, a co-receptor tropism assay should be performed **(AI)** (see [Co-Receptor Tropism Assays](#)).

Use of HIV Proviral DNA Resistance Assays for Optimizing Antiretroviral Regimen in People With Viral Suppression

In the past decade, simpler, more potent, and better-tolerated ARV drugs have become available, and new ARV drugs will likely continue to emerge. Switching individual or multiple ARV drugs in a regimen is sometimes considered for people with HIV with suppressed viral load to simplify a regimen, avoid drug interactions or toxicity, or for other reasons. If a person's viral load is suppressed, standard drug-resistance testing will not be successful.

The next-generation sequencing genotypic resistance assay that analyzes proviral DNA can be considered for people with viral suppression, particularly if complex or semi-complex pre-existing resistance is suspected. In individuals who have experienced no prior virologic failures and who are on their first or second regimen, or who have genotypic testing results from when they had prior virologic failures, the use of the proviral DNA genotypic test is unlikely to provide additional useful information. However, in individuals who have experienced multiple prior failures, have a prolonged history of prior ARV regimens, or for whom prior genotypic resistance test results are not available, it may be appropriate to utilize proviral DNA genotypic testing **(CIII)**. When such testing is obtained, results should be combined with all prior genotypic and phenotypic test results to construct a cumulative genotype, which incorporates all current and previously detected drug-resistance mutations. Results from proviral DNA genotypes should be interpreted with caution, as these assays might miss some or all previously existing drug-resistance mutations.^{49,50} The usefulness of these assays in the clinic is still under investigation and has yet to be fully determined.

Use of Resistance Assays in Pregnancy

In pregnancy, the goal of ART is to rapidly and maximally reduce plasma HIV RNA in order to provide optimal therapy for the pregnant woman and to prevent perinatal transmission of HIV. Genotypic resistance testing is recommended for all pregnant women with HIV before initiation of therapy **(AIII)** and for those entering pregnancy with detectable HIV RNA levels while on therapy **(AI)**. Phenotypic testing in those found to have complex drug-resistance mutation patterns may provide additional information **(BIII)**. Optimal prevention of perinatal transmission requires prompt initiation of ART pending resistance testing results. Once the results are available, the ARV regimen can be changed as needed.

Table 5. Recommendations for the Use of Drug-Resistance Assays

Clinical Setting and Recommendation	Rationale
In Early (Acute and Recent) HIV Drug-resistance testing is recommended (AII). A genotypic assay is generally preferred (AIII). Treatment should not be delayed while awaiting resistance testing results (AIII). See Early (Acute and Recent) HIV Infection for discussion on ART selection.	Drug-resistance testing can determine whether drug-resistant virus was transmitted or acquired while using PrEP. The initial ARV regimen can be modified, if necessary, once resistance test results are available. Genotypic testing is preferred to phenotypic testing because of lower cost, faster turnaround time, and greater sensitivity for detecting mixtures of wild-type and resistant virus.
If ART is deferred, repeat genotypic resistance testing may be considered when therapy is initiated (CIII). A genotypic assay is preferred (AIII).	Repeat testing when ART is initiated may be considered because the person may have acquired a drug-resistant virus (i.e., superinfection).
Before ART Initiation in People With Chronic HIV Drug-resistance testing is recommended at entry into HIV care to guide the selection of initial ART (AII). A genotypic assay is generally preferred (AIII). Treatment should not be delayed while awaiting resistance testing results (AIII).	Transmitted HIV with baseline resistance to at least one drug has been reported, and suboptimal virologic responses may be seen in people with baseline resistance mutations to ARVs in the prescribed regimen. Some drug-resistance mutations can remain detectable for years in untreated people with chronic HIV.
If transmitted or acquired INSTI resistance (including among people who had received INSTI for post-exposure prophylaxis) is a concern, providers should supplement standard resistance testing with a specific INSTI genotypic resistance assay, which may need to be ordered separately (AIII). Given the prolonged half-life of CAB-LA, INSTI-resistance testing should be considered in all people with HIV who previously received CAB-LA for PrEP, regardless of the time since drug discontinuation (AIII). See What to Start for discussion on ART selection.	Genotypic assays provide information on resistance to NRTIs, NNRTIs, PIs, and INSTIs. In some circumstances, INSTI-resistance tests need to be ordered separately (clinicians should check with the testing laboratory). Currently, transmitted INSTI resistance is infrequent, but the risk of people acquiring INSTI-resistant strains may be greater in certain known exposure settings. INSTI-resistance testing should be ordered for all people with prior exposure to CAB-LA for PrEP or an INSTI-based regimen for post-exposure prophylaxis.
For pregnant women or people who will initiate ART on the day of or soon after HIV diagnosis, treatment can be initiated prior to receiving resistance testing results.	If necessary, the ARV regimen can be modified once resistance test results are available.
If therapy is deferred, repeat genotypic resistance testing may be considered before initiation of ART (CIII). A genotypic assay is generally preferred (AIII).	Repeat testing before initiation of ART may be considered, because the person may have acquired a drug-resistant virus (i.e., a superinfection). Genotypic testing is preferred to phenotypic testing because of lower cost, faster turnaround time, and greater sensitivity for detecting mixtures of wild-type and resistant virus.

Table 5. Recommendations for the Use of Drug-Resistance Assays

Clinical Setting and Recommendation	Rationale
In People With Virologic Failure Drug-resistance testing is recommended in people on ART with HIV RNA >200 copies/mL (AI for >1,000 copies/mL, AIII for 501–1,000 copies/mL) and a confirmed HIV RNA 201–500 copies/mL (CIII). In people with confirmed HIV RNA levels between 201–500 copies/mL, testing may not be successful but should still be considered.	Drug-resistance testing can help determine the role of resistance in virologic failure and maximize the ability to select active drugs for the new regimen. Resistance testing for HIV RNA levels 201–500 copies/mL may need to be conducted within a research setting.
Resistance testing should be done while the person is taking ART or, if that is not possible, within 4 weeks after discontinuation of non–long-acting ARV drugs (AII). If >4 weeks have elapsed, resistance testing may still be useful to guide therapy; however, previously selected mutations can be missed due to lack of drug-selective pressure (CIII).	The absence of detectable resistance in such people with HIV must be interpreted with caution when designing subsequent ARV regimens, as mutations may decay with time.
For people who previously received LA CAB/RPV and present with virologic failure, resistance testing (including INSTI genotypic testing) should be performed regardless of the time since the last dose of LA CAB/RPV (AIII).	Because of the long half-lives of LA CAB and RPV, resistance mutations to INSTI and NNRTI may emerge even months after the last doses of these drugs.
Reverse transcriptase and protease genotypic resistance testing should be performed on people with virologic failure. Integrase resistance testing should be performed on individuals who have virologic failure and have a history of prior use of an INSTI (for prevention or treatment) or are currently receiving an INSTI-based regimen (AII).	Genotypic testing is preferred to phenotypic testing because of lower cost, faster turnaround time, and greater sensitivity for detecting mixtures of wild-type and resistant HIV. INSTI-resistance tests may need to be ordered separately. (Clinicians should check with the testing laboratory.)
All prior and current drug-resistance testing results should be reviewed and considered when designing a new ARV regimen for a person experiencing virologic failure (AIII).	Drug-resistance mutations may decay with time, and mutations detected in prior resistance tests may not be detected in current tests, though they remain clinically relevant.
Adding phenotypic testing to genotypic testing is generally preferred in people with known or suspected complex drug-resistance patterns (BIII).	Phenotypic testing can provide additional useful information in people with complex drug-resistance mutation patterns.
If use of a CCR5 antagonist is being considered, a co-receptor tropism assay should be performed (AI).	See Co-Receptor Tropism Assays section.
In People With Suboptimal Suppression of Viral Load Drug-resistance testing is recommended in people with suboptimal viral load suppression after initiation of ART (AII).	Testing can determine the role of resistance in suboptimal viral suppression, and it can help the clinician identify the number of active drugs available in the current ARV regimen and assess the need for a new regimen.

Table 5. Recommendations for the Use of Drug-Resistance Assays

Clinical Setting and Recommendation	Rationale
In Pregnant Women With HIV Genotypic resistance testing is recommended for all pregnant women before initial ART (AIII) and for those entering pregnancy with detectable HIV RNA levels while on therapy (AI).	The goals of ART in pregnant women with HIV are to achieve maximal viral suppression for treatment of HIV in the pregnant woman and to prevent perinatal transmission of HIV. Genotypic resistance testing will assist the clinician in selecting the optimal ARV regimen. However, treatment should not be delayed while awaiting resistance testing results. The initial regimen can be modified once resistance test results are available, if needed.
In People With Undetectable Viral Load or Low-Level Viremia Who Are Planning to Change Their ARV Regimen HIV-1 proviral DNA resistance assays may be useful if HIV RNA is below the limit of detection or with low-level viremia, where an HIV RNA genotypic assay is unlikely to be successful (CIII).	HIV-1 proviral DNA resistance assays may provide information about previously circulating resistant viral variants that are archived within proviral DNA. These assays may miss some or all prior resistance mutations that have occurred within the viral quasi-species and, therefore, they should be interpreted with caution. The clinical utility of HIV-1 proviral DNA assays has not been fully determined.

Key: ART = antiretroviral therapy; ARV = antiretroviral; CAB-LA = long-acting cabotegravir; CCR5 = cysteine-cysteine chemokine receptor 5; INSTI = integrase strand transfer inhibitor; LA = long-acting; **LA CAB/RPV = long-acting cabotegravir/rilpivirine**; NNRTI = non-nucleoside reverse transcriptase inhibitor; NRTI = nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; PrEP = pre-exposure prophylaxis; RPV = rilpivirine

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Co-Receptor Tropism Assays

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Panel's Recommendations
• A co-receptor tropism assay should be performed whenever the use of a CCR5 co-receptor antagonist is being considered (AI). • Co-receptor tropism testing is recommended for patients who exhibit virologic failure on a CCR5 antagonist (BIII). • A phenotypic tropism assay is preferred to determine HIV-1 co-receptor usage (AI). • A genotypic tropism assay should be considered as an alternative test to predict HIV-1 co-receptor usage (BII). • A proviral DNA tropism assay can be utilized for patients with undetectable HIV-1 RNA when a CCR5 antagonist is considered for use in a new regimen (e.g., as part of a regimen switch or simplification) (BII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

HIV enters cells by a complex process that involves sequential attachment to the CD4 T lymphocyte (CD4) receptor followed by binding to either the CCR5 or CXCR4 molecules and fusion of the viral and cellular membranes.¹ CCR5 co-receptor antagonists prevent HIV entry into target cells by binding to the CCR5 receptors.² Phenotypic and genotypic assays have been developed that can determine or predict the co-receptor tropism (i.e., use of CCR5, CXCR4, or both as either dual-tropic virus or a mixed population of viruses referred to for purposes of assay results as dual/mixed [D/M]) of the patient's dominant virus population. An older generation assay (Trofile,[®] Monogram Biosciences, Inc., South San Francisco, CA) was used to screen patients who were participating in clinical trials that led to the approval of maraviroc (MVC), the only CCR5 antagonist currently available. The assay has been improved and is now available with enhanced sensitivity. In addition, genotypic assays to predict co-receptor usage are commercially available.

During acute/recent infection, the vast majority of patients harbor a CCR5-utilizing virus (R5 virus), which suggests that the R5 variant is preferentially transmitted; however, up to 19% of individuals with acute/recent infection can harbor CXCR4-tropic virus.³⁻⁵ Viruses in many untreated patients eventually exhibit a shift in co-receptor tropism from CCR5 usage to either CXCR4 usage or D/M tropism. This shift is temporally associated with a more rapid decline in CD4 counts,^{6,7} but whether this tropism shift is a cause or a consequence of progressive immunodeficiency remains undetermined.¹ Antiretroviral-treated patients with extensive drug resistance or persistently high-level viremia are more likely to harbor CXCR4- or D/M-tropic variants than untreated patients with comparable CD4 counts.^{8,9} The prevalence of CXCR4- or D/M-tropic variants increases to more than 50% in treated patients who have CD4 counts <100 cells/mm³.^{8,10} Since CXCR4-tropic viruses may be present at initial presentation or a patient may shift to CXCR4-tropism over the course of infection, co-receptor tropism should always be assessed prior to the use of CCR5 antagonists for treatment. Once a patient has ever been documented with detectable CXCR4- or D/M-tropic virus, it is assumed that such viruses will always be present. CCR5 co-receptor antagonists will no longer be active for that patient and should not be used.

Phenotypic Assays

Phenotypic assays characterize the co-receptor usage of plasma-derived virus. These assays involve the generation of laboratory viruses that express patient-derived envelope proteins (i.e., gp120 and gp41). These pseudoviruses, which are replication-defective, are used to infect target cell lines that express either CCR5 or CXCR4.^{11,12} Using the Trofile[®] assay, the co-receptor tropism of the patient-derived virus is confirmed by testing the susceptibility of the virus to specific CCR5 or CXCR4 inhibitors *in vitro*. This assay takes about 2 weeks to perform and requires a plasma HIV RNA level $\geq 1,000$ copies/mL.

The performance characteristics of these assays have evolved. Most, if not all, patients enrolled in premarketing clinical trials of MVC and other CCR5 antagonists were screened with an earlier, less sensitive version of the Trofile[®] assay.¹² This earlier assay failed to routinely detect low levels of CXCR4-utilizing variants. As a consequence, some patients enrolled in these clinical trials harbored low levels of such variants at baseline, which were below the assay limit of detection, and these patients exhibited rapid virologic failure after initiation of a CCR5 antagonist.¹³ The assay has been improved and is now able to detect lower levels of CXCR4-utilizing viruses. *In vitro*, the assay can detect CXCR4-utilizing clones with 100% sensitivity when those clones represent 0.3% or more of the virus population.¹⁴ Although this more sensitive assay has had limited use in prospective clinical trials, it is now the only assay that is commercially available. For unclear reasons, a minority of samples cannot be successfully phenotyped with either generation of the Trofile[®] assay.

In patients with an undetectable viral load or detectable plasma HIV RNA $< 1,000$ copies/mL, phenotypic co-receptor usage can be determined using proviral DNA obtained from peripheral blood mononuclear cells (e.g., Trofile[®] DNA, Monogram Sciences); however, the clinical utility of this assay remains to be determined.¹⁵

Genotypic Assays

Genotypic determination of HIV-1 co-receptor usage is based on sequencing of the V3-coding region of HIV-1 *env*, the principal determinant of co-receptor usage. A variety of algorithms and bioinformatics programs can be used to predict co-receptor usage from the V3 sequence.¹⁶ When compared to the phenotypic assay, genotypic methods show high specificity (~90%) but only modest sensitivity (~50% to 75%) for the presence of a CXCR4-utilizing virus. Studies in which V3 genotyping was performed on samples from patients screened for clinical trials of MVC suggest that genotyping performed as well as phenotyping in predicting the response to MVC.¹⁷⁻¹⁹ An important caveat is that the majority of patients who received MVC were first shown to have R5 virus by a phenotypic assay (Trofile[®]). Consequently, the opportunity to assess treatment response to MVC in patients whose virus was considered R5 by genotype but D/M or X4 by phenotype was limited to a relatively small number of patients. Other studies have also demonstrated relatively high concordance between genotypic- and phenotypic-assessed tropism;^{20,21} however, there is variability between different genotypic platforms.²²

Given these performance characteristics, genotypic tropism assays may not be sufficiently robust to completely rule out the presence of an X4 or D/M variant;²³ therefore, the Panel preferentially recommends phenotypic testing. Based on accessibility, capacity, logistics, and cost, European guidelines currently include genotypic testing as an equivalent option to phenotypic testing when determining co-receptor usage among patients with HIV RNA $> 1,000$ copies/mL and preferentially for those with HIV RNA $\leq 1,000$ copies/mL.²⁴

HIV-1 proviral DNA genotypic tropism testing is available for patients with HIV RNA $< 1,000$ copies/mL. These assays evaluate the HIV-1 proviral DNA integrated within infected cells for CXCR4-utilizing viral strains.²⁵ As discussed above, caution is advised when using such assays, as

their detection limit, concordance with plasma HIV RNA tropism, and clinical utility are not yet fully determined.

Use of Assays to Determine Co-receptor Usage in Clinical Practice

An assay for HIV-1 co-receptor usage should be performed whenever the use of a CCR5 antagonist is being considered **(AI)**. This is true even in the setting of prior tropism testing showing CCR5 usage, as viral evolution may occur over the course of infection. In addition, because virologic failure may occur due to a shift from CCR5-using to CXCR4-using virus, testing for co-receptor usage is recommended in patients who exhibit virologic failure on a CCR5 antagonist **(BIII)**. Virologic failure may also be caused by resistance of a CCR5-using virus to a CCR5 antagonist, but such resistance is uncommon. Compared to genotypic testing, phenotypic testing has more evidence supporting its utility. Therefore, a phenotypic test for co-receptor usage is generally preferred **(AI)**. However, because phenotypic testing is more expensive, requires more time to perform, and may have logistic challenges, a genotypic test to predict HIV-1 co-receptor usage should be considered as an alternative test **(BII)**.

As with HIV resistance testing, the results of all prior tropism tests should be obtained. If CXCR4-utilizing or D/M-tropic viruses have ever been detected previously, then repeat testing is not necessary and a CCR5 co-receptor antagonist **should not be used**.

If a CCR5 co-receptor antagonist is being considered in a patient with an undetectable HIV RNA (e.g., in cases of regimen simplification or a toxicity-related switch), a proviral DNA tropism assay can be utilized **(BII)**.²⁶⁻²⁸ If CXCR4-utilizing or D/M-tropic viruses are detected, then the CCR5 co-receptor antagonist **should not be used**.

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HLA-B*5701 Screening

Updated: December 1, 2007

Reviewed: January 10, 2011

Panel's Recommendations
• The Panel recommends screening for HLA-B*5701 before starting patients on an abacavir (ABC)-containing regimen to reduce the risk of hypersensitivity reaction (HSR) (AI). • HLA-B*5701-positive patients should not be prescribed ABC (AI). • The positive status should be recorded as an ABC allergy in the patient's medical record (AII). • When HLA-B*5701 screening is not readily available, it remains reasonable to initiate ABC with appropriate clinical counseling and monitoring for any signs of HSR (CIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak
Rating of Evidence: I = data from randomized controlled trials; II = data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = expert opinion

The abacavir (ABC) hypersensitivity reaction (HSR) is a multiorgan clinical syndrome typically seen within the initial 6 weeks of ABC treatment. This reaction has been reported in 5% to 8% of patients participating in clinical trials when using clinical criteria for the diagnosis, and it is the major reason for early discontinuation of ABC. Discontinuing ABC usually promptly reverses HSR, whereas subsequent rechallenge can cause a rapid, severe, and even life-threatening recurrence.¹

Studies that evaluated demographic risk factors for ABC HSR have shown racial background as a risk factor, with white patients generally having a higher risk (5%–8%) than black patients (2%–3%). Several groups reported a highly significant association between ABC HSR and the presence of the major histocompatibility complex (MHC) class I allele HLA-B*5701.^{2, 3} Because the clinical criteria used for ABC HSR are overly sensitive and may lead to false-positive ABC HSR diagnoses, an ABC skin patch test (SPT) was developed as a research tool to immunologically confirm ABC HSR.⁴ A positive ABC SPT is an ABC-specific delayed HSR that results in redness and swelling at the skin site of application. All ABC SPT–positive patients studied were also positive for the HLA-B*5701 allele.⁵ The ABC SPT could be falsely negative for some patients with ABC HSR and, at this point, is not recommended for use as a clinical tool. The PREDICT-1 study randomized participants with HIV before starting ABC either to be prospectively screened for HLA-B*5701 (with HLA-B*5701–positive patients not offered ABC) or to standard of care at the time of the study (i.e., no HLA screening, with all patients receiving ABC).⁶ The overall HLA-B*5701 prevalence in this predominately white population was 5.6%. In this cohort, screening for HLA-B*5701 eliminated immunologic ABC HSR (defined as ABC SPT positive) compared with standard of care (0% vs. 2.7%), yielding a 100% negative predictive value with respect to SPT and significantly decreasing the rate of clinically suspected ABC HSR (3.4% vs. 7.8%). The SHAPE study corroborated the low rate of immunologically validated ABC HSR in black patients and confirmed the utility of HLA-B*5701 screening for the risk of ABC HSR (100% sensitivity in black and white populations).⁷

On the basis of the results of these studies, the Panel recommends screening for HLA-B*5701 before starting an ABC-containing regimen in a person with HIV **(AI)**. HLA-B*5701–positive patients should not be prescribed ABC **(AI)**, and the positive status should be recorded as an ABC allergy in the patient's medical record **(AII)**. HLA-B*5701 testing is needed only once in a patient's lifetime; thus, efforts to carefully record and maintain the test result and to educate the patient about its implications are important. The specificity of the HLA-B*5701 test in predicting ABC HSR is lower than the sensitivity (i.e., 33%–50% of HLA-B*5701–positive patients would likely not develop

confirmed ABC HSR if exposed to ABC). HLA-B*5701 should not be used as a substitute for clinical judgment or pharmacovigilance, because a negative HLA-B*5701 result does not absolutely rule out the possibility of some form of ABC HSR. When HLA-B*5701 screening is not readily available, it remains reasonable to initiate ABC with appropriate clinical counseling and monitoring for any signs of ABC HSR **(CIII)**.

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TREATMENT GOALS

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Antiretroviral therapy (ART) has reduced HIV-related morbidity and mortality at all stages of HIV infection¹⁻⁴ and has reduced HIV transmission.⁵⁻⁸ Maximal and durable suppression of plasma viremia delays or prevents the selection of drug-resistance mutations, preserves or improves CD4 T lymphocyte (CD4) cell numbers, and confers substantial clinical benefits, all of which are important treatment goals.^{9,10} HIV suppression with ART may also decrease inflammation and immune activation thought to contribute to higher rates of cardiovascular and other end-organ damage reported in cohorts with HIV (see [Initiating Antiretroviral Therapy](#)). Despite these benefits, eradication of HIV infection cannot be achieved with available antiretrovirals (ARVs). Treatment interruption has been associated with rebound viremia, worsening of immune function, and increased morbidity and mortality.¹¹ Thus, once initiated, ART should be continued, with the following key treatment goals:

- Maximally and durably suppress plasma HIV RNA;
- Restore and preserve immunologic function;
- Reduce HIV-associated morbidity and prolong the duration and quality of survival; and
- Prevent HIV transmission.

Achieving viral suppression currently requires the use of combination ARV regimens that generally include three active drugs from two or more drug classes. Baseline patient characteristics and results from drug resistance testing should guide design of the specific regimen (see [What to Start: Initial Combination Regimens for People With HIV](#)). When initial HIV suppression is not achieved or not maintained, changing to a new regimen with at least two active drugs is often required (see [Virologic Failure](#)). The increasing number of ARV drugs and drug classes makes viral suppression below detection limits an achievable goal in most patients.

After initiation of effective ART, viral load reduction to below limits of assay detection usually occurs within the first 12 to 24 weeks of therapy. Predictors of virologic success include the following:

- Low baseline viremia;
- High potency of the ARV regimen;
- Tolerability of the regimen;
- Convenience of the regimen; and
- Excellent adherence to the regimen.

Strategies to Achieve Treatment Goals

Selection of Initial Combination Regimen

Several ARV regimens are recommended for use in ART-naïve patients (see [What to Start](#)). Most of the recommended regimens have comparable efficacy but vary in pill burden, potential for drug interactions and/or side effects, and propensity to select for resistance mutations if ART adherence is suboptimal. Regimens should be tailored for the individual patient to enhance adherence and support long-term treatment success. Considerations when selecting an ARV regimen for an

individual patient include potential side effects, patient comorbidities, possible interactions with concomitant medications, results of pretreatment genotypic drug-resistance testing, and regimen convenience (see [Table 7](#)).

Improving Adherence

Suboptimal adherence may result in reduced treatment response. Incomplete adherence can result from complex medication regimens; patient-related factors, such as active substance abuse, depression, or the experience of adverse effects; and health system issues, including interruptions in patient access to medication and inadequate treatment education and support. Conditions that promote adherence should be maximized before and after initiation of ART (see [Adherence to Antiretroviral Therapy](#)).

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INITIATION OF ANTIRETROVIRAL THERAPY

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Reviewed: September 25, 2025

Panel's Recommendations
• The Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends antiretroviral therapy (ART) for all people with HIV to reduce morbidity and mortality (AI) and to prevent transmission of HIV to others (AI). • The Panel recommends initiating ART immediately (or as soon as possible) after HIV diagnosis in order to increase the uptake of ART and linkage to care, decrease the time to viral suppression for individual patients, and improve the rate of virologic suppression among people with HIV (AII). • When initiating ART, people with HIV should be counseled on the benefits, lifelong need, and importance of adherence to ART; clinicians should also identify and address barriers to care engagement and ART adherence (AIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak
Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Introduction

The primary goals of antiretroviral therapy (ART) are to prevent HIV-associated morbidity and mortality and to prevent transmission of HIV to others. These goals are accomplished by using effective ART to achieve and maintain plasma HIV-1 RNA levels (viral load) below the quantification limits of commercially available assays. Durable viral suppression lowers the risk of both AIDS-defining and other HIV-related complications, improves immune function and overall health, and allows people with HIV to live a lifespan approaching that of people without HIV.^{1,2} High plasma viral load is a major risk factor for HIV transmission; effective ART suppresses viremia and, consequently, substantially reduces the risk of sexual³⁻⁷ and perinatal HIV transmission.^{8,9} Modeling studies and ecological studies of populations with high ART uptake and high viral suppression rates suggest that expanded use of ART lowers the incidence of HIV and, eventually, the prevalence of HIV on a community or population level.¹⁰⁻¹²

Early clinical trials of combination ART focused on people with advanced HIV and low CD4 T lymphocyte (CD4) cell counts.¹³⁻¹⁵ Consequently, treatment guidelines at that time recommended the initiation of ART based on CD4 count thresholds or the development of AIDS-defining conditions. The benefits of ART for people with high CD4 counts were unclear, along with concerns about the toxicities of some antiretroviral (ARV) drugs. However, two large randomized controlled trials, START¹⁶ and TEMPRANO¹⁷, both published in 2015, addressed the optimal time to initiate ART. Both studies demonstrated reductions in morbidity and mortality among people with HIV who had CD4 counts >500 cells/mm³ and who initiated ART immediately when compared with people who delayed initiation of ART.

Deferring ART until CD4 counts decline puts people with HIV at risk of both AIDS-defining conditions and certain serious non-AIDS-defining conditions. Furthermore, the magnitude of CD4 recovery is directly correlated with CD4 count at ART initiation. Consequently, some individuals who start treatment with CD4 counts <350 cells/mm³ do not achieve CD4 counts >500 cells/mm³ after up to

10 years on ART,^{18,19} and they have a shorter life expectancy than those who initiated therapy at higher CD4 count thresholds.¹⁸⁻²⁰

Fundamental to the recommendation for earlier initiation of ART is the assumption that HIV will be diagnosed early in the course of the disease. The Centers for Disease Control and Prevention (CDC) recommends HIV screening to be performed for all people aged 13 to 64 years, including during pregnancy, using an opt-out screening approach.²¹ More frequent HIV screening is recommended for certain populations with higher HIV prevalence.²² The U.S. Preventive Services Task Force (USPSTF) also recommends HIV testing for people aged 15 to 65 years and for all pregnant women. This recommendation has been designated a Grade A recommendation by the USPSTF, meaning that third-party payers should cover this service without cost to patients.²³ HIV testing should also be performed for people younger than 15 and older than 65 years of age when indicated or requested.

Unfortunately, in some individuals, the diagnosis of HIV is not made until the later stages of the disease. In a survey conducted between 2016 and 2017, it was noted that fewer than 40% of American adults had ever had an HIV test.²⁴ Delayed diagnosis of HIV has been reported more often in people who inject drugs, people who live in rural communities, older adults, and among Black and Hispanic persons.²⁵⁻²⁷ Many people with HIV access health care years before their HIV diagnosis but are not offered HIV testing.²⁸

To ensure that the current treatment guidelines have maximum impact, routine HIV screening per current CDC or USPSTF recommendations is essential.^{23,29} Early diagnosis also has clinical and economic benefits, including prolonging life, improving the quality of life, and decreasing the costs related to the management of AIDS and its comorbidities.^{30,31} Additionally, HIV screening is a key step in the Ending the HIV Epidemic initiative to prevent the transmission of HIV to others.³² It is critical that everyone who receives an HIV diagnosis be educated about HIV disease; initiated on ART; and linked to care for full evaluation, follow-up, and management as soon as possible. In order for people with HIV and their sexual partners to fully benefit from early diagnosis, clinicians should initiate ART as soon as possible and provide support to enhance engagement in care and full ART adherence (see [Adherence to the Continuum of Care](#) and [Treatment as Prevention](#)).

Initiating Antiretroviral Therapy

The Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends ART for all people with HIV to reduce the morbidity and mortality associated with HIV infection **(AI)** and to prevent HIV transmission to sexual partners and infants **(AI)**. ART should be initiated as soon as possible after HIV diagnosis **(AII)**. Clinicians should refer to [What to Start](#) for guidance on ART regimen selection for all adults and adolescents with HIV, including those who received pre-exposure prophylaxis (PrEP) before HIV diagnosis; the [Perinatal Guidelines](#) provide guidance on ART regimen selection during pregnancy. When initiating ART, people with HIV should be counseled on the benefits, lifelong need, and importance of adherence to ART; clinicians should also identify and address barriers to care engagement and ART adherence **(AIII)**. People with HIV should also understand that the currently available ART does not cure HIV. To improve and maintain immunologic function and maintain viral suppression, ART should be continued indefinitely without interruption. Initiating ART as soon as possible is particularly important for people with AIDS-defining conditions, those with acute or recent HIV infection, and during pregnancy. Delaying therapy in these subpopulations may increase risks for morbidity, mortality, and HIV transmission. [Laboratory Testing for Initial Assessment and Monitoring of People With HIV Receiving Antiretroviral Therapy](#) provides recommendations for baseline and follow-up laboratory monitoring for people starting ART.

Early (Acute/Recent) HIV Infection

The term “early HIV” refers to both acute and recent infection. “Acute HIV” occurs soon after transmission (onset of symptoms typically within 2–4 weeks after infection) and is characterized by the presence of HIV RNA alone or with positive p24 antigen and nonreactive or indeterminate HIV antibody results on the confirmatory immunoassay. “Recent HIV” refers to the first 6 months after infection, during which anti-HIV antibodies and HIV RNA become detectable.^{33,34} Detailed discussion on the diagnosis of acute HIV can be found in [Early \(Acute and Recent\) HIV Infection](#).

Early HIV is considered a period of high infectivity,³⁵ and early ART substantially reduces the risk of HIV transmission.⁴⁻⁷ Additionally, clinical trial data indicate that individuals who are treated during early HIV may experience immunologic and virologic benefits.³⁶⁻⁴¹ Therefore, ART should be initiated as soon as possible after a positive qualitative test result or a quantitative HIV RNA test result **(AII)**. Before starting ART, and without delaying its initiation, a blood sample should be sent for genotypic resistance testing (see [Drug Resistance Testing](#) and [What to Start](#)). Same-day or rapid ART initiation in people with acute HIV has been shown to be safe, acceptable, and effective.⁴² When initiating ART, clinicians should identify and address barriers to ART adherence and engagement in care for each person with HIV (see [Adherence to the Continuum of Care](#)).

To verify the diagnosis of acute HIV, a new blood specimen should be collected for a confirmatory HIV antibody test and a quantitative plasma HIV RNA test. Given the sensitivity of current HIV RNA assays,⁴³ a positive result by quantitative or qualitative plasma HIV RNA testing in the setting of a negative or indeterminate antibody test result indicates that acute HIV infection is highly likely. If acute HIV is highly suspected, ART should be initiated without delay while awaiting confirmation of diagnosis⁴⁴; if the pending test results later rule out HIV infection, then ART for the treatment of HIV should be discontinued. People who tested negative but remain at risk of sexual HIV transmission should be offered ARV PrEP per [recommendations from the CDC](#).

Some individuals with early HIV may decline ART initially. Individuals who do not begin ART immediately should be maintained in care, and every effort should be made to initiate therapy as soon as they are ready.

Immediate Antiretroviral Therapy Initiation on the Day of HIV Diagnosis

ART initiation on the same day of HIV diagnosis in the outpatient setting may be used as a strategy to increase ART uptake and engagement in care, to accelerate the time to ART-mediated viral suppression, and, consequently, to reduce the potential for HIV transmission. Benefits of rapid ART initiation are supported by randomized controlled trials that were performed in resource-limited settings outside of the United States⁴⁵⁻⁴⁷ and observational trials in the United States that included both immediate initiation of ART (on the day of diagnosis)⁴⁸⁻⁵⁰ and rapid ART initiation (within days or weeks of diagnosis).^{50,51} Two randomized trials were performed in South Africa (n = 377)⁴⁵ and Haiti (n = 703).⁴⁶ In these trials, same-day ART initiation increased the proportion of participants who had suppressed viral loads about a year after HIV diagnosis compared with those who initiated ART between 3 weeks and 3 months after diagnosis. In addition, these studies demonstrated that rapid ART initiation can be feasible and beneficial in resource-limited settings. While no randomized controlled trials have been conducted in the United States, several prospective observational studies have demonstrated the feasibility of same-day ART initiation. City-wide implementation of the San Francisco RAPID program among 225 patients who were newly diagnosed with HIV resulted in ART initiation on the day of diagnosis for more than half of the patients and a median time from ART initiation to viral suppression (defined as <200 copies/mL) of 41 days. Over a median follow-up of 1.1 years (range 0–3.9 years), 92.1% of patients achieved virologic suppression. In the RAPID

study, a substantial proportion of participants had a major substance use disorder (51.4%), a major mental health disorder (48.1%), or unstable housing (30.6%).⁴⁹

It should be emphasized that ART initiation on the same day of HIV diagnosis can be resource intensive. This strategy may require additional staff, multidisciplinary coordination, provision of ART starter packs, and consolidation of “usual care” patient services (e.g., clinical evaluation, education, counseling, initiation or optimization of insurance coverage, intake laboratory testing) into a 2- or 3-hour visit.⁴⁹ While the infrastructure and resources necessary to implement an immediate ART program may not be available in all health care settings, facilitating rapid ART initiation may improve treatment outcomes. The Panel recommends initiating ART at the time of diagnosis (when possible) or soon afterwards to increase the uptake of ART, decrease the time required to achieve linkage to care and virologic suppression, and improve the rate of virologic suppression **(AII)**. The rating for this recommendation reflects the fact that randomized trials have only been conducted in low-resource countries, while evidence for benefit in the United States **is based** on observational studies.

Antiretroviral Therapy for Persons With Acute Opportunistic Infections and Malignancies

Initiation of ART in the setting of an acute, AIDS-associated opportunistic infection (OI) or malignancy can improve immune function and potentially enhance treatment success for the OI. Clinicians should refer to the [Adult and Adolescent Opportunistic Infection Guidelines](#) for a more in-depth discussion on specific OIs. Below are some important factors to consider when initiating ART in these situations.

- **When no effective therapy exists for the treatment of the OI (e.g., cryptosporidiosis, microsporidiosis, progressive multifocal leukoencephalopathy):** In these situations, ART is the only treatment that can improve immune function and clinical outcomes. ART should be initiated without delay.
- **Concerns about immune reconstitution inflammatory syndrome (IRIS):** For some OIs, such as cryptococcal and tuberculosis (TB) meningitis, immediate ART initiation may increase the risk of serious IRIS. A short delay before initiating ART may be warranted.⁵²⁻⁵⁵ After ART initiation, the patient should be closely monitored for signs and symptoms associated with IRIS.
- **Nonmeningeal TB:** In these patients, initiating ART during treatment for TB confers a significant survival advantage⁵⁶⁻⁶⁰; therefore, ART should be initiated as recommended in [Tuberculosis/HIV Coinfection](#).
- **For people with mild to moderate cutaneous Kaposi sarcoma:** Prompt initiation of ART alone without chemotherapy has been associated with improvement of cutaneous Kaposi sarcoma lesions, even though initial transient progression of Kaposi sarcoma lesions as a manifestation of IRIS can also occur.⁶¹
- **For people with other OIs:** After starting treatment for the OI, ART should be initiated as soon as feasible to improve immune function.
- **For patients with malignancies that require chemotherapy:**
 - o A diagnosis of malignancy should not delay initiation of ART, nor should initiation of ART delay treatment for the malignancy.
 - o Although an IRIS-like presentation of non-Hodgkin's lymphoma after initiation of ART has been described,⁶² ART-mediated viral suppression is associated with longer survival among individuals undergoing treatment for AIDS-related lymphoma.⁶³

Importantly, drug interactions should be considered when selecting ART, as there is the potential for significant interactions between some ARV drugs (particularly ritonavir- or cobicistat-boosted regimens) and drugs for treatment of OIs (particularly with rifamycins and azole antifungal drugs) and chemotherapy drugs.

Antiretroviral Therapy Initiation in Hospitalized People With HIV

Some people receive their first HIV diagnosis while hospitalized for HIV-related or unrelated conditions.⁶⁴ The test-and-treat concept of initiating ART has been successful in the outpatient setting as described above; however, newly diagnosed people with HIV who are hospitalized seldom start ART before hospital discharge.⁶⁵ Some concerns about initiating ART during hospitalization include the potential for drug–drug interactions or ART-related adverse effects that may complicate management of the acute condition that prompted hospitalization, the risk of paradoxical or unmasking IRIS in the context of OIs, and limited HIV care experience among hospital staff. Additionally, securing ART prescription coverage for a person with newly diagnosed HIV requires a multidisciplinary approach, including coordination with social workers/case managers to navigate through some complex insurance/patient assistance programs. As a result, some clinicians prefer to leave this task to the primary HIV clinic.

Although receiving an HIV diagnosis during an acute illness can be overwhelming, several reasons make this an opportune time to initiate ART. For example, being hospitalized allows more time for the person with HIV to receive education and ask questions about HIV and treatment goals and to receive ART-related counseling. Upon patient approval, having family or caregivers participate in these discussions will allow them to assist with adherence to ART (and concomitant medications) and clinic follow-up after discharge. Administering the first doses of ART during hospitalization allows clinicians to assess ART tolerance while under observation. Using a multidisciplinary team approach (e.g., HIV specialists, pharmacists, social workers) during the hospital stay to assist with ART initiation can facilitate transition to outpatient care. Clinical pharmacists can assess for potential drug–drug and drug–nutrient interactions and provide counseling on ART and adherence. Social workers/patient navigators can assist with exploring options for outpatient ARV coverage and facilitate linkage to HIV primary care. A multicenter study that enrolled hospitalized people with HIV from 2012 to 2014 demonstrated that starting ART in hospitalized patients who had a recent history of substance use was associated with shorter time to linkage to their first HIV clinical appointment after discharge and more clinic visits in the first 12 months after ART initiation than those who were not started on ART before discharge; however, no difference was observed in the rate of viral suppression and retention in care over 12 months.⁶⁶

Certain situations, such as in the setting of TB meningitis or cryptococcal meningitis, may warrant a short delay in ART initiation due to concerns of severe consequences related to paradoxical IRIS; however, in most other situations, the benefits of starting ART during hospitalization outweigh the concerns listed above. Therefore, the Panel recommends initiating ART during hospitalization whenever possible **(BIII)**.

Before starting ART, and without delaying its initiation, a blood sample should be sent for genotypic resistance testing (see [Drug Resistance Testing](#) and [What to Start](#)). If the resistance testing results are not available before hospital discharge, they should be forwarded to the outpatient HIV provider as soon as they become available. If ART is initiated before hospital discharge, clinicians should coordinate with their multidisciplinary team to ensure sufficient ART supply (e.g., at least 2–4 weeks'

supply) is available until the outpatient clinic appointment. See [Adherence to the Continuum of Care](#) for more discussions on the transition of care between different health care settings.

Evidence Supporting the Benefits of Antiretroviral Therapy in Preventing Morbidity and Mortality

Randomized Controlled Trials of Early Versus Deferred Antiretroviral Therapy

Two large randomized controlled trials, START and TEMPRANO, published in 2015, provided the evidence for the Panel's recommendation to initiate ART in all patients regardless of CD4 count **(AI)**. The results of these two studies are summarized below.

START was a large, multinational, randomized controlled clinical trial designed to evaluate the role of early ART initiation in asymptomatic adults with high CD4 counts in reducing AIDS-defining illnesses, serious non-AIDS events, or death. In this study, ART-naïve adults (aged >18 years) with CD4 counts >500 cells/mm³ were randomized to initiate ART at randomization (immediate arm) or to wait to initiate ART until their CD4 counts declined to <350 cells/mm³ or until they developed a clinical indication for therapy (deferred arm).

The study enrolled 4,685 participants. Initial results were unblinded early (after a mean follow-up of 3 years) because of the superiority of the immediate ART strategy. The primary endpoint of serious AIDS or non-AIDS events was reported in 42 participants (1.8%, or 0.60 events per 100 person-years) in the immediate ART arm and 96 participants (4.1%, or 1.38 events per 100 person-years) in the deferred ART arm (hazard ratio [HR] 0.43, favoring early ART; 95% confidence interval [CI], 0.30–0.62, $P < 0.001$). The majority of clinical events (59%) in the deferred ART arm occurred in participants whose CD4 counts were still above 500 cells/mm³, evidence for a benefit of initiating ART even before CD4 count declines below this threshold. Furthermore, the benefit of immediate ART was consistent across all participant subgroups, including sex, age, plasma HIV RNA levels, and country income level. The benefit appeared to be particularly strong for AIDS events (HR 0.28), TB (HR 0.29), malignancies (HR 0.36), and severe bacterial infections (HR 0.39). The benefit at lower CD4 counts was primarily a reduction in the number of AIDS events, while the benefit at higher CD4 counts was primarily a reduction in the number of serious non-AIDS events. In addition to AIDS-related events, early ART initiation also significantly reduced the rate of pooled serious non-AIDS events (HR 0.61).^{16,67}

The TEMPRANO ANRS 12136 study was a randomized controlled trial conducted in Cote d'Ivoire. Using a two-by-two factorial design, participants with HIV who had CD4 counts <800 cells/mm³ and who did not meet the criteria for starting ART according to World Health Organization guidelines at that time were randomized to start ART early (upon enrollment) or defer ART based on the national guidelines criteria for starting treatment. Half of the participants in each group received isoniazid for prevention of TB for 6 months and half did not. The primary study endpoint was a combination of all-cause deaths, AIDS diseases, non-AIDS malignancies, and non-AIDS invasive bacterial diseases. More than 2,000 participants enrolled in the trial, with a median follow-up of 30 months. Among the 849 participants who had baseline CD4 counts >500 cells/mm³, early ART reduced the rate of primary events compared with deferred ART, with an HR of 0.56 (95% CI, 0.33–0.94).¹⁷

The TEMPRANO and START trials had very similar estimates for the protective effect of ART among individuals with HIV who had CD4 counts >500 cells/mm³. Guidelines around the world now recommend starting ART irrespective of CD4 count at the time or soon after HIV diagnosis.

Use of Antiretroviral Therapy to Prevent HIV Transmission

Prevention of Sexual Transmission

A randomized clinical trial⁴ and several large observational cohort studies⁵⁻⁷ have provided strong evidence that achieving sustained viral suppression prevents sexual transmission of HIV. Thus, a key goal of ART is to prevent transmission of HIV to seronegative sexual partners **(AI)**. All people with HIV should be informed that maintaining a plasma HIV RNA (viral load) of <200 copies/mL (including any measurable value below this threshold value) with ART prevents sexual transmission of HIV to partners **(AII)**. People with HIV may recognize this concept as Undetectable = Untransmittable, or U=U. The results of these studies are summarized in [Antiretroviral Therapy to Prevent Sexual Transmission of HIV](#). Clinicians should also discuss with these individuals the potential harm of non-adherence to ART, including that it can lead to viremia, which may affect their own health, lead to drug resistance, and potentially transmission to sexual partners.

Prevention of Perinatal Transmission

The first well-established example of ART reducing the risk of HIV transmission is the use of ART during pregnancy to prevent perinatal transmission of HIV. Effective suppression of HIV replication is a key determinant in reducing the risk of perinatal transmission. In the setting of maternal viral load suppressed to <50 copies/mL near delivery, the use of combination ART during pregnancy has reduced the rate of perinatal HIV transmission from approximately 20%–30% to 0.1%–0.5%.^{8,9} ART is thus recommended during all pregnancies in the context of HIV, for both maternal health and for the prevention of HIV transmission to the newborn. In ART-naïve pregnant women, ART should be initiated as soon as possible, with the goal of suppressing plasma viremia throughout pregnancy. HIV testing should be performed upon confirmation of pregnancy, with testing repeated throughout pregnancy as needed for those at risk of HIV acquisition (see [Pregnancy and Postpartum HIV Testing and Identification of Perinatal and Postnatal HIV Exposure](#) in the [Perinatal Guidelines](#)).

Considerations When Initiating Antiretroviral Therapy

The ART regimens that are currently recommended as initial therapy in these guidelines (see [What to Start](#)) can suppress and maintain viral loads below the level of quantification in most people who fully adhere to their regimens. Most of the recommended regimens have a low pill burden (one or two pills once daily) and are well tolerated. People with HIV should be assessed for coexisting medical conditions and medications that may influence the selection of ART regimens or require dose adjustments or additional monitoring (see [What to Start](#)). Once started on treatment, patients must continue ART indefinitely, although future switches to alternative ART regimens may occur due to adverse events, drug–drug or drug–food interactions, cost, regimen simplification, or other reasons.

Optimizing Adherence, Antiretroviral Therapy Access, and Care Engagement

The key to successfully maintaining viral suppression is continuous access to ART and full adherence to the prescribed regimen. Lack of adherence or intermittent access to ART can result in treatment failure and the emergence of drug resistance mutations that may compromise future treatment options. While optimizing adherence and linkage to care and ensuring continuous access

are critical regardless of the timing of ART initiation, the evidence thus far indicates that drug resistance occurs more frequently in individuals who initiate therapy later in the course of infection than in those who initiate ART earlier.⁶⁸⁻⁷⁰ It is important to discuss strategies to optimize adherence, care engagement, and ART access with all patients.

Several clinical, behavioral, and social factors have been associated with poor adherence. These factors include untreated psychiatric disorders, neurocognitive impairment, substance use disorder, unstable housing, patient concerns about side effects, difficulty remembering, demanding or irregular daily routines, experiences of HIV-related stigma, and ongoing adjustment to HIV diagnosis. Clinicians should identify areas in which additional intervention is needed to improve adherence both before and after initiation of therapy. Some strategies to improve adherence are discussed in [Adherence to the Continuum of Care](#). However, mental illness, substance use disorder, and psychosocial challenges are not reasons to withhold ART from a patient. Rather, these issues indicate the need for additional interventions to support adherence, and they may influence the ART regimen that is recommended (see [What to Start](#)).

Considerations for Special Populations

Elite HIV Controllers

A small subset of people with HIV maintains plasma viral load below the level of quantification for years without ART. These individuals are often referred to as elite HIV controllers.^{71,72} [Data on the role of ART](#) and the optimal management of elite controllers are sparse, and clinicians should engage in shared decision-making with the elite controller.

Despite the absence of detectable viremia, ongoing HIV replication occurs even in HIV elite controllers. In addition, even with normal CD4 counts, elite controllers show evidence of abnormally high immune activation, [chronic inflammation, immune aging](#), and surrogate markers of atherosclerosis, which may contribute to an increased risk of non-AIDS–related diseases.^{71,73-77} One observational multisite study in the United States suggested that elite controllers not receiving ART are hospitalized more often for cardiovascular and psychiatric disease than well-controlled patients on ART.⁷⁸ [A subsequent observational study did not show an increase in the incidence of hospitalization, but the study population was younger and had a lower incidence of hepatitis C virus \(HCV\) coinfection than that of the previous study.](#)⁷⁹ In additional cohort studies of HIV elite controllers, loss of viral control, older age, and HCV coinfection were independently associated with [a higher incidence of non-AIDS–defining events.](#)⁷⁹⁻⁸²

Elite controllers with preserved CD4 counts appear to experience a decline in immune activation [and the size of the HIV reservoir](#) after ART initiation, suggesting that treatment may be beneficial.^{77,83} Whether this potential immunologic benefit of ART in elite controllers outweighs the potential risks of ART-associated adverse effects is unclear.⁷⁷ Unfortunately, given the very low prevalence of elite controllers, it is unlikely that randomized controlled trials will be able to address this question.⁸⁴ Nevertheless, there is a clear rationale for prescribing ART to some elite controllers even in the absence of detectable plasma viral load.

[Based on discussions above, the Panel strongly recommends ART for elite controllers with evidence of HIV-related complications, declining CD4 counts, intermittent detectable viral load, or comorbidities \(e.g., cardiovascular disease, cancer, hepatitis B virus/HCV coinfection\), or for those who are pregnant \(AIII\). The Panel also recommends initiation of ART for all other HIV elite controllers \(BII\). If](#)

ART is deferred, elite controllers should be followed closely, as some may experience CD4 count decline, loss of viral control, or complications related to HIV infection.

Adolescents and Young Adults With HIV

Neither the START trial nor the TEMPRANO trial included adolescents aged <18 years. The Panel's recommendation to initiate ART in all people with HIV is extrapolated to adolescents based on the expectation that they will derive benefits from early ART initiation similar to those observed in adults. However, adolescence is marked by socioemotional, cognitive, and developmental changes that can increase the risk of treatment nonengagement. Compared with adults, adolescents have demonstrated significantly lower levels of ART adherence and viral suppression, as well as higher rates of viral rebound following initial viral suppression.⁸⁵ In recent years, more adolescents have been prescribed once-daily regimens, which has increased the rate of viral suppression in this population.⁸⁶ Because youth often face psychosocial and other barriers to adherence, their ability to adhere to therapy should be carefully considered when making decisions about ART initiation. Although some adolescents and young adults may not be immediately ready to initiate therapy, clinicians should offer ART, preferably fixed-dose, once-daily regimens with a high barrier to drug resistance, alongside effective interventions to assess and address existing and potential barriers to care and adherence. To optimize the benefits of ART for youths, multidisciplinary care teams should individualize psychosocial and adherence support to adolescents and young adults (see [Adolescents and Young Adults With HIV](#)).⁸⁷

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ANTIRETROVIRAL THERAPY TO PREVENT SEXUAL TRANSMISSION OF HIV (TREATMENT AS PREVENTION)

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Panel's Recommendations
- All people with HIV should be informed that maintaining a plasma HIV RNA (viral load) of <200 copies/mL, including any measurable value below this threshold value, with antiretroviral therapy (ART) prevents sexual transmission of HIV. This concept may be recognized as Undetectable = Untransmittable or U=U (AII). - People with HIV who are starting ART should use another form of prevention with sexual partners (e.g., condoms, sexual abstinence, pre-exposure prophylaxis [PrEP] for the sexual partner who is HIV-negative) for at least the first 6 months of ART and until a viral load of <200 copies/mL has been documented (AII). Many experts would recommend confirming sustained suppression before assuming that there is no further risk of sexual transmission of HIV (AIII). - When the viral load is ≥200 copies/mL, at least one additional HIV prevention method (e.g., condoms, sexual abstinence, or PrEP for the HIV-negative partner) is recommended to prevent sexual HIV transmission until resuppression to <200 copies/mL is confirmed (AIII). - People with HIV who intend to rely on ART as prevention need to maintain high levels of ART adherence (AIII). They should be informed that transmission is possible during periods of poor adherence or treatment interruption (AIII). - At each visit for HIV care, clinicians should assess adherence to ART and counsel people with HIV on the importance of ART for their own health as well as its role in preventing sexual HIV transmission (AIII). - Providers should inform people with HIV that maintaining a viral load of <200 copies/mL does not prevent acquisition or transmission of other sexually transmitted infections (STIs) (AII). - Providers should also routinely screen all people with HIV who are sexually active for STIs, both for their own health and to prevent transmission of STIs to others (AIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Antiretroviral therapy (ART) not only reduces morbidity and mortality for people with HIV but has been definitively shown to prevent sexual transmission of the virus when the plasma HIV RNA (viral load) is consistently suppressed to <200 copies/mL, which includes any measurable viral load that is lower than this threshold value. Providers who manage people with HIV need to be aware of the data supporting treatment as prevention (TasP, which may be recognized as Undetectable = Untransmittable or U=U), its implications, and how to operationalize this prevention strategy in clinical practice. For people with HIV who intend to rely on TasP to prevent sexual transmission, providers should make an individualized assessment that considers the person's risk tolerance, personal health, history of maintaining viral suppression on treatment, and access to health care services and ART, as well as other factors that may affect their ability to maintain a high level of adherence to ART.

Evidence That Viral Load Suppression Prevents Sexual HIV Transmission

Suppressing the HIV viral load to <200 copies/mL with ART prevents sexual transmission of HIV. Observational data collected in the early 1990s from heterosexual couples demonstrated that sexual transmission from untreated people with HIV was rare at viral loads of <1,000 copies/mL to 1,500 copies/mL and that the risk of transmission increased in a dose–response fashion with increasing viral load.^{1,2} Additional reports^{3–7} and a meta-analysis⁸ supported the observation that sexual HIV transmission risk in heterosexual persons was correlated with plasma viral load, and transmission was infrequent below the lowest limits of quantification for the viral load assays used at the time (range <50 to <500 copies/mL).

The first prospective clinical trial designed specifically to address this question was HPTN 052, which randomized people with HIV with CD4 T lymphocyte cell (CD4) count between 350 cells/mm³ and 550 cells/mm³ who were in mixed–HIV status relationships (previously referred to as serodiscordant couples) to initiate ART early or to delay initiation. Initial results from this study were reported in 2011,⁹ with final results reported in 2016.¹⁰ The 2016 analysis reported that no phylogenetically linked sexual transmissions of HIV occurred among 1,763 couples who were followed a median of 5.5 years, in which the partners with HIV were on ART and had a viral load <400 copies/mL for at least 6 months. Notably, four phylogenetically linked infections occurred within the 90 days after the partner with HIV had started ART and had presumably not yet achieved viral suppression. Four other transmissions occurred after the partner with HIV experienced virologic failure. A number of transmission events occurred that were not phylogenetically linked, indicating acquisition from someone other than the partner enrolled in the study.¹¹ HPTN 052 was conducted almost exclusively among heterosexual couples who lived in Africa and Asia and did not track the number or type of sexual exposures. In addition, ART was used as an adjunct to a comprehensive prevention package that provided condoms and encouraged condom use, as well as frequent testing for HIV and other sexually transmitted infections (STIs).

Three prospective observational studies—PARTNER 1,¹² PARTNER 2,¹³ and Opposites Attract¹⁴—provided data from more diverse populations of mixed–HIV status couples in which condomless sex was common. Clinical follow-up in these studies closely mimicked that of routine clinical care. Conducted in 14 European countries (PARTNER 1 and PARTNER 2) as well as Australia, Thailand, and Brazil (Opposites Attract), the investigators followed 548 heterosexual and 1,481 male–male mixed–HIV status couples that engaged in 144,631 episodes of condomless vaginal or anal sex while the partner with HIV had a suppressed viral load on ART, defined as <200 copies/mL. In these studies, no phylogenetically linked transmissions were observed; however, as in HPTN 052, there were numerous non-phylogenetically linked transmissions attributed to partners outside the enrolled study couple relationship.

Integrating the Principles of Treatment as Prevention Into Clinical Care

The Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends that providers inform all persons with HIV that maintaining an HIV viral load <200 copies/mL with ART prevents sexual transmission of HIV **(AII)**. This information may help reduce stigma and provide additional motivation for individuals to get tested, enter into and engage in care, initiate and adhere to ART, and ultimately achieve and maintain viral suppression.¹⁵ Although the PARTNER 1, PARTNER 2, and Opposites Attract trials were designed to follow participants in the study as they would typically be followed in clinical care for HIV, the participants reported high levels of ART adherence at study entry and many reported at least 1 year of condomless sex with an established

sexual partner without transmission. As the principles of TasP are integrated into the clinical management of people with HIV who are on ART, implementation research will be critical to maximize the effectiveness of TasP in practice.

Frequency of Viral Load Assessment

The Panel has issued recommendations for viral load monitoring to manage the health of persons with HIV (see [Plasma HIV-1 RNA \[Viral Load\] and CD4 Count Monitoring](#)). However, current data are insufficient to determine whether these recommendations represent the optimal monitoring schedule for the purpose of preventing sexual transmission of HIV. In the PARTNER and Opposites Attract studies, viral loads were generally assessed every 3 to 6 months during study follow-up, usually during the course of regular HIV care. The Panel recommends a similar frequency for monitoring viral load (every 3–6 months) as in the section of the guidelines **(BII)**.

Time to Adequate Suppression After Starting Antiretroviral Therapy

A subgroup analysis from the Partners PrEP Study provided data regarding the risk of HIV transmission during and after the first 6 months on ART for the partner with HIV.¹⁶ This analysis only included the 1,573 heterosexual East African couples in which the partners without HIV were randomized to the placebo arm of the study and were tested monthly for HIV, while the viral load of the partner with HIV was assessed every 6 months. Three phylogenetically linked infections were diagnosed in the 6 months prior to the first follow-up visit for the partners with HIV. The observed incidence rate of 1.79 per 100 person-years during this initial 6-month period after the partner with HIV started ART was slightly less than the 2.08 per person-years incidence rate observed in couples in which the person with HIV was not receiving ART. Viral suppression in this study was defined as <40 copies/mL, and the three infections were diagnosed at 0 days, 56 days, and 149 days after the partner with HIV started ART and before documented viral suppression. After the partners with HIV had been taking ART for ≥6 months, no further transmissions were observed.

At this time, the Panel recommends that people with HIV who are starting ART should use another form of prevention with sexual partners (e.g., consistent condom use, sexual abstinence, pre-exposure prophylaxis [PrEP] for sexual partners without HIV) for at least the first 6 months of treatment and until a viral load of <200 copies/mL has been documented **(AII)**. Confirming sustained viral suppression is recommended before assuming that sexual transmission of HIV is no longer a risk **(AIII)**.

Adherence to Antiretroviral Therapy

Adherence to ART is paramount for persons who intend to prevent HIV transmission by achieving and maintaining a suppressed viral load. Viral rebound typically occurs within days to weeks after ART cessation and has been observed as early as 3 to 6 days after stopping **oral ARV medications**.¹⁷⁻³¹ The minimum level of adherence that is required to prevent sexual transmission has not been determined and may vary depending on the ART regimen **and the route of administration (orally vs. parenterally)**. In the key studies that defined the efficacy of TasP, adherence levels prior to study entry and during follow-up were very high. In clinical practice, most people who start ART will achieve a viral load <200 copies/mL within 6 months, but maintaining viral suppression can be a challenge for some people, especially for those who have difficulty accessing ART and other HIV care. The Centers for Disease Control and Prevention (CDC) estimates that during **2023, 76.3% of persons aged 13 years or older with HIV were engaged in clinical care, and only 67.2%** had viral loads <200 copies/mL at their most recent assessment.³² Observational cohort data have demonstrated that within the first year of starting ART, up to 10% of persons with HIV can

experience loss of viral suppression; however, the likelihood of maintaining a suppressed viral load generally improves over time. After a few years, typically no more than 5% to 10% of people on ART experience loss of viral suppression.³³⁻³⁵

Adherence can be especially challenging for certain groups, such as adolescents and young adults, people with unstable housing, people with active substance use disorder(s), and people who are justice-involved. Recommendations to help manage and maximize ART adherence can be found in [Adolescents and Young Adults With HIV](#). Persons for whom there is concern about adherence also merit counseling on how to properly use other prevention methods, especially barrier methods that prevent STIs.

The Panel recommends that persons with HIV who intend to rely upon TasP be made aware of the need for high levels of ART adherence **(AIII)**. The Panel further recommends that adherence be assessed and counseling be provided at each visit for HIV care to reinforce the importance of adherence for the individual's health as well as its role in preventing HIV transmission **(AIII)**. People with HIV should be informed that transmission is possible during periods of poor adherence or treatment interruption **(AIII)**.

Managing Transient Viremia, or “Blips”

Occasionally, some people who are highly adherent may experience intermittent or transient viremia, commonly termed “viral blips.” Blips are defined in the context of effective treatment as a single, measurable HIV RNA level, typically <200 copies/mL, that is followed by a return to a viral load below the limit of detection or quantification. With some earlier ART regimens, up to 16% of persons per year who are adherent to ART may experience a blip.³⁶⁻³⁹ Most blips likely represent normal biological fluctuation (i.e., variation around a mean undetectable viral load) or laboratory artifact and not inadequate adherence.⁴⁰⁻⁴² Persistent viremia ≥200 copies/mL has been associated with increasing risk of virologic failure^{36,39,43-46} that, in the context of TasP, can lead to increased risk of sexual transmission of HIV.¹⁰ The PARTNER and Opposites Attract studies excluded observation time when the viral load of the participant with HIV was ≥200 copies/mL. The frequency of blips <200 copies/mL was not reported in Opposites Attract; however, in PARTNER 1 and PARTNER 2, transient elevations in viral loads >50 copies/mL but <200 copies/mL were observed for 6% and 4% of the total follow-up time, respectively, during which time no phylogenetically linked infections were observed.

One of the clinical challenges with blips is that they can only be defined retrospectively once the viral load has returned to a suppressed value. The Panel recommends that any time the viral load is ≥200 copies/mL, people with HIV and their sexual partners should be counseled to use another form of prevention (e.g., condoms, pre-exposure prophylaxis for sexual partners without HIV, sexual abstinence) to protect against HIV transmission until a viral load <200 copies/mL is achieved and sustained **(AII)**. This recommendation applies both to persons who are starting ART (as noted earlier) and to those who have been taking ART and have achieved viral suppression but develop viral loads ≥200 copies/mL.

In cases where a person achieves resuppression to <200 copies/mL after a detectable viral load ≥200 copies/mL (e.g., blips, viremia), or when a person with a viral load <200 copies/mL switches regimens (e.g., for regimen simplification or to avoid certain side effects), providers should follow situation-specific viral load monitoring recommendations **(AIII)** (see [Plasma HIV-1 RNA \[Viral Load\] and CD4 Count Monitoring](#)). There are presently no data to guide how long, if at all, a person might need to continue to use another form of prevention in these two circumstances. Individualized assessment is recommended based on the length and quality of adherence and time with viral load <200 copies/mL preceding the viral load ≥200 copies/mL.

Effect of Sexually Transmitted Infections on Treatment as Prevention

The presence of STIs in a person with HIV does not appear to meaningfully alter the risk of sexual transmission when the person's viral load is <200 copies/mL. The PARTNER studies and the Opposites Attract study regularly assessed participants for STIs, which were diagnosed in 6% of heterosexual participants and 13% to 27% of men who have sex with men. Although the authors of the studies noted that their findings could not rule out the possibility that STIs in participants with viral loads <200 copies/mL might affect the risk of HIV transmission, when viewed collectively, these data suggest that any effect is very small, since STIs were common and no linked HIV transmissions were observed. The Panel recommends that patients using TasP be informed that maintaining a viral load of <200 copies/mL does not prevent acquisition or transmission of other STIs, and that it is not a substitute for condoms or behavioral modifications **(AII)**. Providers should also routinely screen all sexually active persons with HIV for STIs, both for their own health and to prevent transmission of STIs to others **(AIII)**. Refer to the CDC's [Sexually Transmitted Infections Treatment Guidelines](#) for details.

Treatment as Prevention Applies Only to Sexual Transmission of HIV

Available clinical data only support the use of TasP to prevent sexual HIV transmission in patients with viral loads <200 copies/mL. The effectiveness of this strategy in preventing transmission from blood exposure has not been determined. In addition, while suppression of maternal viral load substantially reduces the risk of perinatal transmission and transmission through breastfeeding, it does not eliminate these risks, and transmission has occurred via breastfeeding despite continuous viral suppression (refer to the [Perinatal Guidelines](#) for details).

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WHAT TO START: INITIAL COMBINATION ANTIRETROVIRAL REGIMENS FOR PEOPLE WITH HIV

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Introduction

More than 30 antiretroviral (ARV) drugs in **nine** mechanistic classes are U.S. Food and Drug Administration (FDA)–approved for treatment of HIV infection. These **nine** classes include nucleos(t)ide reverse transcriptase inhibitors (NRTIs), non-nucleoside reverse transcriptase inhibitors (NNRTIs), protease inhibitors (PIs), integrase strand transfer inhibitors (INSTIs), a fusion inhibitor, a CCR5 antagonist, a CD4 T lymphocyte (CD4) cell post-attachment inhibitor, a gp120 attachment inhibitor, **and a capsid inhibitor**. In addition, two drugs—ritonavir (RTV) and cobicistat (COBI)—are used as pharmacokinetic (PK) enhancers (or boosters) to improve the PK profiles of PIs and the INSTI elvitegravir (EVG).

Over time, incremental improvements in potency, tolerability, safety, convenience, drug interactions and genetic barriers to the emergence of drug resistance have led to streamlined recommendations for initial ARV regimens for most people with HIV. The Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) now recommends initial ARV regimens based on an oral second-generation INSTI plus two NRTIs—bictegravir (BIC)/tenofovir alafenamide (TAF)/emtricitabine (FTC) **(AI)** or dolutegravir (DTG) plus TAF/FTC or tenofovir disoproxil fumarate (TDF)/FTC or TDF/lamivudine (3TC) **(AI)**—for most people with HIV. In some people with HIV, the two-drug regimen DTG/3TC can be used **(AI)**. When INSTI resistance is possible, such as after exposure to long-acting injectable cabotegravir (CAB-LA) as pre-exposure prophylaxis (PrEP) and/or if INSTI genotype results are not yet available, a boosted PI (boosted darunavir [DRV]) in combination with two NRTIs (TAF or TDF with FTC or 3TC) is recommended **(AIII)**. The Panel's recommendations are summarized in Tables 6a and 6b.

Table 6a. Recommended Initial Regimens for Most People With HIV

Selection of antiretroviral therapy (ART) should be based on the regimen's virologic efficacy, potential adverse effects, pill burden, dosing frequency, drug–drug interaction potential, cost, access, resistance test results, and the comorbid condition of the person with HIV. A pregnancy test should be performed in women of childbearing potential, and choice of ART for pregnant women should be guided by recommendations from the [Perinatal Guidelines](#). Drug classes and regimens within each class are arranged first by evidence rating and, when ratings are equal, in alphabetical order. **Additional initial ARV regimen options for certain clinical scenarios are listed in Table 6b below.**

Table 6a. Recommended Initial Regimens for Most People With HIV

Table 6a. Recommended Initial Regimens for Most People With HIV Recommended regimens are those with demonstrated durable virologic efficacy, favorable tolerability and toxicity profiles, and ease of use. Choice of ART during pregnancy should be guided by recommendations from the Perinatal Guidelines.
For people who do not have a history of using CAB-LA as PrEP, one of the following regimens is recommended^a: • BIC/TAF/FTC (AI) • DTG plus (TAF or TDF)^b plus (FTC or 3TC) (AI) • DTG/3TC (AI), except for individuals with HIV RNA >500,000 copies/mL, HBV coinfection, or in whom ART is to be started before the results of HIV genotypic resistance testing for reverse transcriptase or HBV testing are available. For people who have a history of CAB-LA use as PrEP, INSTI genotype resistance testing should be performed before starting ART. If ART is to be started before results of genotypic testing results, the following regimen is recommended: • DRV/c^c or DRV/r with (TAF or TDF)^b plus (FTC or 3TC)—pending the results of the genotype test (AIII)
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials, observational cohort studies with long-term clinical outcomes, relative bioavailability/bioequivalence studies, or regimen comparisons from randomized switch studies; III = Expert opinion

^a Because of the current low rates of transmitted INSTI resistance in the United States, even when there is suspicion that HIV was acquired from a partner with virologic failure while on an INSTI, an INSTI-based regimen can be started pending the results of the INSTI genotype.

^b TAF and TDF are two forms of TFV approved by the U.S. Food and Drug Administration. TAF has fewer bone and kidney toxicities than TDF, while TDF is associated with lower lipid levels. Safety, cost, and access are among the factors to consider when choosing between these drugs.

^c COBI should be avoided in pregnancy because lower concentrations of COBI and DRV have been reported during the second and third trimesters. For further information, refer to the [Perinatal Guidelines](#).

Note: The following are available as coformulated drugs: BIC/TAF/FTC, DRV/c/TAF/FTC, DTG/3TC, TAF/FTC, TDF/3TC, and TDF/FTC.

Key: 3TC = lamivudine; ART = antiretroviral therapy; BIC = bictegravir; CAB-LA = long-acting cabotegravir; COBI = cobicistat; DRV = darunavir; DRV/c = darunavir/cobicistat; DRV/r = darunavir/ritonavir; DTG = dolutegravir; FTC = emtricitabine; HBV = hepatitis B virus; INSTI = integrase strand transfer inhibitor; PrEP = pre-exposure prophylaxis; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate; TFV = tenofovir

Table 6b. Other Initial Antiretroviral Regimens for Certain Clinical Scenarios

Several antiretroviral regimens are found to be effective and tolerable as initial regimens but have some disadvantages or have fewer supporting data from randomized clinical trials compared with the recommended regimens listed in Table 6a. However, one of these regimens may be preferred for an individual with HIV in certain clinical situations (also see [Table 7](#)). These regimens are listed below.

Type of Regimen	ARV Regimen	For Certain Clinical Scenarios	Other Considerations
INSTI Plus Two NRTIs	DTG/ABC/3TC (BI) (if HLA-B*5701-negative)	When concern about renal- or bone-associated AEs precludes the use of TDF or TAF	Test for HLA-B*5701 before prescribing ABC; do not prescribe if HLA-B*5701-positive. Consider avoiding ABC for people with multiple CV risk factors or known CV disease. Do not use in people with HBV coinfection unless an HBV-active drug, such as entecavir, TAF, or TDF is also used. Do not use following exposure to CAB-LA unless INSTI genotype shows sensitivity.
Boosted PI Plus Two NRTIs	(DRV/c ^a or DRV/r) plus (TAF or TDF ^b) plus (FTC or 3TC) (BI)	To avoid an INSTI-based regimen (e.g., documented INSTI resistance).	Assess for potential RTV- or COBI-related DDIs.
	(DRV/c ^a or DRV/r) plus ABC/3TC (BII) (if HLA-B*5701-negative)	To avoid an INSTI-based regimen (e.g., with suspected or documented INSTI resistance), <i>and</i> When concern about renal or bone-associated AEs precludes the use of TDF or TAF	Test for HLA-B*5701 before prescribing ABC; do not prescribe if HLA-B*5701-positive. Consider avoiding ABC for people with multiple CV risk factors or known CV disease. Do not use in people with HBV coinfection unless used with an HBV-active drug other than 3TC. Assess for potential RTV- or COBI-related DDIs.

Table 6b. Other Initial Antiretroviral Regimens for Certain Clinical Scenarios

Type of Regimen	ARV Regimen	For Certain Clinical Scenarios	Other Considerations
NNRTI Plus Two NRTIs	DOR/TDF/3TC ^b (BI) or DOR plus TAF/FTC ^b (BIII)	To avoid an INSTI-based regimen (e.g., with suspected or documented INSTI resistance), <i>and</i> To avoid a PI-based regimen (e.g., with significant DDIs with concomitant medications)	
	RPV/TAF/FTC (BII) Only if HIV RNA <100,000 copies/mL and CD4 count >200 cells/mm ³	To avoid an INSTI-based regimen (e.g., with suspected or documented INSTI resistance), <i>and</i> To avoid a PI-based regimen (e.g., with significant DDIs with concomitant medications), <i>and</i> When a single-tablet regimen containing an NNRTI and TAF is desired	Cannot take with PPI; space apart from H ₂ antagonist. Needs to be taken with a meal.
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials, observational cohort studies with long-term clinical outcomes, relative bioavailability/bioequivalence studies, or regimen comparisons from randomized switch studies; III = Expert opinion			

^a COBI should be avoided in pregnancy because lower concentrations of COBI and DRV have been observed during the second and third trimesters. For further information, refer to the [Perinatal Guidelines](#).

^b TAF and TDF are two forms of TFV approved by the U.S. Food and Drug Administration. TAF has fewer bone and kidney toxicities than TDF, while TDF is associated with lower lipid levels. Safety, cost, and access are among the factors to consider when choosing between these drugs.

Note: The following are available as coformulated drugs: ABC/3TC, DOR/TDF/3TC, DRV/c, DRV/c/TAF/FTC, DTG/ABC/3TC, RPV/TAF/FTC, TAF/FTC, TDF/3TC, and TDF/FTC.

Key: 3TC = lamivudine; ABC = abacavir; AE = adverse event; ARV = antiretroviral; CD4 = CD4 T lymphocyte; COBI = cobicistat; CV = cardiovascular; DDI = drug–drug interaction; DOR = doravirine; DRV/c = darunavir/cobicistat; DRV/r = darunavir/ritonavir; DTG = dolutegravir; FTC = emtricitabine; H₂ = histamine type 2; HBV = hepatitis B virus; HLA = human leukocyte antigen; INSTI = integrase strand transfer inhibitor; NNRTI = non-nucleoside reverse transcriptase inhibitor; NRTI = nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; PPI = proton pump inhibitor; RPV = rilpivirine; RTV = ritonavir; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate; TFV = tenofovir

Supporting Evidence and Rationale Used for the Panel's Recommendations

The Panel's recommendations are primarily based on clinical trial data published in peer-reviewed journals and on data prepared by drug manufacturers for FDA review. In select cases, the Panel considers data from abstracts presented at major scientific meetings. The Panel considers published

information from randomized prospective clinical trials with adequate sample size that demonstrate that an ARV regimen produces high rates of viral suppression, increases CD4 count, and has a favorable safety profile to be the strongest evidence on which to base recommendations. Comparative clinical trials of initial treatments generally are not designed to show significant differences in HIV-related clinical endpoints (such as progression to AIDS-defining conditions) or survival. Thus, the assessment of regimen efficacy and safety is primarily based on surrogate marker endpoints (i.e., rates of HIV RNA suppression) and the incidence and severity of adverse events.

In some instances, the Panel recommends regimens that include ARV drugs approved by the FDA based on bioequivalence or relative bioavailability studies that demonstrate that the exposure of the drug(s) in the new formulation or combination is comparable to the exposure of a reference drug(s) that has demonstrated safety and efficacy in randomized clinical trials. When developing recommendations, the Panel may also consider data from randomized switch studies in which a drug in an ARV initial regimen that suppressed patients' viral loads is replaced by a new drug from the same class. Switch trials do not evaluate the ability of a drug or regimen to induce viral suppression; they only examine the drug or regimen's ability to maintain suppression. Therefore, results from switch trials may not be directly applicable to the selection of an initial regimen and should be considered in conjunction with other data, including data from bioavailability/bioequivalence studies and from trials conducted in people taking their first ARV treatment regimen. In this section of the guidelines, the definition of an evidence rating of **II** is expanded to include supporting data from bioavailability/bioequivalence studies or randomized switch studies.

When developing recommendations, the Panel also considers tolerability and toxicity profiles, pill burden and dosing frequency, drug interaction potential, cost and access, postmarketing safety data, observational cohort data published in peer-reviewed publications, the experience of clinicians who are actively engaged in patient care, and the views of community members.

The Panel reviewed the available data to arrive at two classifications for initial ARV treatment regimens: (1) *Recommended Initial Regimens for Most People With HIV* (Table 6a) and (2) *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* (Table 6b). *Recommended Initial Regimens for Most People With HIV* are those regimens with demonstrated durable virologic efficacy, favorable tolerability and toxicity profiles, and ease of use. The Panel also recognizes that in certain clinical situations, other regimens may be preferred; these options are included in Table 6b. See Table 7 for examples of clinical scenarios in which certain drugs in these regimens may be particularly advantageous.

Many other ARV regimens are effective for initial therapy but have disadvantages when compared with the regimens listed in Tables 6a and 6b. These disadvantages include greater toxicity, higher pill burden, less supporting data from large comparative clinical trials, and limitations for use in certain populations. These regimens are no longer included in Tables 6a and 6b. For people with HIV who have a suppressed viral load and are not experiencing any adverse effects while on a regimen that is not listed, changing to a regimen listed in Table 6a or 6b is not necessary. Clinicians should refer to [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#) for further guidance if switching to a new regimen is desired.

Several tables in these guidelines provide clinicians with guidance on selecting and prescribing an optimal regimen for a person with HIV. Table 7 provides information on considerations based on specific clinical scenarios and ARV drug characteristics. Table 9 lists the potential advantages and disadvantages of the different ARV drug components. Appendix A, Tables 3–10, lists characteristics of individual ARV agents for initial therapy (e.g., formulations, dosing recommendations, PK, common adverse effects). Appendix B provides ARV dosing recommendations for people who have renal or hepatic insufficiency.

Changes Since the Last Revision of the Guidelines

Since the last revision of these guidelines, the Panel has made several important changes to the recommendations for initial therapy in people with HIV. Among these changes, the following deserve emphasis:

Changes in the Panel's Recommendations for Initial Regimens in Most People With HIV:

- The single-tablet regimen (STR) DTG/abacavir (ABC)/3TC has moved from *Recommended Initial Regimens for Most People With HIV* ([Table 6a](#)) to *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* ([Table 6b](#)) due to the necessity of obtaining an HLA-B*5701 assay to avoid ABC-associated hypersensitivity, data suggesting increased risk for cardiovascular events associated with ABC, and the availability of several other options for initial therapy (see the [Nucleoside Reverse Transcriptase Inhibitor Options as Part of Initial Therapy](#) section for more information).
- Because of the current low rates of transmitted INSTI resistance in the United States, even when there is suspicion that HIV was acquired from a partner with virologic failure while on an INSTI, an INSTI-based regimen can be started pending the results of the INSTI genotype test.

No Longer Recommended as Initial Antiretroviral Regimens:

- Raltegravir (RAL)-, elvitegravir/cobicistat (EVG/c)-, boosted atazanavir (ATV)-, and efavirenz (EFV)-based regimens and rilpivirine (RPV)/TDF/FTC are no longer recommended as initial therapy due to the following disadvantages compared with other regimens:
 - o RAL: Low genetic barrier to resistance and higher pill burden
 - o EVG/c: Drug–drug interactions with COBI and low genetic barrier to resistance of EVG
 - o Boosted ATV: Toxicities, including hyperbilirubinemia, and drug–drug interactions
 - o EFV: Toxicities, including neuropsychiatric effects and suicidality; low barrier to resistance; and drug–drug interactions
 - o RPV/TDF/FTC: Availability of doravirine (DOR)/TDF/3TC as an STR, more drug–drug interactions than DOR, food restrictions, and HIV RNA and CD4 restrictions. RPV/TAF/3TC remains an option for those who wish to use an STR containing RPV and TAF
- The two-drug regimens darunavir/ritonavir (DRV/r) plus RAL and DRV/r plus 3TC are no longer recommended for initial therapy.
 - o DRV/r plus RAL is not recommended due to the higher rate of virologic failure in people with HIV RNA >100,000 copies/mL, the low genetic barrier to resistance, and the high pill burden of RAL.¹⁻³
 - o DRV/r plus 3TC is not recommended because it requires a PK booster with accompanying drug interactions, and published clinical trial data are lacking.⁴ For further information, see [Other Antiretroviral Regimens for Initial Therapy](#).

Selecting an Initial Antiretroviral Regimen

The goal of ART is to improve health and prolong life for people with HIV and to prevent the transmission of HIV to others by maximizing virologic suppression. This is achieved by initiating therapy with a potent, safe, tolerable, and easy-to-adhere-to ARV regimen as soon as possible after diagnosis. [Table 6a](#) provides a list of Panel-recommended ARV regimens for most people with HIV. Some factors listed below may influence the selection of a regimen. See [Table 6b](#) and [Table 7](#) for additional regimen recommendations to use in specific clinical scenarios.

Initial Characteristics to Consider in All People With HIV

- Pre-treatment HIV RNA level (viral load)
- Pre-treatment CD4 count
- History of prior exposure to CAB-LA or oral TDF/FTC or TAF/FTC as PrEP, or use of INSTI-based post-exposure prophylaxis (PEP)
- Suspected drug resistance (prior to availability of genotypic testing results)
- HIV genotypic drug resistance test results
 - Genotypic drug resistance testing in people without prior ARV exposure should focus on testing for mutations in the reverse transcriptase and protease genes.
 - If transmitted INSTI resistance is a concern, providers should also test for resistance mutations to this class of drugs.
- Individual preferences
- Anticipated adherence to the regimen

Presence of Specific Conditions

- Comorbid conditions: Cardiovascular disease; hyperlipidemia; renal disease; liver disease; osteopenia, osteoporosis, or other conditions associated with bone mineral density loss; psychiatric illness; neurologic disease; substance use disorder requiring narcotic replacement therapy
- Coinfections: Hepatitis B virus (HBV), hepatitis C virus, tuberculosis
- Pregnancy and potential for pregnancy: See below in [General Considerations for Women of Childbearing Potential Initiating Antiretroviral Therapy](#)

Regimen-Specific Considerations

- Regimen's barrier to resistance
- HLA-B*5701 status (only for those considering initiation of ABC). Those who are HLA-B*5701 positive should not receive ABC. See [Table 7](#) for regimens to initiate.
- Potential adverse effects and drug toxicities, including risk for the development of comorbid diseases
- Known or potential drug interactions with other medications (see [Drug–Drug Interactions](#))
- Convenience (e.g., pill burden, dosing frequency, availability of a fixed-dose combination or of STR formulations, food requirements)
- Cost and access (see [Cost Considerations and Antiretroviral Therapy](#))

Considerations for People With Prior Use of Pre- or Post-Exposure Prophylaxis

- For people who acquired HIV after taking oral PrEP with TAF/FTC or TDF/FTC, there is concern for transmitted resistance, especially related to FTC or 3TC with the M184V/I mutation. In a cohort of people in New York City with newly diagnosed HIV, 2% had the M184V/I mutation. People who had used prior oral PrEP were four to seven times more likely to harbor resistance

than people who had never used oral PrEP.⁵ This reinforces the Panel's recommendation not to initiate DTG/3TC without the results of genotypic resistance testing.

- For people who acquired HIV after exposure to CAB-LA as PrEP, there is concern for INSTI resistance. CAB-LA may remain detectable after treatment discontinuation for up to 3 years in men and 4 years in women.⁶ This long PK tail may contribute to the selection of drug-resistant variants in the setting of incident infection. In the HPTN 083 trial of CAB-LA as PrEP in cisgender men and transgender women, 10 of 32 people who acquired HIV after exposure to CAB-LA were found to harbor major INSTI resistance mutations.⁷ This is the basis for the Panel's recommendation to obtain INSTI genotypic drug resistance testing in people with prior CAB-LA exposure and not to initiate INSTIs before these results are available and show INSTI sensitivity. If therapy is initiated before INSTI sensitivity is confirmed, the Panel recommends initiating a boosted DRV regimen with TAF/FTC or TDF/FTC while awaiting INSTI genotype results; once INSTI sensitivity is confirmed by genotypic resistance testing, a switch to an INSTI-based regimen can occur **(AIII)**.
- For people with no history of using CAB-LA for PrEP and who acquire HIV despite INSTI-based PEP use, an INSTI genotype should be obtained prior to beginning an INSTI-based regimen. However, because selection of INSTI-resistant virus is likely to be uncommon in this setting, an INSTI-based regimen could be started prior to the return of genotype results **(CIII)**. This recommendation is based largely on theoretical concerns that INSTI resistance could occur in the setting of INSTI-based PEP failure, but it is likely to be uncommon. In addition, transmitted INSTI resistance remains low in the United States. For these reasons, the Panel supports using INSTI-based regimens in this setting.

General Considerations for Women of Childbearing Potential Initiating Antiretroviral Therapy

- A pregnancy test should be performed before initiating ART.
- Clinicians should discuss intentions regarding pregnancy with all women of childbearing potential.
- People with HIV should attain maximum viral suppression before attempting conception in order to protect their own health, prevent sexual HIV transmission to partners without HIV, and minimize the risk of perinatal HIV transmission to the infant.
- For individuals who are trying to conceive, the Panel recommends initiating a regimen designated as a *Preferred* regimen during pregnancy, as detailed in the [Perinatal Guidelines](#).

General Considerations for INSTI-, PI-, or NNRTI-Based Regimens

Except when HIV is acquired after exposure to CAB-LA as PrEP and results from INSTI genotypic resistance testing are not available, INSTIs (specifically BIC and DTG) are recommended for initial therapy in most people because of their demonstrated efficacy, high barrier to resistance, tolerability, low potential for drug–drug interactions, convenience, and better adverse effects profile compared with NNRTIs and boosted PIs (see [Tables 7](#) and [9](#)).

INSTI-Based Regimens

The Panel's *Recommended Initial Regimens for Most People With HIV*, as listed in [Table 6a](#), include one of two INSTIs: BIC coformulated with TAF/FTC **(AI)**; or DTG plus TAF or TDF plus FTC or 3TC **(AI)**; or coformulated DTG/3TC **(AI)** for people who have not had exposure to CAB-LA as PrEP. In those with prior exposure to CAB-LA as PrEP, these regimens should not be initiated unless a

recent genotype test result showing no INSTI resistance mutations is available. If an INSTI-based regimen is initiated and viral suppression is not achieved in 8 to 12 weeks, genotypic resistance testing should be repeated, including for INSTIs.

For most people, these INSTI-containing regimens will be highly effective and have relatively infrequent treatment-limiting adverse effects and few drug interactions. In several head-to-head comparisons between boosted PI- and INSTI-containing regimens, the INSTI-based regimens were better tolerated and caused fewer treatment discontinuations.⁸⁻¹⁰

The Panel recommends a two-drug regimen of DTG/3TC for initial therapy if certain criteria are met. Data from two randomized trials showed that, in terms of virologic efficacy, DTG plus 3TC was non-inferior to a three-drug regimen of DTG plus TDF/FTC. No treatment-emergent resistance was seen in either the two-drug or the three-drug group.¹¹ Based on these data, DTG/3TC is not currently recommended for rapid ARV initiation as initial therapy before the availability of HBV serology, HIV RNA level, and an HIV genotypic test demonstrating sensitivity to 3TC.

Among the INSTI-based regimens, BIC- and DTG-containing regimens have a higher barrier to resistance than the first-generation INSTI-based regimens containing EVG or RAL and do not require a PK booster. Transmitted resistance to BIC or DTG is rare. Treatment-emergent resistance has been reported in individuals who failed three-drug DTG-based therapy¹²⁻¹⁵ and BIC-based regimens.¹⁶⁻¹⁹ Because of this high barrier to resistance and tolerability, BIC- and DTG-containing regimens may be used in people who have not previously used CAB-LA for PrEP and plan to start ART before resistance testing results are available (e.g., with rapid initiation of ART after diagnosis). BIC-based regimens have been shown to be non-inferior to DTG-based regimens in clinical trials.²⁰⁻²²

There are data suggesting greater weight gain after initiating therapy with certain INSTI-based regimens and TAF than with other ARV drugs. The reasons for differences in weight gain are unclear and should not be a reason to withhold an INSTI- or TAF-based regimen.²³⁻³⁰

EVG- and RAL-based regimens have the disadvantage of having lower barriers to resistance than DTG- or BIC-containing regimens and therefore are no longer recommended as initial therapy. Additionally, the pill burden with RAL is higher than for other INSTI-based regimens whereas EVG-based regimens have a greater potential for drug interactions because EVG is combined with COBI, a strong cytochrome P3A4 inhibitor (see [Table 7](#)).

PI-Based Regimens

In the setting of HIV acquisition following CAB-LA exposure, an RTV- or COBI-boosted DRV regimen containing TAF or TDF plus FTC or 3TC is recommended as initial therapy if a genotypic drug resistance test indicating INSTI sensitivity is not available at the time of ART initiation (**AIII**), if resistance to INSTI is confirmed (**BI**), or in certain clinical situations, such as when INSTIs must be avoided (**BI**). DRV/c/TAF/FTC is available as an STR. Large observational cohorts found an association between some PIs, including DRV/r, and an increased risk of cardiovascular events; however, further study is needed.³¹⁻³⁶ COBI-boosted regimens should not be initiated during pregnancy because of inadequate drug levels. Boosted ATV is no longer recommended as initial therapy due to frequent adverse events (e.g., hyperbilirubinemia) and high rates of drug–drug interactions, including with tenofovir and acid-suppressive therapy.

Boosted ATV is no longer recommended as initial therapy due to frequent adverse events, such as hyperbilirubinemia and higher rates of drug–drug interactions (including with tenofovir and acid-suppressive therapy) compared to boosted DRV.

NNRTI-Based Regimens

NNRTI-based regimens are not recommended for initial therapy in most people with HIV, but selected regimens may be useful in some circumstances. The NNRTI-based regimens that are currently recommended by the Panel to be used in certain clinical scenarios include DOR/TDF/3TC **(BI)**, DOR plus TAF/FTC **(BIII)**, and RPV/TAF/FTC **(BII)**. The emergence of drug resistance at the time of virologic failure has been reported with all NNRTIs, which generally have a lower barrier to resistance than INSTIs or boosted PIs.

DOR is available both as a single-drug tablet to be used with two NRTIs and as part of an STR with TDF/3TC. In randomized trials, DOR was non-inferior to both EFV and DRV/r when either of these drugs was taken in combination with two NRTIs,^{37,38} but DOR has not yet been compared against INSTIs. DOR has fewer central nervous system (CNS) side effects than EFV and more favorable lipid effects than both DRV/r and EFV. DOR also has fewer potential drug interactions than EFV or RPV, and unlike RPV, the virologic efficacy of DOR is not compromised in people with high HIV RNA levels and low CD4 counts.

RPV has fewer adverse effects than EFV, and RPV/TAF/FTC is available as one of the smallest tablet sizes among STRs. However, RPV has lower virologic efficacy in people with baseline HIV RNA levels >100,000 copies/mL and CD4 counts <200 cells/mm³ and is subject to numerous drug–drug interactions.³⁹ EFV is no longer recommended for initial therapy due to a relatively high rate of CNS-related side effects, reported suicidality, high rates of drug discontinuation, and numerous drug–drug interactions.

Regimens When Abacavir, Tenofovir Alafenamide, and Tenofovir Disoproxil Fumarate Cannot Be Used or Are Not Optimal

In people in whom ABC, TDF, or TAF cannot be used or are not optimal, only DTG/3TC is recommended by the Panel **(AI)**. Several other NRTI-sparing/limiting two-drug regimens have been evaluated in clinical trials but are not currently recommended for initial therapy due to insufficient data. For more information on these regimens, see [Other Antiretroviral Regimens for Initial Therapy](#). Two-drug ARV options **should not be used** in pregnancy due to insufficient data, or in those with known pre-existing resistance to any of the ARVs in the combination. Tenofovir-sparing regimens **should not be used** in individuals with HBV coinfection unless a drug with HBV activity (i.e., entecavir) is also used.

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Table 7. Antiretroviral Regimen Considerations for Initial Therapy Based on Specific Clinical Scenarios

Updated: September 12, 2024

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This table guides clinicians in choosing an initial antiretroviral (ARV) regimen according to various patient and regimen characteristics and specific clinical scenarios. ARV drugs/regimens that are listed in [Table 6a](#) as *Recommended Initial Regimens for Most People With HIV* and in [Table 6b](#) as *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* are included in this table (see [Initial Combination Antiretroviral Regimens for People With HIV](#)). When more than one scenario applies to a person with HIV, clinicians should review considerations for each relevant scenario and use their clinical judgment to select the most appropriate regimen. Please see [Table 9](#) for additional information regarding the advantages and disadvantages of particular ARV medications recommended to be used as initiation therapy.

Patient or Regimen Characteristics	Clinical Scenario	Consideration(s)	Rationale/Comments
Pre-ART Characteristics	CD4 count <200 cells/mm ³	Do not use RPV-based regimens.	Higher rates of virologic failure have been observed in those with low pre-treatment CD4 counts.
	HIV RNA >100,000 copies/mL (also see next row if HIV RNA >500,000 copies/mL)	Do not use RPV-based regimens.	Higher rates of virologic failure have been observed in those with high pre-treatment HIV RNA levels.
	HIV RNA >500,000 copies/mL	Do Not Use the Following Regimens: • RPV/TAF/FTC • DTG/3TC	For DTG/3TC, limited data are available in patients with viral loads above this threshold.
	HLA-B*5701 positive or result unknown	Do not use ABC-containing regimens.	ABC hypersensitivity is a potentially fatal reaction that is highly associated with the HLA-B*5701 allele.
	Prior exposure to oral TDF/(3TC or FTC) or TAF/3TC PrEP	Use DTG or BIC plus two NRTIs. DTG/3TC could be considered if testing confirms no 3TC resistance mutations.	DTG/3TC should be avoided if resistance testing results are not available, as presence of 3TC resistance mutations may lead to use of DTG monotherapy.

Table 7. Antiretroviral Regimen Considerations for Initial Therapy Based on Specific Clinical Scenarios

Patient or Regimen Characteristics	Clinical Scenario	Consideration(s)	Rationale/Comments
	Prior exposure to CAB-LA for PrEP	INSTI genotype resistance testing should be performed. If INSTI Resistance Is Present or If ART Needs to Be Started Before Genotype Test Results • (DRV/r or DRV/c) plus (TAF or TDF)^a plus (3TC or FTC) If No INSTI Resistance Is Identified • BIC/TAF/FTC, or • DTG plus (TAF or TDF)^a plus (3TC or FTC)	Mutations conferring resistance to INSTIs have been seen in association with CAB-LA PrEP. CAB-LA has a very long half-life, and drug exposure may persist at levels suboptimal to prevent infection and may select for INSTI-resistant virus.
	People with no prior exposure to CAB-LA for PrEP and ARV regimen should be started rapidly and before HIV drug resistance results are available.	Avoid ABC, DTG/3TC, and NNRTI-based regimens. Use • BIC/TAF/FTC, or • DTG plus (TAF or TDF)^a plus (3TC or FTC) In People Who Used INSTI-Based ART for PEP or Who Are Suspected to Have Acquired HIV From Someone Failing an INSTI-Based Regimen • Obtain INSTI genotypic resistance test and start one of the following regimens: ◦ BIC/TAF/FTC, or ◦ DTG plus (TAF or TDF)^a plus (3TC or FTC)	Transmitted mutations conferring NNRTI and NRTI resistance are more likely than mutations associated with PI or INSTI resistance. HLA-B*5701 results may not be available rapidly; thus, ABC is not recommended. Because of the current low rates of transmitted INSTI resistance in the United States, even when there is suspicion that HIV was acquired from a partner with virologic failure while on an INSTI, an INSTI-based regimen can be started, pending the results of the INSTI genotype.

Table 7. Antiretroviral Regimen Considerations for Initial Therapy Based on Specific Clinical Scenarios

Patient or Regimen Characteristics	Clinical Scenario	Consideration(s)	Rationale/Comments
ART-Specific Characteristics	A one-pill, once-daily regimen is desired.	STR Options as Initial ART Include the Following: • BIC/TAF/FTC • DOR/TDF/3TC • DRV/c/TAF/FTC • DTG/ABC/3TC • DTG/3TC • RPV/TAF/FTC	Do not use DTG/ABC/3TC if the patient is HLA-B*5701 positive. DTG/3TC is not recommended if HIV RNA is >500,000 copies/mL. Do not use DTG/ABC/3TC or DTG/3TC in the setting of HBV coinfection without another HBV agent. Do not use RPV/TAF/FTC if HIV RNA is >100,000 copies/mL and CD4 count is <200 cells/mm³.
	Food effects	Regimens That Can Be Taken Without Regard to Food • BIC-, DOR-, or DTG-based regimens	Oral bioavailability of these regimens is not significantly affected by food.
		Regimens That Should Be Taken With Food • DRV/r- or DRV/c-based regimens • RPV/TAF/FTC	Food improves absorption of these regimens. RPV-containing regimens should be taken with ≥390 calories of food.
Presence of Other Conditions	Chronic kidney disease (defined as CrCl <60 mL/min)	In general, avoid TDF. For patients with progressively declining renal function, consider avoiding all TFV-containing (TAF or TDF) regimens. Refer to Appendix B for specific ARV drug dosing recommendations in patients with renal impairment.	TDF has been associated with proximal renal tubulopathy. Higher rates of renal dysfunction have been reported in patients using TDF in conjunction with RTV-containing regimens. TAF has less impact on renal dysfunction than TDF. Avoid the use of TDF- or TAF-sparing regimens in the setting of HBV coinfection or unknown HBV status unless also receiving a fully active HBV regimen (see Hepatitis B Virus/HIV Coinfection).

Table 7. Antiretroviral Regimen Considerations for Initial Therapy Based on Specific Clinical Scenarios

Patient or Regimen Characteristics	Clinical Scenario	Consideration(s)	Rationale/Comments
	Liver disease with cirrhosis	Some ARVs are contraindicated or may require dosage modification in patients with Child-Pugh class B or C disease.	Refer to Appendix B for specific dosing recommendations. Patients with cirrhosis should be carefully evaluated by an expert in advanced liver disease.
	Concern for weight gain	For many people with HIV, gaining weight after starting ART is part of a “return to health.” However, some ARV regimens are associated with greater weight increase than others.	Reasons for differences in weight gain among ART regimens are unknown. Note: Weight gain should not be a reason to avoid taking an INSTI-based regimen.
	Osteoporosis	Avoid TDF.^a	TDF is associated with decreases in BMD, along with renal tubulopathy, urine phosphate wasting, and resultant osteomalacia. TAF ^a and ABC are associated with smaller declines in BMD than TDF.
	Psychiatric illnesses	Consider avoiding RPV-based regimens. Patients on INSTI-based regimens who have preexisting psychiatric conditions should be closely monitored. Some ARVs are contraindicated, and some psychiatric medications need dose adjustments when coadministered with certain ARVs.	RPV can exacerbate psychiatric symptoms and may be associated with suicidality. Some INSTIs have been associated with adverse neuropsychiatric effects in some retrospective cohort studies and case series. See the drug–drug interaction tables (Tables 24a , 24b , 24d , and 24g) for dosing recommendations when drugs used for psychiatric illnesses are used with certain ARVs.
	Cardiac QTc interval prolongation	Consider avoiding RPV-based regimens if the patient is taking other medications with known risk of Torsades de Pointes or in patients at higher risk of Torsades de Pointes.	High RPV concentrations may cause QTc prolongation.

Table 7. Antiretroviral Regimen Considerations for Initial Therapy Based on Specific Clinical Scenarios

Patient or Regimen Characteristics	Clinical Scenario	Consideration(s)	Rationale/Comments
	High risk for CV events	Consider avoiding ABC-based regimens. Refer to Hyperlipidemia, below, for regimens associated with more favorable lipid profiles.	An increased risk of CV events with ABC has been observed in some, but not all, studies. Certain ARV regimens are associated with more favorable lipid profiles than other regimens.
	Hyperlipidemia	PI/c and PI/r have been associated with hyperlipidemia. BIC, DOR, DTG, and RPV have fewer lipid effects.	TDF has been associated with lower lipid levels than ABC or TAF.
	Patients with history of poor adherence to non-ARV medications or inconsistent engagement in care	Consider using regimens with a boosted PI or BIC or DTG.	These regimens have a high genetic barrier to resistance.
	Pregnancy	Refer to the Perinatal Guidelines for further guidance on ARV use during pregnancy.	
Presence of Coinfections	HBV infection	Avoid regimens that do not contain NRTIs. Use (TDF or TAF) with (FTC or 3TC) as part of the ARV regimen. If TDF and TAF Are Contraindicated For treatment of HBV, use FTC or 3TC with entecavir and a suppressive ARV regimen (see Hepatitis B Virus/HIV Coinfection).	TDF, TAF, FTC, and 3TC are active against both HIV and HBV. 3TC- or FTC-associated HBV resistance mutations can emerge when these drugs are used without another drug that is active against HBV.
	HCV treatment required	Refer to recommendations in Hepatitis C Virus/HIV Coinfection , with special attention to potential interactions between ARV drugs and HCV drugs.	

Table 7. Antiretroviral Regimen Considerations for Initial Therapy Based on Specific Clinical Scenarios

Patient or Regimen Characteristics	Clinical Scenario	Consideration(s)	Rationale/Comments
	Concomitant use with rifamycin antibiotics (e.g., rifabutin, rifampin, and rifapentine)	Recommended regimens may require dose adjustment. See the drug–drug interaction tables (Tables 24a , 24b , 24c , 24d , 24e , 24f , 24g , 25a , and 25b) and Tuberculosis/HIV Coinfection for information on ARV use with rifamycin antibiotics.	Rifamycin antibiotics are inducers of CYP3A4 and UGT1A1 enzymes, causing significant decreases in concentrations of PIs, INSTIs, and RPV.

^a TAF and TDF are two U.S. Food and Drug Administration–approved forms of TFV. TAF has fewer bone and kidney toxicities than TDF, whereas TDF is associated with lower lipid levels. Safety, cost, and access are among the factors to consider when choosing between these drugs.

Key: 3TC = lamivudine; ABC = abacavir; ART = antiretroviral therapy; ARV = antiretroviral; BIC = bictegravir; BMD = bone mineral density; **CAB-LA = long-acting cabotegravir**; CD4 = CD4 T lymphocyte; CrCl = creatinine clearance; CV = cardiovascular; CYP = cytochrome P; DOR = doravirine; DRV/c = darunavir/cobicistat; DRV/r = darunavir/ritonavir; DTG = dolutegravir; FTC = emtricitabine; HBV = hepatitis B virus; HCV = hepatitis C virus; HLA = human leukocyte antigen; INSTI = integrase strand transfer inhibitor; NNRTI = non-nucleoside reverse transcriptase inhibitor; NRTI = nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; PI/c = cobicistat-boosted protease inhibitor; PI/r = ritonavir-boosted protease inhibitor; **PEP = post-exposure prophylaxis**; **PrEP = pre-exposure prophylaxis**; QTc = QT corrected for heart rate; RPV = rilpivirine; RTV = ritonavir; STR = single-tablet regimen; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate; TFV = tenofovir; UGT = uridine diphosphate glucuronosyltransferase

Nucleoside Reverse Transcriptase Inhibitor Options as Part of Initial Therapy

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The following sections provide detailed information on antiretroviral (ARV) drugs that the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends for initial therapy for most people with HIV and for initial therapy in certain clinical scenarios (see [Tables 6a and 6b](#) in the [Initial Combination Antiretroviral Regimens for People With HIV](#) section), including ARV drug characteristics, adverse effects, clinical trial results, and Panel recommendations on their use.

Table 8a. Characteristics of Nucleoside Reverse Transcriptase Inhibitor Options for People Without Prior Antiretroviral Treatment

Note: Listed in order of the Panel's recommendations in Tables 6a and 6b.

Characteristics	TAF/FTC	TDF/FTC	TDF/3TC	3TC	ABC/3TC
Dosing Frequency	Once daily	Once daily	Once daily	Once daily	Once daily
Available Coformulations for People Without Prior ARV Treatment	• TAF 25 mg/FTC • BIC/TAF 25 mg/FTC • DRV/c/TAF 10 mg/FTC • RPV/TAF 25 mg/FTC	• TDF/FTC	• TDF/3TC • DOR/TDF/3TC	• DTG/3TC	• ABC/3TC • DTG/ABC/3TC
Adverse Effects	TAF • Renal insufficiency, proximal renal tubulopathy (less frequent than with TDF) • Decrease in BMD (less than with TDF)	TDF • Renal insufficiency, proximal renal tubulopathy • Decrease in BMD • Renal and bone toxicity are exacerbated by pharmacologic boosters.	TDF • Renal insufficiency, proximal renal tubulopathy • Decrease in BMD • Renal and bone toxicity are exacerbated by pharmacologic boosters.	3TC • No notable adverse effects	ABC • HSR to ABC is associated with the presence of HLA-B*5701 allele.^b • Increase in CV events is associated with ABC use in some but not all cohort studies.

Table 8a. Characteristics of Nucleoside Reverse Transcriptase Inhibitor Options for People Without Prior Antiretroviral Treatment

Characteristics	TAF/FTC	TDF/FTC	TDF/3TC	3TC	ABC/3TC
Other Considerations	- Also used for HBV treatment. Discontinuation may precipitate HBV flare. - See Appendix B for dosing recommendations in people with renal insufficiency. - Some studies reported less weight gain and lower LDL, HDL, TC, and triglycerides with TDF than with TAF.			3TC or ABC/3TC should not be used as treatment for HBV without adding another HBV-active drug.	

^a 3TC is recommended for use with DTG in some people as initial ART. See [Table 6a](#) and the discussion below for more information. Otherwise, dual-NRTI backbones are recommended.

^b Perform HLA-B*5701 testing before initiating ABC; if result is positive, do not start ABC and add ABC to patient's allergy list. See the [HLA-B*5701 Screening](#) section for more information.

Key: 3TC = lamivudine; ABC = abacavir; ART = antiretroviral therapy; ARV = antiretroviral; BIC = bictegravir; BMD = bone mineral density; CV = cardiovascular; DOR = doravirine; DRV = darunavir; DRV/c = darunavir/cobicistat; DTG = dolutegravir; FTC = emtricitabine; HBV = hepatitis B virus; HDL = high-density lipoprotein; HLA = human leukocyte antigen; HSR = hypersensitivity reaction; LDL = low-density lipoprotein; NRTI = nucleoside reverse transcriptase inhibitor; RPV = rilpivirine; TAF = tenofovir alafenamide; TC = total cholesterol; TDF = tenofovir disoproxil fumarate

Summary

The U.S. Food and Drug Administration (FDA)–approved nucleos(t)ide reverse transcriptase inhibitors (NRTIs) include zidovudine (ZDV), stavudine (d4T), didanosine (ddI), abacavir (ABC), tenofovir disoproxil fumarate (TDF), tenofovir alafenamide (TAF), lamivudine (3TC), and emtricitabine (FTC). Older NRTIs (ZDV, d4T, ddI) are no longer recommended for use in clinical practice in the United States because of high rates of serious toxicities. All of these drugs can cause mitochondrial toxicity that may lead to myopathy, hepatic steatosis, lactic acidosis, or lipoatrophy; ZDV may cause bone marrow suppression, whereas peripheral neuropathy is commonly seen with ddI or d4T use. The incidence of these complications is substantially lower with 3TC, FTC, ABC, TDF, and TAF than with older NRTIs.^{1,2}

TAF/FTC, TDF/3TC, and TDF/FTC are NRTI combinations that are part of *Recommended Initial Regimens for Most People With HIV* (see [Table 6a](#)). In addition, 3TC may be used as a single NRTI with dolutegravir (DTG), except in individuals with HIV RNA >500,000 copies/mL or in whom antiretroviral therapy (ART) is to be started before the results of HIV RNA and HIV genotypic resistance tests are available. If used in those with hepatitis B virus (HBV) coinfection, another HBV-active drug should be added. [Tables 6a](#) and [6b](#) provide recommendations and ratings for the individual regimens. These recommendations are based on the virologic potency and durability, short- and long-term toxicity, and dosing convenience of these drugs. TDF has been associated with bone and kidney toxicities, especially when used with a pharmacologic booster.³ TAF is less likely to cause kidney and bone toxicities than TDF, whereas TDF is associated with lower lipid levels than TAF. Like TDF, TAF is a recommended option in pregnancy because of reassuring data from a multinational trial in pregnant women with HIV⁴ and data from the [Antiretroviral Pregnancy Registry](#) that show no evidence of teratogenicity. Please refer to the [Perinatal Guidelines](#) for more information on the use of ARVs during pregnancy.

ABC/3TC, including DTG/ABC/3TC, is no longer recommended for initial therapy in most people with HIV but may be considered in circumstances in which tenofovir (TFV)-containing regimens or DTG/3TC cannot be used. Before starting any regimen with ABC, screening for the HLA-B*5701 allele is necessary because there is a strong link with a potentially life-threatening hypersensitivity

reaction in people who test positive for this allele. In addition, some data continue to support an association between ABC and an increased risk for serious cardiovascular events.^{5,6}

Along with safety and efficacy, cost and access are among the factors to consider when choosing among available options. ABC/3TC, TDF/3TC, TDF/FTC, and 3TC are available as generic formulations.

Clinical Trials Comparing Nucleoside Reverse Transcriptase Inhibitors

Tenofovir Alafenamide Compared to Tenofovir Disoproxil Fumarate

Safety and HIV Efficacy

Two randomized double-blind Phase 3 clinical trials compared the safety and efficacy of elvitegravir/cobicistat (EVG/c)/TDF/FTC and EVG/c/TAF/FTC in 1,733 ART-naïve adults with estimated glomerular filtration rate (eGFR) ≥ 50 mL/min.

- TAF/FTC was virologically non-inferior to TDF/FTC at Week 48 (92% vs. 90% of participants had plasma HIV RNA < 50 copies/mL, respectively),⁷ but TAF/FTC was superior to TDF/FTC at Week 144 (84.2% vs. 80% of participants with plasma HIV RNA < 50 copies/mL), largely driven by a higher rate of treatment discontinuation in the TDF arm.⁸
- Participants in the TAF arm had significantly smaller reductions in bone mineral density (BMD) at the spine and hip than those in the TDF arm through 144 weeks.⁸ Those receiving TAF also had less pronounced changes in eGFR and renal biomarkers and fewer clinically significant renal events through Week 96.⁹ Conversely, levels of fasting low-density lipoprotein (LDL) cholesterol, high-density lipoprotein (HDL) cholesterol, and triglycerides increased more in the TAF group than in the TDF group at Week 96, with no change in total cholesterol to HDL ratio.¹⁰

Two randomized studies have compared the safety and efficacy of TAF/FTC to TDF/FTC, with each combination administered with boosted darunavir (DRV) in ART-naïve participants:

- A Phase 2 study of coformulated darunavir/cobicistat (DRV/c) plus TAF/FTC versus DRV/c plus TDF/FTC in treatment-naïve participants demonstrated similar virologic suppression rates in both arms (75% vs. 74%).¹¹ In the TAF arm, fewer participants developed proteinuria. Changes in BMD were also less pronounced among participants in the TAF group.
- The AMBER study randomized ART-naïve participants to receive either coformulated DRV/c/TAF/FTC or DRV/c plus TDF/FTC. At Week 48, HIV RNA < 50 copies/mL was achieved in 91% of the DRV/c/TAF/FTC participants versus 88% of the DRV/c plus TDF/FTC participants. Participants in the TAF/FTC arm showed less decline in hip and spine BMD and eGFR than participants in the TDF/FTC arm.¹²

One analysis evaluated data from 14 randomized trials that compared the virologic efficacy, frequency of renal events, and bone density changes associated with the use of TDF and TAF when either drug was taken with or without pharmacokinetic (PK) boosters (ritonavir [RTV] or cobicistat [COBI]). No significant differences appeared between unboosted regimens with TDF and TAF in terms of virologic efficacy. When used with PK boosters, TAF resulted in a clinically small but statistically significant greater virologic efficacy than TDF (94% vs. 92%; $P = 0.0004$). No difference was seen in bone-related toxicities and clinical or laboratory adverse events between TAF and TDF, regardless of whether used with a boosting agent. The rate of discontinuation due to renal adverse

events was higher for those on boosted regimens containing TDF compared with those containing TAF, with a small but statistically significant difference ($P = 0.03$).¹³

Although conducted in people without HIV for pre-exposure prophylaxis (PrEP), the DISCOVER trial, with 5,387 treated participants, was the largest trial to directly compare the adverse effects of TAF/FTC with those of TDF/FTC.¹⁴ The following findings were observed after 48 weeks of follow-up:

- Adverse events did not significantly vary between the two groups, including Grade 3 and 4 events, serious adverse events, discontinuations due to adverse events, and overall.
- Changes in renal biomarkers and bone density significantly favored the TAF arm over TDF. One case of proximal tubular disease occurred in the TDF arm.
- LDL, HDL, and total cholesterol were significantly lower in the TDF arm than in the TAF arm, with no significant difference in the total cholesterol to HDL ratio.
- Participants in the TAF arm gained an average of 1.1 kg whereas those in the TDF arm had a mean loss of 0.1 kg.

Efficacy in People With HIV and HBV

To assess the ability of TAF to maintain HIV and HBV suppression, 72 people with HIV/HBV coinfection who had HIV RNA <50 copies/mL and HBV DNA <9 log₁₀ IU/mL on a stable regimen were switched to EVG/c/TAF/FTC.¹⁵ In this study, 96% of participants were on a TDF/FTC-containing regimen before the switch. Key results of the study showed the following:

- Among those who switched to EVG/c/TAF/FTC, HIV suppression was maintained in 91.7% of participants at Week 48, and 91.7% of participants had HBV DNA <29 IU/mL.
- Markers of proximal tubular proteinuria and biomarkers of bone turnover decreased in those who switched to EVG/c/TAF/FTC.¹⁵

Lamivudine Compared to Tenofovir Disoproxil Fumarate/Emtricitabine

A single-tablet regimen (STR) of DTG/3TC has now been approved as an initial ARV regimen. Please refer to the [Integrase Strand Transfer Inhibitor-Based Regimens](#) section for a full discussion.

GEMINI 1 and GEMINI 2 were identically designed randomized, double-blind clinical trials that found DTG plus 3TC non-inferior to DTG plus TDF/FTC in ART-naïve adults with HIV RNA <500,000 copies/mL and eGFR ≥50 mL/min.^{16,17}

Abacavir/Lamivudine Compared to Tenofovir Disoproxil Fumarate/Emtricitabine

Several randomized controlled trials in ART-naïve participants compared ABC/3TC to TDF/FTC, each administered in combination with a third ARV drug¹⁸⁻²⁰ (see the [Integrase Strand Transfer Inhibitor-Based Regimen](#) section).²¹

- The ACTG 5202 study, a randomized controlled trial in >1,800 participants, evaluated the efficacy and safety of ABC/3TC and TDF/FTC when each combination was used with either efavirenz (EFV) or atazanavir/ritonavir (ATV/r). In people with baseline HIV RNA ≥100,000 copies/mL, the time to virologic failure was significantly shorter with ABC/3TC than

with TDF/FTC, regardless of whether the third active drug was EFV or ATV/r. In addition, time to first adverse event was also shorter in the ABC/3TC groups.¹⁸

- In the HEAT study, 688 participants received ABC/3TC or TDF/FTC with once-daily lopinavir/ritonavir. Virologic efficacy was similar in the two study arms, including in a subgroup of participants with HIV RNA $\geq 100,000$ copies/mL.²⁰
- The ASSERT study compared open-label ABC/3TC with TDF/FTC in 385 HLA-B*5701-negative people with HIV who were ART-naïve; all participants also received EFV. The primary study endpoint was renal safety of the regimens. Although eGFR did not differ significantly between the regimens, biomarkers associated with tubular dysfunction (retinol-binding protein and beta-2 microglobulin) increased significantly, to a greater extent in the TDF/FTC arm than the ABC/3TC arm (+50% vs. +24% and no change vs. -47%, respectively). At Week 48, the proportion of participants with HIV RNA < 50 copies/mL was lower among ABC/3TC-treated participants (59%) than among TDF/FTC-treated participants (71%).¹⁹

Nucleoside Reverse Transcriptase Inhibitor Options for Initial Therapy

Tenofovir Alafenamide/Emtricitabine

TAF, an oral prodrug of TFV, is hydrolyzed to TFV in plasma and then converted to TFV-diphosphate (TFV-DP) intracellularly, where it exerts its activity as an NRTI. Unlike TDF, which readily converts to TFV in plasma after oral absorption, TAF remains relatively stable in plasma, resulting in lower plasma and higher intracellular TFV concentrations. After oral administration, TAF 25 mg resulted in plasma TFV concentrations that were 90% lower than those seen with TDF 300 mg. Intracellular TFV-DP concentrations, however, were substantially higher with TAF.

Adverse Effects

Renal and Bone Effects

- In randomized controlled trials that compared TAF and TDF in people without prior ARV treatment experience or those with virologic suppression, TAF had more favorable effects on renal biomarkers and bone density than TDF.

Lipid Effects

- In randomized controlled trials in ART-naïve participants, in switch studies, and in a large study of PrEP, levels of LDL and HDL cholesterol and triglycerides were lower in participants who received TDF than those who received TAF. However, total cholesterol to HDL ratios did not differ between participants receiving TAF and those receiving TDF. The clinical significance of this finding is not clear.^{7,22,23}

Weight Gain

- Initiation of TAF in previously untreated individuals and in people without HIV has been associated with greater weight gain than initiation of TDF²⁴⁻²⁶ and ABC.²⁵ In ADVANCE, an open-label trial conducted in South Africa that compared EFV/TDF/FTC versus DTG plus TDF/FTC versus DTG plus TAF/FTC in ART-naïve participants, a greater increase in body weight was reported with initiation of TAF than with TDF.²⁴ This area is under intense investigation, and the reason for the difference in weight gain between regimens is still uncertain. It is also unclear whether change of ART results in reversal of weight gain.

Other Factors and Considerations

- TAF/FTC is available in fixed-dose combinations (FDCs) with bictegravir (BIC), DRV/c, EVG/c, and rilpivirine (RPV), allowing the regimens to be administered as a single pill taken once daily.
- TAF-containing regimens are approved for people with eGFR ≥ 30 mL/min. Renal function, urine glucose, and urine protein should be assessed before initiating treatment with TAF, and these assessments should be repeated periodically during treatment. EVG/c/FTC/TAF was safe and effective in a single-arm switch study that was conducted in people on hemodialysis.²⁷ Based on the results from this study, TAF/FTC can be used without dosage adjustment in people with HIV who require hemodialysis.
- Both TAF and FTC are active against HBV. In people with HIV/HBV coinfection, TAF/FTC may be used as the NRTI pair in an ARV regimen because these drugs have activity against both viruses (see [Hepatitis B Virus/HIV Coinfection](#)).¹⁵
- TAF is recommended as a preferred drug in pregnancy because of reassuring data from a multinational trial of pregnant women and data from the [Antiretroviral Pregnancy Registry](#) that show no evidence of teratogenicity.⁴

The Panel's Recommendation

- On the basis of clinical trial safety and efficacy data, and its availability as a component of various FDCs, the Panel considers TAF/FTC a recommended NRTI combination for initial ART in most people with HIV when prescribed with BIC or DTG (**AI**), and as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* (see [Table 6b](#)) when given with RPV, DOR, RTV-boosted DRV (DRV/r), or DRV/c.

Tenofovir Disoproxil Fumarate/Emtricitabine and Tenofovir Disoproxil Fumarate/Lamivudine

TDF, with either 3TC or FTC, has been studied in combination with DOR, EFV, RPV, several boosted protease inhibitors (PIs), EVG/c, RAL, and DTG in randomized clinical trials.²⁸⁻³⁷

Adverse Effects

Renal Effects

- New onset or worsening renal impairment has been associated with TDF use.^{38,39} Risk factors may include advanced HIV disease, longer treatment history, low body weight (especially in women),⁴⁰ and preexisting renal impairment.⁴¹ Adverse effects on renal biomarkers, such as proteinuria, especially tubular proteinuria, were more frequent with TDF than with TAF.⁴²
- Adverse renal outcomes are more likely when TDF/FTC is coadministered with PK boosters (RTV or COBI). A meta-analysis of randomized trials found that discontinuation due to renal adverse events is more frequent in people who take TDF/FTC than TAF/FTC with PK boosters.¹³

Bone Effects

- Although initiation of all NRTI-containing regimens has been associated with a decrease in BMD, the loss of BMD is greater with TDF-containing regimens. For example, in two randomized studies that compared TDF/FTC with ABC/3TC, participants who received TDF/FTC experienced a significantly greater decline in BMD than ABC/3TC-treated participants.^{43,44} BMD generally

stabilizes following an early decline after ART initiation. Loss of BMD is also greater with TDF than with TAF.

- Cases of osteomalacia associated with proximal renal tubulopathy have been reported with the use of TDF.⁴⁵ Adverse bone outcomes have been found to be more likely when TDF/FTC is coadministered with PK boosters. However, a meta-analysis found no difference in bone-related toxicities between TAF and TDF, regardless of boosting.¹³

Other Factors and Considerations

- TDF/3TC is available as a coformulated generic drug.
- TDF/3TC is available in an FDC with DOR 100 mg.
- Renal function, urine glucose, and urine protein should be assessed before initiating treatment with TDF and periodically during treatment (see [Laboratory Testing for Initial Assessment and Monitoring of People With HIV Receiving Antiretroviral Therapy](#)). In people who have preexisting renal insufficiency (CrCl <60 mL/min),⁴⁶ use of TDF generally should be avoided. If TDF is used, a dose adjustment is required if the person's CrCl falls below 50 mL/min (see [Appendix B](#) for dose recommendations).
- TDF, FTC, and 3TC are active against HBV. In people with HBV/HIV coinfection, TDF/FTC or TDF/3TC may be used as the NRTI pair of the ARV regimen because these drugs have activity against both viruses (see [Hepatitis B Virus/HIV Coinfection](#)).

The Panel's Recommendations

- On the basis of clinical trial safety and efficacy data, long-term experience in clinical practice, and the combination's availability as a component of FDC drugs, the Panel considers TDF/FTC and TDF/3TC as recommended NRTI combinations for initial ART in most people with HIV when combined with DTG, and as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* ([Table 6b](#)).
- TDF should be used with caution or avoided in people with renal disease and/or osteopenia/osteoporosis.
- When TDF is used, especially in conjunction with a PK booster, clinicians should monitor for renal and bone safety during therapy. Boosters should be avoided when possible in people taking TDF.

Emtricitabine Versus Lamivudine

FTC and 3TC generally are used interchangeably in combination with other ARVs, based on the results of randomized clinical trials and a meta-analysis of 12 trials that compared virologic efficacy and safety.^{47,48} In the ATHENA cohort, virologic efficacy of TDF/FTC was compared to TDF/3TC when either was combined with a non-nucleoside reverse transcriptase inhibitor (NNRTI)—EFV or nevirapine⁴⁹—or with a boosted PI.⁵⁰ No difference was reported in the rates of virologic failure in people who were taking TDF/FTC and people who were taking TDF/3TC when these drug combinations were used with a boosted PI. TDF/3TC was associated with higher rates of virologic failure than TDF/FTC in the NNRTI analysis; however, participants in the NNRTI cohort who were taking 3TC generally had higher viral loads and lower CD4 T lymphocyte cell counts and were more likely to be using injection drugs at the start of the study than those taking FTC.

Adverse Effects

- Both FTC and 3TC have been well tolerated with no significant treatment-limiting adverse effects.
- In early clinical trials, FTC was infrequently associated with mild hyperpigmentation of palms and soles.

Other Factors and Considerations

- 3TC is now generic in the United States and is coformulated with other drugs, such as DOR/TDF/3TC and DTG/ABC/3TC.
- Both 3TC and FTC have activity against hepatitis B but are insufficient for HBV treatment when used alone due to the emergence of resistance. Discontinuation of FTC or 3TC can precipitate a flare in HBV if no other HBV-active drugs are in the regimen.
- See [Appendix B](#) for dosing recommendations in people with renal impairment.
- No significant drug interactions have been identified with FTC. Sorbitol-containing drugs can decrease 3TC concentration, and coadministration should be avoided.
- Both FTC and 3TC select for the M184V mutation when viral suppression is suboptimal.

The Panel's Recommendation

- FTC and 3TC are considered interchangeable in combination with other ARV drugs.

Lamivudine as a Single Nucleoside Reverse Transcriptase Inhibitor

Based on the GEMINI-1 and GEMINI-2 studies¹⁷ that found DTG plus 3TC non-inferior to DTG plus TDF/FTC in ART-naïve people with HIV RNA <500,000 copies/mL, 3TC may be used as a single NRTI with DTG (see [Drug Interactions Between Integrase Strand Transfer Inhibitors and Other Drugs](#) for more information).

Other Factors and Considerations

- 3TC with DTG is available as a STR.
- 3TC is active against HBV but is insufficient for HBV treatment when used alone due to the emergence of resistance. Discontinuation of 3TC can precipitate a flare in HBV if no other HBV-active drugs are in the regimen.
- 3TC is available in two brand-name formulations (one for HIV and the other for HBV), but the doses are different. The dose for HIV treatment is 3TC 300 mg daily.
- See [Appendix B](#) for dosing recommendations in people with renal impairment.
- Sorbitol-containing drugs can decrease 3TC concentration, and coadministration should be avoided.

The Panel's Recommendations

The Panel recommends the use of DTG/3TC **(AI)** as a *Recommended Initial Regimen for Most People With HIV*, with four exceptions. DTG/3TC **is not recommended** for:

- Individuals with HIV RNA >500,000 copies/mL.
- Individuals with HBV coinfection unless receiving another HBV-active drug.
- Individuals who have received long-acting injectable cabotegravir as PrEP and do not have integrase strand transfer inhibitor (INSTI) genotype resistance testing results available demonstrating INSTI sensitivity.
- Individuals starting ART before the results of genotypic resistance testing for reverse transcriptase are available.

Abacavir/Lamivudine

ABC plus 3TC has been studied in combination with EFV, several PIs, and DTG in people who are ART-naïve.^{21,51-53}

Adverse Effects

Hypersensitivity Reactions

- Clinically suspected hypersensitivity reactions (HSRs) were observed in 5% to 8% of individuals who started ABC in clinical trials conducted before the use of HLA-B*5701 testing. The risk of HSRs is highly associated with the presence of the HLA-B*5701 allele; approximately 50% of people who are HLA-B*5701-positive and are given ABC will have a related HSR.^{54,55} HLA-B*5701 testing should be done if the use of ABC is being considered. A person who tests positive for HLA-B*5701 should not be given ABC, and ABC hypersensitivity should be noted on the person's allergy list. People who are HLA-B*5701 negative are far less likely to experience an HSR, but they should be counseled about the symptoms of the reaction. People who discontinue ABC because of a suspected HSR **should never be rechallenged**, regardless of their HLA-B*5701 status.

Cardiovascular Risk

- An association between ABC use and myocardial infarction (MI) was first reported in the Data Collection on Adverse Events of Anti-HIV Drugs (D:A:D) study. This large, multinational, observational study group found that recent (i.e., within 6 months) or current use of ABC was associated with an increased risk of an MI, particularly in participants with preexisting cardiac risk factors.^{5,6}
- Since the D:A:D report, several studies have evaluated the relationship between ABC therapy and cardiovascular events. Some studies have found an association.⁵⁶⁻⁶³ Others, including an FDA meta-analysis of 26 randomized clinical trials that evaluated ABC, have not.⁶⁴⁻⁶⁸
- No consensus has been reached on the association between ABC use and MI risk or the mechanism for such an association.

Other Factors and Considerations

- ABC/3TC is available as a coformulated tablet and as a coformulated STR with DTG.
- ABC and 3TC are available separately and as a coformulated tablet in generic tablet formulations.
- ABC does not cause renal dysfunction and can be used instead of TDF in people with underlying renal dysfunction or in those who are at high risk for renal effects. No dose adjustment is required in people with renal dysfunction.

The Panel's Recommendations

- ABC should be prescribed only for people who are HLA-B*5701 negative.
- Due to the need for HLA-B*5701 testing before use, ongoing concerns about increased risk of cardiovascular events seen in association with ABC (observed in some but not all studies), and the availability of other effective and well-tolerated options, the Panel no longer recommends DTG/ABC/3TC for most individuals initiating ART and now classifies DTG/ABC/3TC as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* (see [Table 6b](#)) **(BI)** (see [Characteristics of Integrase Strand Transfer Inhibitors](#) for discussion regarding the clinical efficacy data for ABC/3TC plus DTG).
- ABC/3TC use with DRV/r or DRV/c is also recommended as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* ([Table 6b](#)) **(BII)**.
- ABC should be used with caution or avoided in people with cardiovascular disease or known high cardiovascular risk.

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Integrase Strand Transfer Inhibitor–Based Regimens as Initial Antiretroviral Therapy

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Table 8b. Characteristics of Integrase Strand Transfer Inhibitors That Are Recommended as Part of Initial Antiretroviral Therapy

	BIC	DTG
Dosing Frequency	Once daily	Once Daily - As initial ART or in people with no INSTI-resistance mutations Twice Daily - If used with certain CYP3A4 and UGT1A1 inducers; <i>or</i> - In people with certain INSTI drug resistance mutations
STR Available as Initial ART	BIC/TAF/FTC	- DTG/ABC/3TC - DTG/3TC
Available as a Single Drug Tablet	No	Yes
Virologic Efficacy Against EVG- or RAL-Resistant HIV	<i>In vitro</i> data indicate activity, but clinical trial data are not available.	Yes, for some isolates; effective with DTG 50 mg twice-daily dose
Adverse Reactions	↑ CPK 4% - CNS side effects were rarely reported in clinical trials. - Diarrhea, nausea, and headache may occur in some cases.	↑ CPK, myositis - CNS side effects such as insomnia and headache have been reported; depression and suicidality are rare, occurring primarily in people with preexisting conditions. - Hypersensitivity, hepatotoxicity
CYP3A4 Drug–Drug Interactions	CYP3A4 substrate	CYP3A4 substrate (minor)
Chelation With Polyvalent Cation Supplements and Antacids	Oral absorption may be reduced by polyvalent cations. See Table 24d for recommendations regarding dosing separations and these drugs.	

Table 8b. Characteristics of Integrase Strand Transfer Inhibitors That Are Recommended for Antiretroviral Therapy–Naïve Patients

	BIC	DTG
Other Key Potential Drug Interaction Mechanisms	P-gp substrate, UGT1A1 substrate, OCT2 and MATE1 inhibitor	P-gp substrate, UGT1A1 substrate
Other Factors	Both BIC and DTG decrease tubular secretion of creatinine without affecting glomerular function. This may result in an increase in serum creatinine of approximately 0.1–0.2 mg/dL.	

Key: 3TC = lamivudine; ABC = abacavir; ART = antiretroviral therapy; BIC = bictegravir; CNS = central nervous system; CPK = creatine phosphokinase; CYP = cytochrome P450; DTG = dolutegravir; EVG = elvitegravir; FTC = emtricitabine; INSTI = integrase strand transfer inhibitor; MATE1 = multidrug and toxic compound extrusion 1; OCT2 = organic cation transporter 2; P-gp = p-glycoprotein; RAL = raltegravir; STR = single-tablet regimen; TAF = tenofovir alafenamide; UGT = uridine diphosphate glucuronosyltransferase

Four **oral** integrase strand transfer inhibitors (INSTIs)—bictegravir (BIC), dolutegravir (DTG), elvitegravir (EVG), and raltegravir (RAL)—are approved for use in people with HIV as their initial ARV treatment. **Intramuscular** cabotegravir (CAB) is approved for use with rilpivirine (RPV) (**with or without an oral CAB + RPV lead-in**) as part of a long-acting injectable complete antiretroviral (ARV) regimen to replace a stable oral regimen in people with HIV and viral suppression. The role of this combination is discussed in the [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#) section. Long-acting injectable cabotegravir (CAB-LA) is also approved for pre-exposure prophylaxis (PrEP). **The first-generation INSTIs EVG and RAL have some disadvantages, which include a low barrier to resistance. In addition, because EVG has to be given with cobicistat (COBI), a pharmacokinetic (PK) booster, it has a high potential for drug–drug interactions, whereas RAL-based regimens have a higher pill burden than other INSTI regimens. Because of these disadvantages, elvitegravir/cobicistat (EVG/c) and RAL are no longer recommended by the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) as initial ART. Because the second-generation INSTIs BIC and DTG have high barriers to resistance, BIC/tenofovir alafenamide (TAF)/emtricitabine (FTC) and DTG plus TAF/FTC or tenofovir disoproxil fumarate (TDF)/FTC are recommended for most people with HIV (see [Appendix A, Table 6](#) for more information on the INSTIs BIC and DTG).**

This section of the guidelines will focus on BIC and DTG, the two INSTIs recommended by the Panel as part of the *Recommended Initial Regimens for Most People With HIV* ([Table 6a](#)) and *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* ([Table 6b](#)) (also see [Table 7](#)).

Panel’s Recommendations for Integrase Strand Transfer Inhibitor–Based Regimens as Initial Therapy

The Panel recommends one of the following INSTI-based regimens as initial ART for people with HIV **who do not have a history of using CAB-LA as PrEP** (see [Table 6a](#)):

- BIC/TAF/FTC **(AI)**
- DTG plus (TAF or TDF) with (FTC or lamivudine [3TC]) **(AI)**
- DTG/3TC **(AI)**, except for those with HIV RNA >500,000 copies/mL, with hepatitis B virus (HBV) coinfection, or in whom ART is to be started before the results of HIV genotypic resistance testing for reverse transcriptase or HBV testing are available.

INSTI resistance has been reported in people who acquired HIV following exposure to CAB-LA as PrEP.¹ For a person who has had exposure to CAB-LA as PrEP, an INSTI-containing regimen

should not be initiated unless an INSTI genotypic resistance test result is available and shows no INSTI-resistance mutations **(AIII)**. If treatment is initiated before genotypic test results are available, boosted darunavir (DRV) plus (TAF or TDF) plus (FTC or 3TC) should be used, pending INSTI resistance results **(AIII)**. See [Table 6a](#) for more details.

Because of the low rates of transmitted INSTI resistance in the United States at present, when there is suspicion that HIV was acquired from a partner with virologic failure while on an INSTI, an INSTI-based regimen can be started pending the results of the INSTI genotype test.

For people who have never used CAB-LA for PrEP and who acquired HIV despite using INSTI-based post-exposure prophylaxis, an INSTI genotype should be obtained prior to beginning an INSTI-based regimen. However, because selection of INSTI-resistant virus is likely to be uncommon in this setting, an INSTI-based regimen could be started prior to the return of genotype results **(CIII)**.

The Panel also recommends using DTG/abacavir (ABC)/3TC **(BI)** when concerns about renal or bone-associated adverse events preclude the use of TAF or TDF. However, ABC should only be given to people who are documented to be HLA-B*5701-negative.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = One or more randomized trials with clinical outcomes and/or validated laboratory endpoints; II = One or more well-designed, nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Adverse Effects

BIC and DTG are generally well tolerated, although there are reports of insomnia in some people. Depression and suicidal ideation, primarily in people with a history of psychiatric illnesses, have been reported rarely in those receiving INSTI-based regimens.²⁻⁵

Among people with HIV who have not yet received ART, initiation of INSTI-containing regimens has been associated with greater weight increases compared to non-nucleoside reverse transcriptase inhibitor (NNRTI) or boosted protease inhibitor (PI) regimens.⁶⁻¹¹ In randomized trials of ARV-naïve individuals, the mean increase in weight from baseline associated with BIC and DTG was similar and greater than with EVG/c or with efavirenz (EFV).^{8,12-14} Weight gain was also greater in those initiating TAF than other nucleoside reverse transcriptase inhibitors (NRTIs).^{7,8,15} These weight increases appear to disproportionately affect women and Black and Hispanic people,^{6-8,16} yet predictors and mechanisms for the weight gain are still unclear. In general, concerns for weight gain should not be a reason to avoid an INSTI- or TAF-based regimen.

Integrase Strand Transfer Inhibitor Use in Women of Childbearing Potential

Clinicians should refer to the [Perinatal Guidelines](#) for detailed recommendations on ARV regimens in treatment-naïve people, including recommendations on the use of INSTI-based regimens during conception and throughout pregnancy.

- Earlier data from a birth outcomes surveillance study in Botswana raised concern about an increased risk of neural tube defects (NTDs) (0.9%) in infants born to women who were receiving DTG at the time of conception.^{17,18} Updated data from the same study showed that the prevalence of NTDs in infants born to women on DTG at the time of conception is not significantly different from those on non-DTG regimens at the time of conception.¹⁹ Based on

these data, DTG is now the preferred INSTI for pregnant women with HIV. See the [Perinatal Guidelines](#) for further discussion.

- BIC is recommended as an alternative INSTI during pregnancy because data about safety, PK, and efficacy in pregnancy are available but are more limited than data about drugs classified as Preferred by the Perinatal Guidelines. See the [Perinatal Guidelines](#) for more details.

Bictegravir

BIC is approved by the U.S. Food and Drug Administration for initial therapy in adults with HIV as a component of a once-daily single-tablet regimen with TAF and FTC.

Efficacy in Clinical Trials

- The efficacy of BIC in ART-naïve adults has been evaluated in two large Phase 3 randomized double-blind clinical trials that compared BIC to DTG administered in combination with two NRTIs. The primary efficacy endpoint was the proportion of participants with plasma HIV RNA <50 copies/mL at Week 48.
 - o The GS-US-380-1490 trial randomized participants 1:1 to receive either BIC/TAF/FTC or DTG with coformulated TAF/FTC. Both regimens were given once daily. At Week 96, 84% of participants in the BIC arm and 86% of those in the DTG arm achieved HIV RNA <50 copies/mL.¹⁵
 - o The GS-US-380-1489 trial randomized participants 1:1 to receive BIC/TAF/FTC or coformulated DTG/ABC/3TC once daily. At Week 96, 88% of participants in the BIC/TAF/FTC arm and 90% of those in the DTG/ABC/3TC arm achieved HIV RNA <50 copies/mL.²⁰
 - o Week 144 follow-up from both trials demonstrated noninferiority of the BIC/TAF/FTC regimen to both DTG-containing regimens, with high levels of virologic suppression and no treatment-emergent resistance.

Adverse Effects

- BIC is generally well tolerated.
- Neuropsychiatric adverse events have been reported with INSTIs. BIC-associated serious neuropsychiatric effects were uncommon (<1%) in clinical trials and mainly occurred in the setting of preexisting depression, other psychiatric illness, or prior suicide attempt.²¹

Drug–Drug Interactions

Note: See [Table 24d](#) for a comprehensive list of potential INSTI-related drug–drug interactions.

- BIC is a cytochrome P450 (CYP) 3A4 substrate and a uridine diphosphate glucuronosyltransferase (UGT) 1A1 substrate; therefore, its metabolism may be affected by concomitant use of CYP3A4 and UGT1A1 inducers or inhibitors. Rifampin or other rifamycins may decrease BIC or TAF concentrations, which may result in loss of therapeutic effect of the ART. For people who require rifamycins, BIC/FTC/TAF should not be used. Use of certain anti-seizure medications and St. John's wort should also be avoided.²¹
- BIC is an inhibitor of the drug transporters OCT2 (organic cation transporter 2) and MATE1 (multidrug and toxic compound extrusion 1); therefore, BIC may increase concentrations of drugs

that are substrates of these transporters. For this reason, dofetilide is **contraindicated** with BIC/TAF/FTC.

- Like other INSTIs, oral absorption of BIC may be reduced when BIC is coadministered with polyvalent cations (e.g., aluminum-, magnesium-, or calcium-containing antacids; calcium or iron supplements).

Other Factors and Considerations

- BIC decreases tubular secretion of creatinine without affecting glomerular function. Increases in serum creatinine are observed typically within the first 4 weeks of BIC therapy (with a median increase of 0.10 mg/dL after 48 weeks). This increase is comparable to that seen with other ARV drugs that have a similar effect on creatinine secretion, including DTG, RPV, and COBI.
- Treatment-emergent mutations that confer BIC resistance have rarely been reported in people receiving BIC for initial therapy.²² BIC has not been studied in prospective trials for people with prior INSTI failure or INSTI-related resistance mutations. One analysis of 20 individuals with preexisting INSTI-related resistance mutations showed maintenance of virologic suppression in 19 people after switch to BIC, and viral suppression after beginning BIC in one person without prior ARV treatment. There are, however, insufficient data at present to recommend BIC use in such cases.

The Panel's Recommendation

The Panel recommends the use of BIC/TAF/FTC as a *Recommended Initial Regimen for Most People With HIV (AI)* (see [Table 6a](#) and [Table 7](#) for more detailed recommendations).

Dolutegravir

As initial ARV regimens, both DTG plus two NRTIs **and DTG/3TC** demonstrated high efficacy in achieving HIV suppression in clinical trials. DTG is given once daily, with or without food.

Efficacy in Clinical Trials

The efficacy of DTG in ART-naïve individuals has been evaluated in several fully powered randomized controlled clinical trials. In these trials, DTG-based regimens were noninferior or superior to a comparator INSTI-, NNRTI-, or PI-based regimen. The primary efficacy endpoint in these clinical trials was the proportion of participants with plasma HIV RNA <50 copies/mL.^{13,23,24}

DTG Plus Two NRTIs Versus Other INSTIs Plus Two NRTIs

- DTG-based regimens (with TAF/FTC or ABC/3TC) have been compared to BIC/TAF/FTC in two randomized controlled trials. These regimens have virologic efficacy that is similar to BIC/TAF/FTC (see discussion in the BIC section above).^{15,20,23,25}

DTG Plus Two NRTIs Versus EFV Plus Two NRTIs

- The SINGLE trial compared DTG 50 mg once daily plus ABC/3TC to EFV/TDF/FTC in 833 participants. At Week 48, DTG plus ABC/3TC was superior to EFV/TDF/FTC, primarily because the study treatment discontinuation rate was higher in the EFV arm than in the DTG arm.²⁶ At Week 144, DTG plus ABC/3TC remained superior to EFV/TDF/FTC.²⁷

- The ADVANCE trial, an open-label, noninferiority trial conducted in South Africa, compared DTG with either TDF/FTC or TAF/FTC to EFV/TDF/FTC. At Week 96, the DTG-based regimens were noninferior to the EFV regimen based on the proportion of participants with HIV RNA levels <50 copies/mL (79% in DTG/TAF/FTC vs. 78% in DTG/TDF/FTC vs. 74% in EFV/TDF/FTC arms). More participants discontinued the trial regimen in the EFV group than in the DTG group. Mean weight gain was 7.1 kg in the DTG/TAF/FTC group, 4.3 kg in the DTG/TDF/FTC group, and 2.3 kg in the EFV/TDF/FTC group and was greater among women than men.¹⁴

DTG Plus Two NRTIs Versus Ritonavir-Boosted Darunavir Plus Two NRTIs

- The FLAMINGO study, a randomized open-label clinical trial, compared DTG 50 mg once daily to the boosted PI—darunavir/ritonavir (DRV/r) 800 mg/100 mg once daily—each administered in combination with investigator-selected ABC/3TC or TDF/FTC. At Week 48, DTG was superior to DRV/r, with 90% and 83% of participants achieving HIV RNA <50 copies/mL, respectively. More participants discontinued their assigned regimen in the DRV/r arm.²⁸ The difference in efficacy between the DTG and DRV/r regimens was more pronounced in people with pre-treatment HIV RNA levels >100,000 copies/mL. At Week 96, DTG remained superior to DRV/r.²⁹

DTG/3TC

- In the GEMINI-1 and GEMINI-2 trials, 1,433 ART-naïve participants with baseline HIV RNA <500,000 copies/mL and no evidence of HBV infection were randomized to receive DTG plus 3TC or DTG plus TDF/FTC. At Week 96, DTG plus 3TC was noninferior to DTG plus TDF/FTC with 86% of participants in the DTG plus 3TC group and 89.5% in the DTG plus TDF/FTC group with viral load <50 copies/mL.³⁰ Virologic nonresponse was uncommon, occurring in 3.1% of participants who received DTG plus 3TC and 2% of participants who received DTG plus TDF/FTC. No treatment-emergent NRTI or INSTI resistance occurred in either treatment group. Among participants who started the study with CD4 counts <200 cells/mm³, the rate of HIV RNA <50 copies/mL at Week 96 was lower in the DTG plus 3TC group than in the DTG plus TDF/FTC group; however, the difference was not related to a higher rate of virologic failure in the two-drug group. Overall mean change in weight from baseline was 3.1 kg in the DTG plus 3TC group and 2.1 kg in the DTG plus TDF/FTC group. At Week 144, DTG plus 3TC maintained noninferiority to DTG plus TDF/FTC with 82% versus 84% of participants maintaining viral load <50 copies/mL, respectively. The proportion of participants with viral load ≥50 copies/mL was 3% in both treatment groups. A lower risk of drug-related adverse events was found with DTG plus 3TC versus DTG plus TDF/FTC (20% vs. 27%; relative risk, 0.76 [95% confidence interval, 0.63–0.92]).³⁰
- Two other small, nonrandomized single-arm studies showed similar rates of viral suppression with DTG plus 3TC.^{31,32}

Adverse Effects

- DTG is generally well tolerated. The most reported adverse reactions of moderate-to-severe intensity were insomnia and headache.
- Some studies have shown greater weight increase among people initiating INSTI-based regimens, including regimens with DTG.^{7,8,33,34} In a pooled analysis of eight randomized controlled trials in ART-naïve individuals, the weight gain at 96 weeks with BIC- and DTG-based regimens was similar (approximately 3.5 kg). **The clinical significance of weight gain in this setting is unclear.**⁸

- Neuropsychiatric adverse events (e.g., sleep disturbances, depression, anxiety, suicidal ideation) associated with the initiation of DTG and other INSTIs have been reported.^{2,3,35,36} However, analyses of data from large randomized controlled trials and a health care database demonstrated similar rates of neuropsychiatric adverse events between DTG-based regimens and regimens that included RAL, EFV, DRV, and ATV.³⁷ Neuropsychiatric events rarely led to DTG discontinuation.

Drug–Drug Interactions

Note: See [Table 24d](#) for a comprehensive list of potential INSTI-related drug–drug interactions.

- DTG has fewer drug interactions **than BIC**.
- DTG oral absorption may be reduced when the ARV is coadministered with polyvalent cations. DTG should be taken at least 2 hours before or 6 hours after cation-containing antacids or laxatives are taken. Alternatively, DTG and supplements containing calcium or iron can be taken simultaneously with food.

Other Factors and Considerations

- DTG decreases tubular secretion of creatinine without affecting glomerular function, with increases in serum creatinine observed within the first 4 weeks of treatment.
- Treatment-emergent mutations that confer DTG resistance have been rarely reported in people receiving DTG as part of a three-drug regimen for initial therapy.^{38–40} The incidence of resistance with DTG is much lower than with EVG or RAL, which suggests that DTG, like BIC, has a higher barrier to resistance than EVG or RAL.

The Panel’s Recommendations

- The Panel classifies the following as *Recommended Initial Regimens for Most People With HIV* (see [Table 6a](#) and [Table 7](#) for more detailed recommendations):
 - o DTG plus (TAF or TDF) plus (FTC or 3TC) **(AI)**
 - o DTG/3TC **(AI)—not recommended** for those with HIV RNA >500,000 copies/mL, with HBV coinfection, or in whom ART is to be started before the results of HIV genotypic resistance testing for reverse transcriptase or HBV testing are available.
- The Panel recommends DTG/ABC/3TC **(BI)** in certain clinical scenarios for people with HIV who are documented to be HLA-B*5701-negative and who have concerns about renal or bone toxicities associated with TAF or TDF. This regimen should not be used in people with HBV coinfection unless an HBV-active drug (i.e., entecavir) other than 3TC is used.

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Non-Nucleoside Reverse Transcriptase Inhibitor–Based Regimens as Initial Antiretroviral Therapy

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Table 8c. Characteristics of Non-Nucleoside Reverse Transcriptase Inhibitors That Are Recommended as Initial Antiretroviral Therapy in Certain Clinical Scenarios

Characteristics	DOR	RPV ^a
Dosing Frequency	Once daily	Once daily
Food Requirement	With or without food	With a meal
STR Available as Initial ART Recommended in Table 6b	DOR/TDF/3TC	RPV/TAF/FTC
Available as a Single-Drug Tablet	Yes	Yes
Adverse Effects	Generally well tolerated	• Depression • Headache • Skin rash • QTc prolongation
CYP3A4 Drug–Drug Interactions	CYP3A4 substrate	CYP3A4 substrate
Other Significant Drug Interactions	None	RPV oral absorption is reduced with increased gastric pH. Use of RPV with PPIs is not recommended; see Drug–Drug Interactions for dosing recommendations when RPV is coadministered with an H2 blocker or antacids.

^a See [Optimizing Antiretroviral Therapy](#) section and [Appendix A, Table 4](#) for information regarding injectable RPV.

Key: 3TC = lamivudine; ART = antiretroviral therapy; CYP = cytochrome P; DOR = doravirine; FTC = emtricitabine; H2 = histamine 2; PPI = proton pump inhibitor; QTc = QT corrected for heart rate; RPV = rilpivirine; STR = single-tablet regimen; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate

Summary

Four non-nucleoside reverse transcriptase inhibitors (NNRTIs)—doravirine (DOR), efavirenz (EFV), nevirapine (NVP), and rilpivirine (RPV)—are currently approved by the U.S. Food and Drug Administration (FDA) for the initial treatment of HIV when used in combination with other antiretroviral (ARV) drugs. A fifth NNRTI, etravirine (ETR), is approved only for people who are ART-experienced.

NNRTI-based regimens have demonstrated virologic potency and durability. The major disadvantages of currently available NNRTIs (especially EFV and RPV) include the prevalence of NNRTI-resistant viral strains in people without prior antiretroviral therapy (ART),¹ a low barrier for the development of resistance, and potential for drug–drug interactions, although drug–drug interactions are less common with DOR. Resistance testing should be performed before initiation of an NNRTI-based regimen in people without prior ART. High-level resistance to all NNRTIs (except ETR or DOR) may occur with a single mutation. In people treated with RPV, the presence of RPV resistance mutations at virologic failure may confer cross-resistance to other NNRTIs, including ETR.^{2,3} DOR-based regimens and one of the RPV-based regimens are now categorized as recommended *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios*. More details about these NNRTIs are provided below.

This section of the guidelines focuses on DOR and RPV, the two NNRTIs recommended by the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) as part of an initial ART regimen for people with HIV in certain clinical scenarios (see [Table 6b](#) in the [Initial Combination Antiretroviral Regimens for People With HIV](#) section and [Table 7](#)). There is no clinical trial comparing the efficacy of these two NNRTIs as initial ARV regimens. The Panel no longer recommends EFV as initial therapy because of the high incidence of serious toxicities, including neuropsychiatric effects and suicidality. NVP is no longer commonly used in clinical practice in the United States due to a relatively high potential for adverse effects compared to other NNRTIs.

Doravirine

Efficacy in Clinical Trials

The efficacy of DOR-based therapy as initial ART was demonstrated in two randomized, double-blind, placebo-controlled trials.

Doravirine-Based Regimen Versus Efavirenz-Based Regimen

- In the [DRIVE-AHEAD](#) trial, 734 participants received either DOR/tenofovir disoproxil fumarate (TDF)/lamivudine (3TC) or EFV/TDF/emtricitabine (FTC), both as a daily fixed-dose combination (FDC) tablet.⁴
 - At 96 weeks, DOR/TDF/3TC was noninferior to EFV/TDF/FTC, with 77.5% of participants who received DOR/TDF/3TC and 73.6% of those who received EFV/TDF/FTC achieving HIV RNA <50 copies/mL. Although virologic responses to ART overall were lower in participants with pre-treatment HIV RNA >100,000 copies/mL or pre-treatment CD4 T lymphocyte (CD4) cell counts of ≤200 cells/mm³, there was no difference between the DOR-treated and EFV-treated participants.
 - Virologic rebound and virologic nonresponse were similar in the DOR/TDF/3TC (9.3%) and EFV/TDF/FTC (7.7%) treatment groups. At 96 weeks, genotype resistance results were reported for 21 participants with protocol-defined virologic failure in the DOR arm and 15 participants in the EFV arm. For the DOR arm, 7 of 21 (33.3%) participants had NNRTI resistance and 6 of 21 (28.6%) had NRTI resistance. For EFV, 10 of 15 (66.7%) participants had NNRTI resistance and 5 of 15 (33.3%) had NRTI resistance.
 - More participants in the EFV arm discontinued their assigned ART due to adverse events than in the DOR arm (6.6% vs. 3.0%). Neuropsychiatric side effects and rash were more common in the EFV arm.

Doravirine-Based Regimen Versus Darunavir/Ritonavir-Based Regimen

- In the [DRIVE-FORWARD](#) trial, 769 participants received DOR or darunavir/ritonavir (DRV/r) once daily along with two investigator-selected nucleoside reverse transcriptase inhibitors (NRTIs), either abacavir (ABC)/3TC or TDF/FTC.⁵
 - o At 48 weeks, DOR was noninferior to DRV/r with 84% of study participants who received DOR versus 80% of those who received DRV/r achieving HIV RNA <50 copies/mL. Participants who received DOR plus ABC/3TC (n = 48) and those who received DOR plus TDF/FTC (n = 316) had similar virologic responses.
 - o At 96 weeks, DOR was superior to DRV/r in terms of virologic suppression (73% vs. 66%).⁵
 - o **Protocol-defined virologic failure** was similar in the DOR and DRV/r groups (9% vs. 11%) by Week 96. Genotype resistance results were reported for 11 and 14 participants with virologic failure in the DOR and DRV/r arms, respectively. Treatment-emergent resistance to any study drug occurred in **2 of 11 participants in the DOR group and 1 of 14 participants** in the DRV/r group.
 - o Treatment-related diarrhea was more frequently reported in the DRV/r arm. Greater increases in fasting low-density lipoprotein cholesterol, triglycerides, non–high-density lipoprotein cholesterol, and total cholesterol were seen in the participants who received DRV/r than in those who received DOR.
- In a pooled analysis of DRIVE-AHEAD and DRIVE-FORWARD participants who entered an open-label extension phase following the initial 96 weeks of randomized treatment, 550 participants continued their DOR-based regimen and 502 participants switched to a DOR-based regimen from the control arms. At 192 weeks, 83% percent of participants who continued on DOR and 81% of participants in the switch arm maintained viral suppression. Discontinuation due to drug-related adverse events was <1% overall and similar between the two arms, with no discontinuations due to renal adverse events.⁶

Other Factors and Considerations

- DOR is available as a single-drug 100-mg tablet⁷ and as part of a single-tablet regimen (STR) that contains DOR/TDF/3TC 100 mg/300 mg/300 mg⁸ and is dosed once daily, with or without food.
- DOR-based regimens have not been directly compared to integrase strand transfer inhibitor (INSTI)–based regimens in clinical trials and have not been studied with tenofovir alafenamide (TAF) in clinical trials.
- A post hoc analysis of three randomized controlled trials examined weight gain among ART-naïve participants receiving DOR versus DRV/r or EFV. At week 96, mean weight gain was similar in the DOR group (2.4 kg), DRV/r group (1.8 kg), and EFV group (1.6 kg). No significant differences between treatment groups were found in the proportion of participants whose body mass index classification increased to overweight or obese at Week 48 or Week 96.⁹
- DOR is primarily metabolized by the CYP3A4 enzyme and should not be coadministered with strong CYP3A4 inducers. DOR concentration may increase in the presence of a CYP3A4 inhibitor (see [Table 24b](#)). DOR is not a CYP3A4 inducer or inhibitor; thus, it is not expected to affect the concentrations of concomitant CYP3A4 substrates.
- Treatment-emergent resistance mutations to DOR may confer cross-resistance to certain other NNRTIs. Most isolates with DOR mutations remain susceptible to ETR.¹⁰
- There are currently no data on the safety of DOR use during pregnancy.

- There are limited clinical trial data with the combination of DOR + ABC/3TC, so the Panel is less certain about the efficacy of this regimen.

The Panel's Recommendations

- Based on the clinical trial data discussed above, the Panel classifies DOR/TDF/3TC (**BI**) and DOR plus two NRTIs (**BI** for TDF/FTC and **BIII** for TAF/FTC) as *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios*.

Rilpivirine

Oral RPV is approved for use in combination with two NRTIs as initial ART for people with HIV who have pre-treatment viral loads <100,000 copies/mL. RPV is also approved as an extended-release injectable suspension as part of a long-acting (LA) injectable complete ARV regimen when used with cabotegravir (CAB), an INSTI. This regimen is approved to replace oral ART in people with virologic suppression and no history of resistance to RPV or INSTIs (see [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#) for discussion of LA CAB/RPV).

Efficacy in Clinical Trials

- Two Phase 3 randomized, double-blind clinical trials—ECHO and THRIVE—compared RPV and EFV, each combined with two NRTIs.¹¹ At 96 weeks, the following findings were reported:
 - o RPV was noninferior to EFV overall.
 - o Among participants with pre-ART viral loads >100,000 copies/mL, more RPV-treated participants experienced virologic failure than EFV-treated participants. NNRTI and NRTI resistance were more frequently found in RPV-treated participants with virologic failure.
 - o Among the RPV-treated participants, the rate of virologic failure was greater in those with pre-treatment CD4 counts <200 cells/mm³ than in those with CD4 counts ≥200 cells/mm³.
- STaR, a Phase 3b, open-label study, compared two STRs—RPV/TDF/FTC and EFV/TDF/FTC—as initial ART in 786 people with HIV. The results at 96 weeks¹² were similar to those reported at 48 weeks.¹³
 - o RPV was noninferior to EFV overall.
 - o RPV was superior to EFV in participants with pre-ART viral loads ≤100,000 copies/mL and noninferior in those with pre-ART viral loads >100,000 copies/mL. Among participants with pre-ART viral loads >500,000 copies/mL, virologic failure was more common in RPV-treated than in EFV-treated participants.
 - o At Week 96, there were more participants with emergent resistance in the RPV/FTC/TDF arm than in the EFV/FTC/TDF arm (5.3% vs. 1.0%, respectively).
- The STR of RPV/TAF/FTC was approved by the FDA based on results from a bioequivalence study. In this study, plasma concentrations of RPV, FTC, and TAF 25 mg in participants taking the coformulated drug were similar to those seen in participants who received RPV as the single-drug tablet and TAF/FTC as part of the STR of EVG/c/TAF 10 mg/FTC.¹⁴

Adverse Effects

- RPV is generally well tolerated. In the ECHO, THRIVE, and STaR trials, fewer instances of CNS adverse events (e.g., abnormal dreams, dizziness, psychiatric side effects), skin rash, and dyslipidemia were reported in the RPV arms than in the EFV arms, and fewer people in the RPV

arms discontinued therapy due to adverse events. However, up to 9% of clinical trial participants experienced depressive disorders, including approximately 1% of participants who had suicidal thoughts or who attempted suicide. People receiving RPV who have severe depressive symptoms should be evaluated to assess whether the symptoms may be due to RPV and if the risks of continuing the same regimen outweigh the benefits.

Other Factors and Considerations

- Oral RPV is formulated both as a single-drug tablet and in STRs with TAF/FTC, with TDF/FTC, and with DTG. Among available STRs, **RPV/TDF/FTC is no longer recommended by the Panel for initial therapy.**
- RPV is also available as part of an LA injectable ARV regimen for use in combination with LA CAB in people with HIV who are virologically suppressed and do not have resistance to these drugs (see [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#)).
- The STR of RPV/TAF/FTC is given once daily and must be administered with a meal (containing at least 390 kcal).
- RPV is also coformulated as a once-daily FDC tablet with DTG that is used as continuation therapy for people with HIV who have achieved viral suppression.¹⁵ However, this combination has not been studied in people with HIV and no prior ART, and it **is not recommended** for initial therapy (see [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#)).
- The oral drug absorption of RPV can be significantly reduced in the presence of acid-reducing agents. RPV is **contraindicated** in people who are receiving proton pump inhibitors and should be used with caution in those receiving H₂ antagonists or antacids (see [Drug–Drug Interactions](#) for dosing recommendations).
- RPV is primarily metabolized in the liver by the CYP3A enzyme; its plasma concentration may be affected in the presence of CYP3A inhibitors or inducers (see [Drug–Drug Interactions](#)).
- At doses above the approved dose of 25 mg, RPV may cause QTc interval prolongation. RPV should be used with caution when coadministered with a drug known to increase the risk of Torsades de Pointes.

The Panel's Recommendations

- Given the availability of other effective regimens that do not have virologic and immunologic prerequisites to initiate treatment, **substantial drug–drug interactions, and a low genetic barrier to resistance, RPV is not recommended for initial therapy in most persons with HIV. RPV/TDF/FTC is no longer recommended as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios*, largely due to the limitations of RPV and the availability of DOR/TDF/3TC, which has fewer drug–drug interactions and no virologic and immunologic prerequisites. The Panel continues to recommend RPV/TAF/FTC as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* (see [Table 6b](#)), largely to retain an STR option containing an NNRTI and TAF.**
- Use of RPV with TAF/FTC (**BII**) should be limited to persons with pre-treatment viral loads <100,000 copies/mL and CD4 counts >200 cells/mm³.
- Data on RPV plus ABC/3TC are insufficient to consider recommending this regimen.

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Protease Inhibitor–Based Regimens

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Table 8d. Characteristics of Protease Inhibitor Options as Initial Antiretroviral Therapy in Certain Clinical Scenarios

Characteristic	DRV
Dosing Frequency	Once daily for persons with no prior PI experience.
PK Boosting	DRV should only be used with a PK booster (i.e., RTV or COBI).
Fixed-Dose Formulation	• DRV/c • DRV/c/TAF/FTC
Available as a Single-Drug Tablet	Yes
Adverse Effects	• Skin rash • Increase in serum transaminase • Hyperlipidemia • Diarrhea, nausea
CYP3A4 Drug–Drug Interactions	CYP3A4 substrate, inhibitor
Other Significant Drug Interactions	N/A

Key: COBI = cobicistat; CYP = cytochrome P; DRV = darunavir; DRV/c = darunavir/cobicistat; FTC = emtricitabine; N/A = not applicable; PI = protease inhibitor; PK = pharmacokinetic; RTV = ritonavir; TAF = tenofovir alafenamide

Summary

The U.S. Food and Drug Administration (FDA)–approved protease inhibitors (PIs) include atazanavir (ATV), atazanavir/cobicistat (ATV/c), darunavir (DRV), darunavir/cobicistat (DRV/c), fosamprenavir (FPV), indinavir (IDV), lopinavir/ritonavir (LPV/r), nelfinavir (NFV), ritonavir (RTV), saquinavir (SQV), and tipranavir (TPV). PI-based regimens using pharmacokinetic (PK) enhancement with either cobicistat (COBI) or RTV (also called PK boosting) increase concentration and prolong the half-lives of the PI. These regimens have demonstrated virologic potency, durability in people who are ART-naïve, and a high barrier to resistance. Because LPV/r, fosamprenavir/ritonavir (FPV/r), ATV (with or without a PK enhancer), and saquinavir/ritonavir (SQV/r) have disadvantages—such as greater pill burden, lower efficacy, or increased toxicity—only boosted DRV in combination with two nucleoside reverse transcriptase inhibitors (NRTIs) is recommended as initial therapy in certain clinical situations (see [Table 6b in Initial Combination Antiretroviral Regimens for People With HIV](#)).

Because transmitted PI resistance is uncommon, boosted DRV-based regimens are preferred over a non-nucleoside reverse transcriptase inhibitor (NNRTI)-based regimen as an option for treatment initiation before resistance test results are available. In addition, a boosted DRV-based regimen can be used for rapid antiretroviral therapy (ART) initiation, in the setting of acute HIV infection, and as the preferred option while awaiting resistance test results for people with a history of long-acting

cabotegravir (CAB-LA) use as pre-exposure prophylaxis (PrEP). Few or no PI mutations are detected when a patient's first PI-based regimen fails, which is not the case with non-nucleoside reverse transcriptase inhibitor (NNRTI)-based regimens and regimens with a first-generation integrase strand transfer inhibitor (INSTI) (e.g., raltegravir or elvitegravir).¹ Because of their high barrier to resistance, PI-based regimens may be useful for people at risk for intermittent therapy because of poor adherence.

DRV requires PK boosting with either RTV or COBI to inhibit the cytochrome P3A4 (CYP3A4) isoenzyme, which may lead to significant drug–drug interactions (see [Drug–Drug Interactions](#)). The specific characteristics of DRV are listed in [Appendix A, Table 5](#).

Darunavir/Ritonavir

Efficacy in Clinical Trials

Darunavir/ritonavir (DRV/r) has been studied in several large, randomized controlled trials in people with HIV without prior antiretroviral (ARV) experience. These trials compared DRV/r-based regimens to PI-, INSTI-, or NNRTI-based regimens. Summaries of the results from some key trials are listed below.

- The FLAMINGO study compared DRV/r with dolutegravir (DTG), each administered in combination with two NRTIs—either tenofovir disoproxil fumarate (TDF)/emtricitabine (FTC) or abacavir (ABC)/lamivudine (3TC)—in 488 participants who were ART-naïve. The rate of virologic suppression at Week 96 was significantly higher among those who received DTG than in those who received DRV/r. The higher rate of virologic failure observed in the DRV/r group was related primarily to the number of failures among those with a viral load >100,000 copies/mL and secondarily to more drug discontinuations in the DRV/r group.²
- The AIDS Clinical Trial Group (ACTG) study A5257 (ARDENT), a large, randomized, open-label trial, compared atazanavir/ritonavir (ATV/r) to DRV/r or raltegravir (RAL) over 96 weeks, each given with TDF/FTC. The trial showed similar virologic efficacy for DRV/r, ATV/r, and RAL, but more participants in the ATV/r group discontinued randomized treatment because of adverse events.³
- The DRIVE-FORWARD study compared DRV/r to doravirine (DOR), both administered with two investigator-selected NRTIs, in 769 ART-naïve participants. At 48 weeks, DOR was found to be non-inferior to DRV/r, with 80% and 84% of participants achieving HIV RNA levels <50 copies/mL, respectively.⁴ At Week 96, DOR was superior to DRV/r in terms of virologic suppression (73% vs. 66%).⁴ Rates of virologic failure were low and similar in the DOR and DRV/r groups (9% vs. 11%). Treatment-emergent resistance to any study drug was infrequent, occurring in <1% of participants in both the DOR group (2 of 383) and the DRV/r group (1 of 383).

Adverse Effects

- People with HIV who take DRV/r may develop a skin rash, which is usually mild-to-moderate in severity and self-limited. Treatment discontinuation is necessary on rare occasions when severe rash occurs with fever or elevated transaminases.
- ACTG A5257 showed similar lipid changes in participants in the ATV/r and DRV/r arms. Bone mineral density decreased to a greater extent in participants in the ATV/r and DRV/r arms compared to participants in the RAL arm.³ The likelihood of developing metabolic syndrome was equivalent among the three arms, although a larger increase in waist circumference was

observed at 96 weeks in participants assigned to the RAL arm than in those assigned to the DRV/r arm ($P = 0.023$).⁵

- In the DRIVE-FORWARD study, treatment-related diarrhea was more frequently reported in the DRV/r arm, and greater increases were observed in fasting low-density lipoprotein cholesterol, triglycerides, non–high-density lipoprotein cholesterol, and total cholesterol compared to the DOR arm.⁴

Other Factors and Considerations

- DRV/r is administered once daily with food in people who are ART-naïve.
- DRV has a sulfonamide moiety and should be used with caution in people with severe sulfonamide allergies. In clinical trials, the incidence and severity of rash were similar in participants with and without a history of sulfonamide allergy. Most people with sulfonamide allergy can tolerate DRV.
- DRV/r is a potent CYP3A4 inhibitor, which may lead to significant interactions with other medications metabolized through this same pathway (see [Drug–Drug Interactions](#)).
- Unlike DRV/c, DRV/r may be used in pregnancy.

The Panel’s Recommendations

- Based on efficacy and safety data from clinical trials and clinical experience, the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) classifies DRV/r with (tenofovir alafenamide [TAF] or TDF) plus (FTC or 3TC) as a *Recommended Initial Regimen for Most People With HIV* who have a history of CAB-LA use as PrEP, pending the results of INSTI genotype testing (**AIII**) (see [Table 6a](#)). DRV/r plus (TAF or TDF) plus (FTC or 3TC) (**BI**) and DRV/r plus ABC/3TC (**BII**) are also part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios* (see [Table 6b](#) and [Table 7](#)).

Darunavir/Cobicistat

Efficacy in Clinical Trials

- The AMBER trial enrolled 725 ART-naïve participants in a Phase 3 randomized controlled trial that compared the single-tablet regimen (STR) DRV/c/TAF/FTC with DRV/c plus TDF/FTC. At 48 weeks, similar virologic suppression rates among participants were achieved in both arms of the study (91% and 88% had HIV RNA <50 copies/mL, respectively). No treatment-emergent mutations associated with DRV or TAF/TDF resistance were observed in either group.⁶ At 96 weeks, 85% of participants on the STR maintained HIV RNA levels <50 copies/mL compared to 84% in the DRV/c plus TDF/FTC arm.⁷
- The DIAMOND study evaluated DRV/c/TAF/FTC as an STR in 109 participants in a rapid-initiation model of care. At Week 48, 97 (89%) participants completed the study and 92 (84%) achieved HIV-1 RNA <50 copies/mL by the FDA snapshot analysis. No protocol-defined virologic failures occurred, and incidences of adverse events at least possibly related to study drugs (33%) were low. No study drug-related serious adverse events occurred, and only one (<1%) participant discontinued because of a study drug-related adverse event.⁸

Adverse Effects

- The most common drug-related adverse events were diarrhea, nausea, fatigue, flatulence, rash, and headache.

Other Factors and Considerations

- DRV/c 800 mg/150 mg is available as a coformulated boosted PI or as an STR with TAF/FTC 10 mg/200 mg.
- Both DRV and COBI exposures are reduced markedly during the second and third trimesters of pregnancy, and should be avoided if possible.⁹ However, if pregnant women with viral suppression while on DRV/c elect to continue on the drug, frequent viral load monitoring is recommended. For further information, please refer to the [Perinatal Guidelines](#).

The Panel's Recommendations

- The Panel recommends DRV/c plus (TAF or TDF) plus (FTC or 3TC) **(BI)** and DRV/c plus ABC/3TC **(BII)** as part of *Other Initial Antiretroviral Regimens for Certain Clinical Scenarios*.
- Per product label recommendation, DRV/c plus TDF/FTC **is not recommended** for people with creatinine clearance (CrCl) <70 mL/min, whereas DRV/c plus TAF/FTC **is not recommended** for people with CrCl <30 mL/min.
- For people who have a history of CAB-LA use as PrEP, INSTI genotype resistance testing should be performed before starting ART. If ART is to be started before genotypic testing results are available, DRV/c with (TAF or TDF) plus (FTC or 3TC) can be started **(AIII)**.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = One or more randomized trials with clinical outcomes and/or validated laboratory endpoints; II = One or more well-designed, nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

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Antiretroviral Regimens for Initial Therapy When Abacavir, Tenofovir Alafenamide, and Tenofovir Disoproxil Fumarate Cannot Be Used or Are Not Optimal

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Most of the currently recommended antiretroviral (ARV) regimens for initial therapy consist of two nucleoside reverse transcriptase inhibitors (NRTIs) plus a third active drug. In addition, dolutegravir (DTG)/lamivudine (3TC) is a two-drug, NRTI-limiting regimen that is a recommended option for people with HIV (see [Table 6a](#) in [Initial Combination Antiretroviral Regimens for People With HIV](#)) and would be preferred over others in situations where it is desirable to avoid abacavir (ABC), tenofovir alafenamide (TAF), and tenofovir disoproxil fumarate (TDF). Several other NRTI-limiting/sparing two-drug regimens have been evaluated in clinical studies, but are not yet recommended by the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) due to insufficient evidence. Note that two-drug regimens **are not currently recommended** during pregnancy and tenofovir-sparing regimens **are not recommended** in people with hepatitis B virus (HBV)/HIV coinfection unless another HBV-active drug (i.e., entecavir) is added. Clinicians should refer to [HBV/HIV Coinfection](#) for guidance on treatment of people with HBV infection when TAF or TDF cannot be used as part of the ARV regimen.

Strategies Supported by Evidence From Clinical Trials

Dolutegravir/Lamivudine

- In the GEMINI-1 and GEMINI-2 trials, 1,433 antiretroviral therapy (ART)–naïve participants with baseline HIV RNA <500,000 copies/mL and no evidence of HBV infection were randomized to receive a two-drug regimen of DTG plus 3TC or a three-drug regimen of DTG plus TDF/emtricitabine (FTC). At Week 96, DTG plus 3TC was non-inferior to DTG plus TDF/FTC based on the proportion of participants with viral loads <50 copies/mL (86% in DTG plus 3TC group and 89.5% in DTG plus TDF/FTC group).¹ Virologic nonresponse was uncommon, occurring in 3.1% of participants who received DTG plus 3TC and 2% of participants who received DTG plus TDF/FTC. No instances of treatment-emergent NRTI or integrase strand transfer inhibitor (INSTI) resistance occurred in either treatment group. Among participants who started the study with CD4 counts <200 cells/mm³, the proportion of participants with HIV RNA <50 copies/mL at Week 96 was lower in the DTG plus 3TC group than in the DTG plus TDF/FTC group; however, the difference was not related to a higher rate of virologic failures in the two-drug group. At Week 144, DTG plus 3TC maintained noninferiority to DTG plus TDF/FTC with 82% versus 84% of participants maintaining viral load <50 copies/mL, respectively. The proportion of participants with viral load ≥50 copies/mL was similar between both treatment groups at 3%. There was a lower risk of drug-related adverse events with DTG plus 3TC versus DTG plus TDF/FTC (19.6% vs. 25.0%; relative risk ratio, 0.78; 95% confidence interval, 0.64–0.95).¹

The Panel's Recommendation

When ABC, TAF, or TDF cannot be used or are not optimal, the Panel recommends DTG/3TC as an initial regimen for people with HIV (**AI**) who—

- Have a viral load $\leq 500,000$ copies/mL,
- Have a genotype available that demonstrates sensitivity to 3TC,
- Do not have a history of long-acting cabotegravir (CAB-LA) use as pre-exposure prophylaxis, or
- Have CAB-LA exposure but with documented INSTI sensitivity on genotypic resistance testing.

HBV status must be determined, and if HBV/HIV coinfection is present, another HBV-active drug should be added (see [HBV/HIV Coinfection](#)).

Nucleoside-Limiting Regimens With Some Supporting Data but Not Recommended as Initial Therapy

The Panel **does not recommend** the following nucleoside-limiting or nucleoside-sparing regimens for initial therapy due to insufficient clinical trial data.

Darunavir/Ritonavir Plus Lamivudine

- In the ANDES trial, 336 participants were randomized 1:1 to receive open-label, once-daily dual therapy with darunavir/ritonavir (DRV/r) plus 3TC or triple therapy with DRV/r plus TDF/3TC. This study was conducted in Argentina, and the researchers used an FDC of DRV/r 800 mg/100 mg that is available in that country. The median baseline HIV RNA was 4.5 log₁₀ copies, and 23% of participants had HIV RNA $>100,000$ copies/mL. At Week 48, 91% of the participants in the dual-therapy group and 93% of the participants in the triple-therapy group achieved an HIV RNA <50 copies/mL, demonstrating that dual therapy was non-inferior to triple therapy.² The rates of virologic suppression among study participants who had pre-therapy HIV RNA levels $>100,000$ copies/mL were similar in the dual- and triple-therapy groups (87% and 90%, respectively). The Panel **does not recommend** this regimen as initial therapy as the results from this study have not yet been published.

Darunavir/Ritonavir Plus Raltegravir

- In the NEAT/ANRS 143 study, 805 treatment-naïve participants were randomized to receive twice-daily raltegravir (RAL) or once-daily TDF/FTC, each with DRV/r (800 mg/100 mg once daily). At Week 96, DRV/r plus RAL was non-inferior to DRV/r plus TDF/FTC based on the primary endpoint, the proportion of people with virologic or clinical failure. Among those with baseline CD4 counts <200 cells/mm³; however, there were more virologic failures in the RAL plus DRV/r arm. A trend towards more failure was also observed among those with pre-treatment HIV RNA $\geq 100,000$ copies/mL.³ High rates of virologic failure in participants with HIV RNA $>100,000$ copies/mL were also seen in two smaller studies of DRV/r plus RAL.^{4,5} The Panel **does not recommend** DRV/r plus RAL as initial ART because of the higher rate of virologic failure in participants with HIV RNA $\geq 100,000$ copies/mL and the higher pill burden of this regimen compared to other Panel-recommended initial ARV regimens.

Darunavir/Ritonavir Plus Rilpivirine

- In a single-arm, open-label pilot study, 36 ART-naïve participants without genotypic evidence of resistance to DRV or rilpivirine (RPV) received DRV/r plus RPV for 48 weeks. Half of the participants (18 of 36) had baseline HIV viral loads >100,000 copies/mL. By Week 36, 97% of participants (35 of 36) achieved HIV RNA <50 copies/mL, and by Week 48, all achieved viral suppression (HIV RNA < 50 copies/mL).⁶ The Panel **does not recommend** this regimen as initial ART given the small sample size of the study described above and the lack of comparative data evaluating DRV/r plus RPV as initial therapy for people with HIV.

Long-Acting Injectable Cabotegravir With Rilpivirine

- The long-acting injectable combination of CAB plus RPV (LA CAB/RPV) has not been studied in ART-naïve participants. In the Phase 3 trial FLAIR and Phase IIb trial LATTE-2,^{7,8} ART-naïve participants were first treated with 20 weeks of DTG/ABC/3TC or oral CAB+ABC/3TC, respectively. Study participants who achieved virologic suppression were eligible for randomization to receive LA CAB/RPV every month or to continue oral daily ART. The Panel **does not recommend** the LA CAB/RPV as initial therapy for people with HIV because of the lack of data supporting the efficacy of this combination in people who are ART-naïve (**AIII**). People desiring to use LA CAB/RPV early in their treatment history should first attain viral suppression on a recommended regimen before transitioning to LA CAB/RPV. See [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#) for more discussion.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = One or more randomized trials with clinical outcomes and/or validated laboratory endpoints; II = One or more well-designed, nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

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Table 9. Advantages and Disadvantages of Antiretroviral Components of Initial Antiretroviral Therapy Listed in Table 6a and Table 6b

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Note: All drugs within an ARV class are listed in alphabetical order. Information based on [Table 6a](#) and [Table 6b](#) in the [Initial Combination Antiretroviral Regimens for People With HIV](#) section.

ARV Class	ARV Agent(s)	Advantage(s)	Disadvantage(s)
Dual-NRTI	ABC/3TC	- Coformulated with DTG - Generic formulations are available for ABC/3TC, ABC, and 3TC.	- May cause life-threatening HSRs in people who test positive for the HLA-B*5701 allele. As a result, HLA-B*5701 testing is required before use. - ABC use has been associated with CV disease and cardiac events in some, but not all, observational studies.
	TAF/FTC	- Coformulated with BIC, DRV/c, or RPV - Active against HBV; a recommended dual-NRTI option for people with HBV/HIV coinfection - Smaller decline in renal function, less proteinuria, and smaller reductions in BMD than TDF/FTC - Approved for people with eGFR ≥ 30 mL/min - Can be used in people on chronic hemodialysis	See text in the NRTI section regarding weight gain with TAF.
	TDF/3TC	- Coformulated with DOR - Generic formulations are available for TDF, 3TC, or TDF/3TC. - Long-term clinical experience - Active against HBV	- Renal toxicity, including proximal tubulopathy and acute or chronic renal insufficiency, especially when combined with pharmacologic boosters - Osteomalacia has been reported as a consequence of proximal tubulopathy. - Decreased BMD has been associated with use of TDF, especially when combined with pharmacologic boosters.
	TDF/FTC	- Active against HBV; a recommended dual-NRTI option for people with HIV/HBV coinfection - TDF is associated with lower lipid levels than TAF.	- Renal toxicity, including proximal tubulopathy and acute or chronic renal insufficiency, especially when combined with pharmacologic boosters - Osteomalacia has been reported as a consequence of proximal tubulopathy.

Table 9. Advantages and Disadvantages of Antiretroviral Components Recommended as Initial Antiretroviral Therapy Listed in Table 6a and Table 6b

ARV Class	ARV Agent(s)	Advantage(s)	Disadvantage(s)
			Decreased BMD has been associated with use of TDF, especially when combined with pharmacologic boosters.
Single NRTI	3TC	- Coformulated with DTG as STR - Avoids potential toxicities associated with TDF, TAF, ABC	DTG/3TC is not recommended for individuals with HIV RNA >500,000 copies/mL, HBV coinfection unless on another HBV active drug, or in whom ART is to be started before the results of HIV genotypic resistance testing for reverse transcriptase or HBV testing are available.
INSTI	BIC	- Coformulated with TAF/FTC - Higher barrier to resistance than EVG and RAL - No food requirement	- Oral absorption of BIC can be reduced by simultaneous administration with drugs or supplements containing polyvalent cations (e.g., Al³⁺, Ca²⁺, or Mg-containing antacids or supplements or multivitamin tablets with minerals). See dosing recommendations in Table 24d. - CYP3A4 and UGT1A1 substrate (but not a CYP3A4 inducer or inhibitor); potential for drug–drug interactions. - See text in the INSTI section regarding weight gain and INSTI use.
	DTG	- Higher barrier to resistance than EVG or RAL - Coformulated with ABC/3TC and 3TC as STR - No food requirement - Minimal CYP3A4 interactions - Favorable lipid profile	- Oral absorption of DTG can be reduced by simultaneous administration with drugs containing polyvalent cations (e.g., Al³⁺, Ca²⁺, or Mg-containing antacids or supplements or multivitamin tablets with minerals). See dosing recommendations in Table 24d. - UGT1A1 substrate; potential for drug interactions (see Table 24d) - Depression and suicidal ideation (rare; usually in people with preexisting psychiatric conditions) - See text in the INSTI section regarding weight gain and INSTI use.
NNRTI	DOR	- Coformulated with TDF/3TC - Fewer CNS side effects compared to EFV and RPV - No food requirement	- Shorter-term clinical experience than with RPV - Potential for CYP450 drug interactions (see Tables 24b, 25a, and 25b) - Treatment-emergent DOR resistance mutations may confer resistance to certain NNRTIs.

Table 9. Advantages and Disadvantages of Antiretroviral Components Recommended as Initial Antiretroviral Therapy Listed in Table 6a and Table 6b

ARV Class	ARV Agent(s)	Advantage(s)	Disadvantage(s)
	RPV	Coformulated with TAF/FTC	Not recommended in people with pre-ART HIV RNA >100,000 copies/mL or CD4 counts <200 cells/mm³ because of higher rate of virologic failure in these people. - Depression and suicidality - QTc interval prolongation; consider using an alternative to RPV in people taking medications with known risk of causing Torsades de Pointes or in those at higher risk of Torsades de Pointes. - Rash - Transmitted resistance is more common than with PIs and INSTIs. - More NNRTI-, TDF-, and 3TC-associated mutations at virologic failure than with regimens that contain EFV and two NRTIs - Potential for CYP450 drug interactions (see Tables 24b and 25a) - Meal requirement (>390 kcal) - Requires acid for adequate absorption - Contraindicated with PPIs. - Use with H2 antagonists or antacids with caution (see Table 24a for detailed dosing information).
PI	DRV/c or DRV/r	- Higher barrier to resistance than NNRTIs - PI resistance at the time of treatment failure is uncommon with PK-enhanced PIs.	- Skin rash - Food requirement - GI adverse effects - CYP3A4 inhibitors and substrates: potential for drug interactions (see Table 24a) - Increased CV risk reported in one observational cohort study^a - Hepatotoxicity has been reported, especially in those with preexisting liver disease.

Table 9. Advantages and Disadvantages of Antiretroviral Components Recommended as Initial Antiretroviral Therapy Listed in Table 6a and Table 6b

ARV Class	ARV Agent(s)	Advantage(s)	Disadvantage(s)
	DRV/c Specific considerations	Coformulated as DRV/c and DRV/c/TAF/FTC	- COBI inhibits active tubular secretion of Cr and can increase serum Cr without affecting renal glomerular function. - Coadministration with TDF is not recommended in people with CrCl <70 mL/min. - COBI (like RTV) is a potent CYP3A4 inhibitor, which can result in significant interactions with CYP3A substrates. - COBI should be avoided in pregnancy because levels of COBI and its boosted drugs are lower in the second and third trimesters. If women who are pregnant with suppressed virus on DRV/c elect to continue on the drug, frequent viral load monitoring is recommended.

^a D:A:D international prospective multicohort study¹

Key: 3TC = lamivudine; ABC = abacavir; Al = aluminum; ART = antiretroviral therapy; ARV = antiretroviral; BIC = bictegravir; BMD = bone mineral density; Ca = calcium; CD4 = CD4 T lymphocyte; CNS = central nervous system; COBI = cobicistat; Cr = creatinine; CrCl = creatinine clearance; CV = cardiovascular; CYP = cytochrome P; DOR = doravirine; DRV/c = darunavir/cobicistat; DRV/r = darunavir/ritonavir; DTG = dolutegravir; EFV = efavirenz; eGFR = estimated glomerular filtration rate; FTC = emtricitabine; GI = gastrointestinal; H2 = histamine 2; HBV = hepatitis B virus; HLA = human leukocyte antigen; HSR = hypersensitivity reaction; INSTI = integrase strand transfer inhibitor; Mg = magnesium; NNRTI = non-nucleoside reverse transcriptase inhibitor; NRTI = nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; PK = pharmacokinetic; PPI = proton pump inhibitor; QTc = QT corrected for heart rate; RAL = raltegravir; RPV = rilpivirine; RTV = ritonavir; STR = single-tablet regimen; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate; UGT = uridine diphosphate glucuronosyltransferase

References

1. Ryom L, Lundgren JD, El-Sadr W, et al. Cardiovascular disease and use of contemporary protease inhibitors: the D:A:D international prospective multicohort study. *Lancet HIV*. 2018;5(6):e291-e300. Available at: <https://pubmed.ncbi.nlm.nih.gov/29731407>.

MANAGEMENT OF PEOPLE WITH HIV AND ANTIRETROVIRAL THERAPY EXPERIENCE

Virologic Failure

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Key Considerations and Recommendations

- Assessing and managing a person with HIV who is experiencing antiretroviral therapy (ART) failure can be complex. Expert advice can be critical and should be sought in many instances.
- Evaluation of virologic failure should include an assessment of ART adherence, drug–drug and drug–food interactions, drug tolerability, HIV RNA level and CD4 T lymphocyte (CD4) cell count trends over time, ART history, and prior and current drug-resistance test results.
- Drug-resistance testing should be performed while the person is taking the failing antiretroviral (ARV) regimen **(AI)** or within 4 weeks of discontinuing a non–long-acting ARV regimen **(AII)**. If more than 4 weeks have elapsed since non–long-acting ARV regimens were discontinued, resistance testing still can provide useful information to guide therapy, although it may not detect previously selected resistance mutations **(CIII)**.
- For people who previously received long-acting (LA) cabotegravir (CAB) plus rilpivirine (RPV), or LA CAB/RPV, and present with virologic failure, resistance testing (including integrase strand transfer inhibitor [INSTI] genotypic testing) should be performed regardless of the time since the last dose of LA CAB/RPV **(AIII)**.
- The goal of treatment for people with HIV with drug resistance who are experiencing virologic failure is to establish virologic suppression (i.e., HIV RNA levels below the lower limits of detection of currently used assays) **(AI)**.
- A new ARV regimen should preferably include two fully active drugs if at least one has a high resistance barrier, such as a second-generation INSTI or a boosted protease inhibitor (PI) **(AI)**.
- A new ARV regimen can also include a second-generation INSTI (i.e., dolutegravir [DTG]) plus a boosted PI (preferably boosted darunavir) without nucleoside reverse transcriptase inhibitors (NRTIs) if both are fully active **(AI)**.
- If the above two options are not feasible, a new ARV regimen that includes at least one drug with a high resistance barrier plus two partially active NRTIs—particularly tenofovir alafenamide (TAF) or tenofovir disoproxil fumarate (TDF) with lamivudine (3TC) or emtricitabine (FTC)—can also be considered, although this is a less well-defined strategy and is based on an extrapolation of existing data. Caution with close monitoring of virologic response is advised **(BII)**, as discussed below.
- If no fully active drug with a high resistance barrier is available, every effort should be made to include three fully active drugs in the new ARV regimen **(AI)**.
- In general, adding a single ARV drug to a virologically failing regimen **is not recommended**, because this would rarely result in full virologic suppression and may risk the development of resistance to all drugs in the regimen **(BII)**.
- In some rare instances, it might not be possible to achieve maximal virologic suppression in people with HIV who are highly ART-experienced and have extensive drug resistance. In this case, ART should be continued **(AI)** with regimens that are designed to maintain CD4 counts, preserve treatment options, delay clinical progression, and minimize toxicity.

- When it is not possible to construct a viable suppressive regimen for a person with multidrug-resistant HIV, the clinician should consider enrolling the person in a clinical trial of investigational agents or contacting pharmaceutical companies that may have investigational agents available.
- In people with HIV and virologic failure, it is crucial to provide continuous adherence support before and after ARV regimen changes.
- When switching an ARV regimen in a person with hepatitis B virus (HBV)/HIV coinfection, the person should remain on an ARV agent that is active against HBV and has a high resistance barrier to HBV (i.e., TAF, TDF, or entecavir) to avoid HBV rebound and hepatocellular damage **(AII)** (see [Hepatitis B Virus/HIV Coinfection](#) for more information).
- Discontinuing or briefly interrupting therapy may lead to a rapid increase in HIV RNA, a decrease in CD4 count, and an increase in the risk of clinical progression. Therefore, discontinuing or interrupting ART **is not recommended** in the setting of virologic failure **(AI)**.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Antiretroviral (ARV) regimens that are currently recommended for initial therapy in people with HIV have a high likelihood of achieving and maintaining plasma HIV RNA levels that are below the lower limits of detection (LLOD) of currently used assays (see [Initial Combination Antiretroviral Regimens for People With HIV](#)). People on antiretroviral therapy (ART) who do not achieve this treatment goal or who experience virologic rebound can develop resistance mutations to one or more components of their ARV regimen with possible cross-resistance to other ARVs. Adherence to ARV regimens can be challenging for some people with HIV, and poor adherence can result in detectable HIV RNA levels. The extent of drug resistance varies based on ARV treatment history, with some people having minimal or no resistance and others with extensive resistance. Managing people with extensive resistance is complex and usually requires consultation with an HIV expert. This section of the guidelines defines virologic failure in people on ART and discusses strategies to manage ART in these individuals.

Virologic Response Definitions

The following definitions are used in this section to describe the different levels of virologic response to ART.

- **Virologic suppression:** A confirmed HIV RNA level below the LLOD of available assays.
- **Virologic failure:** The inability to achieve or maintain suppression of viral replication to HIV RNA level <200 copies/mL.
- **Incomplete virologic response:** Two consecutive plasma HIV RNA levels ≥ 200 copies/mL after 24 weeks on an ARV regimen in a person who has not yet had documented virologic suppression on that regimen. A person's baseline HIV RNA level may affect the time course of response, and some regimens may take longer than others to suppress HIV RNA levels.
- **Virologic rebound:** After virologic suppression, confirmed HIV RNA level ≥ 200 copies/mL.
- **Virologic blip:** After virologic suppression, an isolated detectable HIV RNA level that is followed by a return to virologic suppression.
- **Low-level viremia:** Confirmed HIV RNA level above the LLOD but <200 copies/mL.

Antiretroviral Therapy Goals and Presence of Viremia While on Antiretroviral Therapy

The goal of ART is to suppress HIV replication to a level below which drug-resistance mutations cannot emerge. Although not conclusive, the evidence suggests that selection of drug-resistance mutations does not occur in people with HIV RNA levels that are persistently suppressed below the LLOD of current assays.¹

Virologic blips are not usually associated with subsequent virologic failure.² In contrast, there is controversy regarding the clinical implications of low-level viremia, i.e., persistent HIV RNA levels between the LLOD and <200 copies/mL in people on ART. Viremia at this threshold is detected with some frequency by commonly used real-time polymerase chain reaction (PCR) assays, which are more sensitive than the PCR-based HIV RNA platforms used in the past.³⁻⁵ Several retrospective studies support the supposition that virologic failure is more likely to occur in people with HIV RNA levels ≥ 200 copies/mL than in those with low-level viremia between 50 copies/mL and 199 copies/mL.⁶⁻⁸ However, other studies have suggested that detectable viremia at <200 copies/mL can be predictive of virologic failure⁹ and can be associated with the evolution of drug resistance¹⁰ and non-AIDS morbidities.¹¹

Persistent HIV RNA levels ≥ 200 copies/mL are often associated with evidence of viral evolution and accumulation of drug-resistance mutations.¹² This association is particularly common with HIV RNA levels >500 copies/mL.¹³ Therefore, people who have persistent HIV RNA levels ≥ 200 copies/mL are considered to be experiencing virologic failure.

Causes of Virologic Failure

Virologic failure can occur for many reasons. Data from cohorts in the earlier era of combination ART suggested that suboptimal adherence and drug intolerance/toxicity are key contributors to virologic failure and regimen discontinuations.^{14,15} Current ARVs are easier to tolerate and have less pill burden, which may improve adherence.¹⁶⁻¹⁸ Virologic failure may be associated with a variety of factors, including the following:

HIV-Related Factors

- Presence of transmitted or acquired drug-resistant virus that may or may not be documented by current or past drug-resistance test results
- Prior ARV treatment failure
- Innate drug resistance to prescribed ARV drugs
- Higher pre-treatment HIV RNA levels (e.g., HIV RNA >100,000 copies/mL; some regimens may be less effective at higher levels)

Antiretroviral Regimen–Related Factors

- Suboptimal pharmacokinetics (PK) (e.g., variable absorption, metabolism, penetration into reservoirs)
- Suboptimal virologic potency
- Low barrier to resistance
- Reduced efficacy due to prior exposure to suboptimal regimens (e.g., monotherapy, dual-nucleoside reverse transcriptase inhibitor [NRTI] therapy, or the sequential introduction of drugs)

- Food requirements
- Drug–drug interactions with concomitant medications, which may reduce concentrations of the ARV drugs
- Adverse drug effects
- High pill burden and/or dosing frequency
- Prescription (prescribing or dispensing) errors

Social and Adherence-Related Factors

Note: Each of the social and adherence-related factors listed below are discussed in the [Adherence to the Continuum of Care](#) section.

- Active substance use, mental health disorders, or neurocognitive impairment
- Unstable housing and other psychosocial factors
- Missed clinic appointments
- Interruption of or intermittent access to ART
- Cost and affordability of ARV drugs, which may affect the ability to access or continue therapy (see the [Cost Considerations and Antiretroviral Therapy](#) section)

Managing People With HIV and Virologic Failure

If virologic failure is suspected or confirmed, a thorough assessment of whether one or more of the above factors could have been the cause(s) of failure is indicated. The causes of virologic failure can usually be identified but may not be obvious in some cases. Distinguishing among the causes of virologic failure is important, because the approaches to subsequent therapy may differ, depending on the cause. Potential causes of virologic failure should be explored in depth. Once virologic failure is confirmed, steps should be taken to improve virologic outcomes. If a new ARV regimen is needed, approaches to designing a new regimen are discussed below.

Key Factors to Consider When Designing a New Antiretroviral Regimen After Virologic Failure

General Principles on Antiretroviral Use in Virologic Failure

- When designing a new ARV regimen for a person with virologic failure, it is important to consider the factors outlined above on causes of virologic failure and, if possible, consider better tolerated and adherence-friendly regimens.
- A new ARV regimen should be selected based on ART history, current and previous drug-resistance test results, and whether a fully susceptible ARV drug with a high barrier to resistance and other fully active drugs are available.^{9,19–31}
- ARV drugs with a high barrier to resistance are those in which emergent resistance is uncommon in people experiencing virologic failure. These include boosted darunavir (DRV), dolutegravir (DTG), and bictegravir (BIC).
- Fully active drugs may include—
 - o Drugs in classes for which the person has not previously selected for drug-resistant virus.

- o Newer drugs in existing drug classes that are predicted to be fully active against HIV isolates despite the presence of resistance mutations for some drugs in the same drug class. For example, the non-nucleoside reverse transcriptase inhibitors (NNRTIs) etravirine and possibly doravirine (DOR), the protease inhibitor DRV, and the integrase strand transfer inhibitors (INSTIs) DTG and BIC. However, clinical data supporting the use of DOR or BIC in the setting of virologic failure are limited.
- o Drugs with novel mechanisms of action that the person with HIV has not received before, such as the post-attachment inhibitor ibalizumab (IBA), the gp120 attachment inhibitor fostemsavir (FTR), the capsid inhibitor lenacapavir (LEN), the fusion inhibitor enfuvirtide (T-20), or the CCR5 antagonist maraviroc (MVC) in people with no detectable CXCR4-using virus.
- ARV drugs with partial activity are those predicted to have antiviral activity but to a lesser extent than when there is no underlying drug resistance.
- Administering a drug that a person has never used does not ensure that the drug will be fully or partially active; the potential exists for cross-resistance among drugs from the same class.
- Discontinuing or briefly interrupting therapy in a person with viremia **is not recommended** because it may increase HIV RNA levels, decrease CD4 T lymphocyte (CD4) cell count, and increase the risk of clinical progression **(AI)**^{32,33} (see [Discontinuation or Interruption of Antiretroviral Therapy](#)).
- The presence of preexisting (transmitted) drug resistance also may lead to virologic failure.^{34,35}

Drug-Resistance Testing to Guide New Antiretroviral Regimens

- The Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends that drug-resistance testing should be used to guide ARV regimen design and should be performed while the person with HIV is still taking the failing regimen **(AI)** or within 4 weeks of discontinuation of a non–long-acting regimen **(AII)**. If more than 4 weeks have elapsed since discontinuation of a non–long-acting regimen, drug-resistance testing still may provide useful information to guide therapy, although it may not detect previously selected resistance mutations **(CIII)**.
- Drug-resistance testing is recommended in persons with confirmed virologic failure, i.e., HIV RNA repeatedly >200 copies/mL **(AI)** for >1,000 copies/mL, **AIII** for 501–1,000 copies/mL, **CIII** for confirmed 201–500 copies/mL), although at low HIV RNA levels, testing may be difficult to obtain outside of a research setting. In persons with HIV RNA >200 copies/mL but <500 copies/mL, testing may be unsuccessful, but it still should be considered.
- Drug resistance is cumulative, meaning that once a mutation is detected in a resistance assay, it should be considered present in that person's HIV thereafter (this is sometimes referred to as “archived” resistance), regardless of whether it appears on subsequent drug-resistance assays. Thus, clinicians should evaluate the extent of drug resistance, taking into account all of the person's ART history and, importantly, prior genotypic- or phenotypic-resistance test results.
- Activity of ART based on current and cumulative genotypic mutations can be estimated by tools and interpretation algorithms, such as the [Stanford University HIV Drug-Resistance Database](#). Also see [Drug-Resistance Testing](#).
- Some drug-resistance assays only detect resistance to NRTIs, NNRTIs, or protease inhibitors (PIs); INSTI-resistance testing may need to be ordered separately. INSTI-resistance testing should be ordered in people who experience virologic failure on an INSTI-based regimen. Additional drug-resistance tests for people who experience failure on a fusion inhibitor **(AII)**, and viral tropism tests for people who experience failure on a CCR5 antagonist **(BIII)** are also

available. There is currently no commercially available resistance test for IBA, FTR, or LEN (see [Drug-Resistance Testing](#)).

Strategies for New Antiretroviral Regimen Design

- A new ARV regimen **should preferably include** two fully active drugs if at least one has a high resistance barrier, such as a second-generation INSTI or a boosted PI **(AI)**.³⁶
- A new ARV regimen can also include **a second-generation INSTI (e.g., DTG)** plus a boosted PI (preferably boosted DRV) without NRTIs if both are fully active **(AI)**; ³⁷⁻⁴¹ see more detailed discussion in *Managing Virologic Failure in Different Clinical Scenarios* below. **If this regimen is used in people with hepatitis B virus (HBV)/HIV coinfection, an HBV-active drug with a high barrier to resistance (e.g., tenofovir alafenamide [TAF], tenofovir disoproxil fumarate [TDF], or entecavir) should be continued or started to avoid HBV rebound and hepatocellular damage.**
- **If the above two options are not feasible, a new ARV regimen can also include a fully active drug with a high resistance barrier plus two partially active NRTIs—particularly TAF or TDF with lamivudine (3TC) or emtricitabine (FTC)—though this is less well-defined and close monitoring is advised (BII); this is discussed in more detail in *Managing Virologic Failure in Different Clinical Scenarios* below.**^{36,37,42,43}
- If no fully active drug with a high resistance barrier is available, every effort should be made to include three fully active drugs in the regimen **(AI)**. See the clinical scenarios below for further guidance on the number of fully active drugs a regimen should contain.
- Despite the presence of drug-resistance mutations, some ARV drugs in the regimen may still have partial activity against the person's HIV and may be retained as part of a salvage regimen. These drugs may include NRTIs, PIs, and second-generation INSTIs, although dosing of some drugs (e.g., DRV and DTG) may need to be increased when treating people with relevant resistance mutations to achieve drug concentrations necessary to be at least partially active against a less sensitive virus.⁴⁴⁻⁴⁶
- In contrast, other agents in which resistance may be expected should be discontinued, because their continued use is unlikely to contribute to virologic suppression. These drugs include NNRTIs (especially efavirenz, nevirapine, and rilpivirine [RPV]), the first-generation INSTIs raltegravir (RAL) and elvitegravir (EVG), and T-20.⁴⁷⁻⁴⁹
- **Long-acting (LA) cabotegravir (CAB)/rilpivirine (RPV) is currently approved as a complete ART regimen for individuals engaged in care who have good adherence to an oral regimen, sustained undetectable HIV RNA levels for >3 months, no known or suspected resistance to either drug, and no active HBV infection. To date, no large or randomized clinical trials have been published using any INSTI plus NNRTI, including LA CAB/RPV, as a complete regimen for people without viral suppression. However, a few small observational studies have shown that people with viremia despite intensive efforts on oral ART, without resistance to CAB or RPV, can achieve viral suppression with LA CAB/RPV when intensive supportive services are available.**⁵⁰⁻⁵³ **If considering LA CAB/RPV in this population, it is important to note that data are limited, and this is not an approved indication of the regimen. If the regimen is used in people with inconsistent adherence and virologic failure, there is a high likelihood of developing acquired resistance to CAB and/or RPV, which might limit future treatment options. The approaches and considerations for the use of LA CAB/RPV in people with viremia is further [outlined below](#).**
- When changing an ARV regimen in a person with HBV/HIV coinfection, ARV drugs that are active against HBV (especially TAF or TDF) should be continued as part of the new regimen or, if not possible, entecavir should be initiated **(AII)**. Using 3TC or FTC as the only drug with HBV activity in a regimen **is not recommended (AII)** because HBV resistance to these drugs can

emerge. Discontinuation of these drugs may lead to the reactivation of HBV, which may result in serious hepatocellular damage (see [Hepatitis B Virus/HIV Coinfection](#)).

- Virologic responses should be closely monitored after regimen switch (e.g., HIV RNA level testing performed within 4 to 8 weeks), with prompt drug-resistance testing if virologic response is inadequate.

Managing Virologic Failure in Individuals With Different Levels of Viremia

People with HIV and detectable HIV RNA levels while on ART comprise a heterogeneous group of individuals with different ART exposure histories, degrees of drug resistance, durations of virologic failure, and levels of plasma viremia. Management strategies should be individualized. The first steps are to confirm the level of HIV viremia and to assess and address adherence and potential drug–drug interactions (including interactions with over-the-counter products and supplements) and drug–food interactions. Some general approaches based on the level of viremia are addressed below.

- **Low-level viremia (HIV RNA above the LLOD and <200 copies/mL):** People who have these HIV RNA levels do not typically require a change in treatment (**AII**).⁴ Although there is no consensus on how to manage these individuals, the risk that drug resistance will emerge is believed to be relatively low. Therefore, these individuals should continue their current regimens and have their HIV RNA levels monitored at least every 3 months to assess the need for changes to ART in the future (**AIII**).
- **HIV RNA ≥200 copies/mL and <1,000 copies/mL:** In contrast to people with detectable HIV RNA levels that are persistently <200 copies/mL, people with levels that are persistently ≥200 copies/mL often develop drug resistance, particularly when HIV RNA levels are >500 copies/mL.^{6,7} People who have persistent plasma HIV RNA levels in the range of 200 copies/mL to 1,000 copies/mL are considered to be experiencing virologic failure, and drug-resistance testing should be attempted, particularly in people with HIV RNA levels >500 copies/mL (at <500 copies/mL, testing may not be possible for technical reasons). Management approaches should be the same as for people with HIV RNA >1,000 copies/mL (as outlined below). When drug-resistance testing cannot be performed because of low HIV RNA levels, the decision of whether to empirically change ARV drugs should be made on a case-by-case basis, taking into account whether a new regimen that is expected to fully suppress viremia can be constructed. If genotypic-resistance test results cannot be obtained because of low HIV RNA levels, proviral DNA genotypic testing may be considered. Results from this test should be interpreted with caution because these assays might miss some or all previously existing drug-resistance mutations. However, mutations that are detected using proviral DNA genotypic testing may be significant and can affect the effectiveness of future regimens (see [Drug-Resistance Testing](#)).
- **HIV RNA ≥1,000 copies/mL and no drug-resistance mutations identified using current or previous genotypic-resistance test results:** This scenario is almost always associated with suboptimal adherence. A thorough assessment should be conducted to evaluate the level of adherence, identify and address the underlying cause(s) for incomplete adherence and, if possible, simplify the regimen (e.g., decrease pill count, simplify food requirement or dosing frequency; see [Adherence to the Continuum of Care](#)). Assessment includes the following:
 - o Assessing the person's access to ART, including access to pharmacy, refills, and copays or patient assistance programs, and seeking assistance to overcome any barriers to consistent access to ART.

- o Assessing social determinants that may impact adherence (e.g., substance use, unstable housing, mental health challenges, etc.).
- o Assessing the person's tolerance of the current ARV regimen and the severity and duration of side effects if intolerance is a concern, while considering that even minor side effects can affect adherence.
- o Addressing intolerance by treating symptoms (e.g., with antiemetics or antidiarrheals), switching one ARV agent in a regimen to another agent in the same drug class, or switching from one drug class to another class (e.g., from an NNRTI to a PI or an INSTI; see [Adverse Effects of Antiretroviral Agents](#)).
- o Reviewing food requirements for each medication and assessing whether the person adheres to the requirements.
- o Assessing whether a recent history of gastrointestinal symptoms (e.g., vomiting, diarrhea) may result in short-term malabsorption.
- o Reviewing concomitant medications (including over-the-counter medications and dietary supplements) for possible adverse drug–drug interactions (consult [Drug–Drug Interactions](#) and Tables [24a](#) through [25b](#) for common interactions) and, if possible, making appropriate substitutions for ARV agents and/or concomitant medications.
- o Considering therapeutic drug monitoring if PK drug–drug interactions (e.g., when used with rifamycin) or impaired drug absorption (e.g., using polyvalent cations with an INSTI) leading to decreased ARV drug exposure is suspected.
- o Considering the timing of the drug-resistance test (e.g., was the person mostly or completely nonadherent to oral ART for >4 weeks before testing?) (see [Drug-Resistance Testing](#)).
- o After a thorough assessment, the following approaches may be considered:
 - If the current regimen is well tolerated, with no significant drug–drug or drug–food interactions, it is reasonable to continue the same regimen while focusing on improving adherence.
 - If the agents are poorly tolerated or have important drug–drug or drug–food interactions, changing the regimen to an equally effective but more tolerable regimen should be considered.
 - HIV RNA level testing should be repeated 4 to 8 weeks after treatment adherence is encouraged or treatment is modified (**AII**); if RNA levels remain >200 copies/mL, genotypic drug-resistance testing should be performed to determine whether a resistant viral strain has emerged (**AI** for >1,000 copies/mL, **AIII** for 501–1,000 copies/mL, **CIII** for 201–500 copies/mL), though at low RNA levels, testing may be difficult to obtain outside of a research setting.
- **HIV RNA >1,000 copies/mL and drug resistance identified:** If new or previously detected resistance mutations compromise the regimen, the regimen should be modified as soon as possible to avoid progressive accumulation of resistance mutations.⁵⁴ In addition, several studies have shown that virologic responses to new and fully active regimens are greater in individuals with lower HIV RNA levels and/or higher CD4 counts at the time of regimen changes; thus, the change is best done before viremia worsens or before CD4 count declines.^{9,55} The availability of newer ARV drugs, including some with new mechanisms of action, makes it possible to suppress HIV RNA levels to below the LLOD in most people with HIV. The options in this setting depend on the extent of drug resistance and are addressed in the clinical scenarios outlined below.

Managing Virologic Failure in Different Clinical Scenarios

Note: See [Table 10](#) below for a summary of these recommendations.

Virologic Failure on the First Antiretroviral Regimen

The Panel recommends that drug-resistance testing should be performed upon treatment failure to inform regimen design **(AI)**.

NNRTI plus NRTI regimen failure: Although an NNRTI plus NRTI regimen is no longer considered a preferred first-line ART option in treatment guidelines, data from clinical trials comparing different ARV regimens after NNRTI plus NRTI failure provide the most robust evidence to inform second-line treatment strategies and, therefore, are included here.

In this setting, people with HIV often have viral resistance to the NNRTI, with or without the M184V/I mutation, which confers high-level resistance to 3TC and FTC. Additional NRTI mutations also may be present. Below are some treatment options.

- **Second-generation INSTI (BIC or DTG) plus NRTIs:** The Panel recommends that fully active DTG plus two NRTIs, at least one of which is fully active, can be a treatment option after failure of a first-line NNRTI-based therapy **(AI)**. BIC, which is available only in a combination pill with FTC/TAF, also has a high resistance barrier and may have similar activity to that of DTG in this setting, and **is also an option (AIII)**; however, no clinical trial data for this strategy with BIC is available. If at least one fully active NRTI cannot be assured and a clinician wants to avoid using a boosted PI or a drug from other classes, a regimen that includes fully active DTG plus two NRTIs that are estimated to be only partially active (particularly TAF or TDF with 3TC or FTC) can be considered, **with caution and close monitoring of viral response, as further discussed below (BII)**.

In the DAWNING trial, participants from 13 countries who experienced virologic failure while on a first-line NNRTI-based regimen were randomized to receive either lopinavir/ritonavir (LPV/r) or DTG, each with two NRTIs, one of which had to be fully active based on real-time drug-resistance testing. The study was stopped early after an interim analysis showed that the DTG arm was superior to the LPV/r arm. The superiority of DTG was somewhat counterbalanced by the finding that 2 of 11 participants in the DTG arm who met virologic withdrawal criteria selected for INSTI resistance, with no PI resistance selected for in the LPV/r arm.³⁶

In the NADIA trial, participants in Uganda, Kenya, and Zimbabwe who experienced virologic failure while on a first-line NNRTI plus TDF and 3TC or FTC were randomized to receive either darunavir/ritonavir (DRV/r) or DTG, each with 3TC; participants were assigned by a second randomization to receive a third ARV, either TDF or zidovudine (ZDV). Unlike the DAWNING study, full activity of the NRTIs based on genotype testing at the time of switch was not required.^{42,43} The primary study outcome was virologic suppression <400 copies/mL; at 48 and 96 weeks, >85% of participants had HIV RNA levels <400 copies/mL in all arms, and the DTG-based regimens were non-inferior to the DRV/r-based regimens. However, at 96 weeks, 9 of 235 (4%) participants in the DTG group developed DTG resistance. This represented 45% of participants in the DTG group with HIV RNA levels >400 copies/mL, six of whom were assigned to ZDV. In contrast, no PI resistance was selected for in the DRV/r group. When comparing TDF with ZDV, the two NRTIs demonstrated virologic suppression noninferiority at 48 weeks, but TDF was superior to ZDV at 96 weeks. These results included 84 of 92 (91%) participants in the DTG group who had virologic suppression <400 copies/mL despite no predicted active NRTIs at the time of failure of first-line NNRTI-based regimens, and a large proportion of this group had the K65R and M184V/I mutations. Individual-level drug-resistance data would have enabled further examination of specific mutation patterns and their association with patient characteristics and

treatment outcomes. Although such data are not available, these results suggest that ZDV should not be used over TDF in a public health approach. The decision to use DTG or DRV/r without another fully active drug should balance the overall efficacy data of these regimens, with considerations for potential emergent drug resistance, drug–drug interactions, convenience, and tolerability. The results from these studies should be interpreted with caution, as individual-level drug-resistance data and their linkage to patient characteristics and outcomes were not available, thus preventing full interpretation of these results. Additionally, these results may not be generalizable to settings and populations outside of the trials due to differences in geography, HIV-1 subtype, study population, ART availability, and treatment monitoring practice.

- **Boosted PI plus NRTIs:** The Panel recommends that a boosted PI (preferably boosted DRV) plus two NRTIs, at least one of which is fully active, can be an option after failure of a first-line NNRTI-based therapy **(AI)**. However, if full activity of at least one NRTI in the regimen cannot be assured, fully active boosted DRV plus two NRTIs estimated to be only partially active (particularly TAF or TDF with 3TC or FTC) can be considered, **with close monitoring (BII)**. Notably, boosted PIs as monotherapy **are not recommended (AI)**.^{37,39,40,42,43,56}

Several large, randomized controlled trials (primarily conducted in resource-limited settings where NNRTI-based regimens have been used as first-line therapy) have explored different second-line regimen options. The studies found that regimens that contained LPV/r or DRV/r plus at least two NRTIs were as effective as regimens that contained LPV/r plus RAL or DTG plus two NRTIs. Participants in some of these studies did not undergo drug-resistance testing before randomization. In the NADIA trial (summarized above), virologic efficacy of DTG and DRV/r-based regimens were non-inferior at 48 and 96 weeks, with TDF being non-inferior at 48 weeks and superior at 96 weeks compared with ZDV. Although there were nine participants in the DTG group who developed DTG resistance (six on ZDV and three on TDF), no participant in the DRV/r group developed PI resistance. Additionally, 74 of 80 (93%) participants in the DRV/r group had virologic suppression <400 copies/mL at 96 weeks despite no predicted active NRTIs at the time of failure of first-line NNRTI-based regimens. As outlined above in the discussion about second-generation INSTI plus no estimated fully active NRTIs based on current drug-resistance algorithms, these results should be interpreted with caution within, and particularly beyond, the study populations and settings.

- **Boosted PI plus an INSTI:** The Panel recommends that boosted DRV plus DTG, both with high barriers to resistance, should be the preferred boosted PI plus INSTI option **(AI)**. In a setting where a two-drug boosted PI and INSTI combination are being considered and both boosted DRV and DTG cannot be used, LPV/r plus RAL can be a treatment option for those who experienced virologic failure on an NNRTI-based regimen **(CI)**.

D²EFT is an open-label multinational randomized controlled trial that assessed the efficacies of DRV/r plus DTG versus DTG plus two NRTIs versus standard of care (DRV/r plus two NRTIs) for participants with first-line failure of an NNRTI-based regimen.⁴¹ In D²EFT, TDF plus 3TC or FTC were the NRTIs used. At 48 weeks, the proportion of participants who achieved HIV RNA <50 copies/mL was 84% for the DRV/r + DTG arm, 75% for the DRV/r plus two NRTI arm, and 78% for the DTG plus two NRTI arm, showing that the DRV/r + DTG arm was non-inferior to the other two regimens, with more participants achieving viral suppression. Several other small retrospective studies have found that a regimen of boosted DRV plus DTG is effective in achieving viral suppression in people with documented resistance to multiple drug classes.⁵⁷⁻⁵⁹ Three randomized controlled trials have found that a regimen that consisted of LPV/r plus RAL can be as effective as LPV/r plus at least two NRTIs as second-line therapy for people who failed first-line ART (mostly with an NNRTI-based regimen).^{37,39,40}

If this regimen is used in people with HBV coinfection, an HBV-active drug with a high barrier to resistance (i.e., TAF, TDF, or entecavir) should be continued or started to avoid HBV rebound and hepatocellular damage.

Boosted PI plus NRTI regimen failure: In this scenario, because boosted PI has a high barrier to resistance, most people will have either no resistance or resistance that is limited to 3TC and FTC, though additional NRTI mutations also may be present.^{60,61} Failure in this setting is often attributed to poor adherence, drug–drug interactions, or drug–food interactions. Below are some management options.

- **Switch to a second-generation INSTI plus NRTIs regimen:** Second-generation INSTIs have increasingly become preferred options over boosted PIs due to lower drug–drug interaction potential, improved tolerability, comparable efficacy, and a high barrier to resistance. Therefore, consideration should be given to switching to DTG or possibly BIC plus two NRTIs (if at least one of them is fully active) **(AIII)**. If only one of the NRTIs is fully active or if adherence is a concern, DTG is currently preferred over BIC **(AIII)**. If full activity of at least one NRTI in the regimen cannot be assured, DTG plus two NRTIs estimated to be only partially active **based on current drug-resistance algorithms** (particularly TAF or TDF with 3TC or FTC) can be considered **(CIII)**. As outlined above, results from **studies that may suggest such extrapolations** should be interpreted with caution, **as they did not involve failure of first-line boosted PI plus NRTIs regimens and as** individual-level drug-resistance data and their linkage to patient characteristics and outcomes were not available, thus preventing full interpretation of these results. Additionally, these results may not be generalizable to settings and populations outside of the trials due to differences in geography, **HIV-1 subtype**, patient population, ART availability, and treatment monitoring practice.
- **Maintain the same regimen:** A systematic review of multiple randomized trials that investigated the failures of first-line ritonavir-boosted PI-based regimens showed that taking actions to improve adherence to the original regimen is as effective as changing to new regimens with or without drugs from new classes **(AII)**.⁶² If the regimen is well tolerated with no concerns about drug–drug or drug–food interactions or drug resistance, then the regimen can be continued with adherence support and viral monitoring.
- **Switch to another PI-based regimen:** If a regimen with an INSTI plus two NRTIs is not an option and poor tolerability is contributing to virologic failure, the regimen can be modified with a different boosted PI that has no evidence for cross-resistance, plus a second-generation INSTI **(AIII)**, or plus two NRTIs (at least one of which is fully active) **(AIII)**. If full activity of at least one NRTI in the regimen cannot be assured, another fully active boosted PI plus two NRTIs estimated to be only partially active (particularly TAF or TDF with 3TC or FTC) can be considered, **with close monitoring of viral response (BIII)**.

INSTI plus NRTI regimen failure: Virologic failure in people on a regimen that consists of RAL or EVG plus two NRTIs may be associated with emergent resistance to 3TC or FTC (with/without additional NRTI mutations) and, possibly, the INSTI.⁶³ Viruses with EVG or RAL resistance often remain susceptible to DTG and BIC.⁵⁵ However, in the presence of certain INSTI mutations, DTG dose should be increased from once daily to twice daily.⁴⁴ The effective dose of BIC in these situations is unknown. In contrast, in clinical trials, people who experienced virologic failure while receiving DTG or BIC plus two NRTIs as first-line therapy were unlikely to develop resistance to DTG or BIC.^{63–65} No existing clinical trial data guide therapy for first-line INSTI failures; therefore, treatment strategy should be based on drug-resistance test results and the potential potency of the next regimen. Below are some treatment options, based on drug-resistance pattern considerations.

- **Virologic failure without any resistance mutations:** The person with HIV should be managed as outlined above in the section on virologic failure without drug resistance.
- **Virologic failure without INSTI resistance:** The regimen can be modified to one of the following:
 - o A boosted PI plus two NRTIs (preferably at least one of which is fully active) **(AIII)**; or

- o DTG, or likely BIC, plus two NRTIs (preferably at least one of which is fully active) **(AIII)**; or
- o A boosted PI plus DTG **(AIII)**.
- **Virologic failure with resistance to RAL and/or EVG but susceptibility to DTG:** The regimen can be modified to one of the following:
 - o A boosted PI plus two NRTIs (preferably at least one of which is fully active) **(AIII)**; or
 - o DTG (twice daily) plus two NRTIs (at least one of which is fully active) **(BIII)**; or
 - o DTG (twice daily) plus a boosted PI **(AIII)**.

Although BIC has a high resistance barrier, there are no data on whether the current BIC dose is efficacious in settings with RAL or EVG resistance and, therefore, it **is not currently recommended (BIII)**.

INSTI plus NNRTI regimen failure: Virologic failure in people on a regimen that consists of an INSTI (e.g., DTG or CAB) plus an NNRTI (e.g., RPV) may be associated with resistance to one or both of the medications in the regimen.^{66,67} Experience to guide therapy upon failure of these regimens is limited. Therefore, treatment strategies should be based on past treatment history, drug-resistance test results, and the potential potency of the next regimen, based on the guidance provided above.

Second-Line Regimen Failure and Beyond

Drug resistance with fully active commonly used ARV drug options: Using a person's complete ARV treatment history and drug-resistance data, a clinician can decide whether to include a fully active boosted PI or INSTI in future regimens. For example, those who have no documented PI resistance and who have never been treated with an unboosted PI likely harbor virus that is fully susceptible to PIs. Similarly, people who have no documented INSTI resistance and who have never been treated with an INSTI are likely to have virus susceptible to DTG or BIC. In this setting, virologic suppression should be achievable using a boosted PI plus either two NRTIs (preferably at least one of which is fully active), a boosted PI plus an active INSTI, or DTG or BIC plus two NRTIs (preferably at least one of which is fully active). Drugs should be selected based on the likelihood that they will be fully active, as determined by the person's treatment history, past and present drug-resistance testing, and tropism testing if a CCR5 antagonist is being considered.

Multidrug resistance without fully active commonly used ARV drug options: Use of currently available ARV drugs has resulted in a dramatic decline in the number of people with HIV who have few treatment options because of multiclass drug resistance.^{68,69} Despite this progress, some people have experienced toxicities with and/or developed resistance to most currently available commonly used ARV drugs. Maximal virologic suppression should remain the goal; however, if it cannot be achieved, the goals of ART will be to preserve immunologic function, prevent clinical progression, and minimize the development of further resistance that may compromise future regimens.

Consensus on the optimal management of people with multidrug-resistant HIV is lacking. If neither a fully active boosted PI nor a second-generation INSTI (e.g., DTG or BIC) is available, the new regimen should include at least two, and preferably three, fully active agents. If less than three fully active drugs are available, the regimen should include as many fully active drugs as possible, along with potentially partially active agents **(BII)**. If resistance to NNRTIs, T-20, MVC, BIC, DTG, **CAB**, EVG, or RAL are identified, there is rarely a reason to continue using these drugs, because there is little evidence that keeping them in the regimen helps delay disease progression **(BII)**. Moreover, continuing these drugs (in particular, early-generation INSTIs) may allow selection of additional resistance mutations and development of within-class cross-resistance that may limit

future treatment options. It should be noted that even partial virologic suppression of HIV RNA to $>0.5 \log_{10}$ copies/mL from baseline correlates with clinical benefit.^{68,70} Cohort studies provide evidence that even in the presence of viremia and no improvement in CD4 count, continuing ART reduces the risk of disease progression.⁷¹ Other cohort studies suggest that even modest reductions in HIV RNA levels continue to confer immunologic and clinical benefits.^{72,73} However, these potential benefits must be balanced with the ongoing risk of accumulating additional resistance mutations. In general, adding a single, fully active ARV drug to the regimen **is not recommended** because of the risk of rapid development of resistance (**BII**).

People with HIV and ongoing detectable viremia who lack sufficient treatment options due to an inability to construct a fully suppressive regimen with common ARVs may be candidates for the first-in-class CD4 post-attachment inhibitor IBA,⁷⁴ the gp120-directed attachment inhibitor FTR,⁷⁵ and/or the capsid inhibitor LEN.⁷⁶

- **Ibalizumab (IBA)** is a long-acting CD4 post-attachment inhibitor that is given intravenously every 2 weeks. A single-arm, multicenter clinical trial enrolled 40 heavily ART-experienced participants who had multidrug-resistant HIV-1 and who were experiencing virologic failure on an ARV regimen. Subjects received intravenous IBA infusions every 2 weeks, in addition to an optimized background regimen (OBR) that included at least one additional agent to which the subject's virus was susceptible. At Week 24, 43% of participants achieved HIV RNA <50 copies/mL, and 50% achieved HIV RNA <200 copies/mL.⁷⁷ Of the 27 participants who continued to the 48-week follow-up study, 59% and 63% had HIV RNA <50 copies/mL and <200 copies/mL, respectively. All 15 participants who had HIV RNA <50 copies/mL at Week 24 maintained virologic suppression up to Week 48.⁷⁸
- **Fostemsavir (FTR)** is a gp120 attachment inhibitor that is given orally twice daily. A Phase 3 multicenter trial enrolled 371 heavily ART-experienced participants who had multidrug-resistant HIV-1 and who were experiencing virologic failure. Participants were enrolled into two cohorts, according to their remaining treatment options. The randomized cohort ($n = 272$) included those with at least one fully active approved ARV drug in at least one but no more than two classes. These individuals were randomized to FTR (oral 600 mg twice daily) or placebo for 8 days, followed by open-label FTR plus OBR. In the nonrandomized cohort ($n = 99$), participants with no remaining ARV options were started on open-label FTR (oral 600 mg twice daily) plus OBR on Day 1. The primary endpoint for the randomized cohort was a change in HIV RNA level from baseline at Day 8. In the FTR group, the mean RNA level decrease was $0.79 \log_{10}$ copies/mL compared to $0.17 \log_{10}$ copies/mL in the placebo group ($P < 0.001$). At Week 96, 60% of participants in the randomized cohort and 37% of those in the nonrandomized cohort had HIV RNA levels <40 copies/mL, with mean CD4 increases of 205 cells/mm³ and 119 cells/mm³, respectively.^{79,80} In this study, 15 individuals in the nonrandomized cohort used the CD4 post-attachment inhibitor IBA in combination with FTR and other ARVs. The virological response rate for these participants by snapshot analysis was 53% at Week 48 and 33% at Week 96.
- **Lenacapavir (LEN)** is a LA HIV capsid inhibitor that can be given by one of two initiation schemes (oral plus subcutaneous [SQ] dosing), followed by SQ injections every 6 months (see [Appendix A, Table 10](#) for dosing details).

A Phase 3 multicenter trial (CAPELLA) enrolled 72 heavily ART-experienced participants who had multidrug-resistant HIV-1 and experienced virologic failure into two cohorts.⁸¹ Cohort 1 ($n = 36$) included participants who had a $<0.5 \log_{10}$ HIV-1 RNA decline between screening and baseline (i.e., stable viremia at ≥ 400 copies/mL, confirming lack of response to the failing therapy). The participants were randomized 2:1 to either oral LEN or placebo (on Days 1, 2, and 8) and continued to receive the failing ARV regimen for 14 days to evaluate the virologic effect of LEN functional monotherapy. Starting on Day 15, all participants began on an OBR; those randomized to oral LEN began SQ LEN every 6 months, whereas participants in the placebo arm received oral LEN on Days 15, 16, and 22 followed by SQ LEN on Day 29 (14 days after the first

oral LEN dose) and then every 6 months. On Day 15, 88% of participants in the LEN arm and 17% in the placebo arm had HIV RNA level reduction of $\geq 0.5 \log_{10}$ copies/mL, with least-squares mean change in HIV RNA level of $-2.1 \log_{10}$ copies/mL versus $-0.07 \log_{10}$ copies/mL for the LEN and placebo arms, respectively ($P < 0.001$). At the end of 26 weeks (i.e., after one dose of SQ LEN), 81% of participants had RNA levels < 50 copies/mL, and 89% had RNA levels < 200 copies/mL, with a mean change in RNA level of $-2.58 \pm 1.04 \log_{10}$ copies/mL.⁸¹ At the end of 52 weeks (i.e., after two doses of SQ LEN), 83% of participants had RNA levels < 50 copies/mL, with 94% of those with at least two active OBR drugs and 67% with no active OBR drugs. The mean CD4 cell count change at 52 weeks⁷⁶ was $+82$ cells/mm³.

Cohort 2 ($n = 36$) was a nonrandomized cohort that included participants who either had a $\geq 0.5 \log_{10}$ HIV-1 RNA decline from screening to baseline visit or were enrolled after Cohort 1 reached its planned sample size. All participants were started on an OBR and received oral LEN on Days 1, 2, and 8; on Day 15, SQ LEN was started and given every 6 months. After 26 weeks, 83% of the participants had HIV RNA levels < 50 copies/mL, and 86% had HIV RNA levels < 200 copies/mL. At Week 52, 72% had RNA levels < 50 copies/mL, and the mean CD4 cell count change was $+113$ cells/mm³.

Oral lead-in therapy was well tolerated overall, with nausea reported in 13% of participants who received LEN. Injection site reactions, which were generally mild and transient, were reported in 63% of the participants.⁸¹

Twenty-two of 72 (31%) participants met the criteria for resistance testing at confirmed virologic failure through Week 52.⁷⁶ LEN-associated capsid resistance mutations were found in 9 of the 22 (41%) participants with confirmed virologic failure. The M66I mutation was the most common mutation, reported in six participants. Four of the nine participants with LEN-associated capsid resistance mutations had no active agent in the OBR. Four others had low plasma concentrations of the OBR drugs at Week 26, suggesting poor adherence of the self-administered OBR, resulting in an unfavorable LEN functional monotherapy.⁸²

Taken together, these data^{76,82,83} highlight the importance of selecting a robust OBR to support LEN and counseling about adherence to the OBR. Additionally, LEN is a moderate CYP3A4 inhibitor and may increase concentrations of some coadministered drugs, whereas LEN concentration may be significantly decreased in the presence of a strong CYP3A4 inducer (see [Table 24g. Drug Interactions Between the Capsid Inhibitor Lenacapavir and Other Drugs](#) for further details). Therefore, people with HIV should be routinely counseled to inform all their health care providers of all medications they are taking, including LEN, even though it is not taken daily. Potential drug–drug interactions should be discussed, particularly before a new drug is started, to minimize the risk of toxicities, nonadherence, and drug resistance.

People with HIV who continue to have detectable viremia and who lack sufficient treatment options to construct a fully suppressive regimen also may be candidates for research studies or expanded access programs, or they may qualify for single-person access to an investigational new drug, as specified in the [U.S. Food and Drug Administration's Physician Request for a Single Patient Investigational New Drug for Compassionate or Emergency Use](#). Information about ARV agents that are in clinical studies can be found in the [drug database](#) available on the [Clinicalinfo](#) website.

People With HIV on ART With Suspected Drug Resistance Who Present With Limited Information (Incomplete or No Self-Reported ARV History, Medical Records, or Drug-Resistance Test Results)

Every effort should be made to obtain the person's ARV history and prior drug-resistance test results; however, this may not always be possible. One strategy is to restart the most recent ARV regimen and assess drug resistance in 2 to 4 weeks to guide the selection of the next regimen. Another strategy is to start two or three drugs that are predicted to be fully active based on the

person's treatment history. If no ARV history is available, a clinician may consider using agents with a high barrier to resistance—such as twice-daily DTG, BIC (which is available only in a combination pill with FTC/TAF and for which dosage in this setting is unclear), and/or twice-daily boosted DRV—as part of the regimen. Regardless of which strategy is employed, people should be closely monitored for virologic response (e.g., HIV RNA level testing approximately 4 to 8 weeks after reinitiation of therapy), with prompt drug-resistance testing performed if virologic response is inadequate.

People With HIV Who Are Unable to Achieve Viral Suppression Due to Poor Adherence to Oral ART and Who Do Not Have Resistance to Cabotegravir and Rilpivirine

Based on very limited data discussed below, the Panel recommends the use of LA CAB/RPV on a case-by-case basis in select individuals with persistent virologic failure despite intensive adherence support on oral ART, who have no evidence of resistance to RPV or CAB, and with shared decision-making between providers and people with HIV (CIII).

Some people with HIV cannot reach or maintain viral suppression on oral ART despite intensive adherence support. Although a complete LA injectable regimen may overcome some adherence obstacles for people with viremia, long-term clinical efficacy data are limited in these circumstances and the regimen is only approved for people who are virologically suppressed for at least 3 months. No large, published randomized controlled trial has demonstrated the efficacy of a complete regimen of any INSTI plus NNRTI, including LA CAB/RPV, in people with viremia. However, cohort data are emerging, resulting in two possible approaches to use LA CAB/RPV in this population. If achieving viral suppression with oral ART is possible with intensive adherence support to oral ART, switching to LA CAB/RPV after viral suppression can serve as a goal to incentivize the person to adhere to the oral regimen. Data from the Long-Acting Therapy to Improve Treatment Success in Daily Life (LATITUDE) AIDS Clinical Trial Group (ACTG) A5359 study using this strategy is discussed in Approach #1 below.⁸⁴ If achieving viral suppression with oral ART is not possible despite intensive adherence support to oral ART, switching to LA CAB/RPV without viral suppression may be an option in certain circumstances, as discussed in Approach #2 below.

Approach #1: Intensive efforts to achieve viral suppression prior to switching to LA CAB/RPV

Because all of the large LA CAB/RPV studies enrolled participants who were adherent to ART with sustained viral suppression, every effort should be made to achieve viral suppression before considering a switch to LA CAB/RPV, even in people with poor adherence. The Phase 3 prospective, randomized, open-label LATITUDE study was designed to assess this strategy in people who faced challenges with ART adherence. Preliminary data have been presented from this study that enrolled individuals without resistance to CAB or RPV from 33 sites in the United States and Puerto Rico.⁸⁴ Participants received adherence support and conditional economic incentives to assist in achieving viral suppression on oral ART. Study participants who experienced viral suppression were randomized to continue oral ART or switch to LA CAB/RPV every 4 weeks. At a planned data safety monitoring board (DSMB) interim efficacy analysis, only 294 (68%) of the 434 participants enrolled achieved viral suppression and were eligible for randomization, with 146 assigned LA CAB/RPV and 148 continuing standard of care (SOC) oral ART. At randomization, it was noted that despite the intensive adherence support, 17% in the LA CAB/RPV arm and 7% in the SOC arm had HIV RNA >200 copies/mL. At Week 48 of the randomized phase, 24% in the LA CAB/RPV arm and 39% in the SOC arm met the primary endpoint of earliest confirmed virologic failure or treatment discontinuation, which did not meet the predefined stringent stopping criteria. However, two secondary endpoints—virologic failure alone (7% in LA CAB/RPV vs. 25% in SOC)

and treatment-related failure (defined as first virologic failure or discontinuation due to adverse events) (10% in LA CAB/RPV vs. 26% in SOC)—met the predefined stopping criteria and were statistically significant. Based upon the totality of the data, in February 2024, the DSMB recommended halting randomization and offering all eligible participants LA CAB/RPV. Of the six participants with virologic failure on LA CAB/RPV, two had new drug-resistance mutations (one conferring high-level resistance to CAB and one conferring high-level resistance to both CAB and RPV). The available data suggest that an intensive effort to achieve viral suppression followed by a switch to LA CAB/RPV may be a viable option for individuals who cannot adhere to an oral ARV regimen. Further analyses and formal publication of LATITUDE study results are pending.

Approach #2: Administer LA CAB/RPV in people with viremia who are unable to achieve viral suppression despite intensive adherence support

Emerging data show that select people with HIV who are unable to achieve viral suppression on an oral ARV regimen despite intensive support might benefit from use of LA CAB/RPV, which is currently not approved for use in this population. Available data with this strategy include an initial case report,⁸⁵ followed by outcome data from a compassionate use program,⁸⁶ showing that 16 of 28 (57%) people with viremia achieved virologic suppression after a median of 5 months with LA CAB/RPV. Among those with virologic failure and for whom resistance data were available, all had new NNRTI mutations conferring RPV resistance and 50% had new integrase mutations, two of which conferred CAB resistance. Since then, three small observational studies have suggested that LA CAB/RPV may achieve viral suppression in select people with viremia and poor adherence to oral ART.⁵⁰⁻⁵² In these studies, select people with viremia who were unable to adhere to daily oral ART were initiated on monthly or twice-monthly LA CAB/RPV with or without oral lead-in therapy. At up to 10 months of follow-up, 95% achieved HIV RNA <50 copies/mL. Of those with virologic failure for whom genotypes were available, new reverse transcriptase (e.g., L100I, E138K, Y181I) and/or integrase (e.g., G140S, Q148R, R263K) mutations were detected, conferring intermediate to high resistance to RPV and/or CAB, respectively. Various levels of multidisciplinary support and intensive case management were utilized throughout these studies (e.g., provision of injections in field sites, financial incentives, appointment reminders, and transportation assistance). Despite the intensive support, not all injections were given on time.

In the multisite U.S. Observational Pharmacoepidemiology Research and Analysis (OPERA) cohort,⁸⁷ clinical data on the use of LA CAB/RPV were prospectively captured from electronic health records from 84 clinics in 18 U.S. states and territories. Among the 36 individuals who had HIV RNA >200 copies/mL before starting LA CAB/RPV and had at least one follow-up HIV RNA test, 94% had an HIV RNA <200 copies/mL at their last follow-up visit.

Taken together, until additional data are available, these approaches may provide alternatives for individuals with viremia and difficulties with adherence to oral ART, especially for those at the highest risk for disease progression or death. If either of the above approaches are employed, close monitoring is recommended with drug-resistance testing performed if virologic response is inadequate. Importantly, beyond conventional and intensive multidisciplinary team involvement, case management and outreach support are needed to ensure adherence and adequate monitoring while on LA CAB/RPV (see the [Adherence to the Continuum of Care](#) section for discussion on strategies to improve adherence in these situations). Caution with close monitoring is advised, as data are unavailable for detailed guidance regarding the need for specific adherence or intensive case management services or whether every 4- or every 8-week LA CAB/RPV will yield similar virologic outcomes in people with viremia before switching. Every 4-week dosing was used in the LATITUDE study. In the published case series to date in people with viremia, every 4-week dosing was prescribed for most, with the option of switching to every 8 weeks after achieving viral suppression. Until further data are available, it may be prudent to use similar strategies when LA CAB/RPV is

being considered. People with HIV and providers need to be aware of the significant risk of developing resistance to NNRTIs, and particularly INSTIs if virologic failure occurs on LA CAB/RPV. Such resistance may limit future treatment options and may also lead to its transmission, which should be balanced with the individual's HIV-related risk for disease progression and death.⁸⁸ Closely monitoring for viral responses is advised if LA CAB/RPV is used in people with viremia and who have challenges with adherence.

Summary

The goal of treatment for people with HIV and virologic failure is to establish virologic suppression. The management of people with virologic failure often requires expert advice to construct virologically suppressive regimens. Before modifying a regimen, it is critical to carefully evaluate the potential cause(s) of virologic failure, including incomplete adherence, poor tolerability, and drug–drug and drug–food interactions, as well as review HIV RNA and CD4 count changes over time, complete treatment history, and current and previous drug-resistance test results. If HIV RNA suppression is not possible with currently approved agents, consider the use of investigational agents through participation in clinical trials or expanded/single-patient access programs. If virologic suppression is still not achievable, the choice of regimens should focus on minimizing toxicity and preserving treatment options while maintaining CD4 counts to delay clinical progression.

Table 10. Antiretroviral Options for People With HIV and Virologic Failure

Designing a new regimen for people with HIV who are experiencing treatment failure should always be guided by ARV history and results from current and past resistance testing. This table summarizes the text above and displays the most common or likely clinical scenarios seen in people with virologic failure. For more detailed descriptions, please refer to the texts above and/or consult an expert in HIV drug resistance to assist in the design of a new regimen. It is also crucial to provide continuous adherence support before and after regimen changes.

Clinical Scenario	Type of Failing Regimen	Resistance Considerations	New Regimen Options ^a	Goal
First Regimen Failure	NNRTI plus two NRTIs	Most likely resistant to NNRTI +/- 3TC or FTC (i.e., NNRTI mutations +/- M184V/I). ^b Additional NRTI mutations also may be present.	DTG (or possibly BIC) plus two NRTIs (preferably at least one fully active ^c) (AI) ; or Boosted PI plus two NRTIs (preferably at least one fully active) (AI) ; or Boosted PI plus INSTI (boosted DRV plus DTG [AI]; LPV/r plus RAL [CI])	Resuppression
	Boosted PI plus two NRTIs	Most likely no resistance or resistance only to 3TC or FTC (i.e., M184V/I, without resistance to other NRTIs) ^b	DTG, or possibly BIC, plus two NRTIs (preferably at least one fully active; if only one of the NRTIs is fully active ^c or if adherence is a concern, DTG is currently preferred over other INSTIs) (AIII) ; or Continue same regimen (AI) ; or Boosted PI plus INSTI (boosted DRV plus DTG [AIII]; LPV/r plus RAL [CI]) ; or Another boosted PI plus two NRTIs (at least one fully active ^c) (AIII) .	Resuppression
	INSTI plus two NRTIs	If failure with no INSTI resistance	Boosted PI plus two NRTIs (preferably at least one fully active ^c) (AIII) ; or DTG, or likely BIC, plus two NRTIs (preferably at least one fully active ^c) (AIII) ; or DRV/r plus DTG (AIII)	Resuppression

Table 10. Antiretroviral Options for People with HIV and Virologic Failure

Clinical Scenario	Type of Failing Regimen	Resistance Considerations	New Regimen Options ^a	Goal
		If failure on EVG or RAL, often have INSTI resistance, but potentially susceptible to DTG Can have 3TC or FTC resistance.	Boosted PI plus two NRTIs (preferably at least one fully active ^c) (AIII) ; or DTG ^d twice daily or possibly BIC (if HIV is sensitive) plus two fully active NRTIs (BIII) ; or DTG ^d twice daily or possibly BIC (if HIV is sensitive) plus a boosted PI (preferably DRV/r) (AIII) .	Resuppression
	INSTI plus NNRTI (DTG/RPV or LA CAB/RPV)	INSTI and/or NNRTI resistance possible	Use ART history and past and current resistance testing to design a new regimen. Consult an expert in drug resistance as needed.	Resuppression
Second Regimen Failure and Beyond	Drug resistance with fully active treatment options—	Use past and current genotypic-+/- phenotypic-resistance testing and ART history when designing new regimen.	New regimen according to original treatment type—	Resuppression
	(i) Boosted PI, but not second-generation INSTI, fully active		(i) Boosted PI with two NRTIs (preferably at least one fully active)	
	(ii) Second-generation INSTI, but not boosted PI, fully active		(ii) DTG or BIC with two NRTIs (preferably at least one fully active)	
	(iii) Both PI and INSTI fully active		(iii) The two options above or boosted PI with INSTI	
	Multiple or extensive drug resistance with few treatment options (e.g., fully active boosted PI or second-generation INSTI unavailable)	Use past and current genotypic- and phenotypic-resistance testing to guide therapy. Confirm with a viral tropism assay when use of MVC is considered.	New regimen should include at least two, and preferably three, fully active agents, including those with novel mechanisms of action (e.g., IBA, FTR, LEN). If fewer than three fully active drugs, include as many fully active drugs as possible, along with	Resuppression if possible; otherwise, keep viral RNA levels as low as possible and CD4 count as high as possible.

Table 10. Antiretroviral Options for People with HIV and Virologic Failure

Clinical Scenario	Type of Failing Regimen	Resistance Considerations	New Regimen Options ^a	Goal
		Consult an expert in drug resistance if needed.	potentially partially active drugs (BII). Consider enrollment into clinical trials or expanded access programs for investigational agents if available. Discontinuation of all ARV drugs is not recommended (AI).	
People With Suspected Drug Resistance and Limited or Incomplete ARV and Resistance History	Unknown	Obtain medical records if possible. Resistance testing may be helpful in identifying drug-resistance mutations, even if the person has been off ART. Keep in mind that resistance mutations may not be detected in the absence of drug pressure.	Consider restarting the old regimen with careful monitoring of virologic response and early resistance testing if inadequate virologic suppression. If no ARV history is available, consider initiating a regimen with drugs with high genetic barriers to resistance (e.g., DTG, BIC, and/or boosted DRV) with careful monitoring of virologic response and early resistance testing, if inadequate virologic suppression.	Resuppression

^a When switching an ARV regimen in a person with HBV/HIV coinfection, ARV drugs that are active against HBV and have a high resistance barrier to HBV (i.e., **tenofovir**) should be continued as part of the new regimen, **or another HBV drug (i.e., entecavir) should be started**. Discontinuation of these drugs may lead to the reactivation of HBV, which may result in serious hepatocellular damage.

^b If other NRTI-resistance mutations are present, use resistance test results to guide NRTI usage in the new regimen.

^c See text for details and additional options in special settings.

^d Response to DTG depends on the type and number of INSTI mutations.

Key: 3TC = lamivudine; ART = antiretroviral therapy; ARV = antiretroviral; BIC = bictegravir; CD4 = CD4 T lymphocyte; DRV = darunavir; DRV/r = darunavir/ritonavir; DTG = dolutegravir; EVG = elvitegravir; FTC = emtricitabine; FTR = fostemsavir; HBV = hepatitis B virus; IBA = ibalizumab; INSTI = integrase strand transfer inhibitor; LEN = lenacapavir; LPV/r = lopinavir/ritonavir; MVC = maraviroc; NNRTI = non-nucleoside reverse transcriptase inhibitor; NRTI = nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; RAL = raltegravir

Isolated Central Nervous System Virologic Failure and Neurologic Symptoms

Cerebrospinal Fluid Viral Escape

Presentation with new-onset central nervous system (CNS) signs and symptoms has been reported as a rare form of “compartmentalized” virologic failure. People experiencing this present with new, usually subacute, neurological symptoms that are associated with a breakthrough of HIV replication within the CNS compartment, despite relative plasma HIV RNA suppression. In this case, cerebrospinal fluid (CSF) HIV RNA shows higher concentrations than in plasma.⁸⁹⁻⁹¹ Clinical evaluation frequently shows abnormalities on magnetic resonance imaging (MRI) and abnormal CSF findings with characteristic lymphocytic pleocytosis.⁹² In most (though not all) people, drug-resistant CSF virus is evident.⁹³ Consensus among experts is that this “neurosymptomatic” form of CNS viral escape should be treated through optimization of ARV regimens based on drug-resistance testing results if available **(CIII)**.⁹⁴ Although drug-resistance testing of HIV in CSF can be used to guide changes in the ARV regimen, according to the principles outlined above for plasma HIV RNA resistance, such testing typically needs to be conducted in a research setting. If CSF HIV drug-resistance testing is not available, the regimen may be changed based on the person’s treatment history or on predicted drug penetration into the CNS **(CIII)**.⁹⁵⁻⁹⁸

This “neurosymptomatic” CNS viral escape should be distinguished from “neuroasymptomatic” escape, defined as—

- The incidental detection of asymptomatic and mild CSF HIV RNA elevation, which is similar to plasma viral blips in that it is usually transient with low levels of CSF HIV RNA and has been associated with PI-based regimens⁹⁹⁻¹⁰¹; or
- A transient increase in CSF HIV RNA that is related to other CNS infections that can induce a brief increase in CSF HIV RNA (e.g., herpes zoster).¹⁰²

There is not clear evidence to support a change in an ARV regimen for incidentally detected “neuroasymptomatic” escape, although careful clinical review and follow-up of each person with this condition are recommended to monitor for the emergence of neurologic symptoms or systemic viremia.⁹⁴ There does not appear to be an association between these asymptomatic CSF HIV RNA elevations and the relatively common chronic, usually mild, neurocognitive impairment in people with HIV who show no evidence of CNS viral breakthrough.¹⁰³

Neurological Symptoms in People With HIV on Antiretroviral Therapy

Evidence is currently not available to support empiric intensification or switch of ARV regimens in people on systemically suppressive ART with mild neurological and/or cognitive symptoms who do not have documented CSF escape. Such individuals should be referred for neurological evaluation to determine if further evaluation is indicated. This may include blood laboratory testing, lumbar puncture, neuropsychological testing, and MRI to evaluate for CSF escape, as well as other causes of neurological symptoms. A recent multinational randomized, double-blinded, placebo-controlled trial randomized 191 ART-experienced participants—with cognitive impairment and suppressed plasma HIV RNA level and not taking an INSTI—to one of three arms: dual placebo, addition of DTG plus placebo, or DTG plus MVC. Compared with placebo, ART intensification with DTG or DTG plus MVC did not alter neuropsychological performance or depressive symptoms over time in participants with cognitive impairment.¹⁰⁴

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Suboptimal CD4 Cell Recovery Despite Viral Suppression

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Key Considerations and Recommendations

- **Suboptimal** CD4 T lymphocyte (CD4) cell recovery is associated with increased AIDS- and non-AIDS-related morbidity and mortality among individuals with antiretroviral therapy (ART)-mediated viral suppression.
- Promptly initiating ART in people diagnosed early with HIV provides the best opportunity for maximal CD4 cell recovery.
- Adding antiretroviral (ARV) drugs to a suppressive ART regimen (ART intensification) does not improve CD4 cell recovery and therefore **is not recommended** for this purpose **(AI)**.
- Switching ARV drugs or drug classes in people with suppressed viral load does not improve CD4 cell recovery **substantially** and **is not recommended** for this purpose **(AII)**.
- Interleukin-2 **is not recommended** to increase CD4 counts because clinical trial data demonstrated no clinical benefit **(AI)**.
- Other interventions to increase CD4 counts **are not recommended** outside of a clinical trial, because no current interventions have been proven to decrease morbidity or mortality during ART-mediated viral suppression **(AII)**.
- Efforts to decrease morbidity and mortality during ART-mediated viral suppression should focus on preventive care (e.g., opportunistic infection prophylaxis, vaccinations, cancer screening, statin therapy to reduce cardiovascular risk), addressing modifiable risk factors for chronic disease (e.g., tobacco use, alcohol and substance use, unhealthy diet, sedentary lifestyle), and optimizing management of comorbidities (e.g., hypertension, diabetes).

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Introduction

Most people with HIV who achieve and maintain antiretroviral therapy (ART)-mediated viral suppression experience increases in peripheral blood CD4 T lymphocyte (CD4) cell counts over time¹⁻⁴ and recovery of CD4 counts to within the normal range (>500 cells/mm³) of people without HIV.⁵⁻⁷ However, some people who started ART at low CD4 counts do not experience recovery to the desired immunologic goal of >500 cells/mm³ despite sustained virologic suppression on ART.^{2,3,8} Promptly initiating ART in people diagnosed early with HIV provides the best opportunity for maximal CD4 cell recovery.^{9,10}

There is no clear consensus on terminology to use or thresholds that define “suboptimal CD4 cell recovery.”^{11,12} A common threshold used for “suboptimal CD4 cell recovery” is CD4 ≤ 500 cells/mm³ after several years of suppressive ART. The term “immunologic nonresponse” also has been used to describe a lack of CD4 cell recovery, with common thresholds of <200 cells/mm³ or <350 cells/mm³ after several years of suppressive ART.¹² Some experts consider an increase of <200 cells/mm³ above baseline after 24 months as an inadequate immunologic response for people with CD4 counts ≤ 500 cells/mm³. The relationship between improved clinical outcomes and higher CD4 counts likely spans all ranges of CD4 counts, although incremental benefits are harder to discern once CD4

counts reach moderate-to-high levels,¹³ especially beyond 500 cells/mm³. Rarely, people with HIV may experience substantial declines in CD4 counts while maintaining virologic suppression on ART.¹⁴

Clinical Consequences

Persistently low CD4 counts despite ART-mediated viral suppression are associated with increased risk of morbidity, hospitalizations, and mortality. Data from two large international cohorts of people with HIV initiating ART as early as 1996 showed that individuals with HIV and CD4 counts ≤ 200 cells/mm³ despite at least 3 years of suppressive ART had a 2.6-fold greater risk of mortality than those with higher CD4 counts.¹⁵ A multicenter cohort study from Spain of individuals who started ART with low CD4 counts between 2005 and 2019 found that those whose CD4 counts remained < 200 cells/mm³ after 2 years of ART had a 4.6-fold greater risk of mortality than those with CD4 > 500 cells/mm³.¹⁶

People with CD4 counts < 200 cells/mm³ despite at least 1 year of suppressive ART are at increased risk of AIDS-related and non-AIDS-related events.¹⁷⁻²³ This population is at risk of severe infections (e.g., pneumonia) and is less likely to mount a robust immune response following vaccination.²⁴ In addition, these individuals have an increased risk of infection-related cancers (e.g., Epstein-Barr virus-associated lymphoma, human papillomavirus-associated malignancies, hepatocellular carcinoma associated with hepatitis B virus [HBV] or hepatitis C virus [HCV]). A cohort study of people with HIV from 1996 to 2020 demonstrated that people with CD4 < 200 cells/mm³ had a 3.5-fold higher risk of infection-related cancers compared to people with CD4 ≥ 500 cells/mm³.²⁵ Another study found that CD4 count < 350 cells/mm³ was associated with non-AIDS-defining, smoking-related, and other malignancies.²⁶ People with HIV who receive chemotherapy or radiation therapy can also experience CD4 count decline. One study reported a 27% increase in mortality for every 100 cells/mm³ decrease in CD4 count.²⁷ Lastly, cardiovascular disease and other comorbidities have also been observed at increased rates in people with low CD4 recovery.^{15,28-35}

Pathogenesis

Suboptimal CD4 recovery after ART-mediated virologic suppression is typically associated with higher levels of inflammation and immune activation³⁶ and is most frequently observed in people with older age, prior AIDS-defining illness, lower nadir CD4 count,^{16,37} and some coinfections (e.g., HCV).³⁸⁻⁴⁰ The association of higher inflammation levels with persistently low CD4 counts is likely multifactorial and, to some extent, may be driven by inadequate immunologic control of other chronic viral infections, based on observed decreases in inflammation after treatment of cytomegalovirus and chronic HCV infection.^{41,42} Numerous factors have deleterious effects on T-cell survival and proliferation, such as chronic coinfections, a persistently activated innate immune system (due to stimuli such as microbial translocation and dysbiosis), more severe fibrosis of lymph node tissue with disruption of naive T-cell homeostasis, potential senescence of hematopoietic progenitors, and possibly autoantibodies.⁴³⁻⁴⁶

CD4 count may occasionally decline despite suppressive ART, which may be due to a new medication, a new underlying malignancy, infection, or autoimmune disorder, including the presence of anti-CD4 autoantibodies,^{44,45} and/or dysregulation of interleukin-7 (IL-7)-mediated naive T-cell homeostasis or more accentuated immune senescence.^{14,47,48}

Alternative Causes of Suboptimal CD4 Cell Recovery

Evaluation of people with suboptimal CD4 cell recovery should begin with identifying modifiable causes, such as concomitant medications that reduce total white blood cells, particularly lymphocytes and CD4 cells. When possible, these drugs should be discontinued and replaced with

alternatives if necessary. Untreated coinfections (e.g., HCV, *Mycobacterium avium* complex, histoplasmosis) and serious medical conditions (e.g., malignancy, autoimmune disease) may also contribute to CD4 lymphopenia, particularly in people with declining CD4 counts (and percentages) while on suppressive ART and in those with CD4 consistently <100 cells/mm³. Treating coinfections and optimizing management of contributing conditions should be pursued when feasible.

Interventions Tested to Improve CD4 Cell Recovery

For people with suboptimal CD4 cell recovery despite suppressive ART, effective interventions to increase CD4 counts and improve clinical outcomes have not been identified.⁴⁹ Strategies to raise CD4 counts by modifying or intensifying ART regimens have been evaluated in clinical trials.

Intensification strategy by adding ARV drugs to a suppressive ART regimen does not improve CD4 cell recovery⁵⁰⁻⁵⁶ and does not reduce morbidity or mortality. Therefore, ART intensification **is not recommended** as a strategy to improve CD4 cell recovery **(AI)**. Similarly, switching ARV drugs or classes has not been shown to improve CD4 cell recovery substantially for individuals with viral suppression and **is not recommended (AII)**.^{57,58}

Therapies targeting immunomodulatory pathways (e.g., interleukin-2 [IL-2], IL-7, growth hormone, thymosin alpha-1) have also been investigated as a strategy to increase CD4 counts.⁵⁹ Two large randomized trials, powered to assess impact on clinical endpoints (AIDS and death), evaluated the role of IL-2 for improving CD4 cell recovery. Although there is strong evidence linking low CD4 cell counts with increased morbidity, IL-2 adjunctive therapy did not decrease the risk of clinical events despite substantial increases in CD4 count.⁶⁰ Therefore, IL-2 **is not recommended** as a strategy to improve CD4 recovery **(AI)**. Dysregulation in IL-7-mediated naïve T-cell homeostasis has been reported among people with suboptimal CD4 recovery.^{14,47,48} While several early-phase trials of human recombinant IL-7 treatment have shown an increase in CD4 counts among people with HIV on ART, no clinical trials designed to evaluate whether IL-7 treatment confers benefit on clinical outcomes have been performed.⁶¹

Additional interventions have been studied in early-phase trials, and some have demonstrated improvements in CD4 counts among people with HIV on ART with suboptimal CD4 cell recovery. These studies were not designed to evaluate clinical endpoints. Examples include the use of novel antivirals,⁶² repurposed medications,^{63,64} probiotics,^{65,66} plants and herbal treatments with medicinal properties,⁶⁷⁻⁶⁹ and other nutritional supplements,⁷⁰ as well as human umbilical cord mesenchymal stem cell infusions^{71,72} or allogeneic natural killer cell immunotherapy.⁷³ As demonstrated by the IL-2 clinical trial results, randomized trials with clinical endpoints will be needed to establish clinical benefit. Because of the lack of clinical efficacy data from these alternative interventions, the use of adjunct therapies to improve immune recovery **is not recommended**, except in the context of a clinical trial **(AII)**.

Prevention of Clinical Consequences

With continuous ART and viral suppression, life expectancy for people with HIV is approaching that of people without HIV. However, persistently low CD4 counts despite ART-mediated viral suppression are associated with higher rates of both AIDS and non-AIDS-related morbidity and mortality over the long term. Reducing AIDS- and non-AIDS-related morbidity and mortality remains a cornerstone of HIV treatment and is especially important for people with suboptimal CD4 recovery, given the increased risk of such outcomes. Risk for opportunistic infections (OIs) is generally low for people on ART with viral suppression, but some primary or secondary prophylaxis may need to be extended for certain OIs among people who do not experience CD4 recovery above certain thresholds (see the [Adult OI Guidelines](#) for specific guidance).

Given the increased risk for severe manifestations and/or poor outcomes from infections such as tuberculosis, HBV, and HCV, guidelines for screening for such infections should be followed closely, and treatments should be initiated promptly when indicated. Clinicians should also be vigilant about ensuring that people with HIV are up to date on vaccinations for preventable infections. It should be noted that live virus vaccines are contraindicated for people with CD4 counts <200 cells/mm³ (see [Immunizations for Preventable Diseases in Adults and Adolescents With HIV](#) in the Adult OI Guidelines for detailed recommendations).

For non-AIDS-related morbidities, providers should offer preventive counseling, evaluate and manage cardiovascular risk factors (see [Cardiovascular Complications](#)), ensure that screening for malignancies (e.g., colorectal cancer, anal or cervical cancer, hepatocellular carcinoma, lung cancer, breast cancer) is up to date (see [Primary Care Guidance](#)), initiate statin therapy when indicated (see [Statin Therapy in People With HIV](#)), and provide prompt care and referrals when appropriate.⁷⁴ As for all people with HIV, regardless of CD4 count, preventive counseling should address lifestyle modifiable risk factors, including but not limited to smoking cessation, abstinence from alcohol and substance use, dietary modification, regular exercise, and treatment optimization for comorbidities such as hypertension and diabetes.

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Optimizing Antiretroviral Therapy in the Setting of Viral Suppression

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Key Considerations and Recommendations

- Advances in antiretroviral (ARV) treatment and a better understanding of HIV drug resistance have made it possible to consider switching a person with HIV from an effective ARV regimen to an alternative ARV regimen in some situations.
- The fundamental principle of ARV regimen optimization is to maintain viral suppression without jeopardizing future treatment options.
- Adverse events, drug–drug or drug–food interactions, pill burden, pregnancy, cost, stigma, inconvenience from taking oral medications, or the desire to simplify a regimen may prompt regimen optimization.
- It is critical to review a person’s full ARV history, including virologic responses, past ARV-associated toxicities and intolerances, and cumulative resistance test results before selecting a new ARV regimen (AI).
- People with HIV who have no history of drug-resistance mutations or virologic failure can likely switch to any regimen that has been shown to be highly effective in people who are ARV-naïve (AI) or to NRTI-sparing options extensively researched in switch studies, such as dolutegravir (DTG) plus rilpivirine (RPV) (AI) or long-acting cabotegravir plus rilpivirine (LA CAB/RPV) (AI).
- For regimen optimization in the setting of existing nucleoside reverse transcriptase inhibitor (NRTI) resistance, if an NRTI is to be included in the new regimen, two NRTIs (tenofovir alafenamide [TAF] or tenofovir disoproxil fumarate [TDF] plus emtricitabine [FTC] or lamivudine [3TC]) should be included, along with a fully active drug with a high resistance barrier, such as DTG (AII), bictegravir (BIC) (BIII), or boosted darunavir (BIII). Alternatively, as noted above, an NRTI-sparing regimen (such as DTG/RPV [AI] or LA CAB/RPV [AI]) is possible if there is no evidence of prior integrase strand transfer inhibitor (INSTI) or RPV resistance.
- Monotherapy with either a boosted protease inhibitor or an INSTI has been associated with unacceptable rates of virologic failure and the development of resistance; therefore, monotherapy as a switch strategy is not recommended (AI).
- When switching an ARV regimen in a person with chronic hepatitis B virus (HBV)/HIV coinfection, HBV-active drugs that are potent and have a high barrier to resistance (i.e., TAF, TDF, or entecavir) should be used (AII). Discontinuation of HBV drugs may lead to reactivation of HBV, which may result in serious hepatocellular damage.
- Consultation with an HIV specialist is recommended when planning a regimen switch for a person with a history of resistance to two or more drug classes (AIII).
- Close monitoring to assess tolerability, viral suppression, adherence, and safety is recommended during the first 3 months after a regimen switch (AIII).

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

With currently available antiretroviral therapy (ART), most people with HIV can achieve and maintain HIV viral suppression. Furthermore, advances in antiretroviral (ARV) treatment and a better understanding of drug resistance have made it possible to consider switching a person with HIV from one effective regimen to another in some situations. When considering optimization—a switch in

therapy to improve certain aspects of therapy—clinicians must keep several key principles in mind to maintain viral suppression while addressing the concerns with the current regimen.

Reasons to Consider Regimen Optimization in the Setting of Viral Suppression

- To simplify a regimen by reducing pill burden and/or dosing frequency
- To enhance tolerability and/or decrease short- or long-term toxicity (see [Table 21](#) in the [Adverse Effects of Antiretroviral Agents](#) section for a more in-depth discussion of toxicities)
- To prevent or mitigate drug–drug interactions (see [Drug–Drug Interactions](#))
- To eliminate food requirements
- To switch to a long-acting (LA) injectable regimen for convenience, to relieve pill fatigue, or to decrease potential stigma or disclosure concerns associated with taking daily oral medications
- To allow optimal use of ART during pregnancy or when pregnancy is desired or may occur (see [Perinatal Guidelines](#))
- To reduce costs (see [Cost Considerations and Antiretroviral Therapy](#))

General Principles of Regimen Optimization

Maintain Viral Suppression

The fundamental principle of ARV regimen optimization is to maintain viral suppression without jeopardizing future treatment options. If a regimen switch results in virologic failure with the emergence of new resistance mutations, the person may require a more complex and/or a less tolerated regimen.

Careful Review of Antiretroviral Treatment and Drug-Resistance History Before Optimization

Before switching treatments, it is critical to thoroughly assess a person’s full ARV history, including virologic responses, inferred resistance (as discussed below), cumulative resistance test results, and any past ARV-related intolerances, toxicities, or adverse reactions **(AI)**.

If a person with pre-ART wild-type HIV achieves and maintains viral suppression after ART initiation, one can safely assume that no new drug-resistance mutations emerged while the person was on the suppressive regimen. In people with a history of virologic failure or pre-treatment drug resistance, a review of cumulative resistance test results and clinical and virologic response to prior regimens is essential when designing a new regimen. Cumulative resistance test results refer to all previous and currently available results from standard genotype, proviral DNA genotype (if available), phenotype, and tropism assays that can be used to guide the selection of a new regimen. Once selected, a drug-resistance mutation—even when it is not detected in the person’s most recent drug-resistance test—can be archived in the HIV reservoir and reemerge under the appropriate selective drug pressure. Resistance often can be inferred from the ARV history. For people with documented failure on a regimen that includes drugs with relatively low barriers to resistance—such as a non-nucleoside reverse transcriptase inhibitor (NNRTI), elvitegravir (EVG), raltegravir (RAL), lamivudine (3TC), or emtricitabine (FTC)—a clinician should assume that there is resistance to these drugs, so-called *inferred resistance*. When uncertain about prior resistance, it is generally not advisable to switch from a suppressive ARV regimen, unless the new regimen is likely to be at least as active against potential resistant virus as the current suppressive regimen. This principle is particularly applicable

when switching ARV-experienced individuals from a regimen with a relatively high barrier to resistance—such as those that include pharmacologically boosted protease inhibitors (PIs), dolutegravir (DTG), or bictegravir (BIC)—to one with a lower barrier to resistance.¹ The Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends that clinicians consult an HIV specialist when contemplating optimization for a person with a history of resistance to **two** or more drug classes (**AIII**).

If optimization is considered in people with suppressed viral loads who do not have prior drug resistance data, proviral DNA genotypic resistance testing can be considered. For those who have no prior virologic failures and who are on their first or second regimen, or who have genotypic test results from prior virologic failures, the use of the proviral DNA genotypic test is unlikely to provide valuable information. In individuals with a history of multiple prior failures or multiple prior ARV regimens, the use of proviral DNA genotypic testing may be useful. However, whenever proviral DNA genotypic testing is used, the results must be interpreted with caution because these assays may not detect any or all drug-resistance mutations, especially those that were selected by a previous ARV regimen.² In addition, these assays may identify mutations that appear inconsistent with a person's response to treatment, making the clinical relevance of the assay results questionable. Overall, the clinical utility of these assays remains an area of active investigation (see [Drug-Resistance Testing](#)).

Considerations for HBV/HIV Coinfection

For People With HBV/HIV Coinfection

When switching an ARV regimen in a person with active hepatitis B virus (HBV)/HIV coinfection (hepatitis B surface antigen [HBsAg] positive), tenofovir disoproxil fumarate (TDF) or tenofovir alafenamide (TAF) (generally given with 3TC or FTC) should be continued as part of the new regimen. Both TDF and TAF are active against HBV and can be used as HBV monotherapy. Although 3TC and FTC are active against HBV, their use as monotherapy against HBV infection is **not recommended (AII)** because emergent HBV resistance to these drugs is highly likely. If TAF or TDF cannot be used, entecavir—a potent antiviral with a high HBV resistance barrier—should be used to treat HBV alongside a fully suppressive ARV regimen (**AII**). Entecavir should not be used without an active ARV regimen to prevent the development of the HIV M184V resistance mutation. Discontinuation of HBV antivirals may lead to reactivation of HBV, which can result in serious hepatocellular damage and death. Refer to the [Hepatitis B Virus/HIV Coinfection](#) section for specific recommendations.

Screening for HBV for People With No History of HBV/HIV Coinfection

People with HIV and unknown or nonimmune HBV status should be screened for HBV infection before switching an ARV regimen to one that does not include TAF or TDF plus 3TC or FTC (e.g., DTG/rilpivirine [RPV], DTG/3TC, LA cabotegravir [CAB]/RPV). If a person with HIV shows no evidence of chronic HBV infection (i.e., negative for hepatitis B surface antibody [HBsAg]) and is not immune to HBV (i.e., negative for HBsAb), vaccination should be initiated while considering the ARV switch (including in those with isolated hepatitis B core antibody [HBcAb]).

People with HIV and prior exposure to HBV infection without evidence of chronic infection (i.e., negative HBsAg, positive HBcAb, and either positive or negative HBsAb) are likely at low risk (<1%) of HBV reactivation and even lower risk of HBV reactivation-associated hepatitis, despite the discontinuation of NRTIs. However, there are no published studies to confidently estimate risk in this population. Within this group, those with positive HBsAb are at the lowest risk for HBV reactivation, although HBV reactivation has been described when HBV active therapy is withdrawn as part of an ART regimen in this situation.³ See more information in the [Hepatitis B Virus Infection](#) section in the

Guidelines for the Prevention and Treatment of Opportunistic Infections in Adults and Adolescents With HIV.

For people previously exposed to HBV, alanine aminotransferase (ALT) should be monitored for 6 months (e.g., at the time of HIV viral load follow-up visits) after switching from ART that includes tenofovir (TDF or TAF) to one that does not include either of these drugs. An increase in ALT levels should prompt HBV DNA testing to assess for the reactivation of HBV. The presence of HBV reactivation would require the immediate addition of TAF, TDF, or entecavir to the ARV regimen.

Assessment for Potential Drug Interactions

Before switching a regimen, it is important to review each ARV drug in the new regimen and concomitant medications to assess whether any potential drug–drug interactions exist. For example, oral rilpivirine (RPV) and atazanavir (ATV) interact with acid-lowering agents, and many ARV drugs may interact with drug transporters or drugs that are inhibitors, inducers, or substrates of cytochrome P450 (CYP) enzymes (see [Drug–Drug Interactions](#)). In addition to new drug interactions, the discontinuation of some ARV drugs also may necessitate adjusting the dosage of concomitant medications. For example, discontinuation of pharmacokinetic (PK) boosters (ritonavir or cobicistat) may reduce the concentrations of some concomitant medications. Concomitant medications, which may have been previously managed with dose adjustments, will need to be reevaluated in the context of the new ARV regimen.

Assessment for Pregnancy or Pregnancy Potential

Women of childbearing potential should have a pregnancy test before switching ART. If a woman with HIV is found to be pregnant or desires pregnancy, clinicians should refer to the [Perinatal Guidelines](#) for recommendations on the safety and efficacy of ARV use at the time of conception or during pregnancy. All pregnancies that occur while a woman is receiving ART should be reported to the [Antiretroviral Pregnancy Registry](#).

Monitoring After Switching Antiretroviral Therapy

After a treatment switch, people with HIV should be evaluated closely for 3 months (e.g., a clinic visit or telephone call 1 to 2 weeks after the change and a viral load test to check for rebound viremia 4 to 12 weeks after the switch) (AIII). The purpose of this close monitoring is to ensure the person clearly understands their prescribed ART regimen, assess medication tolerance, and conduct targeted laboratory testing when there are preexisting laboratory abnormalities or if there are potential concerns with the new regimen. For example, if lipid abnormality prompted the ARV change or is a concern with the new regimen, fasting cholesterol subsets and triglycerides should be assessed within 3 months after the ART change. In the absence of any new complaints, laboratory abnormalities, or evidence of viral rebound over the first 2 to 3 months, clinical and laboratory monitoring may resume on a regularly scheduled basis (see [Laboratory Testing for Initial Assessment and Monitoring](#)).

Specific Regimen Optimization Considerations

As with the recommendation for people who are starting ART, the use of a 2- or 3-drug ARV regimen (as discussed below) is generally recommended when switching ART in people with suppressed viral loads (AI). People who have no history of resistance mutations or virologic failure can likely switch to any regimen that has been shown to be highly effective in people who have not been on ART (AI). People with prior drug resistance can be switched to a new regimen based on their ARV history and cumulative resistance testing results. Monotherapy with either a boosted PI or an integrase strand transfer inhibitor (INSTI) has been explored in several trials or cohort studies.⁴⁻⁷

Both INSTI⁸ and boosted PI⁹ monotherapy have been associated with higher rates of virologic failure than combination regimens. In addition, INSTI monotherapy has been associated with the development of resistance;^{10,11} therefore, ARV monotherapy as a maintenance strategy is **not recommended (AI)**.

Optimization Strategies With Good Supporting Evidence for People Without Known Drug Resistance

Many clinical trials have enrolled participants with stable, suppressed viral loads without known underlying drug resistance and switched them to another regimen. Most of these studies demonstrated maintenance of viral suppression, some of which are referenced below. However, some regimen switches have had limited success in clinical trials but have informed optimization strategies. The SWITCHMRK 1 and 2 studies illustrated the importance of considering the possibility of underlying drug resistance before switching therapy in those with virologic suppression, particularly when the switch is to a drug with a low barrier to resistance.¹ In the two SWITCHMRK studies, those with viral suppression on two NRTIs plus lopinavir/ritonavir (LPV/r) were switched to two NRTIs plus RAL. The studies found that individuals with a history of previous virologic failure had a higher risk of failure after switching to the RAL-based regimen. This finding is likely explained by underlying NRTI resistance, a setting where viral suppression can generally be maintained by drugs with a high barrier to resistance, such as boosted PIs, but is less likely to maintain viral suppression when switched to a drug with a low barrier to resistance, such as RAL. The strategies listed below support these observations and principles of optimizing therapy and include a discussion about switching ART regimens in those with underlying resistance.

Three-Drug Regimens

Within-Class Switches

Within-class switches (e.g., switching from one INSTI to another INSTI) may be prompted by adverse events or the availability of ARVs in the same class that offer a better safety profile, reduced dosing frequency, higher barrier to resistance, lower pill burden, or do not require PK boosting. Within-class switches usually maintain viral suppression, provided there is no drug resistance to the new ARV. Below are examples of within-class switch strategies that are proven to be effective in clinical trials or are likely to be effective in those without underlying drug resistance:

- From TDF^{12,13} or abacavir (ABC)¹⁴ to TAF
- From RAL or elvitegravir/cobicistat (EVG/c)¹⁵ to BIC or DTG
- From DTG¹⁶⁻¹⁸ to BIC or BIC to DTG
- From efavirenz to RPV,^{13,19} doravirine (DOR),²⁰ or RPV to DOR

Between-Class Switches

Between-class switches (e.g., switching from a boosted PI to an INSTI) generally maintain viral suppression, provided there is no resistance to the other components of the regimen. In general, such switches should be avoided if any doubt exists about the activity of the other agents in the regimen. As noted earlier, prior resistance test results will be very informative in guiding this switch. The following are examples of between-class switches that have been proven to be effective in clinical trials or are likely to be effective in those without underlying drug resistance:

- From a boosted PI to a second-generation INSTI (e.g., DTG,²¹ BIC²²)

- From a boosted PI to RPV²³ or DOR²⁰
- From an NNRTI to a **second-generation** INSTI^{24,25} or a boosted PI

Two-Drug Regimens

Some two-drug ARV regimens are effective in maintaining HIV virologic control in people who initiated therapy with three-drug regimens and achieved sustained virologic suppression for at least 3 months. However, none of the two-drug regimens are effective as treatment for HBV and are therefore not recommended for people with HBV coinfection. In people with HBV who cannot take TAF or TDF, these regimens can be considered if used with a potent, high barrier-to-resistance anti-HBV active medication (i.e., entecavir) **(AII)**. See the section above on HBV considerations during optimization. The following are examples of successful strategies studied in clinical trials for switching from 3- to 2-drug regimens in people with suppressed HIV.

Dolutegravir Plus Rilpivirine

Two Phase 3 trials, SWORD-1 and SWORD-2, demonstrated noninferior efficacy of switching to a 2-drug regimen of DTG with RPV compared to continuing a first or second ARV regimen in individuals with suppressed HIV-RNA without prior virologic failure.²⁶ Individuals were excluded from the study if they had active HBV infection (unless the person was also on a specific HBV active regimen), resistance to DTG or RPV, or significant drug interactions. DTG/RPV is suitable for use in people with an isolated K103N resistance mutation and in those with NRTI resistance. Individuals with active HBV infection need to be on **a potent HBV drug with a high barrier to resistance (i.e., TAF, TDF, or entecavir)**.

Dolutegravir Plus Lamivudine or Emtricitabine

A switch from a stable 3-drug ARV regimen to DTG plus 3TC or FTC as a maintenance strategy in people with ongoing virologic suppression and no history of prior virologic failure or resistance to these agents was noninferior to continuing a 3-drug therapy in a large randomized clinical trial (TANGO),²⁷ in smaller clinical trials,²⁸⁻³⁰ and in observational studies³¹⁻³³ **(AI)**. Individuals with active HBV infection need to be on **a potent HBV drug with a high barrier to resistance (i.e., TAF, TDF, or entecavir)**. 3TC or FTC monotherapy is not considered standard of care for HBV treatment because of a high risk of emerging resistance.³⁴

Boosted Protease Inhibitor Plus Lamivudine

A boosted PI plus 3TC is a reasonable 2-drug optimization option in individuals without resistance who are suppressed on a current regimen and who do not have active HBV infection. Pill burden and potential drug interactions are limitations compared to the above 2-drug regimens. Several regimens using a boosted PI plus 3TC have been shown to be effective in clinical trials, including darunavir/ritonavir (DRV/r) **(BI)**,³⁵ darunavir/cobicistat (DRV/c) **(BIII)**, atazanavir/ritonavir (ATV/r) **(CI)**,^{36,37} and LPV/r **(CI)**.³⁸ Individuals with active HBV infection need to be on **a potent HBV drug with a high barrier to resistance (i.e., TAF, TDF, or entecavir)**. 3TC or FTC monotherapy is not considered the standard of care for HBV treatment because of a high risk of emerging resistance.³⁴

Boosted Darunavir Plus Dolutegravir

An open-label, Phase 3b, noninferiority clinical trial evaluated the continuation of DRV/r plus two NRTIs versus a switch to DRV/r plus DTG; the switched regimen was noninferior.³⁹ Because of the small sample size of this study, the regimen of DRV/r plus DTG is recommended only in the absence of other alternative options but can be considered in particular for individuals with resistance and/or

intolerance to 3TC (or FTC), ABC, and TDF (or TAF) **(CI)**. Similar results were observed in two small observational studies.^{40,41} Individuals with active HBV infection need to be on **a potent HBV drug with a high barrier to resistance (i.e., TAF, TDF, or entecavir)**.

Long-Acting Intramuscular Cabotegravir Plus Rilpivirine

Long-acting ARV medications provide the convenience of reduced dosing frequency and may improve quality of life for individuals with pill fatigue or concerns about disclosure of HIV status or stigma associated with daily oral medication. Long-acting (LA) intramuscular (IM) injectable formulations of the INSTI cabotegravir (CAB) and the NNRTI RPV are approved by the U.S. Food and Drug Administration (FDA) to be given once monthly or every 2 months as a complete regimen for individuals **≥12 years old** with sustained (e.g., ≥3 months) virologic suppression (HIV-1 RNA <50 copies/mL) on a stable ARV regimen, with no history of treatment failure, and with no known or suspected resistance to either CAB or RPV.⁴²⁻⁴⁴

Panel's Recommendation

The Panel recommends that monthly or every 2-month LA CAB/RPV can be used to replace an existing oral ARV regimen in people with HIV who fulfill all of the following criteria **(AI)**:

- Sustained viral suppression for at least 3 months
- No history of documented or suspected resistance to either CAB or RPV
- No active HBV infection (unless also receiving TAF, TDF, or entecavir)
- Not pregnant or actively planning pregnancy
- Not receiving medications with significant drug interactions with oral (during lead-in or bridging therapy) or injectable CAB or RPV (see [Table 24b](#) and [Table 24d](#))

Oral lead-in therapy with CAB and RPV is optional and can be done based on provider–patient discussion.

Clinical Trial Data

Two Phase 3 clinical trials (ATLAS and FLAIR) randomized almost 1,200 participants with HIV-1 and viral suppression on oral ART to receive once-monthly LA CAB/RPV or to continue their oral ARV regimens.^{45,46} Participants could not have prior resistance to INSTIs or NNRTIs (except the K103N mutation), previous virologic failure (ATLAS trial), or active or occult HBV infection. Both studies used a 1-month oral lead-in of once-daily CAB 30 mg with RPV followed by injectable LA CAB/RPV monthly (loading dose x 1 followed by maintenance dosing). In the intention to treat the exposed population, HIV RNA >50 copies/mL at Week 48 occurred in 11 (1.9%) participants in the IM arms and 10 (1.7%) participants in the oral therapy arms (combining data from both studies) demonstrating that long-acting therapy was noninferior to continued oral therapy.⁴⁷ Virologic failure was rare; however, when it occurred, resistance to INSTIs, NNRTIs, or both was common.

The ATLAS-2M study compared LA CAB 600 mg/RPV 900 mg every 8 weeks (n = 522) versus LA CAB 400 mg/RPV 600 mg every 4 weeks (n = 523) in ART-experienced adults.⁴⁸ At 48 and 96 weeks, every 8-week dosing was noninferior to every 4-week dosing (HIV RNA ≥50 copies/mL; 2% vs. 1% at 48 weeks).^{48,49}

In the SOLAR study, 687 individuals who were virologically suppressed on BIC/FTC/TAF were randomized 2:1 to either switch to LA CAB/RPV every 2 months or remain on BIC/FTC/TAF.⁵⁰ At 48 weeks LA CAB/RPV was virologically noninferior to continued BIC/FTC/TAF with 1% versus

<1% virologic failure, respectively. There was also no difference in viral suppression with or without oral lead-in when initiating LA CAB/RPV therapy. Based on these data and other existing data, oral lead-in therapy is now considered optional (see the [CABENUVA FDA drug label](#)).

Lastly, the Cabotegravir and Rilpivirine Efficacy and Safety (CARES) trial, an open-label trial conducted in Africa, randomized people with HIV and viral suppression to receive LA CAB/RPV every 8 weeks or to continue with their oral ARV regimen. At 48 weeks, 246 of 255 (96%) in the LA CAB/RPV arm and 250 of 257 (97%) in the oral ARV arm maintained HIV RNA <50 copies/mL. Two LA CAB/RPV participants had virologic failure at Week 48 and developed high-level resistance to both CAB and RPV.⁵¹

Adverse Effects

The most common adverse events associated with LA CAB/RPV are injection site reactions (ISRs) which occurred in more than 80% of participants at least once. ISRs reduce in frequency over time, occurring in about 10% to 30% of participants after the first year. ISRs were generally mild to moderate, with 99% being Grade 1 or 2 and the median duration of symptoms being 3 days. Hypersensitivity reactions, post-injection reactions, hepatotoxicity, and depressive disorders have also been reported.^{45,52}

Practical Considerations When Using Long-Acting Injectable Cabotegravir and Rilpivirine

When prescribing LA CAB/RPV, clinicians should refer to the FDA product label⁴² for guidance with the following practical considerations:

- Injection techniques, including injection sites and needle length in people with body mass index >30 kg/m²
- Several drugs, particularly those interacting with CYP3A or uridine diphosphate glucuronosyltransferase 1A1, are contraindicated with CAB and RPV (oral and/or IM) due to significant drug interactions (Drug–Drug Interaction Tables [24b](#) and [24d](#)).
- Management strategies for missed doses of IM injections and use of oral bridging therapy

HIV Viral Load and Drug-Resistance Testing Monitoring

HIV viral load monitoring should be performed 4 to 8 weeks after a switch to LA CAB/RPV. HIV viral load also should be checked in people with unplanned missed visits and delayed dosing of LA CAB/RPV. When viremia develops during LA therapy, resistance testing, including integrase resistance testing, should be performed. Follow-up dosing in people with missed doses should not be delayed while awaiting viral load and resistance test results. In a pooled analysis from FLAIR, ATLAS, and ATLAS-2M, confirmed virologic failure was more common in the setting of baseline, proviral, RPV resistance-associated mutations; HIV-1 subtype A6 (rare in the United States); higher baseline body mass index; and lower week-8 trough RPV concentration.⁵³ However, some recent studies did not observe worse outcomes in people with a high body mass index (Opera Cohort and CARES Study).^{51,54} In the setting of confirmed virologic failure on LA CAB/RPV, resistance to NNRTIs and INSTIs is common.^{45,46,55}

Pregnancy Considerations

Initiation of LA CAB/RPV is not recommended during pregnancy. Clinicians should refer to the [Perinatal Guidelines](#) for recommendations in managing women who become pregnant or are

planning pregnancy while receiving LA CAB/RPV. Health care providers are strongly encouraged to register women with HIV who become pregnant while receiving LA CAB/RPV in the [Antiretroviral Pregnancy Registry](#).

Considerations Regarding Long-Acting Cabotegravir Plus Rilpivirine Use in People With a History of Adherence Challenges

There are emerging data on the use of LA CAB/RPV in people with a history of adherence challenges. The open-label Long-Acting Therapy to Improve Treatment Success in Daily Life (LATITUDE [ACTG A5359]) Study enrolled people who had HIV viremia, challenges in taking daily oral ART, and no evidence of CAB or RPV resistance.⁵⁶ After adherence support and conditional financial incentive, participants who were able to achieve viral suppression were randomized to monthly LA CAB/RPV or to continue on the oral regimen. Study randomization was stopped early by the Data Safety Monitoring Board because of lower rates of virologic failure in individuals randomized to LA CAB/RPV than those staying on oral therapy (see the [Virologic Failure](#) section for more discussion).

Optimization Strategies for People With Viral Suppression and a History of NRTI and/or NNRTI Resistance

Some existing data demonstrate the safety and efficacy of select within-class and between-class switches for individuals with underlying drug resistance who are on a stable ARV regimen with suppressed HIV RNA (e.g., for 6 months or longer). These findings are further supported by data extrapolated from those experiencing virologic failure.

Within-Class Switch From Dolutegravir to Bictegravir in the Setting of Limited Resistance

The GS-US-380-4030 study enrolled 565 participants who were stably suppressed on DTG plus two NRTIs. The participants were randomized to either remain on their current regimen or switch to BIC/FTC/TAF. After 48 weeks, the groups had similar rates of sustained suppression. The rates of viral suppression were similar for those with a documented history of NRTI resistance (approximately 25% of participants) and those without a history of NRTI resistance.⁵⁷ Results from this trial lend theoretical support to other optimization strategies that include a switch from one high-barrier drug to another in the setting of underlying NRTI genotypic resistance. This study also supports using drugs with a high barrier to resistance when only one, and possibly no NRTIs in the background regimen, need to be fully active. This is further supported by other lines of evidence, some of which are discussed below.

Data of Within and Between Class Switches to Bictegravir in the Setting of Limited Drug Resistance

The BRAAVE study was an open-label, optimization study for Black people with HIV and viral suppression for ≥ 12 months on a standard regimen of two NRTIs plus a third agent (INSTI, NNRTI, or PI). Individuals were randomized to switch to BIC/FTC/TAF or to remain on current therapy (although individuals on TDF were switched to TAF). Switching to BIC-based therapy was noninferior for maintaining viral suppression compared with continuing current oral therapy. Baseline regimens included 61% INSTI, 31% NNRTI, and 9% PI. Baseline resistance did not affect the outcomes of therapy. In particular, NRTI resistance was present in 14% of participants, with 10% harboring the M184V/I mutation.⁵⁸

Switch From a Boosted Protease Inhibitor to Dolutegravir in Individuals With Viral Suppression and Prior First-Line NNRTI Plus NRTI Failure

The second-line switch to DTG (2SD) study was conducted at four sites in Kenya in a population of people with HIV who had prior first-line NNRTI plus NRTI regimen failure and achieved viral suppression on a current ritonavir-boosted PI-based regimen without undergoing genotypic resistance testing.⁵⁹ In this prospective, open-label trial, 795 participants with HIV were randomized 1:1 to switch the boosted PI to DTG or continue the current regimen. NRTIs were not switched, although a switch was allowed if clinically indicated. At Week 48, switching to DTG was noninferior to continuing a boosted PI with about 5% of participants in each arm having viral rebound >50 copies/mL. In the 40 people with virologic rebound, 20 in each arm, no resistance to DTG or PIs was found. Grade 3 or 4 adverse events were similar in the two groups. This study was limited by the absence of genotypic resistance data demonstrating the amount of NRTI resistance present at the time of first-line NNRTI plus NRTI failure. Nevertheless, several related studies in similar settings have demonstrated that extensive NRTI and NNRTI resistance is expected in this situation, often including resistance to tenofovir and 3TC or FTC.⁶⁰⁻⁶² Accepting that substantial NRTI resistance was likely to be present in this study population, 2SD provided evidence that a person who is suppressed on a boosted PI-based regimen is likely to remain suppressed on a second-generation INSTI-based regimen in the setting of NRTI resistance. This is further supported by data showing that people with virologic failure and NRTI resistance can be successfully treated with second-generation INSTIs plus TAF or TDF and FTC or 3TC.⁶⁰

Data From Virologic Failure Studies Inform Optimization With NRTI Resistance in Clinical Practice

Several studies conducted in resource-limited countries where people with first-line NNRTI-based virologic failure and who have NRTI-resistance support treatment recommendations for optimizing ART in these settings (details in the [Virologic Failure](#) section). In the DAWNING study,⁶³ for individuals who failed first-line NNRTI-based ART, DTG (84%) outperformed LPV/r (70%) for 48-week viral suppression <50 copies/mL. Participants who had one active NRTI in their study regimen had similar virologic responses as those with two active NRTIs. In the NADIA study, participants failing first-line NNRTI-based therapy were randomized to either DTG or DRV/r combined with randomized NRTIs. At enrollment, resistance to NRTIs was common with 86% of participants having an M184V mutation and 50% having a K65R mutation. At 48 weeks, DTG was noninferior to DRV/r, with both groups achieving viral suppression to <400 copies/mL in about 90% of participants. There were no differences in rates of viral suppression between participants with resistance to one or both NRTIs in their regimen compared to those with no NRTI resistance.⁶⁰ At Week 96, TDF was superior to zidovudine when combined with 3TC.⁶⁴

Although these were studies of ART use in the setting of initial NNRTI-based virologic failure, we expect that regimens that are effective during viremia would also be effective in the setting of viral suppression in people with similar ARV resistance patterns (NRTI resistance, but no known or suspected PI or INSTI resistance). These studies support that for optimization in the setting of NRTI resistance, two NRTIs (TAF or TDF plus 3TC or FTC) should be included in the regimen with a fully active drug with a high resistance barrier, such as DTG **(AII)**, BIC **(BIII)**, or boosted DRV (DRV/c or DRV/r) **(BIII)**. The use of regimens without any fully active NRTIs **is not routinely recommended** when there are other viable treatment options. However, in some clinical situations—such as when prior resistance testing is not fully available, to avoid major drug–drug interactions, to keep regimens simpler, or for other reasons—a regimen with a fully active drug with a high resistance barrier plus two partially active NRTIs can be considered. Both tenofovir and cytidine analogs may retain partial activity even when resistance is present.

Optimization Strategies for People With Viral Suppression and a History of Complex Underlying Resistance

Before optimizing an ARV regimen of a person with viral suppression who has a history of treatment failure and complex drug resistance, a careful review of the individual's full ARV history and cumulative drug resistance profile should be undertaken. Consultation with a clinician with expertise in HIV drug resistance is recommended **(AIII)**.

As described in the Optimization Strategies for People With Viral Suppression and a History of NRTI and/or NNRTI Resistance section above, any regimen that includes a fully active drug with a high barrier to resistance (e.g., DTG, BIC, or boosted DRV) plus two NRTIs (even if resistance is present), is likely to achieve suppression in those experiencing virologic failure and maintain suppression in the setting of optimization.^{59,60} Reliably maintaining suppression when there is not a fully active drug with a high resistance barrier is more complicated and requires careful consideration before any switch is made. In people with prior virologic failure(s) and/or limited prior resistance testing, proviral DNA genotypes can be considered. Clinicians should keep in mind that proviral DNA resistance mutations found will be useful, but some archived resistance mutations may not be found with this testing. Previous studies demonstrated that it is best to include at least 2 (preferably 3) fully active drugs in the setting of virologic failure if a fully active drug with a high resistance barrier is not included in the regimen due to resistance and/or intolerance.⁶⁵ In this setting, it will often require incorporating active drugs in classes other than NRTIs, NNRTIs, PIs, and INSTIs, such as entry inhibitors (i.e., CCR5 antagonists, post-attachment inhibitors, attachment inhibitors, and fusion inhibitors) or capsid inhibitors (e.g., lenacapavir), recognizing that entry and capsid inhibitors have primarily been studied and are indicated for those with virologic failure.

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Discontinuation or Interruption of Antiretroviral Therapy

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Discontinuation or interruption of antiretroviral therapy (ART) may result in viral rebound, immune decompensation, and/or clinical progression.¹⁻⁵ Thus, discontinuation or planned interruption of ART is not recommended outside the context of a clinical trial (**AI**). However, unplanned interruption of ART may occur under certain circumstances as discussed below.

Unanticipated Interruptions of Oral Antiretroviral Drugs

Reasons for short-term interruption (days to weeks) of ART vary and may include intercurrent illnesses that preclude oral intake (e.g., gastroenteritis, pancreatitis), surgical procedures, drug toxicity, or interrupted access to antiretroviral (ARV) drugs. Stopping ART for a short time (i.e., less than 1 day to 2 days) usually can be done by holding all drugs in the regimen. **Whether unplanned interruptions occur by accident or by necessity (e.g., because of drug toxicities), all efforts should be made to minimize their duration.** Recommendations for some specific scenarios are listed below.

When a Patient Experiences Unexpected Inability to Take Solid Oral Medications

For patients who require tube feeding, some ARV drugs are available in liquid formulations, and some pills may be crushed. The [Oral Antiretroviral/HCV DAA Administration](#)⁶ provides information on crushing pills and formulating liquid ARV drugs. Additional information also may be available in drug product labels. Clinicians should consult an HIV specialist and/or pharmacist to assess the best way for a patient with a feeding tube to continue an effective ARV regimen.

For patients unable to take medications by any enteral route (e.g., in the context of severe gastrointestinal disease), all components of the oral drug regimen should be stopped simultaneously, regardless of half-lives of the drugs. After resolution, all components of the ARV regimen should be restarted simultaneously.

Several ARV drugs are available as parenteral formulations; these include zidovudine, enfuvirtide, ibalizumab (IBA), and the long-acting (LA) injectable formulations of cabotegravir (CAB LA) and rilpivirine (RPV LA). The combination of CAB LA and RPV LA is approved as a complete regimen for the treatment of HIV. However, this regimen has not been studied as an alternative for patients who cannot take oral medications. Clinicians should consult with an HIV specialist before prescribing any of these agents.

When a Patient Experiences a Severe or Life-Threatening Toxicity to an Antiretroviral Agent

All components of the ARV drug regimen should be stopped simultaneously, regardless of drug half-life. After resolution, a different complete regimen that does not include the offending agent should be started.

Interruption of Long-Acting Antiretroviral Drugs

The combination of CAB LA and RPV LA is approved as a complete regimen for the treatment of HIV. CAB LA and RPV LA are given as intramuscular (IM) injections and have extended half-lives. Therefore, patients who miss doses or discontinue therapy without bridging with an oral ARV regimen are at increased risk of virologic failure with development of drug resistance. Clinicians should refer to prescribing information for CAB LA and RPV LA for the management of missed doses or discontinuations.⁷ For planned missed injection doses of CAB LA and RPV LA, oral formulations of CAB and RPV should be made available to patients as a bridging therapy for up to 2 months. Oral formulation of RPV is available by prescription in community pharmacies, but oral formulation of CAB is available only through the manufacturer. When stopping long-acting injectable ART, transition to a suppressive oral ARV regimen should occur within 4 weeks of the last planned IM doses. Patients who have missed or delayed clinic visits repeatedly should be reassessed to determine if resumption of injections is appropriate or if they may need to be transitioned back to an oral regimen. Plasma viral load testing should be performed before the transition, and drug-resistance testing should be considered if plasma viremia is present.

Patients with drug-resistant HIV may receive IBA as part of a salvage regimen. IBA is initiated with a 2,000-mg loading dose given as an intravenous (IV) infusion, then followed by 800 mg given as an IV infusion every 14 days as maintenance therapy. If a dose is missed by ≥ 3 days, a repeat loading dose of 2,000 mg IV infusion is recommended before resumption of maintenance therapy.

Analytical Treatment Interruption

Several research studies are evaluating approaches to achieve sustained ART-free viral remission or a functional cure for HIV.⁸ Viral eradication (i.e., elimination of HIV entirely from an individual) remains a more challenging, longer-term goal. Currently, the only way to reliably test the effectiveness of these strategies is to interrupt ART and closely monitor for viral rebound in the setting of a clinical trial, an approach referred to as “analytical treatment interruption” or ATI.⁹ The duration of treatment interruption, the dynamics of viral rebound, and the criteria for restarting ART are part of ATI clinical trial designs with the goal to conduct these clinical trials safely.

Before ART is interrupted, participants of ATI trials should be made aware of and understand the risks of viral rebound,¹⁰ acute retroviral syndrome, increased risk of HIV transmission, decline of CD4 count, HIV disease progression, development of minor HIV-associated manifestations (e.g., oral thrush) or serious non-AIDS complications (e.g., renal, cardiac, hepatic, or neurologic complications), and the development of drug resistance. Patients should be counseled about the need for close clinical and laboratory monitoring during ART interruptions and provided counseling and linkage to pre-exposure prophylaxis services should they wish to refer sexual partners at risk for acquiring HIV.

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CARDIOVASCULAR AND METABOLIC COMPLICATIONS IN PEOPLE WITH HIV

Introduction

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With continuous antiretroviral therapy (ART) and viral suppression, life expectancy for people with HIV is approaching that of people without HIV. Despite a narrowing mortality gap between people with and without HIV, some estimates suggest a gap of up to 10 years, which remains clinically relevant.¹ With prolonged survival, age-related comorbidities have increased at higher rates among people with HIV than among those without HIV, including atherosclerotic cardiovascular disease (ASCVD), metabolic complications (e.g., weight gain, insulin resistance, dyslipidemia, and metabolic syndrome), cancer linked to pro-oncogenic viruses, and other aging-related conditions.²⁻⁸ Three key factors that contribute to this burden of age-related noncommunicable diseases among people with HIV include: (1) an over-representation of traditional risk factors in this population, especially higher rates of smoking, diabetes, and dyslipidemia; (2) possible cardiometabolic toxicities from some antiretroviral drugs; and (3) HIV-associated chronic inflammation and immune activation, which are not completely abated by suppressive ART.⁹⁻¹⁸

In this context, the U.S. Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents has developed this section to provide guidance on the prevention and optimal management of cardiovascular disease (CVD) and metabolic complications in people with HIV. This section is divided into four subsections: (1) HIV-associated inflammation and immune activation as a contributor to CVD and metabolic complications, (2) CVD risks among people with HIV, (3) recommendations on the use of statin therapy for the prevention of ASCVD in people with HIV, and (4) weight gain among people with HIV receiving ART. Given the relevance of HIV-associated immune activation and inflammation as a mediating factor across the comorbidities discussed, this topic is also included within this section of the guidelines.¹⁹⁻²⁴

Other important CVD manifestations and metabolic complications are not currently covered in detail here, such as insulin resistance and risk for diabetes mellitus, metabolic syndrome, metabolic dysfunction–associated steatotic liver disease, heart failure, sudden cardiac death, and bone disease. At this time, guidance on the management of these conditions remains similar for people with and without HIV infection. Still, these conditions represent areas of active research, and as future evidence arises that informs management specific to people with HIV, this section may be expanded.

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Immune Activation and Inflammation Among People With HIV Receiving Antiretroviral Therapy

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Key Considerations and Recommendations
• HIV-related immune activation and systemic inflammation, despite viral suppression with antiretroviral therapy, are associated with increased risk for cardiovascular disease and metabolic complications, as well as a broad spectrum of comorbidities. • Switching or adding antiretroviral (ARV) drugs solely to reduce immune activation or inflammation is not recommended, except in a clinical trial (All). • The use of immunomodulatory or anti-inflammatory therapy to reduce HIV-related immune activation or inflammation is not recommended, except in a clinical trial (All). • Routine monitoring of markers of immune activation or inflammation (e.g. high-sensitivity C-reactive protein [CRP] or interleukin-6 [IL-6]) to inform the clinical management of HIV is not recommended (BII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak
Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

The Influence of Immune Activation and Inflammation on Disease Risk in People With HIV

Compared to people without HIV, people with HIV have demonstrated higher overall immune cell activation and higher levels of systemic inflammation (as measured by biomarkers such as interleukin-6 [IL-6]) and acute phase reactants (e.g., high-sensitivity C-reactive protein [hsCRP]).

Although immune activation declines with suppressive antiretroviral therapy (ART), it often persists at elevated levels in virologically suppressed people with HIV.^{1,2} Extensive literature over the past decades has demonstrated a link between HIV-related systemic inflammation and an increased risk of age-related comorbidities, as well as mortality from non-AIDS conditions.³⁻¹⁶ Specifically, blood markers of systemic inflammation among people with HIV have been associated with increased risk for a wide range of chronic conditions, including atherosclerotic cardiovascular disease (ASCVD), diabetes mellitus, obesity, cancer, osteopenia/osteoporosis, and chronic obstructive pulmonary disease.⁸⁻¹⁸ Even in individuals with CD4 T lymphocyte (CD4) cell counts >500 cells/mm³, immune activation and inflammation are associated with increased morbidity and mortality from age-related comorbidities.^{19,20}

Antiretroviral Therapy as a Strategy to Reduce Inflammation

Early HIV diagnosis combined with prompt ART initiation can effectively achieve a lower level of persistent immune activation. Most inflammatory markers decline during the first several months of ART and may achieve a stable “setpoint” within 1 to 2 years.^{21,22} In observational studies, people with HIV who initiated ART during acute HIV infection appeared to achieve a lower immune activation setpoint during ART-mediated viral suppression than those who started ART at later disease stages.²³ These data reinforce the recommendation to start ART as soon as possible after HIV diagnosis, regardless of CD4 count (see [Initiation of Antiretroviral Therapy](#)).

Currently, data do not support intensifying or modifying ART as a strategy to reduce immune activation after viral suppression is already achieved. For example, adding antiretroviral (ARV) drugs to an already suppressive regimen (or ART intensification) does not consistently improve immune activation.²⁴⁻²⁸ Although some studies have suggested that switching an ARV regimen to one with a more favorable lipid profile may improve some markers of immune activation and inflammation, these studies have limitations, and results are not consistent across markers or among studies.²⁹⁻³¹ Thus, adding or switching ARVs solely to reduce immune activation or inflammation **is not recommended**, except in a clinical trial **(All)**.

Adjunct Treatments to Reduce Immune Activation and Inflammation

Treatment strategies targeting immune activation and inflammation have been a focus of ongoing investigation. In this context, research strategies to mitigate inflammation among people with HIV on suppressive ART have typically taken one of three approaches: (1) target the root drivers of immune activation specific to HIV disease, (2) downregulate inflammatory pathways broadly, or (3) leverage anti-inflammatory properties of medications targeting traditional risk factors.

The HIV-related mechanisms driving immune activation have been posited to include viral persistence, permanent injury impairing immune recovery (i.e., due to lymphatic tissue fibrosis), presence of copathogens, as well as dysbiosis and microbial translocation.³²⁻³⁶ Among these mechanisms, investigational approaches have targeted HIV reactivation from latently infected cells, fibrotic pathways, the gut microbiome, and cytomegalovirus coinfection.^{27,37-41} However, thus far, none of these strategies have improved clinical outcomes. Therefore, these interventions should be pursued only in the context of clinical trials.

Data have emerged demonstrating the potential clinical benefit of broadly anti-inflammatory and immunosuppressive strategies among people without HIV who are at high risk for ASCVD events. The CANTOS trial (Canakinumab Anti-Inflammatory Thrombosis Outcomes Study) studied canakinumab, a human monoclonal antibody targeting cytokine interleukin-1b, a driver of the IL-6 signaling pathway, as a secondary prevention strategy among people with known ASCVD who do not have HIV. CANTOS demonstrated a treatment effect for reducing inflammation, as well as cardiovascular events and cancer death, in the general population.⁴² In people with HIV, canakinumab and tocilizumab, inhibitors of the receptor for IL-6, have been shown to reduce blood levels of markers of inflammation and immune activation.^{43,44} However, it is unclear whether the potential risks of these immunomodulatory therapies, including the increased sepsis risk seen in the CANTOS trial,⁴² outweigh their possible benefits for people with HIV, particularly with long-term use. Therefore, the clinical use of immunomodulatory or anti-inflammatory therapy solely to reduce HIV-associated immune activation or inflammation **is not recommended** among people with HIV, except in a clinical trial **(All)**.

Finally, beyond the well-established clinical benefit for reducing ASCVD events, hydroxymethylglutaryl coenzyme A (or HMG-CoA) reductase inhibitors, also known as statins, have been shown to improve biomarker levels of inflammation (e.g., hsCRP) and immune activation in the general population.^{45,46} In this context, data from REPRIEVE (Randomized Trial to Prevent Vascular Events in HIV) have demonstrated a reduction in cardiovascular events with pitavastatin use among people with HIV (see [Statin Therapy in People With HIV](#)). Other commonly used medications with anti-inflammatory properties—such as aspirin, angiotensin-converting enzyme inhibitors, methotrexate, and angiotensin receptor blockers—have failed to consistently reduce biomarkers of immune activation and/or inflammation in people with HIV in randomized controlled trials.^{39,47-50}

Monitoring Inflammation

As evidence is insufficient to inform differential screening practices or clinical management strategies (e.g., for ASCVD) based solely on the level of individual blood biomarkers, there is no clear rationale to routinely monitor levels of immune activation and inflammation in people with HIV. Thus, routine monitoring of markers of immune activation or inflammation (e.g., hsCRP, IL-6) to inform the clinical management of HIV **is not currently recommended (BII)**.

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Cardiovascular Complications in People With HIV

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Key Considerations and Recommendations

Antiretroviral Management

- All people with HIV should maintain viral suppression with continuous antiretroviral therapy (ART) to reduce the risk of atherosclerotic cardiovascular disease (ASCVD) **(AII)**.
- Due to concerns regarding an increased risk of cardiovascular events, alternative antiretroviral (ARV) medications should be considered in place of abacavir and lopinavir/ritonavir in people with higher underlying ASCVD risk or known ASCVD **(AII)**.

ASCVD Risk Stratification

- For people with HIV aged 40–75 years, 10-year ASCVD risk score should be calculated using the Pooled Cohort Equations (or PCE) at least annually to inform management, including whenever lipids are measured or in certain clinical scenarios that may affect ASCVD risk* **(BIII)** (see Atherosclerotic Cardiovascular Disease Risk Scores in the text).
- Lipid screening should be performed in people with HIV **(AIII)**, with the following suggested frequency: (1) at entry into care or at the time of ART initiation, (2) 3–6 months after ART initiation once viral suppression is achieved, (3) annually for those aged ≥40 years or receiving statin therapy, (4) every 1–3 years for those aged <40 years **and** not on statin therapy, and (5) with changes in cardiovascular disease (CVD) risk factors (see text for additional details on monitoring).
- There is insufficient evidence to support the use of inflammatory biomarkers for ASCVD risk stratification among people with HIV.
- When compared to the general population, women with HIV have both a greater increase in relative risk of ASCVD and a greater underprediction of ASCVD by risk scores than men with HIV. For women with HIV, it is critical to ensure optimal ASCVD risk factor modification regardless of ASCVD risk score, implement ASCVD prevention strategies, and perform diagnostic evaluation if they present with possible cardiac symptoms.

ASCVD Prevention and Risk Factor Management

- Statin therapy is recommended for all people with HIV aged 40–75 years and in additional clinical scenarios for other age groups to reduce the risk of ASCVD and major adverse cardiovascular events. See [Statin Therapy in People With HIV](#) for recommendations on statin use.
- The use of immunomodulatory therapy to reduce HIV-associated ASCVD risk **is not recommended** except in a clinical trial **(AII)**.
- Some drugs prescribed to prevent or treat CVD may have clinically significant interactions with certain ARV drugs. Before initiating or changing ARV drugs or concomitant medications, clinicians should carefully assess for potential interactions **(AIII)**.
- All people with HIV should be counselled about ways to reduce CVD risk through behavioral modifications such as healthy diet, weight control, regular exercise, smoking cessation, and limiting alcohol consumption.
- Optimal management of comorbidities (e.g., hypertension, diabetes, obesity) remains important to further reduce CVD risk among people with HIV.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

* HIV-related CVD risk-enhancing factors include prolonged duration of HIV infection, delayed ART initiation, long periods of HIV viremia and/or treatment nonadherence, low current or nadir CD4 T

lymphocyte cell count (e.g., <350 cells/mm³), exposure to some older ARV drugs associated with cardiometabolic toxicity, and/or coinfection with hepatitis C virus.

Overview

Most people with HIV who are using continuous antiretroviral therapy (ART) and achieve viral suppression have a life expectancy that is approaching that of people without HIV.¹ However, a gap in mortality remains due to comorbidities such as atherosclerotic cardiovascular disease (ASCVD). As mortality from AIDS events among people with HIV has declined, the proportion of ASCVD-related deaths has increased. When compared to people without HIV, people with HIV have an approximately twofold higher risk of developing ASCVD, even after accounting for traditional cardiometabolic risk factors.² The age at incident ASCVD diagnosis among people with HIV is often younger than that of people without HIV. In this context, the 2018 Multisociety Cholesterol Guidelines identified HIV as an ASCVD risk enhancer.³ Factors influencing ASCVD risk among people with HIV include a higher prevalence of traditional cardiometabolic risk factors compared with the general population,⁴ ongoing systemic inflammation associated with HIV despite viral suppression,⁵ and cardiometabolic toxicity associated with certain ARV drugs.^{6,7} Data also suggest that excess ASCVD risk relative to the general population is greater among women with HIV than among age-matched men with HIV.^{8,9} In addition, differential access to screening and treatment of ASCVD risk factors also contributes to excess ASCVD risk among certain people with HIV in the United States.^{10,11}

While there has been extensive focus on the epidemiology and pathogenesis of ASCVD among people with HIV, there are also data demonstrating increased risk of other cardiac and vascular diseases. People with HIV have an increased risk of developing stroke (including ischemic and hemorrhagic), heart failure, arrhythmia or sudden cardiac death, and thrombosis.¹²⁻¹⁵ Risk factors for these outcomes are thought to be similar to those for ASCVD, with an etiologic role for both traditional risk factors and HIV-related factors. Currently, this section of the guidelines is focused on primary prevention strategies for people with HIV, with an emphasis on ASCVD. As data emerge concerning primary and secondary prevention strategies for other cardiovascular outcomes specific to people with HIV, this section may be expanded.

HIV-Related Risk Factors for Atherosclerotic Cardiovascular Disease

Factors related to HIV and its treatment have been shown to be important contributors to excess cardiovascular disease (CVD) risk. Specifically, HIV-related inflammation has been associated with elevated CVD risk. This inflammation is increased in the presence of active viremia and decreases with ART, but still remains elevated when compared to those without HIV (see [Introduction](#)). Data from the SMART trial demonstrated that continuous ART not only improves HIV-related outcomes but also significantly reduces non-AIDS events, including CVD, an observation thought to be related to the beneficial effects of ART on inflammation.¹⁶ In the large Veterans Aging Cohort Study (VACS), myocardial infarction (MI) risk was independently associated with detectable or increased HIV RNA; compared to people without HIV, MI risk was increased by 39% in those with HIV RNA <500 copies/mL and by 75% in those with HIV RNA ≥500 copies/mL.¹⁷ In REPRIEVE, even low-level HIV viremia (median 62 copies/mL) was linked to increased rates of major adverse cardiac events (MACE), as viremic individuals had a higher risk than those with viral load below the limit of detection (adjusted hazard ratio 1.4; 95% confidence interval, 1.00–1.97)^{4,18}; however, another recent study did not show an association between HIV viremia and CVD.¹⁹ Based on these findings, continuous ART is recommended for all people with HIV to maintain viral suppression and reduce ASCVD risk **(All)**. Immunomodulatory therapy as an intervention to reduce HIV-associated ASCVD risk **is not recommended** except in a clinical trial **(All)**. See [Immune Activation and Inflammation Among People With HIV Receiving Antiretroviral Therapy](#) for additional details.

Additional factors related to HIV may further impact ASCVD risk. Lower CD4 T lymphocyte (CD4) cell count and nadir CD4 count have been shown to be associated with CVD events independent of traditional CVD risk factors or specific ART,^{20,21} with VACS demonstrating CD4 <200 cells/mm³ as an independent predictor of MI with an increased risk of nearly 90%.^{17,22} HIV-related CVD risk-enhancing factors include prolonged duration of HIV infection, delayed ART initiation, long periods of HIV viremia and/or treatment nonadherence, low current or nadir CD4 count (e.g., <350 cells/mm³), exposure to older antiretroviral (ARV) drugs associated with cardiometabolic toxicity, and coinfection with hepatitis C.²³

Women with HIV have also been shown to have a greater relative risk of ASCVD compared to men with HIV. Although data indicating the absolute risk of MI is similar between genders—12.71 versus 10.48 per 1,000 person-years for women and men with HIV, respectively—women with HIV have exhibited a nearly 3- to 4-fold increase in relative MI risk versus a 1.5- to 2-fold increase for men compared to sex-specific rates among people without HIV.^{9,24-26} A substudy from REPRIEVE demonstrated the absence of an expected protective effect of female sex for the development of a MACE.⁴ The presence of HIV-related ASCVD risk-enhancing factors, including female sex, may strengthen the rationale for ASCVD prevention strategies, including statin initiation (see [Statin Therapy in People With HIV](#)), and may be used to inform decisions to avoid certain ARV medications that have been associated with increased ASCVD risk.

Antiretroviral Therapy and Cardiometabolic Complications

Specific ARV medications and classes have been associated with potential cardiometabolic toxicity, which has informed the U.S. Department of Health and Human Services Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel)'s recommendations on ART regimen selection to reduce risk for cardiometabolic complications. See [Interactions Between ARV Drugs and Drugs for Treatment or Prevention of Cardiovascular Diseases](#) below and [Table 7](#) of What to Start for more information. Specifically, for individuals with elevated traditional ASCVD risk or known CVD, the selection of an ART regimen should consider options that minimize the introduction of new cardiometabolic adverse effects.

Antiretroviral Therapy and Atherosclerotic Cardiovascular Disease Risk

Individual ARV medications and classes have been linked to increased risk of CVD events. An association between abacavir (ABC) use and MI was first reported in the Data Collection on Adverse Events of Anti-HIV Drugs (D:A:D) study, with the increased risk noted only with recent/current ABC exposure (i.e., within 6 months) and particularly among participants with pre-existing cardiac risk factors.²⁷ Since the D:A:D report, multiple studies, systematic reviews, and meta-analyses have evaluated the relationship between ABC therapy and cardiovascular events, with some confirming the initially observed association and others failing to identify an association.²⁷⁻³² Retrospective analyses from the REPRIEVE study corroborated the D:A:D results, showing an increased hazard of a first MACE with ART regimens, including ABC versus tenofovir-containing regimens,³³ and an increased MACE hazard of 62% in participants with previous use and 41% with current use of ABC compared with participants who were never exposed.³⁴ An association between ABC and ASCVD risk reported in multiple large studies provides a rationale to consider underlying ASCVD risk when using ABC as a component of ART. Based on these data, the Panel recommends considering alternative ARV medications in place of ABC in people with HIV and higher underlying ASCVD risk or known CVD (**All**).

An early study from the D:A:D cohort demonstrated a 16% increased risk of MI per year of cumulative protease inhibitor (PI) exposure,³⁵ specifically with older PIs (e.g., indinavir and zidovudine-boosted lopinavir [LPV/r]). A case-control study within the French HIV ANRS cohort CO4 also found

a 33% increase in MI risk with LPV/r use.³⁶ A later study from the D:A:D cohort showed a 59% increased risk of CVD per 5 additional years of use with ritonavir-boosted darunavir (DRV/r) but not ritonavir-boosted atazanavir (ATV/r),³⁷ yet subsequent studies did not observe an association between boosted darunavir and increased CVD risk.³⁸⁻⁴⁰ Based on these data, the Panel recommends considering alternative ARV medications in place of LPV/r in people with HIV and higher underlying ASCVD risk or known CVD **(All)**. If use of DRV/r is indicated, clinicians should weigh the potential benefits of therapy versus the potential cardiovascular risk.

The effect of integrase strand transfer inhibitors (INSTIs) on the risk of CVD is an area of active research, with recent studies showing inconsistent results. In a multicenter observational study of the international RESPOND consortium, INSTI exposure was associated with a nearly twofold greater incidence of a composite CVD endpoint within the first 6 months after exposure compared to no INSTI exposure,⁴¹ and the increased risk persisted throughout the first 24 months. Subsequently, two target trial emulation studies found no association between INSTI use and a composite CVD outcome that included MI, stroke, or invasive cardiovascular procedures, regardless of previous ART experience.^{42,43} Based on current data, there is insufficient evidence to recommend modifying suppressive INSTI-based ART regimens in people with weight gain, higher underlying ASCVD risk, or known CVD.

Antiretroviral Therapy and Dyslipidemia

Higher levels of low-density lipoprotein cholesterol (LDL), high-density lipoprotein cholesterol (HDL), and triglycerides have been observed with tenofovir alafenamide (TAF), efavirenz (EFV), all boosted PIs, and elvitegravir/cobicistat (EVG/c).^{44,45} Among the PIs, similar lipid changes have been shown in participants from a randomized trial of ATV/r- and DRV/r-based regimens.⁴⁶ Importantly, differences in lipid levels between two ART regimens can be due to a lipid-raising effect of a given medication and/or a lipid-lowering effect of the other medication. For example, levels of total cholesterol, LDL, HDL, and triglycerides are higher in participants who received TAF when compared with those who received tenofovir disoproxil fumarate (TDF) in multiple randomized controlled trials.^{47,48} However, data from a placebo-controlled pre-exposure prophylaxis trial support that TDF exposure has an independent lipid-lowering effect (for total cholesterol, LDL, and HDL).⁴⁹ Further, data support that total cholesterol to HDL ratios did not differ between participants receiving TAF and those receiving TDF,^{50,51} and the clinical significance of CVD risk from increases across all lipid indices without changes in total to HDL ratios is uncertain.

Antiretroviral Therapy and Diabetes/Insulin Resistance

Direct effects on diabetes mellitus (DM) and insulin resistance have been observed with older ARVs, including ZDV and LPV/r.⁵² Several studies have demonstrated an increased risk of DM with INSTI use,⁵³⁻⁵⁵ an association that may be mediated in part by weight gain.⁵⁶ An association between ART and DM has not been consistently observed with other contemporary ARVs, including those recommended as initial therapy in the guidelines.

Antiretroviral Therapy and Other Cardiac Complications

EFV, as well as higher than recommended doses of fostemsavir and rilpivirine, have been associated with QTc (QT corrected for heart rate) prolongation.⁵⁷⁻⁵⁹ These ARV drugs should be used with caution in people with pre-existing heart disease, QTc prolongation, or concomitant use of medications that may increase drug concentration or prolong the QTc interval. Both ATV/r and LPV/r have been associated with PR prolongation, particularly among people with risk factors, including pre-existing heart disease and concomitant use of medications that may cause PR prolongation.^{60,61}

Strategies for Atherosclerotic Cardiovascular Disease Risk Estimation and Screening For People With HIV

Cardiovascular risk assessment and prevention strategies developed for the general population do not reflect HIV-specific factors, such as inflammation and immune activation or potential contributions from certain ARV drugs, which are key mechanistic contributors to HIV-associated CVD. This section will address ASCVD risk estimation and lipid screening for people with HIV. ASCVD prevention strategies specific to people with HIV are further discussed in the [Introduction](#) and [Statin Therapy in People With HIV](#) sections.

Lipid Monitoring

Lipid screening should be performed in people with HIV (**AIII**), given the increased risk for ASCVD and the increased rate of dyslipidemia associated with certain ARV drugs. A fasting or random lipid profile should be obtained for all people with HIV at entry into care or at the time of ART initiation or modification. If a random profile is abnormal, a fasting lipid profile should be obtained. Lipid levels should be rechecked 3 to 6 months after starting ART or later if viral suppression is not achieved within 6 months. Lipid levels measured after initiation of ART should then be used to calculate an ASCVD risk score and determine indications for statin therapy (see [Statin Therapy in People With HIV](#)). Among people with HIV who are aged 40 years or older or who are any age and are receiving statin therapy, lipid monitoring should be performed annually to reassess ASCVD risk and inform decisions on statin management and/or other risk modification strategies. Further, specific to people on statin therapy, the approach to monitor lipids at least annually to assess adherence and the need for treatment intensification is consistent with recommendations by the *2018 American Heart Association (AHA)/American College of Cardiology (ACC)/Multisociety Guideline on the Management of Blood Cholesterol* among people in the general population.⁶² For people with HIV who are under age 40 and not receiving statin therapy, lipid levels should be monitored every 1 to 3 years to reassess ASCVD risk. For all people with HIV, lipid monitoring may occur more frequently based on lifestyle factors or clinical scenarios, including changes in traditional ASCVD risk factors (e.g., weight gain, new diagnosis of hypertension, concomitant medications that affect lipids, or change to an ARV regimen with increased ASCVD risk).

Atherosclerotic Cardiovascular Disease Risk Scores

ASCVD risk prediction algorithms enable clinicians to quantify the risk of a future cardiovascular event and are used in clinical practice to stratify risk and to guide the intensity of ASCVD preventative measures and risk factor management, including statin use and antihypertensive therapy. Yet established ASCVD risk prediction scores validated in the general population typically do not include risk factors specific to HIV, and they have been shown to systematically underestimate risk among people with HIV. Data from multiple studies and a meta-analysis have shown that the ACC/AHA Pooled Cohort Equations (PCE) underpredict ASCVD risk in people with HIV in high-income countries.⁶³⁻⁶⁸ Studies also suggest the accuracy of established ASCVD risk scores for people with HIV may vary by patient demographics and clinical characteristics, with underprediction of ASCVD risk demonstrated among women,^{63,66,67} Black or African-American individuals,^{63,67} and individuals with low or moderate predicted CVD risk versus high risk.^{63,65,66} ASCVD risk score accuracy may also differ by region globally, with a study of REPRIEVE data showing underprediction of risk in high-income countries yet overprediction in low-income and middle-income countries.⁶⁷ Additional research is needed to better tailor ASCVD risk scores to people with HIV overall, as well as by geographic region and demographic and clinical characteristics.

In 2023, the AHA developed a new set of ASCVD risk equations for the general population. The Predicting Risk of CVD EVENTS (PREVENT) equations differ from the PCE on several levels,

including the inclusion of a new derivation dataset of more than 6 million individuals, new variables reflecting a broader cardiovascular-kidney-metabolic framework, a new outcome including heart failure, and the removal of race from the models.⁶⁹ Because PREVENT was developed for the general population, this updated risk score does not incorporate risk factors specific to HIV or its treatment. Predicted ASCVD risk has been shown to be lower using PREVENT in the general population.⁷⁰ For people with HIV,⁷¹ 97% had a lower predicted risk, and 58% fewer would be strongly recommended for statins using the PREVENT (vs. PCE) risk score. Further study is needed to determine if and how the PREVENT risk estimates may inform the clinical management of people with HIV.

The PCE is recommended for predicting ASCVD risk among people with HIV, as it has been more extensively studied in this population and used in the REPRIEVE trial to inform recommendations on the use of statins in people with HIV (see [Statin Therapy in People With HIV](#)). For people with HIV aged 40 to 75 years, the 10-year ASCVD risk score should be calculated using the PCE at least annually, including whenever lipids are measured or in certain clinical scenarios that may affect ASCVD risk (e.g., changes in traditional CVD risk factors, such as weight gain, new diagnosis of hypertension, concomitant medications that affect lipids, or change to an ARV regimen with increased ASCVD risk) **(BIII)**. Presently, there is insufficient evidence to recommend strategies to account for the underprediction of ASCVD risk by the PCE and other established ASCVD risk scores among people with HIV, including use of inflammatory biomarkers.

Interactions Between Antiretroviral Drugs and Drugs for Treatment or Prevention of Cardiovascular Diseases

People with HIV may receive a statin to prevent ASCVD or medications for the treatment of CVD risk factors or underlying CVD. Some of these drugs may have clinically significant interactions with some ARV drugs. Before initiating or changing these medications, clinicians should carefully assess for potential interactions **(AIII)**. Tables [24a](#) to [24g](#) and the [Statin Therapy in People With HIV](#) section of the guidelines, as well as the [University of Liverpool HIV Drug Interaction](#) site, provide guidance for managing these interactions. Major mechanisms of pharmacokinetic (PK) interaction between ARV drugs and cardiac drugs are noted in the sections below.

Cytochrome P450 Enzymes

Many cardiac drugs (such as certain antiarrhythmics, anticoagulants, beta-blockers, calcium channel blockers, statins, and others) are substrates of cytochrome P450 (CYP) enzymes and may interact with ARV drugs that are inhibitors or inducers of these enzymes. Concomitant use with:

- CYP inhibitors (such as the PK boosters ritonavir and cobicistat, and the capsid inhibitor lenacapavir) may—
 - o Increase the systemic exposure and potentially toxicities of cardiac drugs that are CYP substrates
 - o Decrease the efficacy of the antiplatelet drug clopidogrel (a prodrug) by inhibiting the formation of its active metabolite
- CYP inducers (such as EFV and etravirine) may—
 - o Reduce concentrations and efficacy of the cardiac drugs that are CYP substrates

Drug Transporters

Interactions related to drug transporters may also play a role in drug disposition. For example, the INSTIs bictegravir, dolutegravir, and elvitegravir are inhibitors of organic cation transporter 2 (OCT2)

and may reduce renal elimination of certain antiarrhythmics (such as dofetilide and disopyramide), thereby increasing concentrations of the cardiac drug and the risk for adverse events.

Because some interactions may result in serious toxicities or lack of efficacy, certain drug combinations are not recommended or are contraindicated. In these cases, clinicians should assess if alternative ARV drugs and/or alternative cardiac drugs can be used. In other cases, dosage modification and therapeutic drug monitoring of cardiac drugs may be needed to guide therapy, with close follow-up to assess efficacy and potential toxicity.

Management of Comorbidities and Traditional Cardiovascular Disease Risk Factors

Although HIV-specific factors influence CVD risk, traditional risk factors also play an important role among people with HIV. Although factors such as advancing age and other demographic characteristics cannot be changed, many CVD risk factors are modifiable with behavioral and pharmaceutical interventions. The AHA has outlined eight components of cardiovascular health for everyone: healthy diet; participation in physical activity; avoidance of nicotine; healthy sleep; healthy weight; and healthy levels of blood lipids, blood glucose, and blood pressure.⁷²

Traditional CVD risk factors predict incident CVD events and occur at higher rates among people with HIV compared with the general population.⁴ A 2025 global meta-analysis assessed common CVD risk factors among people with HIV and found higher odds of CVD incidence associated with DM, hypertension, dyslipidemia, smoking, and a prior history of CVD.⁷³ A U.S. population-based study among people with HIV found conventional risk factors, such as tobacco use, hypertension, and obesity, to be persistently associated with CVD incidence over follow-up periods for as long as 15 years.⁷⁴ Nontraditional risk factors for CVD that are not specific to HIV, including alcohol use disorders, depression, and hepatitis C virus coinfection, are also more common in people with HIV.

The aforementioned data highlight the importance of preventing and managing traditional risk factors among people with HIV. A longitudinal study comparing CVD risk factors among people with and without HIV found that associations between CVD and HIV status were attenuated when dyslipidemia and DM were well controlled.⁷⁵ Recommendations for screening and management of CVD risk factors are outlined in the HIVMA/IDSA 2024 update, *Primary Care Guidance for Providers Who Care for Persons With Human Immunodeficiency Virus*.⁷⁶ For women with HIV, ensuring optimal CVD risk factor modification regardless of CVD risk score and performing diagnostic evaluation for possible cardiac symptoms is critical given the greater relative risk of ASCVD.

Other Management Considerations and Unanswered Questions

Estrogen, Progestins, or Testosterone Therapy and Cardiovascular Risk

Studies have produced conflicting results on associations between estrogen, progestins, or testosterone therapy and CVD risk, varying by study design, study population, duration and timing of therapy, and other clinical factors.⁷⁷⁻⁸⁰ Research on relationships between these drugs and CVD among people with HIV is even more limited.⁸¹⁻⁸⁴ Data on testosterone use among people with HIV have been limited to the treatment of testosterone deficiency and lack longitudinal information on CVD-related outcomes.^{85,86} Studies of estrogen use suggest a potential increase in CVD risk; however, this association may vary by dose, mode of delivery, and underlying physiological stress.^{79-81,87} A longitudinal study of combined estrogen and progestin use for oral contraception among people with HIV suggested increased CVD risk, though the association was not statistically significant.⁸⁸

As the population with HIV ages, a growing number of women with HIV are entering menopause. CVD is the leading cause of morbidity and mortality in menopausal women, and this risk is increased in women with HIV.^{89,90} Studies of the cardiovascular risk associated with menopausal hormone therapy have led to changes in guidelines over the years.⁹⁰ Systemic hormone therapy is not currently recommended for CVD prevention or treatment.⁹¹ However, hormone therapy remains the most effective treatment for vasomotor symptoms.⁹² Some research suggests that women with HIV are undertreated for menopausal symptoms.⁹³ However, data are lacking on menopause and hormone therapy in women with HIV, including whether menopause amplifies the effects of HIV on CVD risk.⁹⁴ Current HIV primary care guidelines recommend that menopause-related hormone therapy for women with HIV should not differ from the care of women without HIV.⁷⁶

Cardiovascular Risks of Adults Who Acquired HIV Perinatally or During Adolescence

An increasing number of adults acquired HIV perinatally or during adolescence, but the cardiovascular effects of perinatal or adolescent HIV acquisition and long-term ART use are unclear. For individuals who acquired HIV perinatally, it is hypothesized that exposure to HIV and ART *in utero* while the cardiovascular system is developing, use of older ART drugs with cardiometabolic and mitochondrial toxicity, lifelong duration of infection, and a longer period of chronic inflammation may further exacerbate the already heightened CVD risk observed in people with HIV. Investigation of the cardiovascular complications of perinatal HIV infection has largely focused on youth and adolescents. Abnormalities in cardiac structure and function have been well documented and include systolic and diastolic dysfunction, left ventricular hypertrophy and dilatation, and myocardial fibrosis.⁹⁵⁻⁹⁹ Studies of perinatally acquired HIV have also demonstrated elevated rates of surrogate markers of atherosclerosis (including carotid intima-media thickness, arterial stiffness, and endothelial dysfunction),^{100,101} impaired glucose metabolism and insulin resistance,¹⁰² lipodystrophy and metabolic syndrome,¹⁰³ and increased Pathobiological Determinants of Atherosclerosis in Youth (or PDAY) cardiovascular risk scores.^{104,105} Adults with perinatally acquired HIV also appear to be at risk for cardiometabolic comorbidities, with a large North American study of young adults aged 18 to 30 years demonstrating a high incidence of DM, hypercholesterolemia, hypertriglyceridemia, hypertension, and chronic kidney disease.¹⁰⁶ Current gaps in knowledge regarding CVD risk among those who acquired HIV perinatally or during adolescence include (1) the clinical significance of cardiac structural and functional abnormalities and surrogate markers of atherosclerosis with respect to coronary plaque and cardiac endpoints; (2) whether the epidemiology and pathophysiology of CVD differ for people with similar durations of perinatally versus horizontally acquired HIV, including HIV acquired during adolescence; (3) whether the epidemiology and pathophysiology of CVD differ for people in low- and middle-income versus high-income countries; (4) the optimal approach to estimate ASCVD risk and frequency of monitoring in adults who acquired HIV perinatally or during adolescence; and (5) whether primary prevention approaches (e.g., statins) for adults with HIV aged 40 and older apply to younger adults who acquired HIV perinatally or during adolescence.

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Statin Therapy as Primary Prevention of Atherosclerotic Cardiovascular Disease in People With HIV

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Panel's Recommendations
For People With HIV Who Have Low-to-Intermediate (<20%) 10-Year Atherosclerotic Cardiovascular Disease (ASCVD) Risk Estimates Note: Estimates based on the ASCVD Risk Estimator from the American College of Cardiology (ACC). <i>Aged 40–75 Years</i> - When 10-year ASCVD risk estimates are 5% to <20%, the Panel for the Use of Antiretroviral Agents in Adults and Adolescents With HIV (the Panel) recommends initiating at least moderate-intensity statin therapy (AI). - Recommended options for moderate-intensity statin therapy include the following: - Pitavastatin 4 mg once daily (AI) - Atorvastatin 20 mg once daily (AII) - Rosuvastatin 10 mg once daily (AII) - When 10-year ASCVD risk estimates are <5%, the Panel favors initiating at least moderate-intensity statin therapy (CI). The absolute benefit from statin therapy is modest in this population; therefore, the decision to initiate a statin should take into account the presence or absence of HIV-related factors that can increase ASCVD risk.^a - Same options for moderate-intensity statin therapy as recommended for 10-year ASCVD risk estimates of 5% to <20% (see above) <i>Aged <40 Years</i> - Data are insufficient to recommend for or against statin therapy as primary prevention of ASCVD in people with HIV. In the general population, lifestyle modifications are recommended for people aged <40 years, with statin therapy considered only in select populations (see American Heart Association (AHA)/ ACC/Multisociety Guidelines).
Key Recommendations for the General Population (Including People With HIV)
Note: Recommendations based on AHA/ACC/Multisociety Guidelines. For People Aged 40–75 Years With High (≥20%) 10-Year ASCVD Risk Estimates - Initiate high-intensity statin therapy (see Table 11b below). For People Aged 20–75 Years With Low-Density Lipoprotein Cholesterol (LDL-C) ≥190 mg/dL - Initiate high-intensity statin therapy at the maximum tolerated dose (see Table 11b below). For People Aged 40–75 Years With Diabetes Mellitus - Initiate at least moderate-intensity statin therapy (see Table 11b below). Perform further risk assessment to consider using a high-intensity statin.

Key Considerations
• Coadministration of certain statins and antiretroviral drugs may result in significant drug–drug interactions. In some cases, the interaction may require statin dose adjustment, switching to another statin, or increased monitoring for statin-related adverse effects (see the Drug–Drug Interaction section below for details). • As in the general population, initiation of statin therapy should be deferred during pregnancy; statin therapy should be discontinued if pregnancy occurs, and breastfeeding is not recommended while using statin therapy.
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials, observational cohort studies with long-term clinical outcomes, relative bioavailability/bioequivalence studies, or regimen comparisons from randomized switch studies; III = Expert opinion
^a HIV-related factors that may increase ASCVD risk but are not considered in traditional risk estimation tools may strengthen the rationale for initiating statin therapy in this population. Examples include prolonged duration of HIV infection, delayed antiretroviral therapy initiation, long periods of HIV viremia and/or treatment nonadherence, low current or nadir CD4 T lymphocyte cell count (e.g., <350 cells/mm³), exposure to older antiretroviral drugs associated with cardiometabolic toxicity, and/or hepatitis C coinfection (see text below for rationale of the Panel's recommendations).

The above recommendations inform the use of statin therapy in primary prevention of atherosclerotic cardiovascular disease (ASCVD) in people with HIV receiving care in the United States. These recommendations were developed by the Panel for the Use of Antiretroviral Agents in Adults and Adolescents With HIV (the Panel) in collaboration with representatives from the American College of Cardiology (ACC), American Heart Association (AHA), and the HIV Medicine Association (HIVMA). The recommendations and supporting text have been endorsed by these organizations.

Background

Traditional risk factor modification remains an important component of primary ASCVD prevention strategies for people with HIV, including modifications related to diet, exercise, smoking cessation, blood pressure control, and lipid lowering for those with hyperlipidemia. Hydroxymethylglutaryl-coenzyme A reductase inhibitors (i.e., statins) may offer advantages for people with HIV over other prevention strategies, given that statin therapy is associated with a well-described reduction in ASCVD events, as well as treatment effects on inflammatory pathways.¹⁻³

Findings from the Randomized Trial to Prevent Vascular Events in HIV (REPRIEVE) have provided evidence to support specific recommendations for statin therapy as primary ASCVD prevention in people with HIV aged 40 to 75 years in the United States. The [2018 AHA/ACC/Multisociety Guidelines](#) for the general population recommend statin therapy for all people at high 10-year ASCVD risk (defined as $\geq 20\%$). For people at intermediate risk (i.e., $\geq 7.5\%$ to $< 20\%$) or borderline risk (i.e., 5% to $< 7.5\%$), it is recommended to prescribe statin therapy based on shared decision-making. These recommendations are based on risk stratification that uses the pooled cohort risk equations to estimate 10-year ASCVD risk, along with clinician–patient discussions about risk, benefits, and patient preference.^{4,5} Given that current ASCVD risk prediction tools typically underestimate risk among people with HIV, the AHA/ACC/Multisociety Guidelines identify HIV as a potential “risk enhancer” that should influence discussions on the use of statin therapy among those at borderline or intermediate risk.⁵ However, prior to the REPRIEVE trial, no formal recommendations had been issued for people with HIV at low-to-intermediate ASCVD risk (i.e., $< 20\%$). Results from REPRIEVE now provide clear evidence to inform the use of statin therapy

as primary prevention among people with HIV who are aged ≥ 40 years, on continuous antiretroviral therapy (ART), and at low-to-intermediate ASCVD risk.

Primary Study Results From the REPRIEVE Trial

REPRIEVE was a Phase 3 global randomized controlled trial of oral daily pitavastatin 4 mg versus placebo to prevent ASCVD. The study enrolled a total of 7,769 people with HIV who were aged 40 to 75 years and at low-to-intermediate risk based on 10-year ASCVD risk estimates.^{6,7} People with known ASCVD were excluded, and further eligibility criteria were based on low-density lipoprotein cholesterol (LDL-C) thresholds that varied based on 10-year risk estimates for ASCVD (up to a risk of 15%). The primary outcome was the occurrence of major adverse cardiovascular events (MACE), defined as a composite of cardiovascular disease death, myocardial infarction, hospitalization for unstable angina, stroke, transient ischemic attack, peripheral arterial ischemia, revascularization of coronary, carotid, or peripheral artery, or death of undetermined cause. Study participants were 31% female, 53% from high-income countries, 41% Black, and 35% White, with a median age of 50 years (interquartile range [IQR] 45–55) and 10-year ASCVD risk of 4.5% (IQR 2.1–7.0). Additionally, median CD4 T lymphocyte (CD4) cell count was 621 cells/mm³ (IQR 448–827), with prior CD4 nadir < 200 cells/mm³ in 49%, and HIV RNA below quantification in 88%.

Compared to placebo, pitavastatin was associated with a 36% reduction in MACE (hazard ratio 0.64; 95% confidence interval [CI], 0.48–0.84), with event rates of 7.77 and 4.95 per 1,000 person-years, respectively, over a median follow-up of 5.6 years. A similar treatment effect was present across the individual components of MACE. In REPRIEVE, the absolute risk reduction was substantially greater for people with ASCVD risk estimates of $\geq 5\%$ compared to those with risk of $< 5\%$, as discussed in the next section. Median LDL-C levels at baseline and 12 months were 107 mg/dL and 74 mg/dL, respectively, in the pitavastatin group, compared to 106 mg/dL and 105 mg/dL in the placebo group. During the study, a statin was initiated as part of clinical care in 5.7% of the pitavastatin group and 9.6% of the placebo group, leading to discontinuation of the blinded study drug. Incident diabetes (incident rate ratio [IRR] 1.29; 95% CI, 1.07–1.57) and Grade ≥ 3 or treatment-limiting muscle-related symptoms (IRR 1.58; 95% CI, 1.14–2.19) were also higher in the pitavastatin group.

Rationale for the Panel's Recommendations for People With HIV at Low-to-Intermediate ($< 20\%$) 10-Year ASCVD Risk

The overall findings from REPRIEVE, combined with the observation that equations based on traditional risk factors underestimate ASCVD risk among people with HIV, informed the Panel's decision to recommend the use of at least moderate-intensity statin therapy as primary prevention among people with HIV aged 40 to 75 years.^{8,9}

To determine whether some participants demonstrated greater benefit from pitavastatin, subgroup analyses were performed in REPRIEVE.⁶ Relative risk reductions across subgroups defined by key demographic and clinical characteristics did not demonstrate a clear interaction with the treatment effect. However, when incident MACE was stratified by 10-year ASCVD risk score, the absolute reduction in events was greatest for people with a 10-year ASCVD risk $\geq 5\%$. The estimated number needed to treat over 5 years (NNT₅) to avoid incident MACE events with pitavastatin treatment is reported below (Table 11a). Of note, the NNT₅ for people with ASCVD risk $\geq 5\%$ was one-quarter to one-third that for those with ASCVD risk $< 5\%$. These data motivated the Panel's decision to issue a strong recommendation for initiating statin therapy among those with ASCVD risk $\geq 5\%$. It should be noted that some of these NNT₅ for subgroups defined by ASCVD risk are estimated based on low numbers of events and should be interpreted with caution.

Table 11a. Estimated NNT₅ for MACE Prevention With Pitavastatin in People With HIV Based on REPRIEVE^a

10-Year ASCVD Risk Score	N	NNT ₅
>10%	563	34
5% to 10%	2,995	53
2.5% to <5.0%	2,055	130
0% to <2.5%	2,156	187
Overall	7,769	100

^a REPRIEVE⁷

Key: ASCVD = atherosclerotic cardiovascular disease; NNT₅ = number needed to treat over 5 years

The relationship between a higher absolute benefit from statin therapy (i.e., lower NNT₅) with higher baseline ASCVD risk estimates is well described and should be used to inform shared decision-making discussions on the use of statins.⁵ These discussions should take into account the benefits along with side effects (see [Common Adverse Events of Statin Therapy](#) below), costs, drug–drug interaction potential, and other patient-centered factors. For people at low (<5%) 10-year ASCVD risk who may experience a more modest absolute benefit from statin therapy, it is particularly important to weigh potential benefits for the individual against risks.

The findings from REPRIEVE reporting higher absolute benefit of statin therapy among people with HIV aged 40 to 75 years with greater ASCVD risk informed the Panel's decision to recommend the use of at least moderate-intensity statin therapy with a strong recommendation for those with a 10-year ASCVD risk score ≥5% to <20% (**AI**). Among participants aged 40 to 75 years with low estimated ASCVD risk (<5%), additional HIV-related factors may be important to consider. Using a large body of observational data, the AHA identified HIV-related ASCVD risk-enhancing factors that include the following: history of prolonged HIV viremia and/or delayed ART initiation, low current or nadir CD4 count (<350 cells/mm³), HIV treatment failure or nonadherence, and lipodystrophy/lipoatrophy, as well as metabolic syndrome, fatty liver disease, and coinfection with hepatitis C.¹⁰ Additional HIV-related factors associated with greater ASCVD risk that may be considered include, but are not limited to, a longer total duration of HIV infection and prior or current exposure to certain antiretroviral (ARV) drugs with cardiometabolic toxicity (e.g., abacavir, ritonavir-boosted lopinavir, ritonavir-boosted darunavir, and older protease inhibitors [PIs] no longer in use) (see [Cardiovascular Complications](#) for more detailed discussion). Conversely, people with HIV who initiated ART with contemporary regimens at higher CD4 counts (e.g., >500 cells/mm³) are likely to be at lower absolute ASCVD risk. Finally, non-HIV risk factors that may favor initiating statin therapy are discussed within the AHA/ACC/Multisociety Guidelines.⁵ In this context, clinician–patient discussions that weigh individual HIV-related factors along with the potential benefits and risks of statins are particularly important for people with HIV at low (<5%) 10-year ASCVD risk, where the NNT₅ for benefit is substantially higher. In summary, the Panel favors initiating at least moderate-intensity statin therapy for those with a 10-year ASCVD risk score <5% (**CI**), and the decision to initiate a statin should take into account the presence or absence of HIV- and non-HIV-related factors that can increase ASCVD risk.

Initiation of statin therapy should be deferred until after pregnancy for those with HIV and low-to-intermediate ASCVD risk, and statin therapy should be discontinued if pregnancy occurs in the context of HIV. As in the general population, breastfeeding is not recommended while taking statin therapy.

REPRIEVE did not enroll people with HIV under 40 years of age. In the general population, lifestyle modifications targeting traditional risk factors (e.g., diet, exercise, smoking cessation, and blood pressure control) are recommended for people under 40 years of age.⁵ Still, some of these individuals also may benefit from statin therapy, and recommendations for the general population suggest risk factors such as familial hypercholesterolemia or a family history of premature ASCVD may favor statin therapy. There are insufficient data to inform whether risk enhancers, such as HIV-related factors, would favor statin therapy among people with HIV under 40 years of age. However, some younger people with HIV may be at increased ASCVD risk—particularly those with a very long duration of HIV infection (e.g., due to perinatal exposure); in these cases, decisions on statin therapy should be individualized.

Key Recommendations for the General Population (Including People With HIV) Based on AHA/ACC/Multisociety Guidelines

The [2018 ACC/AHA/Multisociety Guidelines](#) recommend the use of statins as primary prevention for all people at high risk, defined as those aged 40 to 75 years with a 10-year ASCVD risk estimate of $\geq 20\%$, as well as those aged 20 to 75 years with an LDL-C ≥ 190 mg/dL.⁵ In this context, an LDL-C reduction of $\geq 50\%$ should be achieved, which typically requires high-intensity statin therapy (see the [Intensity of Statin Therapy](#) table below). For people with diabetes mellitus, the 2018 ACC/AHA/Multisociety Guidelines recommend at least a moderate-intensity statin, with additional risk assessment to inform consideration of initiating a high-intensity statin. These general population guidelines are based on high-quality randomized trial data and should inform the management of people with HIV who meet these criteria.⁵

Rationale for the Panel's Recommendation on Choice and Dose of Statin

In REPRIEVE, pitavastatin was chosen in part due to lower drug–drug interaction potential with certain ARV medications compared with other statins.¹¹ Although there are no clinical outcome comparative effectiveness trials analyzing different statins among people with HIV, some studies support the treatment effect of other moderate-intensity statins for lipid lowering or reductions in inflammation and immune activation, as well as a treatment effect on surrogate measures of ASCVD. Pitavastatin, atorvastatin, and rosuvastatin are all associated with greater reductions in LDL-C among people with HIV than pravastatin. In addition, pitavastatin (4 mg daily), rosuvastatin (10 mg daily), and high-dose atorvastatin (80 mg daily) have demonstrated reductions in inflammatory, monocyte, and T-cell immune activation biomarkers among people with HIV.¹²⁻¹⁴ Finally, in two separate Phase 2 placebo-controlled randomized trials of statin therapy in people with HIV, atorvastatin (initiated at 20 mg daily) was associated with reductions in coronary noncalcified plaque by computed tomography angiography, and rosuvastatin (10 mg daily) was associated with slower progression of common carotid artery intima-media thickness.^{15,16} Cumulatively, these data among people with HIV who are aged 40 to 75 years old and with ASCVD risk between 5% and $<20\%$ motivated the Panel's recommendation for use of at least moderate-intensity statins to include pitavastatin 4 mg daily (**AI**), atorvastatin 20 mg daily (**AII**), or rosuvastatin 10 mg daily (**AII**).

The recommendation for which statin agent to select and the intensity of therapy is based on the person's overall risk profile. As outlined in the 2018 AHA/ACC/Multisociety Guideline on the Management of Blood Cholesterol, statin intensity consists of three categories—high, moderate, and low—based on the LDL-C lowering effect.⁵ Table 11b below outlines the statin agent and dose that qualify for each intensity, as well as the anticipated LDL-C lowering effect. More specifically, for every 39 mg/dL reduction in LDL-C, there is approximately a 20% reduction in ASCVD events and a

10% reduction in all-cause mortality.¹⁷ However, it should also be noted that the magnitude of LDL-C lowering is variable in clinical practice.

Table 11b. Intensity of Statin Therapy

High Intensity (≥50% LDL-C Lowering Effect)	Moderate Intensity (30% to 49% LDL-C Lowering Effect)	Low Intensity (<30% LDL-C Lowering Effect)
Atorvastatin ^a 40–80 mg Rosuvastatin ^a 20–40 mg	Pitavastatin 4 mg (AI)^b Atorvastatin 20 mg (All)^{a,b} Rosuvastatin 10 mg (All)^{a,b} Fluvastatin XL 80 mg Fluvastatin 40 mg twice daily Lovastatin ^c 40–80 mg Pravastatin 40–80 mg Simvastatin ^c 20–40 mg	Pravastatin 10–20 mg Simvastatin ^c 10 mg Fluvastatin 20–40 mg Lovastatin ^c 20 mg

^a Atorvastatin and rosuvastatin have known drug–drug interactions with ritonavir- and cobicistat-boosted antiretrovirals (ARVs); see drug–drug interactions between statin therapies and antiretroviral medications in Table 11c below.

^b Bolded statins are included in recommendations from the Panel for the Use of Antiretroviral Agents in Adults and Adolescents With HIV; see Rationale for the Panel’s Recommendation on Choice and Dose of Statin above.

^c Simvastatin and lovastatin are contraindicated with ritonavir- and cobicistat-boosted ARVs.

Key: LDL-C = low-density lipoprotein cholesterol; XL = extended-release

Drug–Drug Interactions Between Statin Therapies and Antiretroviral Medications

Drug–drug interaction potential varies among the statins based on susceptibility to cytochrome P450 (CYP) and transporter-mediated interactions. Pitavastatin, pravastatin, and rosuvastatin are not significantly metabolized by CYP enzymes. Atorvastatin, lovastatin, and simvastatin are primarily metabolized by CYP3A4, whereas fluvastatin is a major substrate of CYP2C9. All statins are susceptible to interactions related to drug transporters (e.g., OATP1B1).¹⁸

Ritonavir (RTV)- and cobicistat (COBI)-boosted ARVs are particularly prone to drug–drug interactions with statins. When RTV- or COBI-boosted ARVs are coadministered with atorvastatin or rosuvastatin, statin dose reduction, a switch to an alternative statin, or increased monitoring for statin-related adverse effects may be required. Lovastatin and simvastatin are **contraindicated** when coadministered with COBI or RTV.

In general, some degree of drug–drug interactions may be expected between certain statins and HIV PIs, COBI-boosted elvitegravir, older non-nucleoside reverse transcriptase inhibitors, fostemsavir, and lenacapavir. The magnitude of the interaction and management strategy vary based on the individual statin and ARV combination. For more information on drug–drug interactions between pitavastatin, atorvastatin, and rosuvastatin with ARV drugs, see Table 11c below. Interactions

between other statins and ARV drugs can be found in the Drug Interaction tables (Tables [23](#), [24a](#), [24b](#), [24c](#), [24d](#), [24e](#), [24f](#), [24g](#), [25a](#), [25b](#)).

Table 11c. Concomitant Use of Antiretroviral Drugs and Statins Recommended by the Panel as Primary ASCVD Prevention

This table includes recommendations for pitavastatin, atorvastatin, and rosuvastatin when used with different ARV drugs based on their drug–drug interaction potential. Because all statins can be used with nucleoside reverse transcriptase inhibitors (NRTIs) without dosage adjustment, NRTIs are not listed in this table. Drug interaction information for statins not listed in this table can be found in Tables [24a](#), [24b](#), [24d](#), [24f](#), and [24g](#) of the [Adult and Adolescent Antiretroviral Guidelines](#).

Panel Recommended Statins and Doses	ARV Drugs	Recommendations
Pitavastatin 4 mg once daily (AI)	INSTI: BIC, CAB, DTG, RAL NNRTI: DOR, EFV, ETR, RPV PI/r: ATV/r, DRV/r Other: LEN, MVC	No dosage adjustment
	INSTI: EVG/c PI/c: ATV/c, DRV/c Other: FTR	No data; use standard dose and monitor for AEs.
Atorvastatin 20 mg once daily (AII)	INSTI: BIC, CAB, DTG, RAL NNRTI: DOR, RPV Other: LEN, MVC	No dosage adjustment
	INSTI: EVG/c PI/b: DRV/c, DRV/r	↑ atorvastatin concentrations observed. Do not exceed 20 mg per day ^a ; monitor for AEs.
	NNRTI: EFV, ETR	↓ atorvastatin concentrations observed
	PI/c: ATV/c	Do not coadminister.
	PI: ATV, ATV/r Other: FTR	↑ atorvastatin concentrations observed or possible. Monitor for AEs.
Rosuvastatin 10 mg once daily (AII)	INSTI: BIC, CAB, DTG, RAL NNRTI: DOR, EFV, ETR, RPV Other: LEN, MVC	No dosage adjustment
	INSTI: EVG/c PI/r: DRV/r Other: FTR	↑ rosuvastatin concentrations observed. Monitor for AEs.

Table 11c. Concomitant Use of Antiretroviral Drugs and Statins Recommended by the Panel as Primary ASCVD Prevention

Panel Recommended Statins and Doses	ARV Drugs	Recommendations
	PI/c: DRV/c	↑ rosuvasatin concentrations observed. Do not exceed 20 mg per day ^a ; monitor for AEs.
	PI: ATV, ATV/r, ATV/c	↑ rosuvasatin concentrations observed or expected. Do not exceed 10 mg per day ^a ; monitor for AEs.

^a Based on recommendations from the U.S. Food and Drug Administration product label.

Key: AE = adverse effect; ARV = antiretroviral; ATV = atazanavir; ATV/c = atazanavir/cobicistat; ATV/r = atazanavir/ritonavir; BIC = bictegravir; CAB = cabotegravir; DOR = doravirine; DRV/c = darunavir/cobicistat; DRV/r = darunavir/ritonavir; DTG = dolutegravir; EFV = efavirenz; ETR = etravirine; EVG/c = elvitegravir/cobicistat; FTR = fostemsavir; INSTI = integrase strand transfer inhibitor; LEN = lenacapavir; MVC = maraviroc; NNRTI = non-nucleoside reverse transcriptase inhibitor; PI = protease inhibitor; PI/b = boosted protease inhibitor; PI/c = cobicistat-boosted protease inhibitor; PI/r = ritonavir-boosted protease inhibitor; RAL = raltegravir; RPV = rilpivirine

Common Adverse Effects of Statin Therapy

Statin therapy is well tolerated and safe; however, some side effects have been observed. The 2018 AHA/ACC/Multisociety Guidelines introduced the terminology *statin-associated side effects*, as opposed to *statin intolerance*, when these adverse effects occur.⁵ The reason for this nomenclature is due to the fact that most people are able to tolerate statin rechallenge with an alternative statin agent or regimen. The most frequent adverse events are statin-associated muscle symptoms, with reported occurrence in 5% to 25% of people who receive statins.¹⁹

All statins have been implicated in small increases in both relative and absolute risk of diabetes. The increased risk of diabetes with statin use has been associated with older age and the presence of ≥2 risk factors, such as elevated baseline fasting glucose level, elevated fasting triglycerides, elevated body mass index, or history of hypertension.^{20,21} Pitavastatin has previously been shown to have a neutral effect on glucose levels among people with metabolic syndrome, and this effect was consistent in REPRIEVE where there was no difference in glucose levels between pitavastatin and placebo groups over follow-up (median levels at Month 84 were 92 mg/dL and 90 mg/dL, respectively).^{7,22} However, in REPRIEVE, there was a small increase in new-onset diabetes (6.0% in the pitavastatin group; 4.7% in the placebo group). Data are lacking on the risk of new-onset diabetes with other statins among people with HIV. However, a randomized controlled trial in people with HIV demonstrated increases in insulin resistance and impaired fasting glucose levels with rosuvasatin use.²³

Mild increases in liver enzymes are seen in some individuals, though these are usually transient without clinical complications. However, the overall clinical benefits of statin use outweigh the overall risks of adverse effects, especially for people with greater estimated 10-year ASCVD risk.²⁴ Data related to any concerns of cognitive decline have been weak or contradictory and, as such, do not warrant statin avoidance or cessation.⁵ Any time statins are initiated, clinicians should perform a comprehensive evaluation of musculoskeletal symptoms, an assessment of risk factors for diabetes, and a clinician–patient shared decision-making discussion focused on indications, benefits, risks, and patient concerns and preferences. When mild adverse events occur, close clinical follow-up is important.

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Weight Gain in People With Treated HIV

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Panel's Recommendations and Key Considerations
• All people with HIV should receive antiretroviral therapy (ART) to reduce morbidity and mortality (AI) and to prevent transmission of HIV to others (AI). • Weight gain after ART initiation is common. ART initiation should not be delayed because of concerns for weight gain (AIII), and ART should not be interrupted or discontinued because of weight gain (AIII). • ART selection, whether for initiation or switch, should be based on optimization of virologic suppression. Specific antiretrovirals should not be selected to prevent or reduce weight gain (AII), as available evidence suggests this strategy is ineffective. • Providers should include weight monitoring in conjunction with counseling on strategies for weight control as part of comprehensive care for people with HIV.
Rating of Recommendations: A = Strong; B = Moderate; C = Weak
Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Introduction

Weight gain is an increasingly recognized and now expected consequence of antiretroviral therapy (ART) initiation in many people with HIV. Although the mechanisms of weight gain after ART initiation remain unclear, the pathogenesis is complex and multifactorial.^{1,2}

Historically, weight gain following ART initiation was thought to be related to a “return to health” phenomenon, given that early in the HIV epidemic, many people suffered from AIDS-wasting syndrome before receiving lifesaving ART. Although “return to health” may explain some weight gain for people with advanced HIV, weight gain occurs regardless of the time between HIV acquisition and ART initiation, including among people who initiate ART shortly after HIV seroconversion.³⁻⁵ Similarly, weight gain with ART initiation occurs with most modern regimens and regardless of baseline body mass index (BMI).⁶⁻⁹

People with HIV on ART may experience greater weight gain than the expected age-related changes in people without HIV.¹⁰ The greatest weight gain often occurs within the first 2 years after ART initiation. Notably, pooled data from eight randomized controlled trials suggest that approximately 13% of people with HIV experience excessive (>10%) weight gain within the first year of ART initiation.¹¹ The average weight gain trajectory frequently slows after 2 years on ART.^{10,11} Overall, studies differ on the amount of weight gain seen in people with HIV, depending on the ART regimen and study population.

Weight gain among people with HIV is further complicated by the obesity epidemic in the general population in the United States.^{10,12} Importantly, some people with HIV experience a differential weight gain after ART initiation, with increased central fat and/or visceral fat gain, generally termed “lipohypertrophy.”² Generalized weight gain and obesity can obscure the diagnosis of HIV-associated lipohypertrophy, which can carry additional cardiometabolic risk. Yet, no clear definition or consensus exists to distinguish HIV-associated lipohypertrophy from generalized weight gain following ART or from weight gain unrelated to HIV.

Although weight gain after ART initiation is common, the lack of a clear understanding of its pathogenesis has limited the development of HIV-specific prevention and/or treatment guidance. Thus, what follows is a synopsis of contributors to weight gain in people with HIV and general management considerations, based on available data.

Contributors to Weight Gain

Antiretroviral Therapy

Because weight gain is common with almost all ART regimens, antiretrovirals (ARVs) have been implicated as contributors to weight gain in people with HIV, although data on the mechanisms of ART-induced weight gain are limited. The differential effects observed between ART classes, as well as within-class variability, suggest the possibility of multiple mechanisms by which ARVs contribute to weight gain.

Contemporary Antiretroviral Drugs

Weight gain occurs with nearly all ART regimens, although some are associated with more weight gain than others. For example, the least weight gain occurs with some non-nucleotide reverse transcriptase inhibitor (NNRTI)-based and tenofovir disoproxil fumarate (TDF)-based regimens.⁶⁻⁹ Weight gain is most pronounced with integrase strand transfer inhibitors (INSTIs), especially bictegravir and dolutegravir, and the nucleotide reverse transcriptase inhibitor (NRTI) tenofovir alafenamide (TAF),^{11,13} especially when compared to protease inhibitors (PIs) and older NRTIs.^{6,11,13-19} The reasons for differential weight gain among ARVs are unknown. Importantly, differences in weight changes between ART regimens that are described in switch studies may be explained either by a weight-gain effect from exposure to a specific ARV (e.g., INSTI or TAF) and/or the removal of a weight-reduction effect from the ARV(s) that was discontinued (e.g., efavirenz or TDF). Some experts have suggested that medication side effects and/or unknown factors limit weight gain with certain older ARVs, when compared to the newer ARVs that have been linked to greater weight gain. Additional details and specific weight changes associated with individual ARVs can be found in the [INSTI](#) and [NRTI](#) sections of the guidelines.

Legacy Effects of Older Antiretroviral Drugs

The impact of prior ARV toxicity on an individual's current weight should also be considered. People with HIV who received thymidine analog NRTIs (e.g., zidovudine, stavudine) and older PIs (e.g., indinavir) were at risk for developing lipodystrophy. Lipodystrophy encompasses both lipohypertrophy (i.e., accumulation of visceral adipose tissue [VAT] and subcutaneous adipose tissue accumulation in the dorsocervical region, trunk, and/or breasts) and lipoatrophy (i.e., the loss of subcutaneous adipose tissue predominantly from the face, buttocks, and extremities).^{2,20} Lipodystrophy can persist despite stopping the implicated ARV drugs. Particularly in individuals with lipoatrophy, subcutaneous adipose tissue fibrosis can persist and limit normal re-expansion of the fat depot when weight gain (from any cause) occurs, promoting VAT accumulation and its sequelae.²¹ Thus, weight gain distribution and consideration of its effects on cardiometabolic risk may need to be contextualized by both prior and current ART exposure.

Individual Risk Factors and Population Differences

Although studies are limited, the following populations and clinical characteristics may contribute to greater ART-associated weight gain: female sex, older age, Black race/African descent, greater HIV disease severity (including lower CD4 T lymphocyte cell count and/or higher HIV-1 RNA),¹¹ and those who switched from a PI- or NNRTI-based ART regimen to an INSTI-based ART regimen while

virologically suppressed.^{6,11,22} Data on the influence of weight prior to ART initiation or switch are conflicting, with some studies showing lower and others higher weight as a risk factor for subsequent ART-associated weight gain.^{11,22}

Consequences of Weight Gain

Weight gain in people with HIV undoubtedly contributes to obesity-related comorbidities. In two Advancing Clinical Therapeutics Globally (ACTG) network protocols of longitudinal cohorts of people with HIV, individuals who experienced >10% weight gain within a year of ART initiation had an increased risk of type 2 diabetes mellitus (DM), metabolic syndrome, and cardiometabolic events 48 weeks after ART initiation.²³ Moreover, weight gain in people with HIV may confer a greater risk of cardiometabolic disease compared to people without HIV. The Veterans Aging Cohort Study found that each 5-pound weight gain was associated with a 14% greater risk of DM for people with HIV compared to 8% greater risk in people without HIV.²⁴ The Data Collection on Adverse events of Anti-HIV Drugs (D:A:D) cohort observed an 18% to 20% increased relative risk of cardiovascular disease (CVD) per 1.0 kg/m² BMI increase among people with HIV who had a normal weight at ART initiation.²⁵ In comparison, a cohort of people without HIV from Finland demonstrated a 4% to 5% increase in CVD risk per 1.0 kg/m² BMI increase.²⁶ Thus, even small amounts of weight gain in people with HIV could potentially contribute to increased cardiometabolic risk, and any weight gain >10% from baseline weight has been strongly associated with incident DM, metabolic syndrome, and other cardiometabolic outcomes.²³

Management of HIV-Associated Weight Gain

General Antiretroviral Therapy Considerations

Data on effective strategies to prevent or reverse ART-associated weight gain remain limited. Although weight gain among people with HIV likely increases risk for multiple cardiometabolic comorbidities and can negatively affect body image and psychosocial functioning, early and sustained ART treatment is critical to optimize short- and long-term clinical outcomes in people with HIV. ART initiation should not be delayed because of concerns for weight gain (**AIII**), and ART should not be interrupted or discontinued because of weight gain (**AIII**). ART selection should be based on optimization of virologic suppression. Specific ARVs should not be selected (for initiation or switch) to prevent or reduce weight gain (**AII**), as available evidence suggests that this strategy is ineffective. Specifically, greater average weight gains associated with INSTI- or TAF-based regimens should not influence decisions on regimen selection, especially in cases when these agents are likely to be the best option for effective virologic suppression, as the reasons for these differential effects on weight among specific ARVs are not clear.

Sustainable Lifestyle Modifications and Behavioral Weight Loss Programs

Lifestyle modification remains the starting point and gold standard weight management intervention. As in the general population, people with HIV should be educated on long-term healthy eating habits and encouraged to exercise regularly. Some individuals may benefit further from consultation with a licensed dietitian, nutritionist, and/or formal behavioral weight loss program, where available. Of note, few studies have evaluated behavioral weight loss programs specifically in people with HIV, although available data suggest they achieve outcomes similar to those seen in the general population.²⁷⁻²⁹ Referral to an obesity medicine specialist and/or bariatric surgery center should be considered for select individuals.

Weight-Loss Pharmacotherapy in HIV

Weight management may also include pharmacotherapy on a case-by-case basis. Tesamorelin, a synthetic growth hormone–releasing hormone analog, is the only U.S. Food and Drug Administration–approved therapy to treat excessive abdominal fat in people with HIV. The efficacy and safety of tesamorelin were largely established by two large Phase 3 trials conducted in people with HIV with central obesity. Tesamorelin 2 mg given as a daily subcutaneous injection was associated with a 13.1% reduction in VAT from baseline over 26 weeks, representing a 15.4% relative reduction compared to placebo. Through 52 weeks, the reduction from baseline reached 17.5%, demonstrating that most VAT loss occurred during the first 6 months of treatment.³⁰ Although the participants enrolled in the initial trials were primarily treated with the older NNRTIs and PIs used at the time, another small randomized controlled trial demonstrated similar results among people with HIV taking INSTI-based regimens.³¹ The use of tesamorelin, however, is limited by modest response rates (two-thirds of the participants with clinically significant VAT reduction), the need for weekly reconstitution, daily subcutaneous injections, and reversal of effects with drug discontinuation.

Glucagon-like peptide-1 (GLP-1) and dual GLP-1/glucose-dependent insulinotropic polypeptide (GIP) receptor agonists (RAs) have gained considerable popularity in the general population due to the significant weight loss effects along with reduced risk for multiple CVD outcomes. However, few studies exist among people with HIV. The only randomized controlled trial to date in HIV was conducted in 108 participants with HIV-associated lipohypertrophy and overweight or obese BMI who received subcutaneous semaglutide 1.0 mg or placebo weekly over 24 weeks (plus an 8-week titration phase). People with DM were excluded from study entry. Semaglutide treatment resulted in an estimated weight loss of 10.4%, with significant reductions in total body fat and abdominal total and VAT.³² In ACTG A5371, a 24-week, open-label, single-arm study of semaglutide 1.0 mg weekly for the treatment of metabolic dysfunction–associated steatotic liver disease in people with HIV who did not have DM, significant improvements in liver fat and other cardiometabolic parameters occurred that strongly correlated with weight loss (average of 8.1%).³³

Although the few studies conducted in people with HIV showed an adverse event profile similar to the general population, the 1.0-mg dose used is lower than the maximum currently approved 2.0- and 2.4-mg doses for DM and obesity in the general population, respectively. Thus, the safety and side effect profile of GLP-1 and GLP-1/GIP RAs at higher doses or longer duration in people with HIV are currently unknown. Until more data become available, the Panel on Antiretroviral Guidelines for Adults and Adolescents advises prescribing GLP-1 and GLP-1/GIP RAs among people with HIV based on current general population indications and prescribing information.

Other weight loss drugs are available with and without a prescription in the United States and should similarly be used in accordance with general population recommendations. It is important to recognize the potential for known or possible drug–drug interactions with ARV drugs when considering the use of such therapeutic agents. For example, some case reports link orlistat, which promotes weight loss by reversibly inhibiting lipase and reducing dietary fat absorption, to loss of virologic suppression due to decreased ARV drug absorption.^{34,35} In another example, the use of naltrexone-bupropion requires increased drug monitoring when coadministered with certain ARVs that may affect expected pharmacokinetic parameters.³⁶ Thus, it is critical to evaluate for any drug–drug interactions or possible adverse effects that may preferentially affect people with HIV on ARVs.

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CONSIDERATIONS FOR ANTIRETROVIRAL USE IN SPECIAL POPULATIONS

Adolescents and Young Adults With HIV

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Key Considerations and Recommendations
- Adolescents and young adults (AYA) with HIV largely belong to two distinct groups: those who acquired HIV in the first decade of life and who may be heavily antiretroviral therapy (ART)–experienced (early-acquired HIV); and those who acquired HIV in or after the second decade of life who may be mostly ART-naïve. - ART is recommended for all AYA with HIV (AI) to reduce HIV-related morbidity, mortality, and transmission. - For AYA with HIV who are at risk for poor clinical outcomes, it is critically important to assess the behavioral and psychosocial context, and their ability to adhere to ART. Efforts should be made to provide youth-friendly support and infrastructure to reduce barriers to adherence and maximize success in achieving sustained viral suppression (AIII). - Pediatric and adolescent care providers should prepare AYA with HIV for the transition into adult care settings. Adult providers should be knowledgeable about this unique patient population and the challenges that frequently accompany the transition into the adult care setting. Consulting and collaborating with pediatric and adolescent HIV care providers is critical to ensure the successful transition of AYA with HIV to adult providers and continued engagement in care (AIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak
Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Introduction

Adolescents (13–19 years old) and young adults (20–24 years old) (AYA) with HIV consistently account for about one-fifth of new infections in the United States.¹ AYA with HIV have lower rates of testing, diagnosis, treatment engagement, and viral suppression than adults with HIV. Importantly, unique developmental, psychosocial, behavioral, and infrastructural factors affect this vulnerable population. Without having their specific challenges and needs addressed, AYA with HIV remain at risk for poorer HIV-related outcomes, including persistent viremia, drug resistance, morbidity, mortality, and secondary transmission.

Epidemiology

Globally, approximately 5 million young people aged 15 to 25 years live with HIV.² AYA with HIV are mostly individuals who acquired HIV in the first decade of life, mainly perinatally and rarely via blood transfusion or from sexual abuse. During adolescence and young adulthood, most individuals acquire HIV through sexual activity. Among the latter group, the Centers for Disease Control and Prevention (CDC) estimates that consistently one-fifth of the approximately 40,000 individuals newly diagnosed with HIV in the United States annually are adolescents and young adults.¹ The majority of new infections in this age group are among Black/African Americans and Hispanic/Latino men who have sex with men (MSM). For example, in 2018, more than 50% of the infections in this age group

were among young Blacks and African Americans, and 27% were among Hispanics and Latinos. Most (87%) were male, with 92% identifying as MSM. Compared to adults with HIV, AYA with HIV are less likely to have acquired HIV from injection drug use, and trends in HIV and AIDS prevalence indicate that the disproportionate burden of HIV among racial and ethnic minorities is even greater among youths 13 to 24 years of age than among those older than 24. Together, AYAWH account for approximately 34,000 of the individuals being followed at pediatric and adult clinics in the United States.³

Heterogeneity of Adolescents and Young Adults With HIV

AYA with HIV represent a diverse population in terms of socio-demographics, mode of HIV acquisition, sexual and substance use history, clinical and immunologic status, psychosocial development, and ability to adhere to medications. These distinctions have implications for HIV treatment, including the best ways to support AYA with HIV to optimize outcomes. AYA with HIV largely belong to two distinct groups:

- **Adolescents and Young Adults With Early-Acquired HIV:** These individuals, who acquired HIV in the first decade of life, are more likely to be treatment-experienced and have antiretroviral (ARV) drug resistance that may limit their options for ART regimens. This may be even more relevant for AYA with HIV who emigrated from parts of the world where routine viral load and genotypic resistance testing are not as readily available so that recognition of virologic failure and drug resistance may be delayed or missed. Also, individuals in this group generally have undergone a longer duration of disease chronicity and may have greater disease burden and more complications, less functional autonomy, and higher mortality risks.⁴
- **Adolescents and Young Adults With HIV Acquired Later in Life:** Individuals in this group are a more heterogeneous group of mostly young men (especially MSM) and other people who acquired HIV through sexual activities. A minority in this group acquired HIV through injecting drugs. Some of these AYA were victims of sexual abuse. The intersection of adolescence and young adulthood with other key risk populations (e.g., MSM, people who use drugs) magnifies the risk of poor clinical outcomes in this younger population. A statistically higher percentage of individuals in this group reported experiencing an even more extensive number of negative life experiences compared to those who acquired HIV early in life (38.8% versus 16.3%, $P < .012$).⁵

Unique Characteristics and Considerations for Adolescents and Young Adults With HIV

Although a diverse group, many AYA with HIV share unique characteristics that distinguish them from adults with HIV. Furthermore, AYA with HIV have certain commonalities that, while not necessarily unique to youth, disproportionately affect their chances of successful HIV treatment. Compared to adults 25 years old and older, AYA with HIV have poorer outcomes on each step of the HIV care continuum⁶ (i.e., HIV diagnosis, linkage to HIV care, receipt of care, retention in care, achievement of viral suppression, maintenance of viral suppression). Most notably, only 40% of AYA with HIV are aware of their diagnosis and only 6% to 30% are virally suppressed.⁷ These numbers are significantly worse than documented in older adults with HIV.⁸ In one study of MSM of all ages, the percentage of those linked to HIV care within one month of diagnosis was lowest among AYA with HIV aged 13 to 19 (69%) and 20 to 24 (70%) years.⁹ This group also had the lowest rates of retention in care and viral suppression. Older young adults (through age 29), who currently have the highest incidence of HIV infection among all age groups, have similar challenges, with 30% unaware of their infection status. They, too, require focused attention.

There are myriad reasons AYA with HIV do not perform as well on the HIV care continuum and are at greater risk of poorer clinical outcomes than adults with HIV. Perhaps most important, AYA with

HIV often do not have the same developmental capacity or ability to secure resources as their adult counterparts. These key features, discussed below, should be considered carefully because they can negatively affect HIV treatment and may alter clinical decision-making.

Adolescents and Young Adults With HIV as a Developmentally Distinct Patient Population

Developmental maturity in AYA with HIV generally can be grouped into several, often overlapping, areas, including physical, cognitive, communication and language, and social and emotional maturity. Several overarching factors—especially HIV-related stigma, discrimination, and a fear of familial and/or social rejection—can contribute to impaired development in all areas and adversely affect HIV treatment and clinical outcomes for many AYA with HIV. Additional important psychosocial factors discussed below commonly are seen in this population and also can affect development and successful HIV treatment.

Key developmental factors that may impact HIV treatment are highlighted below:

- **Cognitive development:** Evolving cognitive processes, which normally continue well into the third decade of life, are particularly relevant to HIV treatment in AYA with HIV. Their developing decision-making capacity often is driven by concrete thinking processes, risk-taking behaviors, preoccupation with self-image, and the need to fit in with their peers, all of which can affect HIV treatment negatively, including medication adherence and long-term clinical outcomes.^{4,10,11} Likewise, AYA with HIV are at risk for neurocognitive impairment and mental health comorbidities,¹²⁻¹⁵ including psychiatric, behavioral, and substance use disorders, which can further affect cognitive development and compromise effective HIV treatment.
- **Physical development:** The rapid physiologic changes (e.g., puberty, rapid growth) that occur in adolescence may result in altered ARV pharmacokinetics, underscoring the importance of adolescent-specific studies.⁴ AYA with HIV, particularly those with infection acquired early in life and/or while sexually immature, also are at risk for impaired physical development, especially delayed sexual maturation and impaired normal bone development, which may have long-term consequences like reduced final height and peak bone mass, the latter being a key risk factor for developing osteoporosis.¹⁶ Both delayed maturation and short stature may increase anxiety, depression, and stigma, which may, in turn, affect treatment adherence.¹⁷ AYA with HIV who acquire HIV later in life still may be affected because peak bone mass is not achieved until around age 30. A small study showed lower bone mass in Tanner Stage 5 young men aged 20 to 25 years who acquired HIV during adolescence than in HIV-uninfected age-matched controls.¹⁸

Thus, developmental maturity should be considered in AYA with HIV, because associated clinical implications may alter HIV treatment decisions, including ARV selection and dosing.

Psychosocial and Related Risk Factors Affecting Adolescents and Young Adults With HIV

Several psychosocial, behavioral, and environmental risk factors affect many AYA with HIV and can undermine successful HIV treatment disproportionately in this population. Common key risk factors are summarized here:

- **Mental and behavioral health:** The percentage of AYA with HIV with behavioral and mental health conditions is very high and can undermine engagement in care and medication adherence. The most common conditions include anxiety and behavioral disorders, mood disorders (including depression), and attention deficit hyperactivity disorder. Among adolescents

with early-acquired HIV, nearly 70% meet criteria for a psychiatric disorder at some point in their lives.^{13,19-21} Similarly, depression and anxiety were identified by symptom inventory 43% and 31% of the time, respectively, among AYA with HIV presenting for care at treatment sites in the Adolescent Trials Network.²²

- **Substance use:** Substance use is prevalent among AYA with HIV. Among more than 2,000 AYA with HIV (72% acquiring HIV later in life) surveyed by the Adolescent Trials Network, weekly or more frequent use of tobacco (33%), marijuana (28%), alcohol (21%), and other illicit drugs (23%) was reported.^{23,24} Young MSM had higher odds of each substance use behavior. Suboptimal ART was associated with increased risk of substance use behaviors,²⁴ underscoring the need to screen for and address substance use to improve treatment outcomes.
- **Homelessness and unstable housing:** Among the 4.2 million homeless adolescents and young adults in the United States (of whom 700,000 are unaccompanied minors),²⁵ the estimated HIV prevalence ranges from 3% to 16%. Youth who are lesbian, gay, bisexual, or queer; those with mental health concerns; and those who engage in substance use are disproportionately represented among homeless youth.²⁶ Homeless and unstably housed AYA with HIV have greater difficulties securing and sustaining resources and engaging in and being retained in care and treatment.²⁷
- **Additional social and environmental factors:** A number of social and environmental factors commonly found among AYA with HIV negatively affect their HIV treatment, including limited familial and/or social support, lack of health insurance and/or experience with health care systems, unstructured and chaotic lifestyles, transportation barriers, food insecurity, limited educational opportunities, limited employment opportunities and/or unstable employment, and a history of trauma and/or sexual abuse.²⁸

Optimizing Treatment Effectiveness and Supporting Adherence in Adolescents and Young Adults With HIV

Given the unique physiologic, developmental, and psychosocial characteristics discussed above, AYA with HIV require comprehensive systems of care with culturally competent providers and tailored treatment to serve all their specific medical and psychosocial needs. To maximize their chances of success, it is also imperative to routinely assess each AYA with HIV for individual factors that may need to be addressed or considered in treatment decisions or that may affect adherence.

Table 14 summarizes common adherence barriers among AYA with HIV, along with recommended support strategies. Refer to the [Guidelines for the Use of Antiretroviral Agents in Pediatric HIV Infection](#) and the [Adherence to Antiretroviral Therapy in Children and Adolescents with HIV](#) additional approaches. Targeted interventions to improve treatment effectiveness and adherence related to a few key psychosocial factors among AYA with HIV also are highlighted here:

- **Mental health care:** Strategies to improve the provision of mental health care for AYA with HIV are critically important for optimizing treatment for co-occurring HIV and mental health problems.²⁹ These strategies include improving provider education, integrating trauma-informed care practices, increasing access to mental health professionals through colocated services for HIV care, expanding care delivery paradigms like telemedicine, and optimizing treatment approaches. An example of the latter is a combination of tailored psychotherapy and pharmacotherapy.^{30,31} Further guidance for providing appropriate mental health care for AYA with HIV can be found in the [Pediatric Guidelines](#).
- **Substance use disorders:** Providers should assess and recommend treatment for substance use disorders, with consideration of emerging substance use trends, such as the use of electronic vapor products. Further guidance for providing appropriate substance use screening

and treatment for AYA with HIV can be found in the [Pediatric Guidelines](#) and in the [Substance Use Disorders and HIV](#) section.

- **Psychosocial and environmental stressors:** Multimodal interventions that enhance social support and teach adaptive coping skills may help AYA with HIV manage environmental stressors and improve clinical outcomes.

Specific Antiretroviral Therapy Considerations in Adolescents and Young Adults With HIV

All AYA with HIV should initiate ART as soon as possible and stay on ART indefinitely to maximize viral suppression, reduce morbidity and mortality, and prevent secondary HIV transmission (**AI**). As described below, clinicians should consider simplifying ART regimens and using antiretrovirals with high barriers to resistance whenever possible to support adherence.

Strategies to Improve Medication Adherence

Clinicians selecting ART for AYA with HIV must balance the goal of prescribing a maximally potent regimen with a patient-by-patient assessment of existing and potential adherence barriers and available youth-friendly support strategies to facilitate adherence.³² Additional considerations and strategies that may affect adherence among AYA with HIV are highlighted below in **Table 12**.

Table 12. Antiretroviral Therapy–Specific Strategies to Improve Medication Adherence

Antiretroviral Therapy-Specific Strategies to Improve Medication Adherence	
Regimen selection	• Simple ART regimens (e.g., fixed-dose, once daily combinations) with high barriers to resistance are preferable, if possible.³³ • Minimal side effects (e.g., gastrointestinal)
Treatment plan	• Develop the plan in partnership with AYA with HIV, considering daily schedule; tolerance of pill number, size, and frequency; issues affecting absorption; and potential adverse effects and interactions with other medications.^{32,34} • Design adolescent-friendly reminder systems³⁵ (e.g., apps, cell phone reminders, pill boxes) for adherence support.³⁶
Motivators	• Emphasize personal benefits (e.g., viral suppression, improved health). • Undetectable equals untransmittable (U=U) status disclosure to sexual partners without HIV may act as a particularly strong motivator for reducing stigma and improving adherence among AYA with HIV.

Antiretroviral Therapy Regimens for Adolescents and Young Adults With HIV Without Drug Resistance

The boosted protease inhibitor (PI) darunavir (DRV) and the integrase strand transfer inhibitors (INSTIs) dolutegravir (DTG) and bictegravir (BIC) offer once-daily dosing. When coformulated with a dual nucleoside backbone, they also provide single-tablet regimens with high genetic barriers to resistance.

Clinical trials have demonstrated the superiority of DTG over boosted PI-based regimens. BIC coformulated with tenofovir (TAF) and emtricitabine (FTC) also appears to have a low risk of treatment-emergent resistance and is available as a single-tablet regimen with a small pill size and no food requirements. BIC is currently licensed for use in children or adolescents ≥ 25 kg. Adolescent studies are ongoing with an adult fixed-dose combination of BIC/FTC/TAF from 12 years of age and 35 kg with a favorable interim analysis in a stable adolescent switch study.³⁷

A two-drug once-daily single-tablet regimen of DTG/lamivudine is recommended as an initial ART regimen except for individuals with HIV RNA $>500,000$ copies/mL, hepatitis B virus (HBV) coinfection, or in whom ART is to be started before the results of HIV genotypic resistance testing for reverse transcriptase or HBV testing are available.

Long-acting formulation regimens, the first of which (cabotegravir/rilpivirine) was recently FDA-approved in the United States, are considered a viable option for patients who are engaged in care, virologically suppressed on oral therapy, and agreeable to the administration schedule. These agents are being studied for AYA with HIV³⁸ ages 12 to 17 without relevant antiretroviral drug resistance. Case reports of viral suppression with the use of long-acting rilpivirine (RPV) and cabotegravir (CAB)³⁹ in AYA with HIV with a history of poor adherence are encouraging (see [Optimizing Antiretroviral Therapy in the Setting of Virologic Suppression](#)). Studies to evaluate these modalities among nonadherent AYA with HIV are under development.⁴⁰

Antiretroviral Therapy—Experienced Adolescents and Young Adults With HIV and Drug Resistance

AYA with HIV who acquired HIV early in life often have treatment challenges associated with the long-term use of ART that mirrors those of ART-experienced adults, such as extensive resistance, complex regimens, and adverse drug effects. In ART-experienced adolescents, DTG was safe and well tolerated, and it achieved viral suppression rates of 44% to 66% when combined with an optimized background regimen. Acquired treatment-emergent INSTI resistance may occur.⁴¹ For adolescents with dual-class resistance, the introduction of DRV/cobicistat/FTC/TAF in combination with DTG offers the potential of a potent triple-class regimen with a high genetic barrier to resistance with only two pills once daily⁴² (see [Virologic Failure](#) and [Optimizing Antiretroviral Therapy in the Setting of Virologic Suppression](#)).

Antiretroviral Therapy Considerations in Sexually Immature Adolescents and Young Adults With HIV

The physiologic changes (e.g., puberty, rapid growth) that occur in adolescence may result in altered pharmacokinetics. Therefore, although generally it is appropriate for postpubertal adolescents to be dosed with ARV drugs according to adult guidelines, adolescents in early puberty should be dosed according to the [Pediatric Guidelines](#), which factor in dosages by weight and sexual maturity ratings.

Additional Antiretroviral Therapy Considerations in Adolescents and Young Adults With HIV

Additional considerations include an increased risk of side effects, such as bone and renal toxicity with tenofovir-based drugs in the rapidly growing adolescent. These concerns are magnified in low-weight adolescents for whom appropriate lower-dose formulations are not available. Because AYA with HIV have not yet reached peak bone mass, TAF generally should be used instead of TDF, because of a greater bone loss with the latter ARV.

For a more detailed discussion on ART therapy in AYA with HIV, please see the [What to Start](#) section and the [Pediatric Guidelines](#). For additional information on treatment adherence in AYA with HIV, please see [Table 18, Adherence to the Continuum of Care](#), and the [Pediatric Guidelines](#).

Preventive Measures and Supporting Long-Term Health in Adolescents and Young Adults With HIV

People with HIV are at an increased risk of HIV- and ART-related comorbidities, including cardiovascular disease, diabetes, metabolic syndrome, osteoporosis, and neurocognitive impairment. When HIV is acquired at birth or early in life, an individual can live for many decades with the condition. However, engagement in health-risk behaviors (e.g., tobacco smoking, alcohol and drug use, unhealthy diet, physical inactivity) may have greater long-term implications for clinical outcomes in this population.

Preventive health care and promotion of positive health behavior during the critical time of adolescence and young adulthood can shape future habits and clinical outcomes over a lifetime. Incorporating regular screening, preventive health care, and health education is critical for optimizing short- and long-term physical and mental well-being and should be considered part of routine HIV treatment. Careful attention should be paid to modifiable risk factors in these early decades, such as weight gain and obesity, dyslipidemia, vitamin D deficiency, and tobacco use. Aggressive screening and risk factor mitigation early in the life of AYA with HIV not only improves current health but also can decrease their risk of developing comorbidities later in life. See [HIV Medicine Association of the Infectious Diseases Society of America HIV primary care guidance](#) for more details.

Transitioning to Adult HIV Care

Given lifelong infection with HIV and the need for treatment throughout the life course, HIV care programs and providers need flexibility to transition care appropriately for AYA with HIV. A successful transition requires an awareness of fundamental differences between many adolescent and adult HIV care models.

In most adolescent HIV clinics, care is more teen-centered and multidisciplinary, with primary care highly integrated into HIV care.⁴³ Often there is anonymity, with clinics not being devoted specifically to HIV or infectious diseases. Moreover, such services as sexual and reproductive health and mental health care are often found in one clinic setting (i.e., the medical home). Additionally, these clinics are more likely to be “youth-friendly” by including such aspects as waiting areas where AYA with HIV can access computers and other items that may facilitate engagement; flexible schedules that include evening hours or walk-ins; technology like social media and texting to engage patients; staff who are trained specifically in the unique cognitive, developmental, and other psychosocial aspects of AYA with HIV; and lower patient-to-provider ratios. In contrast, some adult HIV clinics may be more HIV-specific and rely more on referral of the patient to separate subspecialty care settings, such as gynecology.⁴⁴ Furthermore, adult clinic settings tend to be larger and can easily intimidate younger, less motivated patients.

Transitioning the care of an AYA with HIV must consider such factors as medical insurance; degree of independence, autonomy, and decision-making capacity; patient confidentiality; and informed consent. Nonetheless, given the structural limitations (adolescent clinics not being able to see patients over a certain age), care transitions must occur, usually between the ages of 21 and 25. The period of transition is a highly vulnerable time for attrition from HIV care. Data on transition outcomes are emerging, showing variable rates of successful transition to adult care ranging between 50% and 85%.⁴⁵⁻⁴⁸

It is important to recognize that the transition for AYA with HIV with early-acquired HIV versus those who acquired HIV later in life may pose distinct challenges. AYA with HIV who acquired HIV early in life—who often have experienced significant instability and prior loss—may experience the transition to adult HIV care as yet another traumatizing event. Alternatively, those who acquired HIV later in life, given their more recent engagement in the medical system, may be less likely to be effectively engaged in pediatric and adolescent care, which may affect their ability to transition successfully. Factors to date that have been associated with successful transition for those with early-acquired HIV include high self-management and perceived emotional and social support.⁴⁹

To maximize the likelihood of a successful transition, interventions to facilitate transition are best implemented early on.^{47,50-52} Strategies and approaches for both the adult and pediatric and adolescent programs are discussed below:

Table 13. Approaches to Optimize Care Transition for AYA With HIV

Pediatric/Adolescent	Adult
Personnel	
- Engage a multidisciplinary team knowledgeable about medical and psychosocial issues of AYA with HIV, including the challenges of transitioning youth to adult care settings. - Utilize combined internal medicine and pediatrics-trained providers if available. - Assign a transition point person and have their contact information readily available. - Educate HIV care teams and staff about transitioning AYA with HIV and their needs.	- Engage a multidisciplinary adult care team knowledgeable about medical and psychosocial issues of AYA with HIV, including the challenges of transitioning youth to adult care settings. - Utilize combined internal medicine and pediatrics providers if available. - Assign a transition point person and have their contact information readily available. - Identify outreach specialists, navigators, social workers, case managers, and providers with a youth-friendly approach. - Educate clinic personnel about AYA with HIV and their challenges to enhance sensitivity and understanding and minimize stigma.
Education and Preparation of AYA With HIV	
- Enhance AYA with HIV health literacy, including understanding of HIV and their medical history. - Address patient and family resistance to transition of care caused by lack of information, concerns about stigma or risk of disclosure, and differences in practice styles. - Help youth develop life skills, including, but not limited to, counseling on appropriate use of a primary care provider and how to manage appointments; the importance of prompt symptom recognition and reporting; and self-efficacy in managing medications, insurance, and assistance benefits.	- Meet AYA with HIV before transition, if possible. - Clearly outline policies and expectations before and during the first visit. - Have an orientation plan to acquaint newly transitioned AYA with HIV to the clinic environment and adult clinical care program. - Implement interventions that may improve outcomes, such as patient navigators, peer support groups, mental health assessment, and inclusion of parents and guardians where available. - Address health literacy and ensure AYA with HIV understand HIV, goals of care, etc.

Table 13. Approaches to Optimize Care Transition for AYA With HIV

Pediatric/Adolescent	Adult
	Continue to work with AYA with HIV toward developing life skills, etc.
Strategies and Approaches	
- Identify adult care providers able to provide youth-friendly care for adolescents and young adults. - Develop a formal, purposeful individualized transition plan to address comprehensive care needs, including medical, psychosocial, and financial aspects of transitioning to adult HIV care. - Optimize provider communication between adolescent and adult clinics, including a warm multidisciplinary, comprehensive medical history hand-off that includes prior regimens and outcomes (e.g., adherence, virologic failure and resistance).	- Develop a realistic clinic model based on specific needs (e.g., simultaneous transition of mental health and/or case management versus a gradual phase-in) and staffing. - Engage in a warm handoff from the pediatric team, which allows the accepting adult team to learn about and understand the multidisciplinary challenges and goals for the patient. Devise a plan for how to continue building the skills on the adult side. - Build in flexibility (e.g., permissive grace period for appointments, leniency for missed appointments, particularly when first transitioning). - Incorporate other aspects of care beyond HIV management, if possible (e.g., family planning, sexually transmitted infection testing and treatment, mental health, substance use).
Communication	
- Foster regular dialogue between pediatric and adolescent and adult teams before and after transition through regular meetings, case conferences, etc. - Solicit feedback from the AYA with HIV - Use technology (e.g., texting, HIPAA-compliant messaging apps, telemedicine).	
Evaluation	
Implement ongoing evaluation to measure the success of the selected model (retention in adult care).	

Discussions regarding transition should begin early and before the actual transition process.⁵³ Attention to the key interventions noted above will be likely to improve adherence to appointments and avoid the potential for youth to fall through the cracks, as this concept is referred to commonly in adolescent medicine. For a more detailed discussion on specific topics related to transitioning care for adolescents and young adults, see *Transitioning into Adult HIV Care*. Please also refer to the [Pediatric Guidelines](#).

Table 14. AYA With HIV ARV Adherence Barriers and Strategies to Support Adherence

ART Adherence Barrier	Adherence Support Strategy	Rationale for Adherence Support Strategy
Prioritization of short-term goals and socialization with peers over daily HIV treatment adherence	Youth-friendly reminder systems (e.g., text, phone, apps)	- Daily adherence to ARV regimens may not take priority in the lives of AYA with HIV. - AYA with HIV benefit from reminder systems to facilitate adherence.
	Novel ART delivery strategies (e.g., long-acting oral or injectable ARVs)	- AYA with HIV show interest in long-acting alternatives for ART delivery. - Long-acting ARVs are a promising tool to facilitate adherence, once approved for AYA with HIV.
Social concerns related to loss of confidentiality	Simple ARV regimens	- Adolescents do not want to be different from peers; adherence to complex regimens is particularly challenging. - Simple ARV regimens are preferable for AYA with HIV.
	User-friendly and discreet regimens	- Avoidance of HIV-related stigma and of unintentional disclosure of HIV status is a priority for AYA with HIV. - Protect confidentiality with user-friendly and discreet adherence supports (e.g., discreet pill bottles, reminder systems, etc.).
Side effects/fear of side effects	ARV regimens that minimize side effects	- Side effects are associated with nonadherence to ARVs. - Regimens with minimal side effects and medications that manage side effects have utility for AYA with HIV.
Denial or dismissal of HIV diagnosis	Motivational interviewing (MI) and motivational enhancement therapy (MET)	- MI and MET acknowledge AYA with HIV's autonomy and potential ambivalence about treatment adherence. - MI and MET have shown promise for improving adherence to chronic disease treatment, including HIV.
	Positive affirmation messages (e.g., text, app)	Electronically delivered positive affirmation messages can improve self-esteem and ARV adherence among AYA with HIV.
Lack of health literacy regarding the benefits of ART	Health literacy support and U=U education	- AYA with HIV may not fully understand the importance of taking ARVs daily, particularly when they are asymptomatic. - Increased health literacy is associated with better adherence to ARV regimens. - U=U education holds promise for AYA with HIV.

Table 14. AYA With HIV ARV Adherence Barriers and Strategies to Support Adherence

ART Adherence Barrier	Adherence Support Strategy	Rationale for Adherence Support Strategy
Mistrust of providers and the medical establishment	Empathetic and patient-centered communication	Communication exploring the needs of AYA with HIV patients can build trust, including exploring needs not directly related to HIV treatment (e.g., school, employment, relationships, etc.).
Mental health and/or substance use	Individualized mental health and substance use services	- Comprehensive mental health and substance use services have shown promise for improving viral suppression among AYA with HIV. - Service should be delivered based on individualized needs assessments.
	Directly observed therapy may be considered	For some AYA with HIV with difficult adherence problems, directly observed therapy may be considered.
Lack of familial and social support	Family and peer support groups	- Family members and peers are a defense against stigma and social isolation, source of emotional support, and partners in medication management. - Family and peer support groups have utility for AYA with HIV living with HIV.
Provider views of AYA with HIV as “risky” and/or not ready for ART	Promote development of a positive rather than risk-centered identity among AYA with HIV	- Adolescence and young adulthood are periods of identity development where HIV stigma is particularly problematic. - Providers should not conceptualize AYA with HIV as “high risk” to reduce stigma and improve ARV adherence.
Provider implicit biases of AYA with HIV	Implicit bias training	Consciously changing biased associations and repeated bias self-regulation training can reduce providers’ implicit biases.
Lack of youth-friendly services	Dedicated youth HIV clinic	- Clinic days or hours dedicated to AYA with HIV patients better address unique adherence needs; youth-friendly services include the following: - flexible hours, easy scheduling, telephone/telehealth appointments; - providers trained in working with AYA with HIV; - youth-friendly waiting rooms and physical spaces; - supplemental services that comprehensively address psychosocial and health needs of AYA with HIV; and - incentives for AYA with HIV care engagement.

Table 14. AYA With HIV ARV Adherence Barriers and Strategies to Support Adherence

ART Adherence Barrier	Adherence Support Strategy	Rationale for Adherence Support Strategy
	Youth-friendly hours, staff, and physical space	- Where dedicated hours and services are not possible, youth-friendly service elements can be integrated into existing clinic structures, e.g.: - offering evening hours; - staff training on service delivery to AYA with HIV; and - youth-friendly waiting rooms and physical spaces.
	Referrals to more youth-friendly HIV providers	- Where youth-friendly services are not possible, referrals to more youth-friendly HIV care providers should be considered. - Referral decisions should be made collaboratively with the patient.
Lack of comprehensive services that address common psychosocial stressors	Supplemental health, behavioral health, and psychosocial support services	- Individualized delivery of comprehensive supplemental services helps address unique needs of AYA with HIV, including the following: - primary care and sexual and reproductive health services; - behavioral health services; and - psychosocial support services (e.g., school support, transportation, support groups, housing and food assistance).
	Collaboration with and referrals to outside support services	Where delivery of comprehensive supplemental services is not possible, collaborations with and referrals to outside support services should be considered.

Key: ART = antiretroviral treatment; ARV = antiretroviral; AYA = adolescent and young adult; U=U = undetectable equals untransmittable

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Early (Acute and Recent) HIV Infection

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Panel's Recommendations
- Antiretroviral therapy (ART) is recommended for all people with HIV, including those with early^a HIV infection (AI). ART should be initiated as soon as possible after HIV diagnosis (AII). - The goals of ART are to suppress plasma HIV RNA to undetectable levels (AI), prevent transmission of HIV (AI), and preserve immune function (AIII). Monitoring of plasma HIV RNA levels, CD4 T lymphocyte cell counts, and antiretroviral (ARV) drug-related adverse effects should be done as recommended for people with chronic HIV infection (AII). - A blood sample for genotypic resistance testing should be sent to the laboratory before initiating ART (AIII). - Standard genotypic drug-resistance testing should be performed for mutations in the reverse transcriptase and protease genes (AIII) for all people with early HIV. - Genotype testing for integrase strand transfer inhibitor (INSTI) resistance should be performed for those who acquire HIV during or after the use of long-acting cabotegravir (CAB-LA) as pre-exposure prophylaxis (PrEP), if transmitted integrase strand transfer inhibitor (INSTI) resistance is suspected, or if HIV diagnosis occurs after receiving an INSTI-based regimen for post-exposure prophylaxis (PEP) (AIII). - ART can be initiated before drug-resistance test results are available. - For those without a history of using CAB-LA as PrEP, one of the following ARV regimens is recommended^b (AIII): - Bictegravir (BIC)/tenofovir alafenamide (TAF)/emtricitabine (FTC) - Dolutegravir (DTG) with (TAF or tenofovir disoproxil fumarate [TDF])^c plus (FTC or lamivudine [3TC]) - For those with a history of CAB-LA use as PrEP, genotype testing before starting ART should include screening for INSTI-resistance mutations (AIII). - A regimen of cobicistat- (COBI)^d or ritonavir-boosted darunavir (DRV) with (TAF or TDF)^b plus (FTC or 3TC) is recommended while awaiting the results of the genotype testing (AIII). - Use of an empiric INSTI-containing regimen is not recommended unless genotype testing shows no evidence of INSTI resistance (AIII). This is because INSTI resistance may be present in those who acquire HIV during and possibly after the use of CAB-LA as PrEP. - In people with HIV RNA levels ≥200 copies/mL and who are taking PrEP, immediate initiation of an effective HIV treatment regimen is recommended while awaiting confirmation of HIV diagnosis (AIII). - Pregnancy testing should be performed in women of childbearing potential before initiating ART (AIII). - When the results of drug-resistance tests are available, the treatment regimen can be modified if needed (AII). - Providers should inform individuals starting ART of the importance of adherence in achieving and maintaining viral suppression (AIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed non-randomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

^a Early infection represents either acute or recent (≤6 months) infection.

^b Because of the low rates of transmitted INSTI resistance in the United States at present, even when there is suspicion that HIV was acquired from a partner with virologic failure while on an INSTI, an INSTI-based regimen can be started while awaiting the results of the INSTI genotype.

^c TAF and TDF are two forms of tenofovir that are approved in the United States. TAF has fewer bone and kidney toxicities than TDF, whereas TDF is associated with lower lipid levels. Safety, cost, and accessibility are among the factors to consider when choosing between these drugs.

^d COBI should be avoided in pregnancy because lower concentrations of COBI and DRV have been reported during the second and third trimesters.

Introduction

Acute HIV infection occurs soon after transmission and is typically characterized by the lack of anti-HIV antibodies and the presence of viremia, which can be detected by an HIV RNA test or p24 antigen test. Recent HIV infection is considered the period of ≤ 6 months after infection during which anti-HIV antibodies become detectable. Throughout this section, the term “early HIV infection” is used to refer to either acute or recent HIV infection. People with acute HIV infection may experience fever, lymphadenopathy, pharyngitis, skin rash, myalgia, arthralgia, and other symptoms; however, illness is generally nonspecific, and people can be asymptomatic or experience relatively mild symptoms.¹⁻⁶ Clinicians may fail to recognize acute HIV infection because its manifestations are similar to those of many other viral infections, such as COVID-19, influenza, and infectious mononucleosis. [Table 15](#) below provides clinicians with guidance to recognize, diagnose, and manage acute HIV infection.

Diagnosing Acute HIV Infection

Health care providers should consider a diagnosis of acute HIV infection in a person who has a suggestive clinical syndrome or in asymptomatic individuals who report recent engagement in activities that may put them at high risk of HIV acquisition (see [Table 15](#) below).⁷ Individuals may not always disclose high-risk activities or perceive that such activities put them at risk for HIV acquisition. Thus, even in the absence of reported high-risk activities, health care providers should have a low threshold for considering a diagnosis of acute HIV infection. Health care visits to emergency departments provide an opportunity for health care providers to screen for acute or established HIV infection, as well as other sexually transmitted infections. Since 2019, the [United States Preventive Services Task Force](#) has recommended routine screening for HIV in adolescents and adults aged 15 to 65 years (Grade A recommendation). Testing of remnant blood specimens from an emergency department identified acute HIV infection in approximately 5 of 499 (1%) people presenting with flu-like symptoms.⁸ Acute HIV infection was also diagnosed in 7 of 563 (1.2%) people presenting for evaluation of possible mononucleosis with negative heterophile antibody tests.⁹ A study of HIV screening in nine emergency departments in six U.S. cities found that a new HIV diagnosis was made in 0.4% of 214,524 adolescents and adults, of whom 14.5% had acute HIV infection.¹⁰ Current statistics on the prevalence of HIV in geographical areas in the United States can be found on the [AIDSVu](#) and Centers for Disease Control and Prevention (CDC) [AtlasPlus](#) websites.

The recommended initial laboratory HIV testing algorithm includes combination immunoassays that detect HIV-1 and HIV-2 antibodies, as well as HIV p24 antigen (Ag/Ab assays),¹¹ primarily due to their enhanced ability to detect acute HIV infection. Specimens that are reactive on an initial Ag/Ab assay should be tested with an immunoassay that differentiates HIV-1 from HIV-2 antibodies. Specimens that are reactive on the initial assay and have either a negative or indeterminate antibody differentiation test result should be tested for quantitative or qualitative HIV RNA; an undetectable HIV RNA test result indicates that the original Ag/Ab test result was a false positive. Detection of HIV RNA in this setting indicates that acute HIV infection is highly likely.¹¹ People presenting to care during the earliest days following HIV infection may have yet to develop a positive p24 Ag response, which typically occurs with viral load levels of $>20,000$ to 30,000 copies/mL. In clinical settings with a

high probability of infection, quantitative or qualitative HIV RNA testing should be considered even if the HIV Ag/Ab test result is negative. HIV infection should be confirmed by repeat quantitative HIV RNA testing or subsequent testing to document HIV antibody seroconversion. People who receive antiretroviral therapy (ART) during acute or very early HIV infection may demonstrate weaker reactivity to screening antibody assays or incomplete HIV antibody evolution; may remain non-reactive to confirmatory antibody assays; and in the setting of sustained virologic suppression, may have complete or partial seroreversion.¹²⁻¹⁶

Providers should be aware that even a low-positive quantitative HIV RNA level (e.g., <200 copies/mL but detectable) in the setting of a negative or indeterminate antibody test result is consistent with acute HIV infection. When a low-positive quantitative HIV RNA test result is present at this level, the HIV RNA test should be repeated on a new blood specimen to confirm the diagnosis. Repeated false-positive HIV RNA test results are unlikely.² HIV RNA levels in acute infection are generally very high (e.g., >100,000 copies/mL)^{1,2,4}; however, levels may be <200 copies/mL in the earliest weeks following infection as viral load continues to rise. In rare cases, however, such low HIV RNA levels also may represent a false-positive result. The previously proposed threshold of <3,000 copies/mL is based on historical data, which used laboratory methods that are now considered obsolete.¹⁷ Improvements in plasma viral load methodology suggest that any positive result on a quantitative plasma HIV RNA test in the setting of a negative or indeterminate antibody test result may be consistent with acute HIV infection. Some health care facilities may still be using HIV testing algorithms that test only for anti-HIV antibodies. In such settings, when acute HIV infection is suspected in a person with a negative or indeterminate HIV antibody test result, a quantitative or qualitative HIV RNA test should be performed. A negative or indeterminate HIV antibody test result and a positive HIV RNA test result (<200 copies/mL)¹⁸ indicate that acute HIV infection is highly likely.

Diagnosing Acute HIV Infection in People Taking Pre-Exposure Prophylaxis

Three antiretroviral (ARV) options—oral emtricitabine (FTC) with either tenofovir alafenamide (TAF) or tenofovir disoproxil fumarate (TDF) and intramuscular long-acting cabotegravir (CAB-LA)—are now available for HIV pre-exposure prophylaxis (PrEP). People who acquire HIV while taking PrEP may sometimes have ambiguous HIV test results. A positive HIV Ag/Ab test result or a positive HIV RNA test result in the setting of a negative HIV antibody test result should prompt immediate confirmation of HIV diagnosis. It is important to collect a new blood specimen to verify the HIV diagnosis before initiating ART. Important considerations include the following:

- In people with HIV RNA level ≥ 200 copies/mL who are taking PrEP, immediate initiation of an effective HIV treatment regimen¹⁸ is recommended while awaiting confirmation of HIV diagnosis **(AIII)**.
- In people taking PrEP who have a negative HIV antibody test result and a low-positive quantitative HIV RNA test result (<200 copies/mL), a confirmatory HIV antibody test and repeat quantitative plasma HIV RNA test should be performed, and results should be available before initiating ART.
- In rare cases, particularly when PrEP is transitioned to an ARV regimen and HIV RNA and antibody diagnostic testing are inconclusive, HIV DNA testing may be of value.¹⁹ Options for confirming HIV infection and managing such cases are areas of evolving science summarized by the CDC.¹⁹ Clinicians seeking urgent advice can contact the [Clinical Consultation Center's PrEP Service](#) at 1-855-HIV-PREP (or 1-855-448-7737).

Acute HIV Infection in People Taking Long-Acting Cabotegravir for Pre-Exposure Prophylaxis

In the HPTN 083 trial, a pivotal trial of CAB-LA versus TDF/FTC for HIV PrEP with more than 2,000 participants enrolled in each arm, 25 incident cases of HIV were identified in the CAB-LA arm compared to 72 cases in the TDF/FTC arm.^{20,21} Selection of a potent ARV regimen in people who develop acute HIV infection while taking CAB-LA for PrEP should consider that injectable cabotegravir (CAB) may remain detectable after treatment discontinuation, for up to 3 years in men and 4 years in women.²² This long pharmacokinetic tail may contribute to the selection of drug-resistant variants in the setting of incident infection. In an extended analysis of HPTN 083, 34 cases of HIV infection were reported in the CAB-LA arm, with 6 of these cases occurring despite on-time CAB-LA injections.^{23,24} Detection of HIV infection was delayed in 15 of 32 cases and was associated with waxing and waning HIV Ag/Ab, HIV antibody, and qualitative and quantitative HIV RNA test results (i.e., fluctuating between reactive/non-reactive, detectable/non-detectable).²³ Major resistance to an integrase strand transfer inhibitor (INSTI) occurred in 10 of the 32 cases evaluated. All 10 participants with INSTI resistance in the CAB-LA arm received their last CAB dose within 6 months of their initial HIV-positive site visit. While the risk of acquired INSTI resistance appears to wane after 6 months, this trend is based on few observations. These data reinforce the importance of screening for INSTI-resistance mutations when acute HIV infection is diagnosed in people taking CAB-LA or with a history of prior CAB-LA PrEP use **(AIII)**.

Treating Early HIV Infection

The goals of ART during early HIV infection are to suppress plasma HIV RNA to undetectable levels **(AI)**, prevent the transmission of HIV **(AI)**, and preserve immune function **(AIII)**.²⁵⁻²⁷ Importantly, as with chronic HIV infection, an individual's barriers to ART adherence and appointments should be assessed at the time of ART initiation (see [Adherence to the Continuum of Care](#)). ART should be initiated as soon as possible after a positive qualitative or quantitative HIV RNA test result **(AII)**. Same-day or rapid ART initiation in people with acute HIV has been shown to be safe, acceptable, and effective.²⁸ It is important to collect a new blood specimen for a confirmatory HIV antibody test and quantitative plasma HIV RNA test to verify the HIV diagnosis. Given the sensitivity of current HIV RNA assays,²⁹ a positive result by quantitative or qualitative plasma HIV RNA testing in the setting of a negative or indeterminate antibody test result indicates that acute HIV infection is highly likely. HIV treatment does not need to be delayed while awaiting confirmation of HIV diagnosis. Some individuals may not accept their diagnosis or may decline ART initially for other reasons. Individuals who do not begin ART immediately should be maintained in care, and every effort should be made to initiate therapy as soon as they are ready.

Clinical trial data indicate that individuals who are treated during early HIV infection may experience immunologic and virologic benefits.^{26,30-41} In addition, early HIV infection is considered a period of high infectivity,⁴² and early ART substantially reduces the risk of HIV transmission.⁴³⁻⁴⁶

Drug-Resistance Testing in the Setting of Early HIV Infection

Prior to the widespread use of INSTIs, data from the United States and Europe demonstrated that transmitted virus strains with resistance to at least one ARV drug were present in up to 16% of people with HIV.^{47,48} In one study, 21% of isolates from people with acute HIV infection demonstrated resistance to at least one ARV drug, most commonly non-nucleoside reverse transcriptase inhibitors (NNRTIs).⁴⁹⁻⁵¹ The rate of transmitted INSTI resistance was reported to be **≤2.5% of the samples tested, and <0.5% with second-generation INSTIs**.⁵²⁻⁵⁵ Before initiating ART in a person with early HIV infection, a blood specimen should be sent for standard genotypic drug-resistance testing for mutations in the reverse transcriptase and protease genes **(AIII)**. Genotype

testing for INSTI resistance should be performed for those who acquire HIV during or after the use of CAB-LA as PrEP, if transmitted INSTI resistance is suspected, **or if HIV diagnosis occurs after receiving an INSTI-based regimen for post-exposure prophylaxis (PEP) (AIII).**²²

Treatment should not be delayed pending resistance-test results. The test results should be used to modify the ARV regimen if necessary **(AII)**. **In people with no history of CAB-LA use or use of an INSTI-based regimen for PEP**, the Panel on Antiretroviral Guidelines for Adults and Adolescents does not currently recommend genotype testing for INSTI resistance given the low rate of transmitted INSTI resistance and the high barrier to resistance of bictegravir (BIC) and dolutegravir (DTG), unless transmitted INSTI resistance is a concern **(AIII)**.

Considerations for Preventing HIV Transmission During Early HIV Infection

People with early HIV infection have a higher likelihood of sexual transmission of HIV to others. Prompt initiation of ART and sustained viral suppression to <200 copies/mL can prevent transmission of HIV to sexual partners. Individuals starting ART should use another form of prevention (e.g., condoms, PrEP for partners who are HIV negative, sexual abstinence) for at least the first 6 months of treatment and until they have a documented viral load of <200 copies/mL **(AII)**. Many experts would recommend confirming sustained viral suppression before assuming no risk of sexual transmission of HIV **(AIII)** (see [Antiretroviral Therapy to Prevent Sexual Transmission of HIV](#)).

Antiretroviral Regimens for Early HIV Infection

ART should be initiated with one of the combination regimens recommended for people with chronic HIV infection **(AIII)** (see [What to Start](#)). Providers should inform individuals starting ART of the importance of adherence in achieving and maintaining viral suppression **(AIII)**. If available, the results of ARV drug-resistance testing or the resistance pattern of the source person's virus should be used to guide selection of the regimen. All women of childbearing potential should undergo a pregnancy test before initiating ART **(AIII)**.

For individuals who have not received CAB-LA as PrEP prior to diagnosis of acute HIV, one of the following INSTI-based regimens is recommended **(AIII)**:

- BIC/TAF/FTC
- DTG with (TDF or TAF) plus (3TC or FTC)

DTG and BIC are good treatment options because transmitted resistance to second-generation INSTIs in the United States is rare at present and both BIC and DTG have a high barrier to resistance.

For individuals who acquire HIV during and after the use of CAB-LA as PrEP:

- While awaiting integrase genotype results, the use of an INSTI-based regimen **is not recommended (AIII)**. The recommended regimen, until resistance testing confirms the absence of INSTI-resistance mutations, is a cobicistat (COBI)- or ritonavir (RTV)-boosted darunavir (DRV) with (TAF or TDF) plus (FTC or 3TC) **(AIII)**.

A pharmacologically boosted protease inhibitor (PI)-based regimen (e.g., boosted DRV) is an option because resistance to PIs emerges slowly, and clinically significant transmitted resistance to PIs is uncommon. Therefore, boosted DRV plus (TAF or TDF) plus (FTC or 3TC) is generally recommended in this setting. Baseline laboratory testing recommended for individuals with chronic

HIV infection should be performed (see [Laboratory Testing for Initial Assessment and Monitoring of People With HIV Receiving Antiretroviral Therapy](#)). Individuals with hepatitis B virus/HIV coinfection should receive TDF/FTC or TAF/FTC as part of their ARV regimen.

Given the increasing use of TDF/FTC⁵⁶⁻⁵⁸ and TAF/FTC as PrEP,^{59,60} early HIV infection may be diagnosed in some people while they are taking TDF/FTC or TAF/FTC. In this setting, drug-resistance test results are particularly important; however, these regimens remain reasonable treatment options pending drug-resistance test results.

Abacavir/3TC **is not recommended** for treatment of acute HIV infection unless the person is known to be HLA-B*5701-negative, and this information is rarely available in people with acute HIV infection. Additionally, due to relatively high rates of transmitted drug resistance for NNRTIs, agents in this drug class **are not recommended** as a component in the regimen of people initiating ART before the results of drug-resistance tests are available.

Treatment Regimens for Early HIV Infection During Pregnancy

All women of childbearing potential who receive a diagnosis of early HIV infection should have a pregnancy test **(AIII)**. Because early HIV infection, especially in the setting of high-level viremia, is associated with a high risk of perinatal transmission, all pregnant women with HIV should start combination ART as soon as possible to prevent perinatal transmission **(AI)**. **COBI should be avoided in pregnancy because lower concentrations of COBI and DRV have been reported during the second and third trimesters. For more information on the safety and efficacy of ARV use in pregnancy, refer to the [Perinatal Guidelines](#).**

Follow-Up After Antiretroviral Therapy Initiation

After ART initiation, monitoring of plasma HIV RNA levels, CD4 T lymphocyte cell counts, and adverse effects should be performed as described in [Laboratory Testing for Initial Assessment and Monitoring of People With HIV Receiving Antiretroviral Therapy](#) (i.e., HIV RNA testing 2–8 weeks after ART initiation, then every 4–8 weeks until viral suppression and every 3–4 months thereafter) **(AII)**.

Table 15. Identifying, Diagnosing, and Treating Acute and Recent HIV Infection

Suspicion of Acute HIV Infection
- Health care providers should consider the possibility of acute HIV infection in people with the signs, symptoms, or laboratory findings described below and in asymptomatic people with a possible acute (within 2–6 weeks) exposure to HIV.^a - Signs, symptoms, or laboratory findings of acute HIV infection may include but are not limited to, one or more of the following: fever, lymphadenopathy, skin rash, myalgia, arthralgia, headache, diarrhea, pharyngitis, oral ulcers, leucopenia, thrombocytopenia, and transaminase elevation. - High-risk exposures include sexual contact with someone who has HIV or is at risk of HIV infection, sharing needles, syringes, or equipment for drug preparation or injection, or any situation where a person's mucous membranes or broken skin come into contact with bodily fluids that may carry HIV. Differential Diagnosis - The differential diagnosis of acute HIV infection may include but is not limited to, viral illnesses such as COVID-19, EBV and non-EBV (e.g., CMV) infectious mononucleosis syndromes, influenza, viral hepatitis, streptococcal infection, or syphilis. Diagnosis of any STI should prompt HIV testing and consideration of acute HIV infection.
Testing to Diagnose or Confirm Acute HIV Infection
- Acute HIV infection is defined as detectable HIV RNA or p24 antigen (the specific antigen used in currently available HIV-1/2 Ag/Ab combination assays) in the setting of a negative or indeterminate HIV antibody test result. - A reactive HIV antibody test result or Ag/Ab combination test result must be followed by supplemental confirmatory testing. - A negative or indeterminate HIV antibody test result in a person with a reactive Ag/Ab test result or in whom acute HIV infection is suspected requires plasma HIV RNA testing to diagnose acute HIV infection. - A positive result on a quantitative or qualitative plasma HIV RNA test in the setting of a negative or indeterminate antibody test result indicates that acute HIV infection is highly likely. In this case, the diagnosis of HIV infection should be confirmed by subsequent documentation of HIV antibody seroconversion. - A positive HIV Ag/Ab test result or a positive HIV RNA test result in the setting of a negative HIV antibody test result in a person taking PrEP should prompt immediate confirmation of HIV diagnosis. It is important to collect a new blood specimen to verify the HIV diagnosis before initiating HIV treatment.
Antiretroviral Therapy After Diagnosis of Early HIV Infection
- ART is recommended for all people with HIV, including those with early HIV infection (AI). ART should be initiated as soon as possible after HIV diagnosis (AII). - Once initiated, the goals of ART are to achieve sustained plasma virologic suppression, prevent HIV transmission (AII), and preserve immune function (AIII). - All women of childbearing potential who receive a diagnosis of early HIV infection should have a pregnancy test (AIII). - Pregnant women with early HIV infection should begin ART as soon as possible for their own health and to prevent perinatal transmission of HIV (AI).

Table 15. Identifying, Diagnosing, and Treating Acute and Recent HIV Infection

- A blood sample for genotypic drug-resistance testing should be obtained before initiating ART to guide the selection of the regimen **(AIII)**, but ART should be initiated as soon as possible, often before resistance-test results are available. If resistance is subsequently identified, treatment should be modified as needed.
- Standard genotypic drug-resistance testing should be performed for mutations in the reverse transcriptase and protease genes **(AIII)** for all people with early HIV. Genotype testing for INSTI resistance should be performed for those who acquire HIV during or after the use of CAB-LA as PrEP, if transmitted INSTI resistance is suspected, **or if HIV diagnosis is made after receiving an INSTI-based regimen for PEP (AIII)**.
- ART can be initiated before the results of drug-resistance testing are known. For individuals who do not have a history of using CAB-LA as PrEP, one of the following ARV regimens is recommended **(AIII)**:
 - o BIC/TAF/FTC **(AIII)**; or
 - o DTG with (TAF or TDF)^b plus (FTC or 3TC) **(AIII)**
- For individuals with a history of using CAB-LA as PrEP, genotypic resistance testing performed before starting ART should include screening for INSTI-resistance mutations **(AIII)**. Recommended regimens include the following:
 - o (DRV/c^c or DRV/r) with (TAF or TDF)^b plus (FTC or 3TC)—pending the results of the genotype **(AIII)**. Empiric INSTI-containing regimens **are not recommended (AIII)**, because INSTI resistance may be present in those who acquire HIV during the use of CAB-LA and possibly up to 4 years after.

^a In some settings, activities that increase the risk of HIV infection may not be recognized or perceived as risky by the health care provider, the person at risk, or both. Thus, even in the absence of reported high-risk activities, symptoms and signs consistent with acute retroviral syndrome should motivate health care providers to consider a diagnosis of acute HIV infection.

^b TAF and TDF are two forms of tenofovir that are approved in the United States. TAF has fewer bone and kidney toxicities than TDF, whereas TDF is associated with lower lipid levels. Safety, cost, and accessibility are among the factors to consider when choosing between these drugs.

^c **COBI should be avoided in pregnancy because lower concentrations of COBI and DRV have been reported during the second and third trimesters.**

Key: 3TC = lamivudine; Ag/Ab = antigen/antibody; ART = antiretroviral therapy; ARV = antiretroviral; BIC = bictegravir; CAB-LA = long-acting cabotegravir; CMV = cytomegalovirus; COBI = cobicistat; DRV = darunavir; DRV/c = darunavir/cobicistat; DRV/r = darunavir/ritonavir; DTG = dolutegravir; EBV = Epstein-Barr virus; FTC = emtricitabine; INSTI = integrase strand transfer inhibitor; PEP = post-exposure prophylaxis; PrEP = pre-exposure prophylaxis; STI = sexually transmitted infection; TAF = tenofovir alafenamide; TDF = tenofovir disoproxil fumarate

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HIV-2 Infection

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Key Considerations and Recommendations
- The clinical course of HIV-2 infection is generally characterized by a longer asymptomatic stage, lower plasma viral loads, and lower mortality rate than HIV-1 infection. However, without treatment, the majority of individuals with HIV-2 will progress to AIDS and death. - No randomized controlled trials have addressed when a person with HIV-2 should start antiretroviral therapy (ART) or which regimens are most effective for initial or second-line ART when treating HIV-2; thus, the optimal treatment strategy is not well defined. - Existing data on the treatment of HIV-2, and extrapolation from data on the treatment of HIV-1, suggest that ART should be started at or soon after HIV-2 diagnosis to prevent disease progression and transmission of HIV-2 to others (AIII). - Quantitative plasma HIV-2 RNA viral load testing for clinical care is available and should be performed before initiating ART (AIII). - For ART-naïve patients who have HIV-2 mono-infection or HIV-1/HIV-2 coinfection, antiretroviral (ARV) regimens should include an integrase strand transfer inhibitor (INSTI) plus two nucleoside reverse transcriptase inhibitors (NRTIs) (AII). A recommended alternative regimen is a boosted protease inhibitor (PI) that is active against HIV-2 (darunavir or lopinavir) plus two NRTIs (BII). - HIV-2 is intrinsically resistant to non-nucleoside reverse transcriptase inhibitors (NNRTIs); therefore, NNRTI-based regimens, including long-acting injectable rilpivirine (given with the INSTI cabotegravir), are not recommended for treatment of HIV-2 (AIII). - Patients with hepatitis B virus (HBV)/HIV-2 coinfection should be prescribed ART that contains drugs with activity against both HIV-2 and HBV (AIII). - HIV-2 RNA, CD4 T lymphocyte (CD4) cell counts, and clinical status should be used to assess treatment response (AII). Unlike people with HIV-1, people with HIV-2 should continue to undergo periodic CD4 testing even if their viral loads are persistently suppressed, because disease progression can occur despite an undetectable viral load (AIII). - Resistance-associated viral mutations to INSTIs, PIs, or NRTIs may develop in people with HIV-2 while they are on ART. However, no validated HIV-2 genotypic or phenotypic antiretroviral resistance assays are approved for clinical use. - In the event of virologic, immunologic, or clinical failure, a new ARV regimen should be constructed in consultation with an expert in HIV-2 management. <i>In vitro</i> studies demonstrate that HIV-2 is intrinsically resistant to the fusion inhibitor enfuvirtide, and limited data also show that HIV-2 is intrinsically resistant to fostemsavir; therefore, these drugs are not recommended for treatment of people with HIV-2 (AIII). - For patients with multidrug-resistant virus, ibalizumab and lenacapavir demonstrate <i>in vitro</i> potency against HIV-2 and may be considered (BIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Overview

HIV-2 infection is endemic in West Africa, with certain countries experiencing a population prevalence of >1%. The possibility of HIV-2 infection should be considered when treating people of West African origin, people who have had sexual contact with or shared needles with people of West African origin, and people who reside in countries with strong socioeconomic ties to West Africa (e.g., France, Spain, Portugal, and former Portuguese colonies, such as Brazil, Angola, Mozambique, and parts of India). Globally, it has been estimated that 1 million to 2 million people have HIV-2—a number that includes people with HIV-1/HIV-2 dual infection. However, current and accurate prevalence data are scarce, and neither the Joint United Nations Programme on HIV and AIDS nor the World Health Organization has formal surveillance systems for HIV-2.¹

Clinical Course of HIV-2 Infection

The clinical course of HIV-2 infection is generally characterized by a longer asymptomatic stage, lower plasma viral loads, and a lower mortality rate than HIV-1 infection.^{2,3} However, without effective antiretroviral therapy (ART), HIV-2 infection will progress to AIDS and death in the majority of individuals.⁴ Concomitant infection with HIV-1 and HIV-2 may occur, and the possibility of this coinfection should be considered when treating people from areas with a high prevalence of HIV-2.

Diagnostic and Monitoring Assays for HIV-2 Infection

In the appropriate epidemiologic setting, HIV-2 infection should be suspected in people who have clinical conditions that suggest HIV infection but who have atypical serologic results (e.g., a positive screening assay with an indeterminate HIV-1 Western blot).⁵ The possibility of HIV-2 infection should also be considered in the appropriate epidemiologic setting in people who have serologically confirmed HIV infection but low or undetectable HIV-1 RNA levels, or in those who have declining CD4 T lymphocyte (CD4) cell counts despite apparent virologic suppression on ART.

The 2014 Centers for Disease Control and Prevention guidelines for HIV diagnostic testing⁶ recommend using an HIV-1/HIV-2 antigen/antibody combination immunoassay for initial testing and using an HIV-1/HIV-2 antibody differentiation immunoassay for subsequent testing. The Geenius HIV 1/2 Supplemental Assay (Bio-Rad Laboratories) is approved by the U.S. Food and Drug Administration (FDA) to differentiate HIV-1 infection from HIV-2 infection. The Multispot HIV-1/HIV-2 Rapid Test is no longer available. Commercially available HIV-1 RNA assays do not reliably detect or quantify HIV-2 RNA.⁷ Quantitative HIV-2 RNA testing is available at the [University of Washington \(UW\)](#)⁸ and the [New York State Department of Health \(NYSDOH\)](#).⁹ HIV-2 nucleic acid amplification test–based (total DNA/RNA) diagnostic testing is available for clinical care at [UW](#).¹⁰ However, it is important to note that up to one-third of people with untreated HIV-2 will have HIV-2 RNA levels below the limits of detection (10 copies/mL for UW testing and 7 IU/mL for NYSDOH testing); some of these people will have clinical progression and CD4 count decline. No validated HIV-2 genotypic or phenotypic antiretroviral (ARV) resistance assays are approved by the FDA for clinical use. HIV-2 genotypic ARV resistance assays are available at UW for research use only.

Treatment of HIV-2 Infection

To date, no randomized controlled trials that address when to start ART or the choice of initial or subsequent ART for HIV-2 have been reported¹¹; thus, the optimal treatment strategy has not been defined. Existing data on the treatment of HIV-2 and extrapolation from data on the treatment of HIV-1 suggest that ART should be started at or soon after HIV-2 diagnosis to prevent disease progression and transmission of HIV-2 to others (**AIII**). However, CD4 cell recovery in people with HIV-2 who are on ART is generally poorer than that observed in people with HIV-1.^{12,13}

Data from *in vitro* studies suggest that HIV-2 is sensitive to the currently available nucleoside reverse transcriptase inhibitors (NRTIs); however, HIV-2 is more likely to develop resistance to NRTIs than HIV-1.¹⁴ HIV-2 is intrinsically resistant to non-nucleoside reverse transcriptase inhibitors (NNRTIs)¹⁵; thus, NNRTI-based regimens, including the long-acting injectable rilpivirine (RPV) (given with the integrase strand transfer inhibitor [INSTI] cabotegravir [CAB]), are **not recommended** for treatment of HIV-2 (AIII). Several small studies in individuals with HIV-2 have reported poor responses to dual-NRTI regimens^{16,17} or regimens that contain an NNRTI plus two NRTIs.^{18,19} Clinical data on the effectiveness of triple-NRTI regimens are conflicting.^{20,21}

INSTI-based regimens or protease inhibitor (PI)-based regimens are treatment options for people with HIV-2. As discussed below, **three** single-arm clinical trials showed favorable outcomes in patients who received INSTI-based regimens. Data regarding the efficacy of PI-based regimens primarily come from observational reports. A randomized controlled trial comparing raltegravir (RAL) plus tenofovir disoproxil fumarate/emtricitabine (TDF/FTC) to lopinavir/ritonavir (LPV/r) plus TDF/FTC is **completed, but the results have not been reported** (FIT-2; NCT02150993).

Integrase Strand Transfer Inhibitor–Based Regimens

All FDA-approved INSTIs—RAL, elvitegravir (EVG), dolutegravir (DTG), bictegravir, **and CAB**—have potent activity against HIV-2 *in vitro*.²²⁻²⁷ INSTI-based regimens have shown favorable treatment responses in observational studies.²⁸⁻³⁰ **Three small**, single-arm, open-label clinical trials have assessed the effectiveness of INSTI-based regimens in ART-naïve individuals with HIV-2. **These studies** evaluated RAL plus TDF/FTC, EVG/cobicistat/TDF/FTC, **and DTG plus either abacavir/lamivudine or TDF/FTC**. **All** the studies demonstrated favorable clinical and immunovirologic results at 48 weeks, providing the best evidence to date for HIV-2 treatment recommendations.³¹⁻³³

Protease Inhibitor–Based Regimens

In general, regimens that contain boosted PIs that are active against HIV-2 (and that also include two NRTIs) have resulted in more favorable virologic and immunologic responses than regimens that consist of only two or three NRTIs.^{12,13,21,34} Darunavir (DRV), lopinavir (LPV), and saquinavir are more active against HIV-2 than other approved PIs.³⁵⁻³⁷ **While LPV/r is more active against HIV-2 than DRV *in vitro*, it is less well tolerated. Some clinicians may use boosted DRV (where available) to enhance adherence.** Older unboosted PI-based regimens, including nelfinavir or indinavir plus zidovudine and lamivudine, and atazanavir-based regimens have shown poor clinical success rates.^{11,16,17,38,39}

Other Antiretroviral Drugs

Among the entry inhibitors, HIV-2 is intrinsically resistant to enfuvirtide.⁴⁰ The CCR5 antagonist maraviroc appears to be active against some HIV-2 isolates⁴¹; however, there are no FDA-approved assays that can determine HIV-2 co-receptor tropism, and HIV-2 is known to use many other minor co-receptors in addition to CCR5 and CXCR4.⁴² **The post-attachment inhibitor ibalizumab (IBA) has potent *in vitro* activity against HIV-2 isolates.⁴³ Limited data suggest HIV-2 is intrinsically resistant to the attachment inhibitor fostemsavir (FTR).**^{44,45}

The capsid inhibitor lenacapavir (LEN) demonstrates nanomolar potency against HIV-2 *in vitro* but is 11- to 25-fold less active against HIV-2 than HIV-1.^{46,47}

Some national and international guidelines have recommended specific preferred and alternative drug regimens for initial and second-line ART for HIV-2 infection⁴⁸⁻⁵¹; however, there are currently no

comparative randomized controlled clinical trial data that support the effectiveness of a specific recommended regimen.

The Panel's Recommendations

Until more definitive data on outcomes are available, the Panel on Antiretroviral Guidelines for Adults and Adolescents provides the following recommendations for the management of individuals with HIV-2 mono-infection or HIV-1/HIV-2 dual infection:

- A regimen that contains one INSTI plus two NRTIs is the recommended initial ART for most individuals with HIV-2 **(AII)**.
- An alternative regimen is a boosted PI (DRV or LPV) that is active against HIV-2 plus two NRTIs **(BII)**.
- HIV-2 is intrinsically resistant to NNRTIs; therefore, NNRTI-based regimens, including long-acting injectable RPV (given with the INSTI CAB), are not recommended for treatment of HIV-2 **(AIII)**.
- HIV-2 demonstrates intrinsic resistance to the fusion inhibitor enfuvirtide *in vitro*, and limited data show intrinsic resistance to FTR; therefore, these drugs are not recommended for treatment of people with HIV-2 **(AIII)**.
- For people with multidrug-resistant HIV-2, IBA and LEN may be considered based on *in vitro* data **(BIII)**.
- Patients with hepatitis B virus (HBV)/HIV-2 coinfection require ART that contains drugs with activity against both HIV-2 and HBV **(AIII)**. See [Hepatitis B Virus/HIV Coinfection](#) for more information.
- HIV-2 plasma RNA levels, CD4 counts, and clinical status should be monitored to assess treatment response, as is recommended for HIV-1 **(AII)**.
- People who have HIV-2 RNA levels that are below the limits of detection before they initiate ART should still undergo routine HIV-2 plasma RNA monitoring in addition to CD4 count and clinical monitoring. Unlike HIV-1, people with HIV-2 require continued CD4 count monitoring, as disease progression can occur in the setting of undetectable HIV-2 viral load **(AIII)**.

Women with HIV-2 who are of childbearing potential require similar considerations when choosing a regimen as women of childbearing potential with HIV-1 (see [What to Start](#)). There are no data on HIV-2 treatment as prevention; however, both data from studies of people with HIV-1 and data on the natural history of HIV-2 transmission suggest that effective ART likely provides a reduced risk of transmission to sexual partners.⁵²⁻⁵⁴

Viral mutations that are associated with resistance to NRTIs, PIs, and/or INSTIs may develop in people with HIV-2 while they are on ART.^{37,55,56} Currently, transmitted drug resistance appears to be rare among people with HIV-2.^{57,58} In several small studies, twice-daily dosing of DTG was found to have some residual activity as a second-line INSTI in some people with HIV-2 who had extensive ART experience and RAL resistance.⁵⁹⁻⁶² Genotypic algorithms that are used to predict drug resistance in HIV-1 may not be applicable to HIV-2 because the pathways and mutational patterns that lead to resistance may differ between the HIV types (see the [HIV2EU Algorithm](#) and the [Stanford University HIV Drug Resistance Database](#)).⁶³ In the event of virologic, immunologic, or clinical failure, a new ART regimen should be constructed in consultation with an expert in HIV-2 management.

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HIV and the Older Person

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Key Considerations and Recommendations When Caring for Older People With HIV

- Diagnosis of HIV at a later stage of disease is more common among older people. Early diagnosis and treatment of HIV and counseling to prevent secondary HIV transmission remain important in the clinical care of older people with HIV.
- Antiretroviral therapy (ART) is recommended for all people with HIV **(AI)**. ART is especially important for older individuals because they have a greater risk of serious non-AIDS complications and potentially a blunted immunologic response to ART.
- Compared to people without HIV, people with HIV have a twofold higher risk of developing atherosclerotic cardiovascular disease (ASCVD), and their age at incident ASCVD diagnosis is about a decade younger. In addition to current American Heart Association (AHA)/American College of Cardiology (ACC)/Multisociety Guidelines, the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) recommends initiating at least a moderate-intensity statin in people with HIV aged 40 to 75 years who have 10-year ASCVD risk estimates of 5% to <20% **(AI)**. See [Statin Therapy in People With HIV](#) for the Panel's additional recommendations on the use of statin therapy as primary prevention for people with HIV.
- Polypharmacy is common in older people with HIV, and all drugs, supplements, and herbal treatments should be assessed regularly for appropriateness, potential for adverse effects, proper dosing, and drug interactions **(AIII)**.
- Potential for drug–drug interactions between antiretroviral drugs and concomitant medications (including statins, supplements, and herbal medicines) should be assessed regularly, especially before a new ART regimen or concomitant medication is started. In this context, it is also important to inquire and counsel about the use of non-daily medications, including long-acting injectables and as-needed medications.
- Adverse drug events from ART and concomitant drugs may occur more frequently in older people with HIV than in younger individuals with HIV. Therefore, the bone, kidney, metabolic, cardiovascular, cognitive, and liver health of older individuals with HIV should be monitored closely.
- HIV infection is associated with immunologic aging and systemic inflammation, which may contribute to the development of comorbidities across multiple organ systems as well as aging phenotypes like frailty. HIV experts, primary care providers, and other specialists should work together to optimize the medical care of older people with HIV, including adhering to treatment and prevention guidelines for different medical comorbidities.
- Age-related decline in neurocognitive function is faster in people with HIV compared to those without. Cognitive impairment in people with HIV—with manifestations including problems with memory, attention, and executive function—is associated with reduced adherence to therapy and poorer health outcomes, including increased risk of death. For people with progressive cognitive impairment, referral to a specialist (e.g., neurologist, neuropsychologist, geriatrician) for evaluation, testing, and management may be warranted **(BIII)**.
- Mental health disorders, including an increased risk of anxiety and depression, are a concern among aging people with HIV. Screening for depression and management of mental health issues are important when caring for older people with HIV.
- Given that the burden of aging-related diseases is significantly higher among people with HIV than in the general population, additional medical and social services may be required to effectively manage both HIV and comorbid conditions.

Rating of Recommendations: A = Strong; B = Moderate; C = Weak

Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Introduction

Effective antiretroviral therapy (ART) has increased survival in people with HIV¹⁻⁴, resulting in an increasing number of older people with HIV. In the United States in 2022, the proportion of people newly diagnosed with HIV aged ≥55 years was 9%, according to the [Centers for Disease Control and Prevention \(CDC\)](#). Among people with HIV in 2022, 38% were aged ≥55 years, 13.2% were aged ≥65 years, and these proportions are expected to increase steadily.⁵ The difference in life expectancy between people with HIV and the general population has significantly narrowed since the use of effective ART and is now estimated to be 5 years or fewer for those with CD4 T lymphocyte (CD4) cell count >500/μL after initiating ART.³ However, the difference in mortality rates increases significantly with advancing age. While there has been a large reduction in AIDS-related mortality with effective ART, noninfectious comorbidities and complications, including atherosclerotic cardiovascular diseases (ASCVD) and non-AIDS malignancies, now account for a growing proportion of the causes of death among people with HIV.^{2,3,6,7}

The discussion in this section of the guidelines refers to individuals aged ≥50 years as older people with HIV. Many older people with HIV have had HIV for years or decades; therefore, accumulating disease risk associated with longstanding HIV disease itself and/or exposure to antiretroviral (ARV) medications (including prior use of older ART regimens with greater toxicity in some cases) should be considered. Specifically, HIV may affect the biology of aging, possibly resulting in earlier manifestations of morbidities generally associated with more advanced age. As a result, older people with HIV may suffer from aging-related illnesses earlier and at higher rates than those without HIV, and will require more non-ART medications than younger people, which may complicate HIV clinical care.^{8,9,10}

It is also important to consider distinct issues for people who are diagnosed with and/or acquire HIV at older ages. For example, older adults may have an increased risk of acquisition and transmission of HIV, in part as a consequence of reduced mucosal and immunologic defenses and changes in risk related-behaviors (e.g., decrease in condom use because of less concern about pregnancy or more high-risk sexual activity with increased use of erectile dysfunction drugs).^{11,12} In addition, HIV screening among older adults remains low because they are generally perceived to be at low risk of acquiring HIV. The sections below review important considerations for the diagnosis and treatment of HIV among older people, as well as the potential clinical implications for people now living to older ages with longstanding HIV disease.

HIV Screening and Diagnosis in the Older Person

Older people with HIV (aged ≥55 years) are significantly more likely to meet the case definition of AIDS at HIV diagnosis than people aged <55 years (threshold defined by CDC surveillance data).¹³ Also, older adults appear to have lower CD4 counts at seroconversion and steeper CD4 count decline over time¹⁴ and tend to present to care with significantly lower CD4 counts.^{15,16} Therefore, education, preventive measures (including pre-exposure prophylaxis or PrEP), and routine screening of asymptomatic older adults for HIV should be emphasized,¹⁷ as well as considering HIV infection in the medical evaluation of older adults.

Antiretroviral Therapy in the Older Person With HIV

Importance of Prompt Initiation of Antiretroviral Therapy After HIV Diagnosis

ART is recommended for all individuals with HIV (AI; see [Initiation of Antiretroviral Therapy](#)). Early treatment may be particularly important in older adults because of blunted immune recovery and

increased risk of serious non-AIDS events in this population.^{18,19} However, ART-associated CD4 cell recovery in older adults is generally slower and lower in magnitude than in younger people,²⁰⁻²³ suggesting that early diagnosis and treatment may result in better immunologic response and possibly improved clinical outcomes. In a modeling study based on data from an observational cohort, the beneficial effects of early ART were projected to be greatest in the oldest age group (people aged 45 to 65 years).²⁴ The benefits of early ART initiation were further demonstrated in an analysis of HIV cohorts from Europe and the Americas showing a lower all-cause and non-AIDS mortality with immediate ART initiation in people aged 50 to 70 years.²⁵ More definitive information was provided by the START study, which randomized ARV treatment-naïve participants with CD4 counts >500 cells/mm³ to either immediate ART or deferred ART until CD4 count dropped below 350 cells/mm³. The highest absolute risk reduction of serious AIDS and non-AIDS events associated with immediate ART was found in participants aged 50 years or older.²⁶ All older people with HIV should be informed that maintaining a plasma HIV RNA (viral load) at <200 copies/mL with ART improves overall health and prevents sexual transmission of HIV (see the [Treatment as Prevention](#) section).

Choice of Antiretroviral Regimens in the Older Person With HIV

The choice of ARV regimen for an older person with HIV should be informed by a comprehensive review of the person's other medical conditions and medications (including non-prescription drugs, supplements, and herbal treatments), and results from cumulative genotypic resistance testing. The What to Start section ([Table 7](#)) of these guidelines provides guidance on selecting an ARV regimen based on a person's characteristics and specific clinical conditions (e.g., kidney disease, elevated risk for cardiovascular disease [CVD], osteoporosis). In older people with HIV and reduced renal function, dosage adjustment of nucleoside reverse transcriptase inhibitors (NRTIs) may be necessary (see [Appendix B](#)). In addition, ARV regimen selection may be influenced by potential interactions between ARV medications and drugs used concomitantly to manage comorbidities (see [Tables 24a](#) through [25b](#)). Adults aged >50 years should be monitored for ART effectiveness and safety as similarly recommended for other populations with HIV (see [Table 3](#)); however, in older people, special attention should be paid to the greater potential for adverse effects of ART on renal, liver, cardiovascular, central nervous system, metabolic, and bone health (see [Table 21](#)). A meta-analysis confirmed that people with HIV have a 1.5-fold higher risk of fragility-related fracture and a fourfold higher risk of hip fracture. The increased risk is not completely explained by differences in bone mineral density (BMD) alone, indicating there may be differences due to bone microarchitecture or other HIV-related factors contributing to the marked fracture risk. ART regimens that contain tenofovir disoproxil fumarate (TDF),²⁷ boosted protease inhibitors (PIs), or both are associated with a significantly greater loss of BMD, with or without osteoporosis, compared to regimens containing other NRTIs and integrase strand transfer inhibitors (INSTIs).²⁸⁻³¹ Abacavir (ABC), NRTI-sparing regimens, and tenofovir alafenamide (TAF) may be considered as alternatives to the use of TDF in older individuals who may be at risk of nephrotoxicity, osteopenia, or osteoporosis; however, with ABC, the benefit should be balanced with its potential for increasing the risk of CVD (see [What to Start](#)). The long-acting (LA) injectable combination of cabotegravir (CAB) and rilpivirine (RPV) is also an option for ART among older people with HIV, with the same considerations as for all people with HIV (see [Optimizing Antiretroviral Therapy in the Setting of Viral Suppression](#)).

Antiretroviral Efficacy and Safety Considerations in the Older Person With HIV

The efficacy, pharmacokinetics (PK), adverse effects, and drug interaction potentials of ART in older adults have not been studied systematically. Data from clinical trials of second-generation INSTIs (dolutegravir [DTG] and bictegravir),³² two-drug regimens (DTG/lamivudine),³³ and LA CAB/RPV

have shown similar virologic efficacy, safety, and tolerability between participants >50 years of age and those aged 50 and younger.³⁴

Hepatic metabolism and renal elimination are the major routes of drug clearance, including the clearance of ARV drugs. Both liver and kidney functions decrease with age and may result in impaired drug elimination and increased drug exposure.³⁵ Most clinical trials have included only a small proportion of participants aged >50 years, and current ARV dosing recommendations are based on PK and pharmacodynamic data derived from participants with normal organ function. Because it is not known whether drug accumulation in older people may lead to greater incidence and severity of adverse effects than in younger people, therapy in older people requires close monitoring and heightened awareness of drug-related adverse effects, especially in those with hepatic or renal impairment.

Adherence Concerns

Suboptimal adherence to ART is the most common cause of treatment failure. Although most older people with HIV are able to achieve viral suppression with a single-tablet regimen, some people who have a long history of ART may have acquired drug resistance, thus requiring more complex multiple-pill regimens. Complex dosing requirements, high pill burden, polypharmacy, inability to access medications because of cost or availability, limited health literacy (including misunderstanding of instructions), depression, substance use, frailty, and neurocognitive impairment are among the key reasons for nonadherence and should be regularly assessed and managed (see [Adherence to the Continuum of Care](#)).³⁶⁻³⁸ Although some of the factors associated with nonadherence may be more prevalent in older people with HIV, older people have demonstrated better adherence to ART than younger people in some studies.³⁹⁻⁴¹ Severe menopausal symptoms are also associated with reduced adherence to ART, which increases the risk of drug resistance and adverse HIV-related health outcomes in menopausal women, highlighting the importance of managing these symptoms among women with HIV.^{42,43} To facilitate ART adherence, it may be useful to discontinue unnecessary non-HIV medications, simplify ART regimens, and apply evidence-based behavioral approaches including the use of adherence aids such as pillboxes or daily calendars, and support from family members and friends (see [Adherence to the Continuum of Care](#)). LA dosing options, such as LA CAB/RPV, may ease pill burden and adherence concerns for some older adults (see [Optimizing Antiretroviral Therapy](#)).

Optimizing Antiretroviral Therapy in Older People With HIV

Given the greater incidence of comorbidities, non-AIDS complications, and frailty among older people with HIV, switching one or more ARVs in an HIV regimen may be necessary to minimize toxicities, drug–drug interactions, or to reduce pill burden.³⁸ Expert guidance now recommends bone density monitoring in men aged ≥50 years and postmenopausal women and suggests switching from TDF or boosted PIs to other ARVs (such as TAF or INSTIs, respectively) in older people at high risk for fragility fractures.⁴⁴ Given the high prevalence and faster progression of chronic kidney disease (CKD) in aging people with HIV—likely from a combination of HIV, ART, and non-HIV risk factors—development of CKD must be closely monitored in an older person on ART.^{45,46} [Table 7](#) provides guidance for ARV drugs to use based on selected clinical scenarios, including comorbidities.

Polypharmacy in Older People With HIV

People with HIV and aging-associated comorbidities may require additional pharmacologic interventions that can complicate therapeutic management.⁸ In addition to taking medications to manage HIV and comorbid conditions, many older people with HIV are also taking medications to relieve discomfort (e.g., pain medications, sedatives) or to manage adverse effects of medications

(e.g., anti-emetics). Older individuals may also self-medicate with over-the-counter medicines or supplements.

Polypharmacy is more common in older people with HIV than similarly aged people in the general population.^{8,47,48} Among people with HIV aged ≥ 65 years, it is predicted that the prevalence of comorbidities and polypharmacy rose with increasing age and duration of HIV infection.^{49,50} In the Swiss HIV Cohort Study, the prevalence of polypharmacy and inappropriate prescribing in people aged ≥ 75 was 66% and 67%, respectively.⁵¹

In people aging with HIV, the effects of polypharmacy, including the use of drugs that affect neurocognitive function, can contribute to serious adverse outcomes, such as serious falls and fractures, delirium, hospitalization (including intensive care unit admissions), and death.⁵²⁻⁵⁷ Polypharmacy may also increase the likelihood of adverse consequences such as medication errors (by prescribers or prescription recipients), medication nonadherence, additive drug toxicities, and drug–drug interactions. Older people with HIV are probably at an even greater risk of polypharmacy-related adverse consequences than younger or similarly aged people without HIV. When evaluating any new clinical concern or laboratory abnormality in people with HIV, especially in older people, clinicians should always consider the possible role of adverse drug reactions from both ARV drugs and other concomitantly administered medications.

Drug–Drug Interaction Concerns

Interactions between ARV drugs and concomitant medications, supplements, and herbal treatments can occur and can be easily overlooked by prescribers.⁵⁸ Potential drug–drug interactions can occur between ARV and non-ARV medications, as well as between non-ARV medications.^{48,59} The available drug interaction information on ARV agents is derived primarily from PK studies performed in small numbers of relatively young participants with normal organ function who do not have HIV (see Tables [24a–25b](#)). Data from these studies provide clinicians with a basis to assess whether a significant interaction may exist. However, the magnitude of an interaction may be greater in older than in younger people with HIV; therefore, it is important for clinicians to remain vigilant to assess for potential drug–drug interactions given the high prevalence of polypharmacy in older people with HIV. The risk of significant drug–drug interactions appears to be higher with PI-based or non-nucleoside reverse transcriptase inhibitors-based ART than with INSTI-based ART, although there are some important interactions with INSTIs that should be recognized and managed (including with polyvalent cations that are frequently taken as supplements).⁶⁰⁻⁶²

Hormonal therapies may be taken to alleviate symptoms of menopause or for other medical indications. Although data on drug–drug interactions between hormonal therapies and ART are limited, existing data suggest little to no impact of hormonal therapies on the efficacy of some ART.⁶³⁻⁶⁵

HIV and Immunologic Aging

Chronic HIV infection is associated with elevated soluble markers of immune activation and inflammation, as well as cellular phenotypes indicative of T cell and myeloid cell activation, even after HIV RNA levels are suppressed with ART.⁶⁶ The levels of inflammation and immune activation also increase with age and age-associated diseases.⁶⁷ In this context, the combined effects of HIV and advancing age on chronic inflammation have implications for risk from a broad spectrum of diseases.^{68,69}

A contributing factor to the early development of aging-related diseases among people with HIV has been posited to be the effect of HIV on immunologic aging. In addition to higher levels of immune activation, aging is associated with a higher frequency of cellular senescence as well as impaired

immune responses and surveillance.⁷⁰ HIV infection is associated with higher frequencies of T cell phenotypes characteristic of immunosenescence, as well as phenotypes reflective of highly differentiated or activated states.⁷¹⁻⁷⁴ Higher frequencies of senescent T cells are then associated with a greater risk for such comorbidities as CVD. In addition, the degree of CD4 T cell recovery after ART initiation is diminished among people ≥ 50 years old versus younger people with HIV.^{75,76} This impaired T cell recovery with advancing age may then have functional consequences, as people with HIV demonstrate reduced immune responses to vaccination, as well as impaired control of copathogens such as cytomegalovirus (CMV).⁷⁷⁻⁷⁹ The persistence of impaired or ineffective immune response toward copathogens then contributes to ongoing chronic immune activation (e.g., CMV-specific responses),⁷⁸ as well as the increased risk of both AIDS and non-AIDS-defining cancers that are driven by pro-oncogenic viruses (e.g., human papillomavirus [HPV]- and Epstein-Barr virus [EBV]-related cancers).⁸⁰

Immunologic aging in the context of HIV also entails chromosomal and genetic changes. Whether a consequence of untreated HIV viremia, direct toxicity from ARV drugs, or both, people with HIV have been shown to have shorter leukocyte chromosome telomere length, itself a marker of poor health and mortality.⁸¹⁻⁸³ DNA methylation is an epigenetic alteration that occurs at a predictable rate with advancing age, predicts lifespan, and increases in the setting of chronic HIV disease.⁸⁴⁻⁸⁷ Finally, a genetic mechanism that may more directly link immunologic aging to comorbid disease risk is clonal hematopoiesis, which entails an age-related expansion of somatic gene mutations among leukocytes (also referred to as age-related clonal hematopoiesis or ARCH). Clonal hematopoiesis is more common among people with HIV, and the degree of difference with people without HIV may be more pronounced with advancing age^{88,89} and in people with low nadir CD4 counts.^{90,91} Several of the leukocyte gene mutations characteristic of clonal hematopoiesis lead to dysregulation of inflammatory pathways, which is then associated with higher levels of inflammatory markers and ASCVD.⁹²⁻⁹⁴ In summary, clonal hematopoiesis is an example of HIV disease influencing age-related genetic somatic mutations that may contribute to ongoing systemic inflammation and associated disease risk among older people with HIV.

Disease-modifying pharmacologic interventions targeting immunologic aging are an area of active study within geroscience, but none are currently recommended specifically for older people with HIV with the exception of statin therapy as primary prevention of ASCVD risk (see Atherosclerotic Cardiovascular Disease in the Older Person With HIV below and [Statin Therapy in People With HIV](#)).

Morbidity, Mortality, and Non-AIDS Complications Among Older People With HIV

The clinical implications of greater immunologic aging and persistent inflammation can manifest in any organ system, diminishing the health span of older people with HIV. The frailty phenotype—defined clinically as a decrease in muscle mass, weight, physical strength, energy, and physical activity—represents an important example of an age-related syndrome that occurs with greater frequency and at earlier ages among people with HIV and is associated with a broad spectrum of adverse effects and disease risk. Specifically, frailty among people with HIV has been associated with incident CVD, diabetes mellitus, recurrent falls and fractures, lower quality of life, cognitive impairment, hospitalization, and mortality.⁹⁵⁻¹⁰⁴ The frailty phenotype remains incompletely characterized among older people with HIV and identifying effective treatment strategies is an important priority for clinical management. However, pharmacologic and non-pharmacologic interventions addressing frailty among people with HIV have been proposed and have the potential to significantly impact their quality of life and survival.^{105,106}

Although the life expectancy of people with HIV is approaching that of people without HIV, the incidence and prevalence of age-associated comorbidities in people with HIV are increasing.^{107,108}

The age-related comorbidity burden is significantly higher in people with HIV compared to the age-matched general population.⁸ These predictions differ by comorbidity and by demographic factors—including race, ethnicity, age, sex, and HIV transmission mode—displaying substantial disparities for some populations.

Data from insured adults suggest that people with HIV spend approximately 15 years more of their lifespan with one or more major comorbidities when compared to people without HIV; this difference is reduced but persists (at approximately 10 years) for people who initiate ART at a CD4 count of >500 cells/mm³.¹⁰⁹ A simulation modeling study projected that by 2030, 45% of people with HIV on ART would be living with two or more physical comorbidities and 64% with at least one mental health comorbidity.¹¹⁰

Non-AIDS–Defining Complications and Comorbidities in the Older Person With HIV

Non-AIDS comorbid conditions constitute an increasing proportion of morbidity and mortality among people with HIV.^{3,111} Heart disease and cancer are the leading causes of death in older people with HIV in the United States. Similarly, other non-AIDS events, such as cognitive impairment and liver disease, have also emerged as major causes of morbidity and mortality in people with HIV receiving effective ART. The prevalence of multimorbidity among people with HIV has increased in the past decade,¹¹² with hypertension and hypercholesterolemia being the most common comorbidities.¹¹³

Three groups of factors likely contribute to this burden of noncommunicable diseases among people with HIV: (1) an over-representation of traditional risk factors in this population, especially higher rates of smoking, diabetes, and dyslipidemia; (2) HIV-associated chronic inflammation and immune activation, which are not completely abated by suppressive ART; and (3) possible cardiometabolic complications of some ARV drugs.¹¹⁴⁻¹²²

As the life expectancy of people with HIV increases with ART, more women with HIV are experiencing menopause. Although menopause may occur earlier in women with HIV than in women in the general population,¹²³ early menopause may also be a consequence of smoking, depression, substance use disorder, and other psychosocial factors that are disproportionately present in women with HIV.¹²⁴ Also, HIV might compound menopause-related metabolic changes. For instance, HIV infection and menopausal stage were independent predictors of lower BMD and had an additive effect on lumbar spine and total hip BMD.¹²⁵

HIV-specific primary care guidelines have been developed and are available for clinicians caring for people with HIV, especially for older people with HIV.^{126,127} Additional specific guidelines have also been developed for the evaluation and management of the following specific comorbidities in people with HIV: bone health,⁴⁴ kidney disease,¹²⁸ and secondary prevention for ASCVD.¹²⁹ In addition, the following guidelines recently developed for the general population can be applied to older people with HIV: management of hypertension¹³⁰, hyperglycemia¹³¹, and hyperlipidemia.¹³² However, it is important to note that the recommendations in many of these guidelines have not all been validated in the context of HIV disease. For instance, cardiovascular risk prediction functions developed for the general population likely underestimate the risk in people with HIV.^{129,133} Findings of the Randomized Trial to Prevent Vascular Events in HIV (REPRIEVE) study have provided significant insight into the potential role of statins in the prevention of major adverse cardiovascular events in people with HIV. See [Statin Therapy in People With HIV](#) for recommendations from the Panel on Antiretroviral Guidelines for Adults and Adolescents (the Panel) on statin therapy as primary prevention of ASCVD among people with HIV. Further analyses of the REPRIEVE data will likely yield additional insights into their benefits (and potential risks) in people with HIV, as well as differences in the predictive value of risk equation scores in demographic subsets of study participants.

Atherosclerotic Cardiovascular Disease in the Older Person With HIV

When compared to people without HIV, data from several studies have shown that people with HIV have about a twofold higher risk of developing ASCVD, and their age at incident ASCVD diagnosis is about a decade younger.^{108,109,134,135} ASCVD risk prediction tools used for the general population tend to underestimate risk among people with HIV.^{133,136} Data also suggest that the relative increase in ASCVD risk is greater among women with HIV than among age-matched men with HIV when compared to people without HIV.^{137,138} Factors influencing ASCVD risk among people with HIV include both higher prevalence of traditional cardiometabolic risk factors and ongoing systemic inflammation associated with HIV,^{114,137} even in individuals with viral suppression while on ART. In addition, structural barriers and health disparities in screening and treatment of ASCVD risk factors likely also contribute to excess ASCVD risk among people with HIV in the United States.^{139,140}

The [2018 American Heart Association \(AHA\)/American College of Cardiology \(ACC\)/Multisociety Guidelines](#) provide recommendations on the use of statins as primary prevention based on risk stratification using estimates of 10-year ASCVD risk from the pooled cohort equations.^{141,142} In the general population, statin therapy is recommended for all people at high 10-year ASCVD risk (defined as $\geq 20\%$). Recent findings from the REPRIEVE trial have provided evidence for the use of statin therapy as primary prevention for people with HIV at low-to-intermediate ASCVD risk (i.e., $< 20\%$). REPRIEVE, a large randomized controlled trial among people with HIV aged 40 to 75 years who were receiving ART, showed that when compared to placebo, pitavastatin 4 mg daily was associated with a 35% reduction in major adverse cardiovascular events over a median follow-up duration of 5 years.¹⁴³ Results from REPRIEVE informed the Panel's recommendation on the use of at least moderate-intensity statin therapy as primary prevention among people with HIV of age 40 to 75 years, with a stronger recommendation for those with ASCVD risk score $\geq 5\%$ who will have a higher absolute benefit from statin therapy. See [Statin Therapy in People With HIV](#) for the Panel's full recommendations on statin therapy.

Neurocognitive Impairment in the Older Person With HIV

Cognitive impairment in people with HIV can manifest as difficulty with memory, attention, speed of information processing, and executive and motor functions. Studies that use neuropsychological testing to define categories of impaired cognition—termed HIV-associated neurocognitive disorder (HAND)—find that up to 30% of people with HIV on virally suppressive ART meet these research-based criteria¹⁴⁴, although these categories have not been validated for use in clinical practice. Research studies also demonstrate a steeper decline in performance on neuropsychological tests with advancing age in individuals with HIV compared to those without HIV over the same age range.¹⁴⁵ Although viral persistence and chronic brain inflammation can contribute, cognitive impairment is multifactorial in people with HIV. It is not solely HIV-associated, but also influenced by factors related to comorbidities, including polypharmacy, mood disorders, vascular disease, social isolation, stress, and independent neurodegenerative conditions.¹⁴⁶⁻¹⁵¹ Hormonal shifts that occur with aging may contribute to neurocognitive impairment, and these changes may manifest as unique differences in clinical manifestations by sex.¹⁵² Finally, the risk of neurodegenerative disease rises with increasing age independent of HIV, and diagnosis of Alzheimer's disease or other forms of progressive dementia in older people with HIV is now an important clinical concern.¹⁵³

Cognitive impairment carries potentially detrimental clinical consequences for aging people with HIV. In a prospective observational study, cognitive impairment was predictive of a lower likelihood of retention in care among older people.¹⁵⁴ Cognitive impairment is also associated with reduced adherence to therapy¹⁵⁵ and poorer health outcomes including increased mortality.¹⁵⁶ Given the importance of cognitive health, screening for neurocognitive impairment is important, though optimal primary care-based screening methods remain unclear. Initial screening with questions about

symptoms of memory loss, slowness, and poor attention can help to guide the need for referral to a specialist (e.g., neurologist, neuropsychologist, or geriatrician).^{157,158} Though such questions can be useful in identifying cognitive impairment, a positive response to these questions is not necessarily specific to impairment related to HIV.¹⁵⁹ Of note, people with substantial impairment may not have enough insight into their condition to answer the questions, and input from a partner or family member corroborating the impaired person's cognitive history may enhance the validity of the assessment. No brief cognitive screening test has been clearly shown to be sensitive or specific for cognitive impairment in HIV; the frequently used Mini-Mental State Exam does not typically capture executive function impairment, which is the main manifestation of subtle cognitive impairment in people with HIV.¹⁶⁰ The Montreal Cognitive Assessment may be more sensitive but is not specific. A lower threshold than the original cut-off score of 26 on the Montreal Cognitive Assessment is likely more optimal for HAND screening, as it lowers false positive rates and improves diagnostic accuracy.¹⁶¹ If a person has persistent concerns over time, has symptoms corroborated by an acquaintance, or has progressively worsening symptoms, referral to a neurologist for evaluation and management or to a neuropsychologist for formal neuropsychological testing may be warranted (BIII).¹⁶⁰

Beyond screening and referral to specialists in neurology, neuropsychology, or geriatrics, an approach to cognitive impairment in aging people with HIV should involve assessing and correcting reversible and treatable factors that contribute to cognitive symptoms, as well as directly addressing modifiable risk factors. Reversible and treatable factors that may contribute to cognitive symptoms include hypothyroidism, low vitamin B12 levels, syphilis, sleep apnea, and cerebrovascular disease.¹⁶² Hearing impairment, diabetes, obesity, hypertension, depression, smoking, alcohol or substance use, and limited physical and social activity have been identified as significant modifiable risk factors for dementia in people without HIV.¹⁶³ Given the prevalence of many of these risk factors in older people with HIV, careful attention to their prevention or amelioration is an important component of primary care and cognitive care in aging people with HIV. Finally, the [U.S. Preventive Service Task Force](#) now recommends screening older adults (>65 years) for depression.¹⁶⁴

Mental Health Concerns in the Older Person With HIV

Mental health-related problems are a growing concern in aging people with HIV, though little is known about their prevalence and consequences in this population specifically. In a study that compared a cohort of individuals aged >60 years with HIV to a historical control group of healthy older people, a heightened risk of mood disorders including anxiety and depression was noted among people with HIV.¹⁶⁵ Social isolation combined with depression is particularly common among older adults with HIV and, in addition to its direct effects on morbidity and mortality, it may contribute to poor medication adherence and retention in care.^{166,167} The risk of suicide remains greater in people with HIV than in the general population, though increasing age might not further heighten the risk.¹⁶⁸ In one analysis of a multicenter observational study in France, suicide emerged as the second-highest cause of death among virologically suppressed people with HIV on ART.¹⁶⁹ Screening for depression and management of mental health issues are critical aspects of HIV primary care; guidelines for people with HIV, as well as for aging individuals without HIV, recommend behavioral approaches including individual psychotherapy, cognitive behavioral therapy, group therapy, and often pharmacological treatment.¹⁶⁹ When resources are available, integrated care models with routine screening by health care support staff, review by primary providers, and referral to on-site mental health specialists are likely to be the most effective approaches in vulnerable aging populations.

Health Care Utilization and Cost Sharing

The substantially increased burden of age-related comorbidities among people aging with HIV, including CVD, CKD, neurocognitive disease, and fractures, can lead to increased healthcare

utilization and higher costs.¹⁷⁰ Out-of-pocket health care expenses (e.g., copayments, deductibles), loss of employment, and other financial-related factors can cause temporary interruptions in treatment, including ART, which should be avoided whenever possible (see [Cost Considerations and Antiretroviral Therapy](#) for more information). The increased life expectancy and higher prevalence of chronic complications in aging populations with HIV can place greater demands upon HIV services¹⁷¹ and require a focused approach to prioritize modifiable health-related problems. Facilitating continued access to health care, including access to insurance-covered care and medications, is essential to minimize treatment interruptions; clinical outcomes can improve with additional multidisciplinary services to manage concomitant chronic disorders and assist with overcoming structural barriers to care. See the Ryan White HIV/AIDS Program's [Available Care & Services](#) page for additional information.

End-of-Life Issues, Including Palliative Care and Interruption of ART

As with all aging people, it is important to discuss living wills, advance directives, and long-term care planning. Palliative care and hospice should be discussed with a person-centered approach that incorporates the person's family situation and cultural, religious, and personal values, among other factors.^{172,173}

Few data exist on the use of ART in people with debilitating chronic, severe, or non-AIDS terminal conditions.^{174,175} Discontinuation of ART usually results in rebound viremia and a decline in CD4 count; an acute retroviral syndrome has also been reported after abrupt ART discontinuation. For these reasons, most clinicians would continue ART if there are no significant adverse reactions to the ARV drugs. In cases where ART negatively affects quality of life, the decision to continue therapy should be made together with the affected person and/or family members after a discussion of the risks and benefits of continuing or stopping ART.

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Substance Use Disorders and HIV

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Key Considerations and Recommendations
• Substance use disorders (SUDs) are prevalent among people with HIV and contribute to poor health outcomes; therefore, screening for SUDs should be a routine part of clinical care (AII). • The most commonly used substances among people with HIV include the following (listed in alphabetical order): alcohol, benzodiazepines, cannabinoids, club drugs, opioids, stimulants (cocaine and methamphetamines), and tobacco. • Health care providers should be nonjudgmental when addressing substance use with people with HIV (AIII). • People with HIV and SUDs should be screened for additional mental health disorders (AII). • People with HIV and SUDs should be offered evidence-based pharmacotherapy (e.g., opioid agonist therapy, tobacco cessation treatment, alcohol use disorder treatment; see Table 16 below) as part of comprehensive HIV care in clinical settings (AI). • Ongoing substance use is not a contraindication to antiretroviral therapy (ART). People who use substances can achieve and maintain viral suppression with ART. • Substance use may increase the likelihood of HIV transmission risk behaviors, the potential for drug–drug interactions, and the risk or severity of substance-associated adverse events (e.g., increased hepatotoxicity or an increased risk of overdose). • Selection of antiretroviral (ARV) regimens for individuals who practice unhealthy substance and alcohol use should take into account potential adherence barriers, comorbidities that could impact care (e.g., advanced liver disease from alcohol or hepatitis viruses), potential drug–drug interactions, and possible adverse events associated with the medications (AII). • For people with SUDs, ARV regimens with once-daily formulations (ideally as a single-tablet regimen), high barriers to resistance, low hepatotoxicity, and low potential for drug–drug interactions are preferred (AIII).
Rating of Recommendations: A = Strong; B = Moderate; C = Weak
Rating of Evidence: I = Data from randomized controlled trials; II = Data from well-designed nonrandomized trials or observational cohort studies with long-term clinical outcomes; III = Expert opinion

Background on Substance Use Disorders Among People With HIV

Ending the HIV epidemic requires addressing substance use among people with HIV, which poses a barrier to optimal engagement in the HIV care continuum. Ongoing substance use may prevent an individual from being tested for HIV, initiating antiretroviral therapy (ART), or adhering to ART. Substance use may also increase the likelihood of risk-taking behaviors (e.g., sexual transmission risk behaviors, needle sharing, injection of substances), the potential for drug–drug interactions, and the risk or severity of substance-related adverse events (e.g., increased hepatotoxicity and increased risk of overdose). In the United States, the death toll for drug overdose (102,123 deaths as of February 2024)¹ far exceeds the death toll for HIV (4,941 deaths in 2022).² As the drug overdose epidemic continues to expand, health care providers need a basic understanding of how to screen for and treat substance use disorders (SUDs) in people with HIV in clinical settings.³